Compact Textbooks in Mathematics

Belkacem Said-Houari

Linear Algebra

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Linear Algebra

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To my daughter NOUR

Preface

Linear algebra is the study of the algebraic properties of linear transformations and matrices and it is an essential part of virtually all areas of mathematics. It is also a fundamental and an extremely powerful tool in every single discipline of sciences and engineering.

This is a self-contained textbook on linear algebra written in an easy way, so that it can be accessible to many readers. It begins in ▶ Chap. 1 with the simplest linear equation and generalizes many notions about this equation to systems of linear equations, and then introduces the main ideas using matrices and their properties. We believe that this is the right approach, since most students take the first course of linear algebra already knowing something about linear equations, lines, and systems of linear equations. Then follows a detailed chapter (▶ Chap. 2) on determinants and their properties where we also study the relationship between determinants and the inverses of matrices and the use of determinants in solving systems of linear equations. We introduce the main ideas with detailed proofs. We also investigate some particular determinants that are very useful in applications. In addition, we explain in a simple way where the ideas of determinants come from and how they fit together in the whole theory.

In ▶ Chap. 3, we introduce the Euclidean spaces using very simple geometric ideas and then discuss various important inequalities and identities. These ideas are present in the theory of general Hilbert spaces in a course of functional analysis, so it is much better for students to learn them and understand them clearly in Euclidean spaces.

The core of ▶ Chap. 4 is a detailed discussion of general vector spaces where rigorous proofs to all the main results in this book are given. This is followed by a chapter (▶ Chap. 5) on linear transformations and their properties.

In ▶ Chap. 6, we introduce notions concerning matrices through linear transformations, trying to bridge the gap between matrix theory and linear algebra.

▶ Chapters 7 and 8 are more advanced, where we introduce all the necessary ideas concerning eigenvalues and eigenvectors and the theory of symmetric and orthogonal matrices.

One of the aspects that should make this textbook useful for students is the presence of exercises at the end of each chapter. We did choose these exercises very carefully to illustrate the main ideas. Since some of them are taken (with some modifications) from recently published papers, it is possible that these exercises appear for the first time in a textbook. All the exercises are provided with detailed solutions and in each solution, we refer to the main theorems in the text when necessary, so students can see the main tools used in the solution. In addition all the main ideas in this book come

with illustrating examples. We did strive to choose solutions and proofs that are elegant and short. We also tried to make this textbook in about 400 pages by focusing on the main ideas, so that students will be able to easily and quickly understand things. In addition, we tried to maintain a balance between the theory of matrices and the one of vector spaces and linear transformations.

This book can be used as a textbook for a first course in linear algebra for undergraduate students in all disciplines. It can be also used as a booklet for graduate students, allowing to acquire some concepts, examples, and basic results. It is also suitable for those students who are looking for simple, easy, and clear textbook that summarizes the main ideas of linear algebra. Finally it is also intended for those students who are interested in rigorous proofs of the main theorems in linear algebra. We believe that if a good student uses this book, then she (he) can read and learn the basics of linear algebra on her (his) own.

We would like to thank Salim A. Messaoudi (from KFUPM) for valuable suggestions and corrections which improved the contents of some parts of this book. We also thank Sofiane Bouarroudj (from NYU Abu Dhabi) for the many discussions that we have had about the proofs of some theorems of linear algebra.

Abu Dhabi, United Arab Emirates — Belkacem Said-Houari
October 06, 2016

Contents

1 Matrices and Matrix Operations 1
Belkacem Said-Houari
1.1 **Systems of Linear Equations** 1
1.1.1 The Group of Matrices 11
1.1.2 Multiplication of Matrices 12
1.2 **Square Matrices** 19
1.2.1 The Ring of Square Matrices 19
1.2.2 The Identity Matrix 21
1.2.3 Inverse Matrices 22
1.2.4 Diagonal Matrices 30
1.2.5 Triangular Matrices 35
1.2.6 Trace 37
1.3 **Solving Linear Systems with Elementary Row Operations** 39
1.3.1 The Gauss–Jordan Elimination Method 41
1.4 **The Matrix Transpose and Symmetric Matrices** 49
1.4.1 Transpose of a Matrix 50
1.4.2 Symmetric Matrices 52
1.5 **Exercises** 53

2 Determinants 69
Belkacem Said-Houari
2.1 **Introduction** 69
2.2 **Determinants by Cofactor Expansion** 70
2.3 **Properties of the Determinants** 76
2.4 **Evaluating Determinants by Row Reduction** 79
2.4.1 Determinant Test for Invertibility 89
2.5 **The Adjoint of a Square Matrix** 91
2.5.1 Cramer's Rule 99
2.6 **Exercises** 104

3 Euclidean Vector Spaces 121
Belkacem Said-Houari
3.1 **Introduction** 121
3.2 **Vector Addition and Multiplication by a Scalar** 121
3.2.1 Vector Addition 121
3.2.2 Multiplication of a Vector by a Scalar 123
3.2.3 Vectors in Coordinate Systems 123
3.2.4 Linear Combination 127
3.3 **Norm, Dot Product, and Distance in $\mathbb{R}^n$** 128
3.3.1 The Norm of a Vector in $\mathbb{R}^n$ 128
3.3.2 Distance in $\mathbb{R}^n$ 131
3.3.3 The Dot Product of Two Vectors in $\mathbb{R}^n$ 132

3.4 **Orthogonality in $\mathbb{R}^n$** 141
3.4.1 Orthogonal Projections in $\mathbb{R}^n$ 142
3.5 **Exercises** 145

4 General Vector Spaces 159
Belkacem Said-Houari
4.1 **Definition and Examples** 159
4.2 **Properties of Vector Spaces** 161
4.3 **Subspaces** 163
4.3.1 Direct Sum Decomposition 170
4.4 **Linear Independence** 172
4.5 **Bases of Vector Spaces** 175
4.6 **Dimension of a Vector Space** 178
4.6.1 Dimension of a Subspace 182
4.6.2 Construction of a Basis 182
4.7 **Exercises** 183

5 Linear Transformations 199
Belkacem Said-Houari
5.1 **Definition and Examples** 199
5.2 **Fundamental Properties of Linear Transformations** 201
5.2.1 The Kernel and the Image of a Linear Transformation 204
5.3 **Isomorphism of Vector Spaces** 211
5.4 **Exercises** 215

6 Linear Transformations and Matrices 227
Belkacem Said-Houari
6.1 **Definition and Examples** 227
6.2 **Change of Basis and Similarity** 234
6.3 **Rank of Matrices** 241
6.3.1 Some Properties of the Rank of a Matrix 243
6.4 **Methods for Finding the Rank of a Matrix** 246
6.4.1 The Method of Elementary Row and Column Operations 246
6.4.2 The Method of Minors for Finding the Rank of a Matrix 251
6.5 **Exercises** 255

7 Eigenvalues and Eigenvectors 269
Belkacem Said-Houari
7.1 **Definitions** 269
7.2 **Properties of Eigenvalues and Eigenvectors** 272
7.3 **Eigenvalues and Eigenvectors of a Matrix** 277
7.4 **Diagonalization** 290
7.4.1 Spectrum of Symmetric Matrices 296

7.5 **Triangularization and the Jordan Canonical Form** 298
7.5.1 Triangularization of an Endomorphism 298
7.5.2 The Jordan Canonical Form 301
7.6 **Exercises** 307

8 Orthogonal Matrices and Quadratic Forms 323
Belkacem Said-Houari
8.1 **Orthogonal Matrices** 323
8.1.1 The Gram–Schmidt Process 328
8.1.2 The QR Factorization 330
8.1.3 The LU Factorization 336
8.2 **Positive Definite Matrices** 341
8.2.1 The Cholesky Decomposition 348
8.3 **Quadratic Forms** 352
8.3.1 Congruence and Sylvester's Law of Inertia 354
8.4 **Exercises** 361

Servicepart 377
References 379
Index 381

List of Figures

Fig. 1.1 The case where the system has a unique solution (x_0, y_0): the solution is the intersection point of two lines 2
Fig. 1.2 The case where the system has no solution 2
Fig. 1.3 The two lines coincide and there are infinitely many points of intersection 3
Fig. 1.4 The addition of two vectors in the xy-plane 8
Fig. 1.5 An easy way to perform the multiplication of two matrices 15

Fig. 2.1 To evaluate the 3×3 determinant, we take the products along the main diagonal and the lines parallel to it with a $(+)$ sign, and the products along the second diagonal and the lines parallel to it wit a $(-)$ sing 75

Fig. 3.1 The vector $v = \vec{AB}$ 122
Fig. 3.2 The vectors v and w are equal since they have the same length and the same direction 122
Fig. 3.3 The sum of two vectors $v + w$ 122
Fig. 3.4 The sum of two vectors $v + w$ 122
Fig. 3.5 The sum of vectors is associative 123
Fig. 3.6 Multiplication of a vector by a scalar 123
Fig. 3.7 Components of a vector if the initial point is the origin 123
Fig. 3.8 Components of a vector if the initial point is not the origin 124
Fig. 3.9 The sum of two vectors $v + w$ in a coordinate system 125
Fig. 3.10 The vector $-v$ 125
Fig. 3.11 The vector v is a linear combination of e_1 and e_2 128
Fig. 3.12 The dot product of u and v 132
Fig. 3.13 The cosine law 133
Fig. 3.14 The sum of two vectors $v + w$ in a coordinate system 133
Fig. 3.15 The parallelogram identity 140
Fig. 3.16 The projection of u on v 143
Fig. 3.17 Pythagoras' theorem in a right triangle 144
Fig. 3.18 Apollonius' identity 150

Matrices and Matrix Operations

Belkacem Said-Houari

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1.1 Systems of Linear Equations

In order to introduce the main ideas of Linear Algebra, we first study matrix algebra. So, the first thing we begin with is the following simple linear equation:

$$ax = b, \tag{1.1}$$

where a and b are two real numbers. We know from elementary algebra that if $a \neq 0$, then Eq. (1.1) has in $\mathbb{R}$ the unique solution

$$x = \frac{b}{a} = a^{-1}b. \tag{1.2}$$

Next, suppose that we want to solve the following system of two linear equations in $\mathbb{R}^2$:

$$\begin{cases} ax + by = p, \\ cx + dy = q, \end{cases} \tag{1.3}$$

where a, b, c, d, p and q are real numbers. There are at least two ways of looking for the solutions of (1.3): geometrically and algebraically.

Geometrically
It is well known that both equations in (1.3) are equations of lines in the xy-plane. Thus, the solutions of the system in the xy-plane are the points of intersection of the two lines. Therefore, we may distinguish the following three cases:

Case 1. The two lines intersect in exactly one point (x_0, y_0) as in Fig. 1.1. This is the case where the slopes of the two lines are not the same. That is,

$$-\frac{a}{b} \neq -\frac{c}{d}, \qquad \text{or equivalently,} \qquad \frac{a}{b} \neq \frac{c}{d}.$$

In this case the system (1.3) has a unique solution (x_0, y_0).

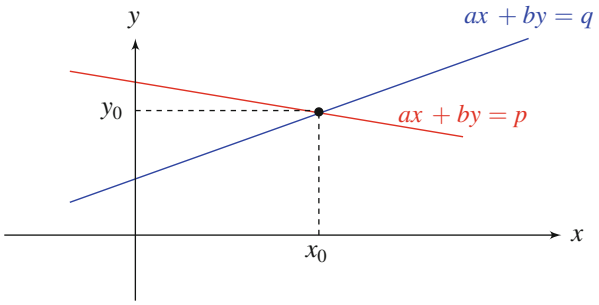


Fig. 1.1 The case where the system has a unique solution (x_0, y_0): the solution is the intersection point of two lines

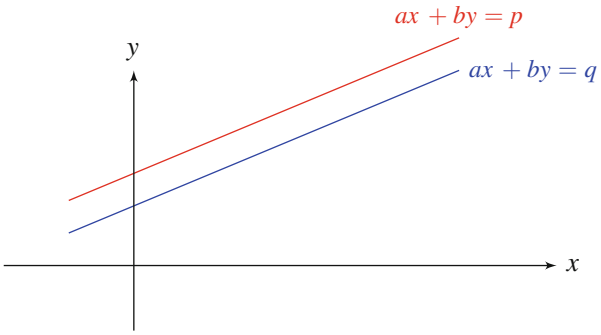


Fig. 1.2 The case where the system has no solution

Case 2. The two lines may be parallel and distinct, which occurs when

$$\frac{a}{b} = \frac{c}{d} \quad \text{and} \quad \frac{p}{b} \neq \frac{q}{d}.$$

In this case there is no intersection and the system (1.3) has no solution. See Fig. 1.2.

Case 3. The two lines may coincide, which occurs when

$$\frac{a}{b} = \frac{c}{d} \quad \text{and} \quad \frac{p}{b} = \frac{q}{d}.$$

In this case there are infinitely many points of intersection and consequently, there are infinitely many solutions to (1.3). See Fig. 1.3.

Algebraically

Algebraically, we may solve system (1.3) by at least two methods: The method of *substitution* and the method of *elimination*. For the substitution method, we express

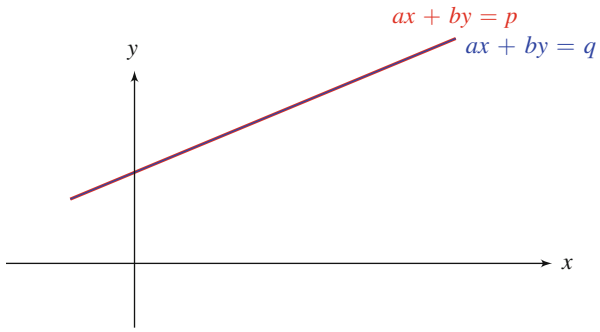


Fig. 1.3 The two lines coincide and there are infinitely many points of intersection

one of the two variables from the first equation in (1.3) and substitute it in the second equation as follows:

$$y = -\frac{a}{b}x + \frac{p}{b}, \tag{1.4}$$

provided that $b \neq 0$ and then plugging expression (1.4) into the second equation of (1.3), we find

$$(ad - bc)x = pd - bq. \tag{1.5}$$

It is clear that if

$$ad - bc \neq 0,$$

then the system (1.3) has a uniques solution. On the other hand, if $ad - bc = 0$ and $pd - bq \neq 0$, then Eq. (1.5) shows that system (1.3) has no solution. Finally, if $ad - bc = 0$ and $pd - bq = 0$, then system (1.3) has infinitely many solutions.

Thus, as we have said earlier, Eq. (1.1) has a unique solution if and only if

$$a \neq 0.$$

On the other hand, system (1.3) has a unique solution if and only if

$$ad - bc \neq 0.$$

So, the question now is *what if we have a* 3×3 *system, or in general an* $n \times n$ *system? Can we always discuss existence of solutions in terms of similar relations between the coefficients? And if so, how can we find such relations in general?* As we will see later, the answer to the above question leads us to the study of Matrix Algebra.

1

Definition 1.1.1 (Linear Equation)

A linear equation in n variables $x_1, x_2, \ldots, x_n$ is an equation of the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b, \tag{1.6}$$

where $a_1, a_2, \ldots, a_n$ and b are real numbers.

Remark 1.1.1 Equation (1.6) can be written as

$$f(X) = a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

where f is a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}$ and X is the vector

$$X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}. \tag{1.7}$$

Also, in a linear equation as (1.6) all variables occur only to the first power and the equation does not involve any products of variables and they do not appear as arguments of trigonometric, logarithmic, exponential, or other functions.

Example 1.1

The following equations are not linear:

$$x_1x_2 = 1, \quad x_1x_2 + x_2 + x_3 = 0, \quad x_1 + 2\cos x + 3x_3 = -1, \quad \sqrt{x_1} - x_2 = 0.$$

■

Definition 1.1.2

We define a system of linear equations as a set of m equations with n unknowns

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2, \\ \vdots \qquad\quad \vdots \qquad\quad \vdots \qquad\quad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m, \end{cases} \tag{1.8}$$

where a_{ij}, $1 \le i \le m$, $1 \le j \le n$ are real numbers.

Remark 1.1.2 As we have seen for the system of two equations (1.3), we will show in the sequel that system (1.8) has either

- no solution, or
- exactly one solution, or
- infinitely many solutions.

As in Remark 1.1.1, we may write the system (1.8) using a linear transform from $\mathbb{R}^n$ to $\mathbb{R}^m$

$$f(X) = \begin{pmatrix} f_1(X) \\ \vdots \\ f_m(X) \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix},$$

where X is the vector in (1.7) and

$$f_i(X) = a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n = b_i, \qquad 1 \le i \le m. \tag{1.9}$$

Now, going back to the system (1.8), let us assume that $n = 3$. Then each equation is the equation of a plane in three-dimensional space. So, the solution of the system is represented by a point in the intersection of m planes in the xyz-space. It is quite hard to find the intersections of those planes. In general, the geometric method becomes hard to apply if $n \ge 3$. So, we rely on the algebraic method to solve such system for $n \ge 3$. The core problem of linear algebra is how to solve the system (1.8).

Definition 1.1.3 (Homogeneous System)
If all the right-hand sides in (1.8) are zero, that is $b_i = 0$, $1 \le i \le m$, then system (1.8) is called *homogeneous*.

We can easily see that every homogeneous system has the zero vector $X = \mathbf{0}$ as a solution.

Now, we introduce the following definition which will help us to rewrite system (1.8) in a more convenient form.

Definition 1.1.4 (Dot Product)
Let

$$X = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \quad \text{and} \quad Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix}$$

(Continued)

Definition 1.1.4 (continued)

be two vectors in $\mathbb{R}^n$. Then the *dot product* or the *inner product* of X and Y is the real number $X \cdot Y$ defined as

$$X \cdot Y = Y \cdot X = x_1y_1 + x_2y_2 + \cdots + x_ny_n.$$

Now, using Definition 1.1.4, we may rewrite Eq. (1.9) as

$$V_i \cdot X = b_i, \qquad 1 \leq i \leq m,$$

where V_i is the vector

$$V_i = \begin{pmatrix} a_{i1} \\ \vdots \\ a_{in} \end{pmatrix}, \qquad 1 \leq i \leq m. \tag{1.10}$$

We may also write the dot product as a *row* vector times a *column* vector (with no dot):

$$(a_{i1}, \ldots, a_{in}) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n.$$

Using this notation, we recast system (1.8) as

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \ldots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \ldots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \ldots & a_{mn} \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} V_1 \cdot X \\ V_2 \cdot X \\ \vdots \\ V_m \cdot X \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}, \tag{1.11}$$

or equivalently as

$$AX = b, \tag{1.12}$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \ldots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \ldots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \ldots & a_{mn} \end{bmatrix}, \quad X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \text{and} \quad b = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}. \tag{1.13}$$

Definition 1.1.5

In the above formulas, the rectangular array of numbers A is called a *matrix*. The numbers a_{ij}, $1 \leq i \leq m$, $1 \leq j \leq n$, are called the *entries* or *coefficients* of the matrix A.[a]

[a] See ▶ Chap. 6 for the definition of matrices through linear transformations.

The matrix A consists of m rows (horizontal lines) and n columns (vertical lines).

Notation A matrix A is sometimes denoted by $A = (a_{ij})$ with $1 \leq i \leq m$ and $1 \leq j \leq n$. The entry a_{ij} lies in the ith row and the jth column.

Now, if we want to solve system (1.8), then it is natural to consider the system in its *matrix form* (1.12), since it looks similar to Eq. (1.1). Therefore, our first intuition is to write the solution as in (1.2), that is

$$X = A^{-1}b. \tag{1.14}$$

In doing so, many questions arise naturally. For instance:

- How can we define the inverse of a matrix?
- We know that the inverse of a real number a exists if and only if $a \neq 0$; when does the inverse of a matrix exists? And if it does exist, how can we find it?
- How do we multiply a matrix by a vector?
- Can we perform the usual algebraic operations (addition, multiplication, subtraction,...) on matrices?

The answers of the above questions are the building blocks of matrix algebra in particular, and linear algebra in general.

One of the interesting cases is when $m = n$. In this case, we say that A is a *square* matrix and we have following definition.

Definition 1.1.6 (Square Matrix)

A matrix A with m rows and n columns is called an $m \times n$ matrix. If $m = n$, then the matrix A is called a *square* matrix. The collection of the entries a_{ij} with $i = j$ is called the *main diagonal*.

Example 1.2

1. The matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & \sqrt{2} & 3 \end{bmatrix}$$

is a 2×3 matrix (two rows and three columns).

2. The matrix

$$A = \begin{bmatrix} 3 & 1 & 0 \\ -2 & 0 & 2 \\ 1 & \pi & 9 \end{bmatrix}$$

is a square matrix and the entries of the main diagonal are 3, 0, and 9.

■

In order to define the addition of matrices, let us first consider two vectors in $\mathbb{R}^2$,

$$X_1 = \begin{pmatrix} x_1 \\ y_1 \end{pmatrix} \quad \text{and} \quad X_2 = \begin{pmatrix} x_2 \\ y_2 \end{pmatrix}.$$

Each vector can be seen as 2×1 matrix. In order to define $X_1 + X_2$, we first need to think geometrically and draw both vectors in the xy-plane.

It is clear from Fig. 1.4 that the vector $X_1 + X_1$ is

$$X_1 + X_2 = \begin{pmatrix} x_1 + x_2 \\ y_1 + y_2 \end{pmatrix}. \tag{1.15}$$

Guided by the case for the 2×1 matrices, we can go ahead and define analogously the addition of two $m \times n$ matrices.

Definition 1.1.7 (Addition of Matrices)

Let A and B be two $m \times n$ matrices (that is A and B have the same size).[a] Then, the *sum* of A and B is the matrix $A + B$ obtained by adding the entries of A to the corresponding entries of B. More precisely, if $A = (a_{ij})$ and $B = (b_{ij})$, with

(Continued)

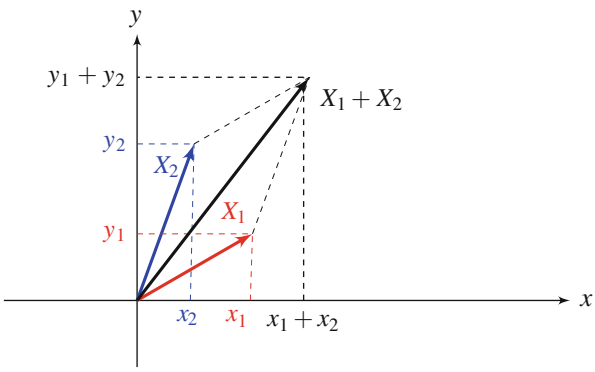


Fig. 1.4 The addition of two vectors in the xy-plane

Definition 1.1.7 (continued)

$1 \le i \le m$ and $1 \le j \le n$, then

$$A + B = (a_{ij} + b_{ij}), \qquad 1 \le i \le m, \quad 1 \le j \le n.$$

[a] The size of a matrix is described in terms of the rows and columns it contains.

It is clear from above that the addition of matrices is commutative, that is,

$$A + B = B + A.$$

Example 1.3

Consider the matrices

$$A = \begin{bmatrix} 1 & 0 & -1 \\ -2 & 0 & 3 \end{bmatrix} \quad \text{and} \quad \mathrm{B} = \begin{bmatrix} 0 & 0 & 1 \\ 3 & 1 & 0 \end{bmatrix}.$$

Then $A + B$ is the matrix given by

$$A + B = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 3 \end{bmatrix}.$$

■

Using the same intuition we multiply a vector by a scalar, then this means geometrically that we multiply each of its components by the same scalar. In a similar manner we define the multiplication of a matrix by a scalar.

Definition 1.1.8 (Multiplication by Scalar)

Let A be an $m \times n$ matrix and λ be a scalar. Then the product λA is the matrix obtained by multiplying each entry of the matrix A by λ. Thus,

$$\lambda A = \lambda(a_{ij}) = (\lambda a_{ij}), \qquad 1 \le i \le m, \quad 1 \le j \le n.$$

Example 1.4

Take

$$A = \begin{bmatrix} 2 & -1 \\ -3 & 0 \end{bmatrix} \quad \text{and} \quad \lambda = 2.$$

Then

$$2A = \begin{bmatrix} 4 & -2 \\ -6 & 0 \end{bmatrix}.$$

■

Notation We denote the set of scalars by $\mathbb{K}$ ($\mathbb{K}$ will ordinarily be $\mathbb{R}$ or $\mathbb{C}$) and the set of $m \times n$ matrices by $\mathscr{M}_{m\times n}(\mathbb{K})$. If $m = n$, then we write $\mathscr{M}_{m\times n}(\mathbb{K})$ simply as $\mathscr{M}_n(\mathbb{K})$.

Remark 1.1.3 It is clear that for A and B in $\mathscr{M}_{m\times n}(\mathbb{K})$, $A + B$ is in $\mathscr{M}_{m\times n}(\mathbb{K})$. Thus, the addition $(+)$ (binary operation)[a] in the set $\mathscr{M}_{m\times n}(\mathbb{K})$ satisfies the *closure* property. Also, we have seen that for any λ in $\mathbb{K}$ and for any A in $\mathscr{M}_{m\times n}(\mathbb{K})$, λA is in $\mathscr{M}_{m\times n}(\mathbb{K})$.

[a] This binary operation is defined from $\mathscr{M}_{m\times n}(\mathbb{K}) \times \mathscr{M}_{m\times n}(\mathbb{K}) \to \mathscr{M}_{m\times n}(\mathbb{K})$ and takes (A, B) from $\mathscr{M}_{m\times n}(\mathbb{K}) \times \mathscr{M}_{m\times n}(\mathbb{K})$ to $A + B$ in $\mathscr{M}_{m\times n}(\mathbb{K})$.

Definition 1.1.9 (Zero Matrix)

We define the zero matrix in $\mathscr{M}_{m\times n}(\mathbb{K})$, and denote it by $\mathbf{0}$ or $\mathbf{0}_{m\times n}$, as the matrix whose entries are all zero.

Example 1.5

The following matrices are zero matrices:

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \quad [0].$$

■

Theorem 1.1.4 (Properties of the Zero Matrix)

Let A be a matrix in $\mathscr{M}_{m\times n}(\mathbb{K})$ and $\mathbf{0}$ be the zero matrix in $\mathscr{M}_{m\times n}(\mathbb{K})$. Then

1. $\mathbf{0} + A = A + \mathbf{0} = A$,
2. $0A = \mathbf{0}$,

where 0 is the zero in $\mathbb{K}$.

The proof of this theorem is a straightforward consequence of the definition of the addition of matrices and of the multiplication of a matrix by a scalar.

1.1.1 The Group of Matrices

In this subsection we introduce an important algebraic structure on the set of matrices $\mathscr{M}_{m\times n}(\mathbb{K})$. We start with the following definition.

Definition 1.1.10

A *group* is a nonempty set G together with a binary operation

$$G \times G \to G$$
$$(a, b) \mapsto a * b,$$

satisfying the following conditions:

G1. *(associativity)* for all a, b, c in G,

$$(a * b) * c = a * (b * c);$$

G2. *(existence of a neutral (identity) element)* there exists an element e in G such that

$$a * e = e * a = a,$$

for all a in G;

G3. *(existence of the inverse)* for each a in G, there exists a' in G such that

$$a * a' = a' * a = e.$$

We usually denote a group by $(G, *)$.
If

$$a * b = b * a$$

for all a, b in G, then the group is called *commutative* or *Abelian*.[1]

Example 1.6

One of the simplest Abelian group is the group of integers $(\mathbb{Z}, +)$ with the addition $(+)$ as the binary operation. The structure of group in this set gives meaning to the negative integers as the inverses of positive integers with respect to the addition law $(+)$. ■

[1] Named after the Norwegian mathematician Niels Henrik Abel.

Theorem 1.1.5
The set $(\mathscr{M}_{m\times n}(\mathbb{K}), +)$ *is a commutative group.*

Proof
We saw earlier that addition in $\mathscr{M}_{m\times n}(\mathbb{K})$ is a binary operation from $\mathscr{M}_{m\times n}(\mathbb{K}) \times \mathscr{M}_{m\times n}(\mathbb{K}) \to \mathscr{M}_{m\times n}(\mathbb{K}) : (A, B) \mapsto A + B$. It is not hard to see from Definition 1.1.7 that if A, B, and C are matrices in $\mathscr{M}_{m\times n}(\mathbb{K})$, then

$$(A + B) + C = A + (B + C),$$

i.e., the binary operation $(+)$ is associative in $\mathscr{M}_{m\times n}(\mathbb{K})$.

Also, if $\mathbf{0}$ is the zero matrix in $\mathscr{M}_{m\times n}(\mathbb{K})$, then

$$A + \mathbf{0} = \mathbf{0} + A = A,$$

for each matrix A in $\mathscr{M}_{m\times n}(\mathbb{K})$. Thus, the zero matrix is the neutral element in $\mathscr{M}_{m\times n}(\mathbb{K})$ with respect to $(+)$.

Now, for each A in $\mathscr{M}_{m\times n}(\mathbb{K})$, $-A$ is also a matrix in $\mathscr{M}_{m\times n}(\mathbb{K})$ and satisfies

$$A + (-A) = (-A) + A = \mathbf{0}.$$

Therefore, the matrix $-A$ is the inverse of A in $\mathscr{M}_{m\times n}(\mathbb{K})$ with respect to the binary operation $(+)$.

Next, since

$$A + B = B + A,$$

for any two matrices A and B in $\mathscr{M}_{m\times n}(\mathbb{K})$, the group $\mathscr{M}_{m\times n}(\mathbb{K})$ is commutative. This completes the proof of Theorem 1.1.5. □

1.1.2 Multiplication of Matrices

The next step is to define the multiplication of matrices. We have seen in (1.11) the multiplication of a matrix by a vector. Now, consider a matrix B in $\mathscr{M}_{p\times r}(\mathbb{K})$. Then B can be written as

$$B = [B_1, B_2, \ldots, B_r],$$

where B_j, $1 \leq j \leq r$ are vectors in $\mathscr{M}_{p\times 1}(\mathbb{K})$, that is,

$$B_j = \begin{pmatrix} b_{1j} \\ \vdots \\ b_{pj} \end{pmatrix}.$$

The matrix A in (1.11) is in $\mathscr{M}_{m\times n}(\mathbb{K})$ and we write as before

$$A = \begin{bmatrix} V_1 \\ \vdots \\ V_m \end{bmatrix}$$

where V_i is the row vector defined as

$$V_i = (a_{i1}, a_{i2}, \ldots, a_{in}), \quad 1 \leq i \leq m.$$

Now, using the same idea as in (1.11), and assuming that $p = n$, we may find, for instance

$$\begin{bmatrix} V_1 \\ \vdots \\ V_m \end{bmatrix} B_1 = C_1,$$

where C_1 is a vector in $\mathscr{M}_{m\times 1}(\mathbb{K})$ whose first component is the dot product $V_1 \cdot B_1$ and generally whose ith component is the dot product $V_i \cdot B_1$. In order for these dot products to be defined, B_1 must have the same number of components as V_i, $1 \leq i \leq m$. That is $p = n$. If we do this for all the vectors B_j, $1 \leq j \leq r$, then we obtain the matrix

$$C = [C_1, C_2, \ldots, C_r],$$

where each C_k, $1 \leq k \leq r$, is a vector with m components. Therefore, the matrix C is in the set $\mathscr{M}_{m\times r}(\mathbb{K})$ and we have the following definition.

Definition 1.1.11 (Multiplication of Matrices)

Let A be a matrix in $\mathscr{M}_{m\times n}(\mathbb{K})$ and B be a matrix in $\mathscr{M}_{p\times r}(\mathbb{K})$. Then:

- If $p = n$, we define the product AB as

$$AB = (a_{ij}) \cdot (b_{jk}) = C = (c_{ik}), \qquad 1 \leq i \leq m, \quad 1 \leq j \leq n, \quad 1 \leq k \leq r,$$

(Continued)

Definition 1.1.11 (continued)

with

$$c_{ik} = a_{i1}b_{1k} + a_{i2}b_{2k} + \cdots + a_{in}b_{nk} = \sum_{j=1}^{n} a_{ij}b_{jk}.$$

- If $p \neq n$, then the product AB is undefined.

In Fig. 1.5 we show an easy way of how to multiply two matrices by positioning them as in the Fig. 1.5. So, to get the entry c_{22}, for instance, we multiply the entries of the second row of the matrix A with the entries of the second columns of the matrix B.

Example 1.7

Consider the two matrices

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix},$$

Then

$$AB = \begin{bmatrix} 8 & 5 \\ 20 & 13 \\ 2 & 1 \end{bmatrix}.$$

■

Example 1.8

Consider the square matrices

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

Find all values of a, b, c and d (if any) for which $AB = BA$. ■

Solution

We first compute

$$AB = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & a \\ 0 & c \end{bmatrix}. \tag{1.16}$$

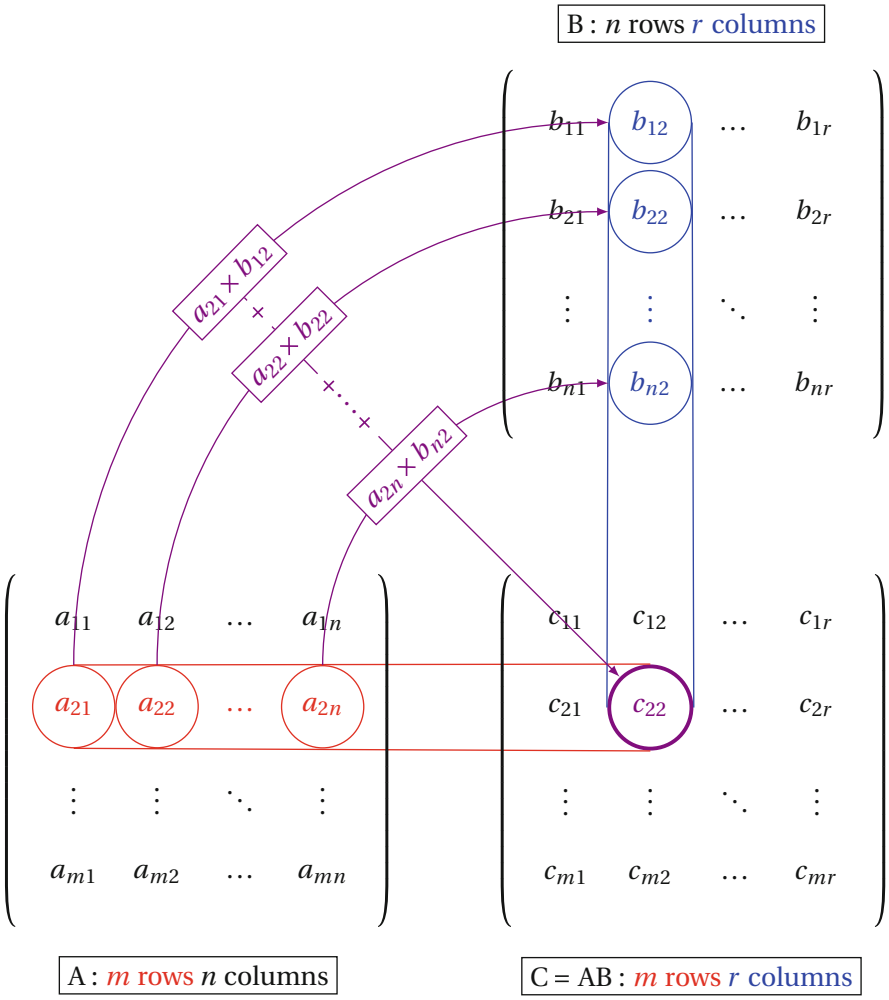


Fig. 1.5 An easy way to perform the multiplication of two matrices

On the other hand,

$$BA = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} c & d \\ 0 & 0 \end{bmatrix}. \tag{1.17}$$

From (1.16) and (1.17), we deduce that $AB = BA$ if and only if

$$\begin{bmatrix} 0 & a \\ 0 & c \end{bmatrix} = \begin{bmatrix} c & d \\ 0 & 0 \end{bmatrix},$$

that is $c = 0$ and $a = d$. Thus, the matrices A satisfying $AB = BA$ are the matrices of the form

$$A = \begin{bmatrix} a & b \\ 0 & a \end{bmatrix},$$

where a and b are any real numbers. Some examples of such matrices are

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}, \quad \begin{bmatrix} -3 & 7 \\ 0 & -3 \end{bmatrix}$$

there are infinitely many of them. ◄

Remark 1.1.6 It is clear that even if we can do the multiplications AB and BA, as in the case of square matrices, for instance, then in general

$$AB \neq BA.$$

To illustrate this, take

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Then

$$AB = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad \text{but} \quad BA = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Theorem 1.1.7 (Multiplication is Distributive over the Addition)
Let A be a matrix in $\mathscr{M}_{m\times n}(\mathbb{K})$ and B and C be matrices in $\mathscr{M}_{n\times r}(\mathbb{K})$. Then

$$A(B + C) = AB + AC. \tag{1.18}$$

The proof is straightforward. We can just compute the two sides in (1.18) and find that they are equal. We omit the details.

Theorem 1.1.8 (Multiplication is Associative)
Let A, B and C, be matrices in $\mathscr{M}_{m\times n}(\mathbb{K})$, $\mathscr{M}_{n\times p}(\mathbb{K})$ and $\mathscr{M}_{p\times r}(\mathbb{K})$, respectively. Then we have

$$A(BC) = (AB)C. \tag{1.19}$$

Proof
Let

$$A = (a_{ij}), \quad B = (b_{jk}), \quad C = (c_{kl}), \quad 1 \le i \le m, \quad 1 \le j \le n, \quad 1 \le k \le p, \quad 1 \le l \le r.$$

Thus, we have

$$AB = (\alpha_{ik}), \qquad \text{with} \quad \alpha_{ik} = \sum_{j=1}^{n} a_{ij} b_{jk}$$

and

$$BC = (\beta_{jl}), \qquad \text{with} \quad \beta_{jl} = \sum_{k=1}^{p} b_{jk} c_{kl}.$$

Therefore,

$$A(BC) = (\gamma_{il}),$$

with

$$\begin{aligned}
\gamma_{il} &= \sum_{j=1}^{n} a_{ij} \beta_{jl} = \sum_{j=1}^{n} a_{ij} \sum_{k=1}^{p} b_{jk} c_{kl} \\
&= \sum_{j=1}^{n} \sum_{k=1}^{p} a_{ij} b_{jk} c_{kl} = \sum_{k=1}^{p} \sum_{j=1}^{n} a_{ij} b_{jk} c_{kl} \\
&= \sum_{k=1}^{p} \Big(\sum_{j=1}^{n} a_{ij} b_{jk} \Big) c_{kl} = \sum_{k=1}^{p} \alpha_{ik} c_{kl} \\
&= (AB)C.
\end{aligned}$$

This finishes the proof of Theorem 1.1.8. □

Example 1.9
Consider the matrices

$$A = \begin{bmatrix} 1 & -1 \\ -2 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 3 \\ -4 & 5 \end{bmatrix}, \quad C = \begin{bmatrix} 2 & 6 \\ -3 & 7 \end{bmatrix}.$$

Then

$$(AB)C = \begin{bmatrix} 6 & -2 \\ -20 & 14 \end{bmatrix} \begin{bmatrix} 2 & 6 \\ -3 & 7 \end{bmatrix} = \begin{bmatrix} 18 & 22 \\ -82 & -22 \end{bmatrix}$$

and

$$A(BC) = \begin{bmatrix} 1 & -1 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} -5 & 33 \\ -23 & 11 \end{bmatrix} = \begin{bmatrix} 18 & 22 \\ -82 & -22 \end{bmatrix}.$$

■

Example 1.10

Consider the matrices:

$$A = \begin{bmatrix} 1 & 4 \\ -2 & 3 \\ 1 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix}, \quad C = \begin{bmatrix} 8 & 6 & -6 \\ 6 & -1 & 1 \\ -4 & 0 & 0 \end{bmatrix}.$$

Find a matrix K such that $AKB = C$. ■

Solution

Since A is a 3×2 matrix and B is a 2×3 matrix, K must be a 2×2 matrix, otherwise, the product AKB is not defined. Thus, we put

$$K = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Since the product of matrices is associative (Theorem 1.1.8), we first compute the product AK as

$$AK = \begin{bmatrix} 1 & 4 \\ -2 & 3 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a + 4c & b + 4d \\ 3c - 2a & 3d - 2b \\ a - 2c & b - 2d \end{bmatrix}.$$

Now, we multiply AK by B and obtain

$$(AK)B = \begin{bmatrix} a + 4c & b + 4d \\ 3c - 2a & 3d - 2b \\ a - 2c & b - 2d \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & -1 \end{bmatrix} = \begin{bmatrix} 2(a + 4c) & b + 4d & -b - 4d \\ 2(3c - 2a) & 3d - 2b & 2b - 3d \\ 2(a - 2c) & b - 2d & 2d - b \end{bmatrix}.$$

The equality, $AKB = C$ now reads

$$\begin{bmatrix} 2(a + 4c) & b + 4d & -b - 4d \\ 2(3c - 2a) & 3d - 2b & 2b - 3d \\ 2(a - 2c) & b - 2d & 2d - b \end{bmatrix} = \begin{bmatrix} 8 & 6 & -6 \\ 6 & -1 & 1 \\ -4 & 0 & 0 \end{bmatrix}.$$

This gives

$$\begin{cases} 2(a+4c) = 8, \\ b+4d = 6, \\ 2(3c-2a) = 6, \\ 3d-2b = -1, \\ b-2d = 0. \end{cases}$$

Solving this system, we find $a = 0, b = 2, c = 1$, and $d = 1$. Thus, the matrix K is

$$K = \begin{bmatrix} 0 & 2 \\ 1 & 1 \end{bmatrix}.$$

◀

1.2 Square Matrices

We have introduced square matrices in Definition 1.1.6. In this section, we show that this class of matrices plays a central role in matrix algebra in particular, and linear algebra in general.

1.2.1 The Ring of Square Matrices

The class of *square* matrices enjoys very important algebraic properties. One of these properties is that the set $\mathscr{M}_n(\mathbb{K})$ has the *closure* property under multiplication. That is, for any two matrices A and B in $\mathscr{M}_n(\mathbb{K})$, the product AB is an element in $\mathscr{M}_n(\mathbb{K})$. (This does not hold for matrices in $\mathscr{M}_{m\times n}(\mathbb{K})$ if $m \neq n$). In other words, we have the binary operation

$$\mathscr{M}_n(\mathbb{K}) \times \mathscr{M}_n(\mathbb{K}) \to \mathscr{M}_n(\mathbb{K}) : \quad (A, B) \mapsto AB. \tag{1.20}$$

Another property is that the multiplication is *distributive over addition* from the right and from the left. That is, for all matrices A, B, and C in $\mathscr{M}_n(\mathbb{K})$, one can easily verify that

$$A(B+C) = AB + AC \tag{1.21}$$

and

$$(B+C)A = BA + CA. \tag{1.22}$$

Remark 1.2.1 (Binomial Formula) Let A and B be two matrices in $\mathscr{M}_n(\mathbb{K})$ such that $AB = BA$. Then we have the binomial formula

$$(A+B)^m = \sum_{k=0}^{m} C_m^k A^k B^{m-k}, \qquad C_m^k = \frac{m!}{k!(m-k)!}.$$

In particular, since the identity matrix commute with all matrices in $\mathscr{M}_n(\mathbb{K})$, then we have

$$(I+A)^m = \sum_{k=0}^{m} C_m^k A^k.$$

Definition 1.2.1

Let R be a nonempty set on which two binary operations called addition $(+)$ and multiplication $(\cdot)$ are defined. Then, R is a *ring* (with respect to the given addition and multiplication) if:

R1. $(R, +)$ is an Abelian group;

R2. multiplication is associative, that is

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c,$$

for all a, b and c in R;

R3. multiplication is distributive over addition from the right and from the left, that is

$$a \cdot (b + c) = a \cdot b + a \cdot c$$

and

$$(b + c) \cdot a = b \cdot a + c \cdot a.$$

The ring R is *commutative* if

$$a \cdot b = b \cdot a$$

for all elements a, b in R, and *noncommutative* otherwise.
If there exists e in R such that

$$a \cdot e = e \cdot a = a,$$

for all elements a in R, then e is called a *unit* or *identity element* of R, and R is called a *unitary ring*.

Example 1.11
The set of real numbers $(\mathbb{R}, +, \cdot)$ with the usual addition and multiplication is a commutative unitary ring. ■

1.2.2 The Identity Matrix

Here we define the identity matrix precisely as the identity element in a ring R.

Definition 1.2.2
Let I be a square matrix in $\mathscr{M}_n(\mathbb{K})$. Then I is an identity matrix if and only if (here we omit "·" in the multiplication)

$$AI = IA = A, \tag{1.23}$$

for any square matrix in $\mathscr{M}_n(\mathbb{K})$.

We can easily check, by using the definition of the product of two matrices (Definition 1.1.11), that (1.23) holds if and only if the matrix I has the form

$$I = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}, \tag{1.24}$$

i.e., means that all entries are zero except the entries a_{ii} on the main diagonal, which are $a_{ii} = 1$.

Notation In the sequel, we will sometimes denote by I_n the identity matrix in $\mathscr{M}_n(\mathbb{K})$.

Example 1.12
The following are examples of identity matrices

$$I = [1], \qquad I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \qquad I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

■

Theorem 1.2.2

The set of square matrices $(\mathscr{M}_n(\mathbb{K}), +, \cdot)$ *with the binary operations* $(+)$ *and* $(\cdot)$ *introduced in Definitions 1.1.7 and 1.1.11, respectively, is a unitary noncommutative ring.*

Proof

We know from Theorem 1.1.5 that $(\mathscr{M}_n(\mathbb{K}), +)$ is an Abelian group. Since (1.19)–(1.22) are also satisfied, $(\mathscr{M}_n(\mathbb{K}), +, \cdot)$ is a ring. It is clear that this ring is noncommutative since the multiplication of matrices is noncommutative. This ring has also an identity element, the identity matrix defined above. □

1.2.3 Inverse Matrices

As we have seen before, the solution of the linear equation (1.1) is given by (1.2). The constant a in (1.1) can be seen as a square matrix in $\mathscr{M}_1(\mathbb{K})$ and a^{-1} is the inverse matrix of a in $\mathscr{M}_1(\mathbb{K})$. So, the solution in (1.2) is defined only if a^{-1} exists. Thus, the natural question is now whether *we can generalize this idea to any square matrix in* $\mathscr{M}_n(\mathbb{K})$*, with* $n \geq 1$*?* In other words, *can we write a solution of system (1.12) in the case* $m = n$ *in the form*

$$X = A^{-1}b \tag{1.25}$$

where A^{-1} *is the inverse matrix of* A *analogously as in (1.2)?* To answer this question, we need first to define A^{-1}. For $n = 1$ and $a \neq 0$ we have $a^{-1} = 1/a$ and satisfies

$$aa^{-1} = a^{-1}a = 1, \tag{1.26}$$

thus a^{-1} exists if and only if

$$a \neq 0. \tag{1.27}$$

So, as we indicated above, if we think about the constants in (1.26) as square matrices in $\mathscr{M}_1(\mathbb{K})$ and 1 as the identity matrix in $\mathscr{M}_1(\mathbb{K})$, then we can define the inverse of the matrix a as a new matrix a^{-1} in $\mathscr{M}_1(\mathbb{K})$ satisfying (1.26). Now it is quite obvious how to extend this definition to matrices in $\mathscr{M}_n(\mathbb{K})$, $n \geq 1$ and introduce the inverse of the matrix A as follows.

Definition 1.2.3 (Inverse of a Square Matrix)

Let A and B be two square matrices in $\mathscr{M}_n(\mathbb{K})$. Then, B is the *inverse* of A if and only if

$$AB = BA = I. \tag{1.28}$$

Notation The inverse of the matrix A is denoted by A^{-1}. Thus, (1.28) reads

$$AA^{-1} = A^{-1}A = I. \tag{1.29}$$

Now, using (1.29) and multiplying the equation

$$AX = b \tag{1.30}$$

by A^{-1}, we get (formally)

$$A^{-1}(AX) = A^{-1}b.$$

Since the multiplication of matrices is associative, we obtain from above that

$$A^{-1}(AX) = (A^{-1}A)X = IX = X.$$

Therefore,

$$X = A^{-1}b. \tag{1.31}$$

Consequently, to find a solution to the system of linear equations (1.8) (with $m = n$) it is enough to find the inverse matrix A^{-1}. Here the following questions arise naturally:

- Does the inverse A^{-1} of the square matrix A does always exist? If yes, we say that A is *invertible* or *nonsingular*.
- In practice it is really important to know whether the solution of the system of linear equations is unique or not. Formula (1.31) indicates that the solution is unique if and only if A^{-1} is unique. So, is the inverse A^{-1} unique?
- If the inverse A^{-1} exists, then how can we find it?

We can immediately answer the first question in the negative. As a simple example, for a matrix a in $\mathscr{M}_1(\mathbb{K})$, a^{-1} exists if and only if $a \neq 0$. Also, the zero matrix $\mathbf{0}$ in $\mathscr{M}_n(\mathbb{K})$ has no inverse since for any matrix A in $\mathscr{M}_n(\mathbb{K})$

$$\mathbf{0}A = A\mathbf{0} = \mathbf{0} \neq I,$$

which violates the definition of the inverse A^{-1}. So, we need a criteria to determine which square matrices in $\mathscr{M}_n(\mathbb{K})$ have inverses in $\mathscr{M}_n(\mathbb{K})$. This will be investigated in the coming sections.

For the second question, we have the following theorem.

Theorem 1.2.3 (Uniqueness of the Inverse Matrix)
Let A be a square matrix in $\mathscr{M}_n(\mathbb{K})$ and assume that A^{-1} exists. Then A^{-1} is unique.

Proof

To prove this statement, we assume that there are two inverses B and C of the matrix A and show that $B = C$. Now, since both B and C are inverses of A, they both satisfy (1.28). That is

$$AB = BA = I, \qquad \text{and} \qquad AC = CA = I. \tag{1.32}$$

Now, since the multiplication of matrices is associative (Theorem 1.1.8), we have that

$$A(BC) = (AB)C = IC = C. \tag{1.33}$$

On the other hand, the first identity in (1.32) yields

$$A(BC) = (AB)C = (BA)C = B(AC) = BI = B. \tag{1.34}$$

Relations (1.33) and (1.34) show that

$$B = C,$$

which ends the proof of Theorem 1.2.3. □

Finally, to answer the third question is really a challenging problem especially if the matrix A is of large size. To understand why this is the case, let us consider a 2×2 matrix and compare the amount of work required with that in the case of an 1×1 matrix.

So, we consider the matrix A in $\mathscr{M}_2(\mathbb{K})$ given by

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

and try to find the inverse A^{-1}. Actually, there are at least two obvious methods of how to proceed. First, we may just assume that A^{-1} exists as a matrix in $\mathscr{M}_2(\mathbb{K})$ and apply (1.29) to find the entries of A^{-1}. The second method is based on the strong connection between the inverse of A and the solution of the linear system (1.30). That is, if we know A^{-1}, then we know the solution by formula (1.31). Conversely, if the solution of system (1.30) exists, then it should be written in the form (1.31). Consequently, our strategy is to solve the 2×2 system where A is the matrix of coefficients and then, once we found the solution, we try to write it in the form (1.31) and thus obtain A^{-1}.

We consider system (1.3), that is

$$\begin{cases} ax + by = p, \\ cx + dy = q. \end{cases} \tag{1.35}$$

As we have seen in (1.5), we can write

$$x = \frac{pd - bq}{ad - bc}.$$

Plugging this expression in the first equation of (1.35), we get

$$y = -\frac{a}{b}x + \frac{p}{b} = \frac{aq - bc}{ad - bc}. \tag{1.36}$$

Therefore, the solution of (1.35) is

$$X = \frac{1}{ad - bc}\begin{bmatrix} pd - bq \\ aq - bc \end{bmatrix}$$

which can be written as

$$X = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}\begin{bmatrix} p \\ q \end{bmatrix}. \tag{1.37}$$

Of course, formula (1.37) makes sense only if

$$ad - bc \neq 0. \tag{1.38}$$

We summarize the above discussion in the following theorem.

Theorem 1.2.4
If $ad - bc \neq 0$, then the inverse of the square matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is given by

$$A^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}. \tag{1.39}$$

We can plainly see how the level of difficulty of finding the inverse changes from the 1×1 matrix to the 2×2 matrix. For the 1×1, matrix, the inverse exists if and only if (1.27) is satisfied and then $a^{-1} = 1/a$. On the other hand, the inverse of the 2×2 matrix exists if and only if (1.38) is verified, and then the inverse is given by (1.39). So, as we have seen above, the difficulty of finding A^{-1} is increasing. We will see in the coming sections other methods for finding the inverse A^{-1} of a matrix A in $\mathscr{M}_n(\mathbb{K})$ with $n \geq 3$.

Example 1.13
Find the inverses of the following matrices:

$$A = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}.$$

■

Solution

For the matrix A, since $ab - cd = 3 \neq 0$, A^{-1} exists and applying (1.39), we get

$$A^{-1} = \frac{1}{3}\begin{bmatrix} 3 & 0 \\ -2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -2/3 & 1/3 \end{bmatrix}.$$

For the matrix B and since $ab - cd = 0$, then B^{-1} does not exists. ◀

We have defined before the product of two matrices, so the question now is how is the inverse related to the product. For matrices in $\mathscr{M}_1(\mathbb{K})$, we have

$$(ab)^{-1} = a^{-1}b^{-1} = b^{-1}a^{-1}.$$

But keep in mind that this is true since the product of matrices in $\mathscr{M}_1(\mathbb{K})$ is commutative, while we already know that the product of matrices in $\mathscr{M}_n(\mathbb{K})$, $n > 1$ is not commutative in general. So, only one of the above equalities is true for matrices, and as we will see, it is the second one.

Theorem 1.2.5 (Inverse of the Product of Two Matrices)
Let A and B be two matrices in $\mathscr{M}_n(\mathbb{K})$ and assume that their inverses exist. Then,

$$(AB)^{-1} = B^{-1}A^{-1}. \tag{1.40}$$

Proof

Since the multiplication of matrices is associative, we can write

$$\begin{aligned}(AB)(B^{-1}A^{-1}) &= A(B^{-1}B)\,A^{-1} \\ &= AIA^{-1} = I.\end{aligned}$$

Similarly,

$$\begin{aligned}(B^{-1}A^{-1})(AB) &= B^{-1}(A^{-1}A)B \\ &= B^{-1}IB = I.\end{aligned}$$

Consequently, by (1.29), $B^{-1}A^{-1}$ is an inverse of AB, and since the inverse of a matrix is unique (Theorem 1.2.3), then $B^{-1}A^{-1}$ is the only inverse of AB. □

Remark 1.2.6 Using induction, we may easily generalize (1.40) for any finite number of matrices as

$$(A_1A_2\cdots A_{\ell-1}A_\ell)^{-1} = A_\ell^{-1}A_{\ell-1}^{-1}\cdots A_2^{-1}A_1^{-1}$$

where A_i, $1 \leq i \leq \ell$, are matrices in $\mathscr{M}_n(\mathbb{K})$.

Example 1.14
Given the two matrices

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix}.$$

Find $(AB)^{-1}$ by two methods. ■

Solution
In the first method, we compute the product AB and then we use Theorem 1.2.4 to find the inverse $(AB)^{-1}$. We have

$$AB = \begin{bmatrix} -1 & 2 \\ 0 & 2 \end{bmatrix},$$

and so

$$(AB)^{-1} = -\frac{1}{2}\begin{bmatrix} 2 & -2 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 0 & 1/2 \end{bmatrix}.$$

In the second method, we use Theorem 1.2.5. Thus, we have, by using (1.39),

$$A^{-1} = \frac{1}{2}\begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix}$$

and

$$B^{-1} = -1\begin{bmatrix} 1 & -2 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ 0 & 1 \end{bmatrix}.$$

Therefore,

$$(AB)^{-1} = B^{-1}A^{-1} = \begin{bmatrix} -1 & 1 \\ 0 & 1/2 \end{bmatrix}.$$

◄

Theorem 1.2.7
Let A be an invertible matrix in $\mathscr{M}_n(\mathbb{K})$ and let $\lambda \neq 0$ be a scalar in $\mathbb{K}$. Then $(A^{-1})^{-1}$ and $(\lambda A)^{-1}$ exist and we have

$$(A^{-1})^{-1} = A, \quad \textit{and} \quad (\lambda A)^{-1} = \frac{1}{\lambda}A^{-1}. \tag{1.41}$$

Proof

The first property in (1.41) is trivial. To prove the second one, we have

$$(\lambda A)\left(\frac{1}{\lambda}A^{-1}\right) = \lambda\frac{1}{\lambda}AA^{-1} = I.$$

Similarly,

$$\left(\frac{1}{\lambda}A^{-1}\right)(\lambda A) = I.$$

The uniqueness of the inverse yields the desired result. □

Now we can collect the above properties of invertible matrices in $\mathscr{M}_n(\mathbb{K})$ and give them an algebraic structure.

Theorem 1.2.8 (The General Linear Group $GL(n,\mathbb{K})$)
The set of square invertible matrices in $\mathscr{M}_n(\mathbb{K})$ is a group with respect to multiplication. This group is non-Abelian for $n \geq 2$. We denote this group by $GL(n,\mathbb{K})$.

Proof

To prove the theorem, we can simply verify the assumptions in Theorem 1.1.10. First, it is clear that the $GL(n,\mathbb{K})$ is not empty, since the identity matrix lies in this set. In addition, it is clear from Theorem 1.2.5 that if A and B are two elements of $GL(n,\mathbb{K})$, then AB is also an element of $GL(n,\mathbb{K})$. Since the multiplication of matrices is associative in $\mathscr{M}_n(\mathbb{K})$, it is also associative in $GL(n,\mathbb{K})$.

Furthermore, it is clear that I is the identity element in $GL(n,\mathbb{K})$. Next, for any A in $GL(n,\mathbb{K})$ there exists A^{-1} in $GL(n,\mathbb{K})$ satisfying (1.29). Thus, $(GL(n,\mathbb{K}),\cdot)$ is a group. It is non-Abelian since, we know that multiplication is noncommutative: for example take

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix}.$$

Then both A and B belong to $GL(2,\mathbb{K})$, but

$$AB = \begin{bmatrix} 1 & 1 \\ 0 & 6 \end{bmatrix} \qquad \text{whereas} \qquad BA = \begin{bmatrix} 1 & 2 \\ 0 & 6 \end{bmatrix}.$$

□

In the next theorem we exhibit the relationship between invertible matrices and homogeneous systems of linear equations.

Theorem 1.2.9

Let A be a square matrix in $\mathscr{M}_n(\mathbb{K})$. Then the following two properties are equivalent:

1. *the matrix A is invertible;*
2. *the homogeneous system associated to the matrix A has the trivial solution $X = \mathbf{0}$ as the unique solution.*

Proof

We first need to show that (1) implies (2). So, assume that A is invertible. Then the homogenous system associated to A has the solution

$$X = A^{-1}b \tag{1.42}$$

where $b = \mathbf{0}$ is the zero vector. Since the inverse of A is unique, the solution X is uniquely defined by (1.42), and so $X = \mathbf{0}$ is the unique solution of the homogeneous system. We leave it to the reader to show that (2) implies (1), which can be done in several ways. □

As we have stated before, since one of the main goals of matrix algebra is to provide necessary conditions for the invertibility of a square matrix A and ways to calculate its inverse A^{-1}. Hence, we want to characterize at least some particular sets of square matrices, where we can easily determine if matrices from these sets are invertible or not and compute the inverses if possible. Among these sets are the set of *diagonal* matrices and *triangular* matrices.

First, we exclude some classes of square matrices that have no inverse.

Theorem 1.2.10

A square matrix that has either a zero row or a zero column is not invertible.

Proof

Let A be a square matrix in $\mathscr{M}_n(\mathbb{K})$ with a zero row. Then, for any matrix B in $\mathscr{M}_n(\mathbb{K})$, the corresponding row in the product AB is also zero. So, AB cannot be the identity matrix. Similarly, if A has a zero column, then the product BA has a zero column, so, BA cannot be the identity matrix. □

Example 1.15

The matrices

$$\begin{bmatrix} 0 & 0 \\ 3 & 2 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 3 \\ 5 & 0 & 2 \end{bmatrix}$$

are not invertible. ■

1

1.2.4 Diagonal Matrices

Here we introduce the set of diagonal matrices that plays an important role in the theory of matrices.

Definition 1.2.4 (Diagonal Matrix)

Let $D = (d_{ij})$, $1 \leq i \leq n$, $1 \leq j \leq n$ be a square matrix in $\mathscr{M}_n(\mathbb{K})$. Then D is a *diagonal* matrix if all the entries outside the main diagonal are zero. That is

$$D = \begin{bmatrix} d_1 & 0 & 0 & \cdots & 0 \\ 0 & d_2 & 0 & \cdots & 0 \\ 0 & 0 & d_3 & \cdots & 0 \\ \vdots & & & \ddots & \\ 0 & 0 & 0 & \cdots & d_n \end{bmatrix}, \tag{1.43}$$

with $d_i = d_{ii}$, $1 \leq i \leq n$. We may also write $D = \operatorname{diag}(d_1, d_2, \ldots, d_n)$.

Example 1.16

The following matrices are diagonal:

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \pi \end{bmatrix}, \quad C = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & \sqrt{2} \end{bmatrix}, \quad I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{0} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

■

The next, theorem provides an easy test, which tells us if a diagonal matrix is invertible or not and gives us right away the inverse.

Theorem 1.2.11 (Inverse of a Diagonal Matrix)

Let D be a diagonal matrix in $\mathscr{M}_n(\mathbb{K})$. Then D is invertible if and only if all its entries (i.e., the entries of its main diagonal) are nonzero. In this case the inverse of D is given by

$$D^{-1} = \begin{bmatrix} 1/d_1 & 0 & 0 & \cdots & 0 \\ 0 & 1/d_2 & 0 & \cdots & 0 \\ 0 & 0 & 1/d_3 & \cdots & 0 \\ \vdots & & & \ddots & \\ 0 & 0 & 0 & \cdots & 1/d_n \end{bmatrix} = \operatorname{diag}\left(\frac{1}{d_1}, \frac{1}{d_2}, \ldots, \frac{1}{d_n}\right).$$

Proof

We first suppose that $d_i \neq 0$. Then, one can easily check that the matrix B defined by

$$B = \begin{bmatrix} 1/d_1 & 0 & 0 & \cdots & 0 \\ 0 & 1/d_2 & 0 & \cdots & 0 \\ 0 & 0 & 1/d_3 & \cdots & 0 \\ \vdots & & & \ddots & \\ 0 & 0 & 0 & \cdots & 1/d_n \end{bmatrix}$$

satisfies

$$DB = BD = I,$$

which means that B is an inverse of D and since the inverse is unique (Theorem 1.2.3),

$$B = D^{-1}.$$

Also, it is clear that D^{-1} exists if and only if $d_i \neq 0$, $1 \leq i \leq n$.

Now assume that D is invertible. We need to show that $d_i \neq 0$, $1 \leq i \leq n$. Indeed, there exists a matrix $K = (k_{ij})$, $1 \leq i,j \leq n$ such that

$$DK = KD = I. \tag{1.44}$$

We have (see Exercise 1.2)

$$DK = \begin{bmatrix} d_1k_{11} & d_1k_{12} & d_1k_{13} & \ldots & d_1k_{1n} \\ d_2k_{21} & d_2k_{22} & d_2k_{23} & \ldots & d_2k_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ d_nk_{n1} & d_nk_{n2} & d_nk_{n3} & \ldots & d_nk_{nn} \end{bmatrix}$$

and

$$KD = \begin{bmatrix} d_1k_{11} & d_2k_{12} & d_3k_{13} & \ldots & d_nk_{1n} \\ d_1k_{21} & d_2k_{22} & d_3k_{23} & \ldots & d_nk_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ d_1k_{n1} & d_2k_{n2} & d_3k_{n3} & \ldots & d_nk_{nn} \end{bmatrix}.$$

Hence, (1.44) gives

$$d_ik_{ii} = 1, \quad 1 \leq i \leq n.$$

This shows that $d_i \neq 0$, $1 \leq i \leq n$. □

Example 1.17

Find the inverses of the following matrices:

$$A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & -2 \end{bmatrix}, \qquad B = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{bmatrix}.$$

■

Solution

For the matrix A, since all the entries of its main diagonal are nonzero, A is invertible and

$$A^{-1} = \begin{bmatrix} 1/4 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & -1/2 \end{bmatrix}.$$

On the other hand, since in the matrix B one entry of the main diagonal is zero, B is not invertible i.e., B^{-1} does not exist. ◄

Among the interesting applications of matrix algebra is the solution of systems of differential equations. This is essentially based on the computation of the exponential of a square matrix A, which is defined as the infinite sum

$$e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots + \frac{A^k}{k!} + \cdots \tag{1.45}$$

for a matrix A in $\mathscr{M}_n(\mathbb{K})$. So, to compute e^A, we need to compute A^k for all $k \geq 1$, and this is a challenging and difficult problem even with more advanced computers, especially if the size of the matrix is large. In addition, since the sum in (1.45) is infinite, then we need some advanced mathematical tools to tackle such a problem (see ► Sect. 7.5.2). So, the problem of finding (1.45) is reduced to the computation of A^k. One of the useful properties of a diagonal matrix D is that we can easily compute D^k for any $k \geq 1$, as shown in the following theorem.

Theorem 1.2.12 (Power of a Diagonal Matrix)

Let D be a diagonal matrix (defined as in (1.43)). Then for any positive integer k, we have

$$D^k = \begin{bmatrix} d_1^k & 0 & 0 & \cdots & 0 \\ 0 & d_2^k & 0 & \cdots & 0 \\ 0 & 0 & d_3^k & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & d_n^k \end{bmatrix} = \operatorname{diag}(d_1^k, d_2^k, \ldots, d_n^k). \tag{1.46}$$

Proof

The proof of (1.46) is simple and can be done by induction. First, it is clear that (1.46) holds for $k = 1$. Now, we assume that (1.46) holds for k and show that it also holds for $k + 1$. That is, we assume that

$$D^k = \begin{bmatrix} d_1^k & 0 & 0 & \cdots & 0 \\ 0 & d_2^k & 0 & \cdots & 0 \\ 0 & 0 & d_3^k & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & d_n^k \end{bmatrix} \tag{1.47}$$

and show that

$$D^{k+1} = \begin{bmatrix} d_1^{k+1} & 0 & 0 & \cdots & 0 \\ 0 & d_2^{k+1} & 0 & \cdots & 0 \\ 0 & 0 & d_3^{k+1} & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & d_n^{k+1} \end{bmatrix}. \tag{1.48}$$

It is straightforward to see that (1.48) can be obtained by simply computing

$$D^{k+1} = D^k \cdot D$$

and using (1.43) and (1.47). □

Example 1.18

Consider the matrix

$$A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & \sqrt{2} \end{bmatrix}.$$

Find A^6. ■

Solution

Since A is diagonal, Theorem 1.2.12 shows that

$$A^6 = \begin{bmatrix} (-1)^6 & 0 & 0 \\ 0 & (2)^6 & 0 \\ 0 & 0 & (\sqrt{2})^6 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 64 & 0 \\ 0 & 0 & 8 \end{bmatrix}.$$

◀

Example 1.19
Consider the matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 3 \end{bmatrix}.$$

Show that A^{-1} exists and find $(A^{-1})^5$. ■

Solution
Since A is diagonal, and the main diagonal does not contain zero, it follows (see Theorem 1.2.11) that A^{-1} exists and can be computed easily as

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1/3 \end{bmatrix}.$$

Also, since A^{-1} is diagonal we have (see Theorem 1.2.12)

$$(A^{-1})^5 = \begin{bmatrix} (1)^5 & 0 & 0 \\ 0 & (-1)^5 & 0 \\ 0 & 0 & (1/3)^5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1/243 \end{bmatrix}.$$

◄

Example 1.20
Find an invertible diagonal matrix A that satisfies

$$A^{-2} = \begin{bmatrix} 16 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

■

Solution
Take A in the form

$$A = \begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix},$$

with $d_i \neq 0$, $i = 1, 2, 3$. The inverse of A is

$$A^{-1} = \begin{bmatrix} 1/d_1 & 0 & 0 \\ 0 & 1/d_2 & 0 \\ 0 & 0 & 1/d_3 \end{bmatrix},$$

Therefore,

$$A^{-2} = (A^{-1})^2 = \begin{bmatrix} (1/d_1)^2 & 0 & 0 \\ 0 & (1/d_2)^2 & 0 \\ 0 & 0 & (1/d_3)^2 \end{bmatrix} = \begin{bmatrix} 16 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Whence

$$\frac{1}{d_1} = 4, \qquad \frac{1}{d_2} = 3, \qquad \frac{1}{d_3} = 1.$$

This yields

$$d_1 = \frac{1}{4}, \qquad d_2 = \frac{1}{3}, \qquad d_3 = 1.$$

Therefore, the matrix A is given by

$$A = \begin{bmatrix} 1/4 & 0 & 0 \\ 0 & 1/3 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

◀

1.2.5 Triangular Matrices

We have introduced the diagonal matrices in Sect. 1.2.4 and showed that these matrices have very important properties. In particular, we have shown that we can immediately know if a diagonal matrix is invertible or not and if it is invertible, then we can easily find its inverse. Now, since the diagonal matrices form a very narrow set in the class of square matrices $\mathscr{M}_n(\mathbb{K})$, it is quite natural to ask the following question: *is there a larger set of square matrices than the set of diagonal matrices which enjoy some properties of the diagonal matrices?* The answer is yes and this class of square matrices consists of the so-called *triangular* matrices.

Definition 1.2.5 (Triangular Matrix)

A square matrix in which all the entries above the main diagonal are zero is called *lower triangular*, and a square matrix in which all the entries below the main diagonal are zero is called *upper triangular*. A matrix that is either upper triangular or lower triangular is called *triangular*.

Example 1.21
The matrix

$$A = \begin{bmatrix} 1 & 4 & -1 & 2 \\ 0 & 3 & -5 & 1 \\ 0 & 0 & -1 & 6 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

is upper triangular while the matrix

$$B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 2 & -4 & 0 & 0 \\ -1 & 3 & -1 & 0 \\ 6 & 2 & 5 & -2 \end{bmatrix}$$

is lower triangular.

Obviously, every diagonal matrix is triangular. ■

In the next theorem we characterize the invertible triangular matrices exactly as we did for the diagonal matrices; however, we do not know the inverse immediately as in the diagonal case. Thus, we have already lost some properties by expanding the set of diagonal matrices.

Theorem 1.2.13
Let A be a triangular matrix in $\mathscr{M}_n(\mathbb{K})$. Then, A is invertible if and only if all the entries of the main diagonal are nonzero.

Proof
We prove the statement for upper triangular matrices. The proof for lower triangular matrices is similar.

Let A be an upper triangular matrix in $\mathscr{M}_n(\mathbb{K})$, then A has the form

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a_{22} & a_{23} & \cdots & a_{2n} \\ 0 & 0 & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix},$$

that is $a_{ij} = 0$ for all $i > j$. The linear homogeneous system associated to the matrix A is

$$\begin{aligned} &a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0,\\ &a_{22}x_2 + \cdots + a_{2n}x_n = 0,\\ &\vdots\\ &a_{(n-1)(n-1)}x_{n-1} + a_{(n-1)n}x_n = 0,\\ &a_{nn}x_n = 0. \end{aligned} \tag{1.49}$$

It is clear that if $a_{nn} \neq 0$, then the last equation in (1.49) has only one solution $x_n = 0$. Inserting this value into the equation just before the last one, we deduce that if $a_{(n-1)(n-1)} \neq 0$, then $x_{n-1} = 0$ is the unique solution. If we apply the same procedure to all the equations in (1.49), we deduce that if $a_{ii} \neq 0$, $1 \leq i \leq n$, then the only solution of (1.49) is the trivial solution $X = \mathbf{0}$. Consequently, applying Theorem 1.2.9, we conclude that A is invertible if and only if $a_{ii} \neq 0$, $1 \leq i \leq n$. □

1.2.6 Trace

As we have seen, for diagonal and triangular matrices, the entries of the main diagonal of those matrices are very important and by examining those entries we can immediately identify the invertible matrices. Now, since the entries of the main diagonal are also important in a general square matrix, so *can we gain something by doing the usual algebraic operations on those entries?* For example, for diagonal and triangular matrices, if the product of all the entries of the main diagonal is not zero, then the matrix is invertible. Now, what about the sum of the entries of the main diagonal in a square matrix, does it give us anything? The answer is affirmative, and as we will see later, it turns out to be very useful. We call this sum the *trace* of the square matrix.

Definition 1.2.6 (Trace)

Let $A = (a_{ij})$, $1 \leq i \leq n$, $1 \leq j \leq n$ be a square matrix in $\mathscr{M}_n(\mathbb{K})$. The *trace* of A, denoted by $\operatorname{tr}(A)$, is defined to be the sum of the entries of the main diagonal of A:

$$\operatorname{tr}(A) = \sum_{i=1}^{n} a_{ii}. \tag{1.50}$$

Example 1.22

Consider the matrices

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 4 & 0 \\ -1 & 5 & -2 \end{bmatrix}, \qquad B = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}.$$

Then we have

$$\operatorname{tr}(A) = 1 + 4 - 2 = 3, \qquad \operatorname{tr}(B) = b_{11} + b_{22}.$$

■

In the next theorem we summarize some properties of the trace.

Theorem 1.2.14

Let A and B be two square matrices in $\mathscr{M}_n(\mathbb{K})$ and k be a scalar. Then:

1. $\operatorname{tr}(A + B) = \operatorname{tr}(A) + \operatorname{tr}(B)$.
2. $\operatorname{tr}(A^T) = \operatorname{tr}(A)$.
3. $\operatorname{tr}(kA) = k \operatorname{tr}(A)$.
4. $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

In fact the last property holds for A in $\mathscr{M}_{m\times n}(\mathbb{K})$ and B in $\mathscr{M}_{n\times m}(\mathbb{K})$. Here A^T denotes the transpose of A (see Definition 1.4.1).

Proof

Properties (1)–(3) are trivial and follow directly from the definition. So, we only need to show property (4). We have, by the definition of the multiplication of matrices,

$$AB = (c_{ik}) \quad \text{with} \quad c_{ik} = \sum_{j=1}^{n} a_{ij} b_{jk}, \quad 1 \le i \le m, \quad 1 \le j \le n, \quad 1 \le k \le m.$$

Hence,

$$\begin{aligned} \operatorname{tr}(AB) &= \sum_{i=1}^{m} c_{ii} = \sum_{i=1}^{m} \Big(\sum_{j=1}^{n} a_{ij} b_{ji} \Big) \\ &= \sum_{i=1}^{m} \Big(\sum_{j=1}^{n} b_{ji} a_{ij} \Big) \\ &= \sum_{j=1}^{n} \Big(\sum_{i=1}^{m} b_{ji} a_{ij} \Big) \\ &= \operatorname{tr}(BA). \end{aligned}$$

□

Example 1.23
Use Theorem 1.2.14 to show that we cannot find two square matrices A and B in $\mathscr{M}_n(\mathbb{R})$ such that

$$AB - BA = I, \tag{1.51}$$

where I is the identity matrix in $\mathscr{M}_n(\mathbb{R})$. ■

Solution
We assume that (1.51) holds and show that this leads to a contradiction. Indeed, if (1.51) holds, then by Theorem 1.2.14 we have

$$\operatorname{tr}(AB) = \operatorname{tr}(BA).$$

Whence

$$\operatorname{tr}(AB) - \operatorname{tr}(BA) = \operatorname{tr}(AB - BA) = 0.$$

On the other hand

$$\operatorname{tr}(I) = n.$$

This is a contradiction. Hence, there are no matrices A and B such that (1.51) holds.

◄

1.3 Solving Linear Systems with Elementary Row Operations

As we have seen above, in order to find the solution of a linear system (of n equations and n unknowns) it is enough to compute the inverse of its associated $n \times n$ matrix A. Moreover, since it is very simple to find the inverse of a diagonal matrices, it is quite simple to solve the systems associated to them. We know from elementary algebra that if we add an equation in the system to another one and then replace the original equation by the sum of the two, then the solution does not change. For example, in system (1.3), if we replace the second equation, by the sum of the two equations, we obtain

$$\begin{cases} ax + by = p, \\ (a + c)x + (b + d)y = p + q. \end{cases} \tag{1.52}$$

Thus, if we assume that $ad - bc \neq 0$, then the solution of (1.52) is the same solution of (1.3). In matrix language, this operation is equivalent to replace the second row in the matrix A

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

to get

$$\begin{bmatrix} a & b \\ a+c & b+d \end{bmatrix}$$

and replacing the vector

$$b = \begin{bmatrix} p \\ q \end{bmatrix}$$

by

$$\begin{bmatrix} p \\ p+q \end{bmatrix}.$$

For simplicity, we may collect these operations in one matrix and transform the matrix

$$B = \left[\begin{array}{cc|c} a & b & p \\ c & d & q \end{array}\right] \tag{1.53}$$

into the matrix

$$\left[\begin{array}{cc|c} a & b & p \\ a+c & b+d & p+q \end{array}\right].$$

The matrix B in (1.53) is called the *augmented* matrix associated to the system (1.3). Similarly, the same thing is true if we replace a row r in the augmented matrix by the product kr, where k is a scalar.

Definition 1.3.1 (Augmented Matrix)

The *augmented* matrix associated to system (1.8) is the matrix in $\mathscr{M}_{m\times(n+1)}(\mathbb{K})$ defined as

$$\left[\begin{array}{ccccc|c} a_{11} & a_{12} & a_{13} & \dots & a_{1n} & b_1 \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} & b_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} & b_m \end{array}\right]. \tag{1.54}$$

The following *elementary* row operations, will not change the solution of (1.8):

- Multiply a row through by a nonzero constant.
- Interchange two rows.
- Add a constant multiple of a row to another row.

1.3.1 The Gauss–Jordan Elimination Method

This method is simply based on some row operations that lead to the simplest diagonal matrix (the identity matrix if possible) for which the inverse matrix can be easily computed if it exists. To apply the method, and for simplicity, we use the augmented matrix described in Definition 1.3.1. Essentially, the idea is to reduce the augmented matrix $[A|b]$, where A is a square matrix in $\mathscr{M}_n(\mathbb{K})$ and b is a vector in $\mathscr{M}_{n\times 1}(\mathbb{K})$, to the form $[D|c]$ where D is a diagonal matrix in $\mathscr{M}_n(\mathbb{K})$ and c is a vector in $\mathscr{M}_{n\times 1}(\mathbb{K})$, or simply to $[I|d]$, where I is the identity matrix in $\mathscr{M}_n(\mathbb{K})$ and d is in $\mathscr{M}_{n\times 1}(\mathbb{K})$. In this case the solution of the system

$$AX = b \tag{1.55}$$

will be simply $X = d$. As an example, consider the system

$$\begin{cases} 2x_1 + x_2 + x_3 = 5, \\ -8x_2 - 2x_3 = -12, \\ 8x_2 + 3x_3 = 14, \end{cases} \tag{1.56}$$

which can be written in matrix form as

$$AX = b,$$

where

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 0 & -8 & -2 \\ 0 & 8 & 3 \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 5 \\ -12 \\ 14 \end{bmatrix}.$$

To apply the row operation method, consider the augmented matrix

$$B = \left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 8 & 3 & 14 \end{array}\right].$$

So, we want to get zeros everywhere except on the main diagonal. Let us denote in each step of the row operation the obtained first, second, and third rows by r_1, r_2, and r_3 respectively. So, first in the matrix B we replace r_3 by $r_3 + r_2$ and obtain

$$\left[\begin{array}{ccc|c} 2 & 1 & 1 & 5 \\ 0 & -8 & -2 & -12 \\ 0 & 0 & 1 & 2 \end{array}\right]. \tag{1.57}$$

Next, in (1.57) we replace r_1 by $8r_1 + r_2$ and obtain

$$\left[\begin{array}{ccc|c} 16 & 0 & 6 & 28 \\ 0 & -8 & -2 & -12 \\ 0 & 0 & 1 & 2 \end{array}\right]. \tag{1.58}$$

Now, in (1.58) we replace r_1 by $r_1 - 6r_3$ and obtain

$$\left[\begin{array}{ccc|c} 16 & 0 & 0 & 16 \\ 0 & -8 & -2 & -12 \\ 0 & 0 & 1 & 2 \end{array}\right], \tag{1.59}$$

and then in (1.59) we replace r_2 by $r_1 + 2r_3$, obtaining

$$\left[\begin{array}{ccc|c} 16 & 0 & 0 & 16 \\ 0 & -8 & 0 & -8 \\ 0 & 0 & 1 & 2 \end{array}\right]. \tag{1.60}$$

Finally, in (1.60) we replace r_1 by $\frac{1}{16}r_1$ and r_2 by $-\frac{1}{8}r_2$ obtaining

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right]. \tag{1.61}$$

Now, since the inverse of the identity matrix is itself, we deduce from (1.61) that

$$X = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

is the solution of (1.56).

Finding the Matrix Inverse

A simple and important application of the Gauss–Jordan method is to compute the inverse A^{-1} of an invertible matrix A in $\mathscr{M}_n(\mathbb{K})$. The idea is as follows: suppose that we want to solve the system (1.55) with b the vector

$$b = e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}.$$

We apply the Gauss–Jordan method to transform the augmented matrix $[A|e_1]$ to the matrix $[I|b_1]$, where I is the identity matrix in $\mathscr{M}_n(\mathbb{K})$ and b_1 is the new resulting vector in $\mathscr{M}_{n\times 1}(\mathbb{K})$. Then the solution of (1.55) is $X_1 = b_1$.

We can repeat the same process for all standard vectors $e_2, e_3, \ldots, e_n$ given by

$$e_i = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix},$$

i.e., all components are zero except the ith component which is 1. In this way, we get the augmented matrices $[I|b_i]$ and the corresponding solutions $X_i = b_i$. For each vector e_i the steps are the same: apply the Gauss–Jordan method to the augmented matrix $[A|e_i]$ to get the new augmented matrix $[I|b_i]$. Hence, we can do all the steps simultaneously and transform the matrix

$$[A|e_1, e_2, \ldots, e_n]$$

into the matrix

$$[I|b_1, b_2, \ldots, b_n].$$

Now, since $e_1, e_2, \ldots, e_n$ are the column vectors of the identity matrix I, if we denote by B the matrix which has $b_1, b_2, \ldots, b_n$ as column vectors then the above procedure is equivalent to transform the matrix

$$[A|I]$$

to the new matrix

$$[I|B].$$

It is readily verified that $B = A^{-1}$. Indeed, since $X_i = b_i$ we have

$$\begin{aligned} AB = A[b_1, b_2, \ldots, b_n] &= [Ab_1, Ab_2, \ldots, Ab_n] \\ &= [AX_1, AX_2, \ldots, AX_n] \\ &= [e_1, e_2, \ldots, e_n] \\ &= I. \end{aligned}$$

1

Example 1.24

Use the Gauss–Jordan method to find the inverse of the matrix

$$A = \begin{bmatrix} 5 & 8 & 4 \\ 2 & 3 & 2 \\ 1 & 2 & 1 \end{bmatrix}.$$

■

Solution

We apply the Gauss–Jordan method to find A^{-1}. Consider the matrix

$$\left[\begin{array}{ccc|ccc} 5 & 8 & 4 & 1 & 0 & 0 \\ 2 & 3 & 2 & 0 & 1 & 0 \\ 1 & 2 & 1 & 0 & 0 & 1 \end{array}\right].$$

We apply elementary row operations and in each step, we denote by r_1, r_2, and r_3 the rows of the new matrix. First, we replace r_2 by $r_2 - 2r_3$ and get

$$\left[\begin{array}{ccc|ccc} 5 & 8 & 4 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & -2 \\ 1 & 2 & 1 & 0 & 0 & 1 \end{array}\right].$$

Next, we replace r_3 by $5r_3 - r_1$ and get

$$\left[\begin{array}{ccc|ccc} 5 & 8 & 4 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & -2 \\ 0 & 2 & 1 & -1 & 0 & 5 \end{array}\right].$$

Continuing, we replace r_3 by $r_3 + 2r_2$ to obtain

$$\left[\begin{array}{ccc|ccc} 5 & 8 & 4 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & -1 & 2 & 1 \end{array}\right],$$

then replace r_1 by $r_1 + 8r_2$ to obtain

$$\left[\begin{array}{ccc|ccc} 5 & 0 & 4 & 1 & 8 & -16 \\ 0 & -1 & 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & -1 & 2 & 1 \end{array}\right].$$

Furthermore, we replace r_1 by $r_1 - 4r_3$ to get

$$\left[\begin{array}{ccc|ccc} 5 & 0 & 0 & 5 & 0 & -20 \\ 0 & -1 & 0 & 0 & 1 & -2 \\ 0 & 0 & 1 & -1 & 2 & 1 \end{array}\right].$$

Finally, replacing r_1 by $\frac{1}{5}r_1$ and r_2 with $-r_2$ we get

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & -4 \\ 0 & 1 & 0 & 0 & -1 & 2 \\ 0 & 0 & 1 & -1 & 2 & 1 \end{array}\right].$$

Consequently,

$$A^{-1} = \begin{bmatrix} 1 & 0 & -4 \\ 0 & -1 & 2 \\ -1 & 2 & 1 \end{bmatrix}.$$

◀

Example 1.25
Consider the matrix

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 5 & 4 & 0 \\ 1 & 0 & 1 \end{bmatrix}.$$

Show that A^{-1} exists and use elementary row operations (Gauss–Jordan method) to find A^{-1}.

■

Solution
Since A is a lower triangular matrix and the entries of its main diagonal are nonzero, the inverse exists (Theorem 1.2.13).

To find A^{-1}, use elementary row operation to transform the matrix

$$[A|I]$$

into a matrix of the form

$$[I|B].$$

If we achieve this, then $A^{-1} = B$.

So, we consider the matrix

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 5 & 4 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right]. \tag{1.62}$$

Let r_1, r_2, and r_3 be the rows of all the matrices obtained by means of row operations. We replace in (1.62) r_2 by $r_2 - 5r_1$ to get

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 4 & 0 & -5 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array}\right], \tag{1.63}$$

then replace r_3 in (1.63) by $r_3 - r_1$ to get

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 4 & 0 & -5 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right], \tag{1.64}$$

and finally replace r_2 by $\frac{1}{4}r_2$ in (1.64) obtaining

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -5/4 & 1/4 & 0 \\ 0 & 0 & 1 & -1 & 0 & 1 \end{array}\right]. \tag{1.65}$$

Consequently,

$$A^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -5/4 & 1/4 & 0 \\ -1 & 0 & 1 \end{bmatrix}.$$

◀

Example 1.26

Find the inverse of the matrix

$$A = \begin{bmatrix} 0 & 0 & 0 & k_1 \\ 0 & 0 & k_2 & 0 \\ 0 & k_3 & 0 & 0 \\ k_4 & 0 & 0 & 0 \end{bmatrix}$$

where k_1, k_2, k_3, and k_4 are all different from zero. ■

Solution

We proceed as above and write

$$\left[\begin{array}{cccc|cccc} 0 & 0 & 0 & k_1 & 1 & 0 & 0 & 0 \\ 0 & 0 & k_2 & 0 & 0 & 1 & 0 & 0 \\ 0 & k_3 & 0 & 0 & 0 & 0 & 1 & 0 \\ k_4 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{array}\right]. \tag{1.66}$$

We may exchange the rows as follows: r_1 and r_4, and then r_3 and r_2, to obtain

$$\left[\begin{array}{cccc|cccc} k_4 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & k_3 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & k_2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & k_1 & 1 & 0 & 0 & 0 \end{array}\right]. \tag{1.67}$$

Now, in (1.67) we replace r_1 by $\frac{1}{k_4}r_1$, r_2 by $\frac{1}{k_3}r_2$, r_3 by $\frac{1}{k_2}r_3$, and r_4 by $\frac{1}{k_1}r_4$, obtaining

$$\left[\begin{array}{cccc|cccc} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 1/k_4 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1/k_3 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1/k_2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1/k_1 & 0 & 0 & 0 \end{array}\right].$$

Consequently, the inverse of A is given by

$$A^{-1} = \begin{bmatrix} 0 & 0 & 0 & 1/k_4 \\ 0 & 0 & 1/k_3 & 0 \\ 0 & 1/k_2 & 0 & 0 \\ 1/k_1 & 0 & 0 & 0 \end{bmatrix}.$$

◀

Example 1.27

Let $k \neq 0$ be a real number. Consider the matrix

$$A = \begin{bmatrix} k & 1 & 0 \\ 0 & k & 1 \\ 0 & 0 & k \end{bmatrix}.$$

Show that A^{-1} exists and use elementary row operations to find A^{-1}. ■

Solution

Since A is an upper triangular matrix, the inverse exists if and only if all the entries of the main diagonal are nonzero. So, since we took $k \neq 0$, A^{-1} exists.

To find it, we use elementary row operations to transform the matrix

$$[A|I]$$

to a matrix of the form

$$[I|B].$$

Once we achieve this, $A^{-1} = B$.

So, we consider the matrix

$$\left[\begin{array}{ccc|ccc} k & 1 & 0 & 1 & 0 & 0 \\ 0 & k & 1 & 0 & 1 & 0 \\ 0 & 0 & k & 0 & 0 & 1 \end{array}\right]. \tag{1.68}$$

As above, let r_1, r_2 and r_3 be the rows of all the matrices obtained from row operations. In (1.68) we replace r_2 by $kr_2 - r_3$ to get

$$\left[\begin{array}{ccc|ccc} k & 1 & 0 & 1 & 0 & 0 \\ 0 & k^2 & 0 & 0 & k & -1 \\ 0 & 0 & k & 0 & 0 & 1 \end{array}\right]. \tag{1.69}$$

Next, in (1.69), we replace r_1 by $k^2 r_1 - r_2$ to obtain

$$\left[\begin{array}{ccc|ccc} k^3 & 0 & 0 & k^2 & -k & 1 \\ 0 & k^2 & 0 & 0 & k & -1 \\ 0 & 0 & k & 0 & 0 & 1 \end{array}\right], \tag{1.70}$$

and then in (1.70), replace r_1 by $\frac{1}{k^3} r_1$, r_2 by $\frac{1}{k^2} r_2$, and r_3 by $\frac{1}{k} r_3$ to find

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1/k & -1/k^2 & 1/k^3 \\ 0 & 1 & 0 & 0 & 1/k & -1/k^2 \\ 0 & 0 & 1 & 0 & 0 & 1/k \end{array}\right].$$

Consequently,

$$A^{-1} = \left[\begin{array}{ccc} 1/k & -1/k^2 & 1/k^3 \\ 0 & 1/k & -1/k^2 \\ 0 & 0 & 1/k \end{array}\right].$$

◄

Example 1.28
Show that the matrix

$$A = \begin{bmatrix} 1 & 6 & 4 \\ 2 & 4 & -1 \\ -1 & 2 & 5 \end{bmatrix}$$

is not invertible. ■

Solution
To show that A is not invertible, it suffices to do some row operations and find one row which has only zeros.

So, we consider the matrix

$$\left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 2 & 4 & -1 & 0 & 1 & 0 \\ -1 & 2 & 5 & 0 & 0 & 1 \end{array}\right]. \tag{1.71}$$

Let r_1, r_2 and r_3 be as before. In (1.71) we replace r_2 by $r_2 + 2r_3$, obtaining

$$\left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 0 & 8 & 9 & -5 & 1 & 2 \\ -1 & 2 & 5 & 0 & 0 & 1 \end{array}\right]. \tag{1.72}$$

Now, in (1.72) we replace r_3 by $r_3 + r_1$ to get

$$\left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 0 & 8 & 9 & -5 & 1 & 2 \\ 0 & 8 & 9 & 6 & -1 & -1 \end{array}\right], \tag{1.73}$$

and then in (1.73), we replace r_3 by $r_3 - r_2$ to finally get

$$\left[\begin{array}{ccc|ccc} 1 & 6 & 4 & 1 & 0 & 0 \\ 0 & 8 & 9 & -5 & 1 & 2 \\ 0 & 0 & 0 & 11 & -2 & -3 \end{array}\right]. \tag{1.74}$$

Since the third row in left-hand side of (1.74) contains only zeros, A is not invertible.

◄

1.4 The Matrix Transpose and Symmetric Matrices

In this section, we introduce two important notions: the transpose of a matrix and symmetric matrices.

1

1.4.1 Transpose of a Matrix

As we have seen before, we usually use two notations for a vector X:

$$X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \qquad \text{or} \qquad X = (x_1, \ldots, x_n).$$

Using the first notation, we can write the system (1.8) in the matrix from (1.12), with A the matrix given in (1.13). The question now is: *can we write the system* (1.8) *in a matrix form using the second notation for the vector X?* To do this, we recast (1.8) as

$$(x_1, \ldots, x_n) \begin{bmatrix} a_{11} & a_{21} & a_{31} & \ldots & a_{m1} \\ a_{12} & a_{22} & a_{32} & \ldots & a_{m2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{1n} & a_{2n} & a_{3n} & \ldots & a_{mn} \end{bmatrix} = (b_1, \ldots, b_m).$$

The $n \times m$ matrix appearing here is called the *transpose* of the matrix A.

Definition 1.4.1 (Transpose of a Matrix)

Let A be a matrix in $\mathscr{M}_{m\times n}(\mathbb{K})$. We define the *transpose* of A, denoted by A^T, as the matrix in $\mathscr{M}_{n\times m}(\mathbb{K})$ obtained by interchanging the rows and columns of A. That is, the first row of A^T is the first column of A, the second row of A^T is the second column of A, and so on.

Example 1.29

Let

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 4 & 0 \end{bmatrix}.$$

Then

$$A^T = \begin{bmatrix} 1 & 3 \\ 0 & 4 \\ 2 & 0 \end{bmatrix}.$$

■

Now, we list some properties of the transpose of matrices.

Theorem 1.4.1 (Properties of Transpose)
Let k be a scalar. We assume that the sizes of the matrices A and B are such that the operations below can be performed. Then:
1. $(A^T)^T = A$.
2. $(A + B)^T = A^T + B^T$.
3. $(kA)^T = kA^T$.
4. $(AB)^T = B^T A^T$.

Proof
The first three properties are direct consequences of the definition of the transposed matrix. We need to prove the last two. The proof of (4) can be done by a direct computation. So, assume that A is a matrix in $\mathscr{M}_{m\times n}(\mathbb{K})$ and B is a matrix in $\mathscr{M}_{n\times r}(\mathbb{K})$. Then,

$$AB = C = (c_{ik}), \qquad 1 \le i \le m, \quad 1 \le k \le r,$$

with

$$c_{ik} = \sum_{j=1}^{n} a_{ij} b_{jk}.$$

Hence,

$$C^T = (AB)^T = (c_{ki})_{\substack{1\le i\le m\\ 1\le k\le r}} = \Big(\sum_{j=1}^{n} b_{kj} a_{ji}\Big)_{\substack{1\le i\le m\\ 1\le k\le r}} = B^T A^T.$$

□

We would also like to know how to find the inverse of the transpose A^T if we know the inverse of A? The answer is given in the following theorem.

Theorem 1.4.2 (The Inverse of the Transpose of a Matrix)
Let A be a square matrix in $\mathscr{M}_n(\mathbb{K})$. If A is invertible, the A^T is invertible and

$$(A^T)^{-1} = (A^{-1})^T.$$

Proof
We can establish the invertibility and obtain the formula at the same time, by showing that

$$A^T (A^{-1})^T = (A^{-1})^T A^T = I, \tag{1.75}$$

and using the uniqueness of the inverse (Theorem 1.2.3) to conclude that $(A^T)^{-1} = (A^{-1})^T$. To show (1.75), we have, by (4) in Theorem 1.4.1,

$$A^T(A^{-1})^T = (A^{-1}A)^T = I^T = I.$$

and

$$(A^{-1})^T A^T = (AA^{-1})^T = I^T = I.$$

Thus, (1.75) is verified and Theorem 1.4.2 is proved. □

1.4.2 Symmetric Matrices

In this section, we discuss an important class of square matrices. We have introduced above the matrix transpose A^T associated to a matrix A and its properties. The interesting question now is the following: *can we gain something if the matrix A is a square matrix and its transpose A^T turns out to coincide with A?* In this case the main diagonal will not change and all the other entries of the matrix A are symmetric with respect to the main diagonal. Accordingly, we call A a symmetric matrix and in fact yes, such matrices enjoy many interesting properties.

Definition 1.4.2 (Symmetric Matrix)

Let A be a square matrix in $\mathscr{M}_n(\mathbb{K})$. The matrix A is said to be symmetric if

$$A^T = A. \tag{1.76}$$

Example 1.30

The following matrices are symmetric:

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}, \quad \begin{bmatrix} 1 & 4 & 5 \\ 4 & -3 & 0 \\ 5 & 0 & 2 \end{bmatrix}, \quad \begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}.$$

■

In the next theorem we exhibit some important symmetric matrices.

Theorem 1.4.3

Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$. Then AA^T, A^TA, and $A + A^T$ are symmetric matrices.

Proof

First, for the matrix $B = AA^T$, Theorem 1.4.1 shows that

$$B^T = (AA^T)^T = (A^T)^T A^T = AA^T = B.$$

Thus, B is symmetric.

Second, by the same method we have for $C = A^T A$

$$C^T = (A^T A)^T = A^T (A^T)^T = A^T A = C.$$

Therefore, C is symmetric.

Finally, for $D = A + A^T$, then, we have, again by Theorem 1.4.1,

$$D^T = (A + A^T)^T = A^T + (A^T)^T = A^T + A = D,$$

so, D is also symmetric. □

1.5 Exercises

Exercise 1.1

We consider, for any real number x, the matrix

$$A = \begin{bmatrix} \cosh x & \sinh x \\ \sinh x & \cosh x \end{bmatrix}.$$

1. For x and y in $\mathbb{R}$, find $A(x)A(y)$.
2. Let n be an integer. Find $[A(x)]^n$.

■

Solution

1. We have

$$\begin{aligned} A(x)A(y) &= \begin{bmatrix} \cosh x & \sinh x \\ \sinh x & \cosh x \end{bmatrix} \begin{bmatrix} \cosh y & \sinh y \\ \sinh y & \cosh y \end{bmatrix} \\ &= \begin{bmatrix} \cosh x \cosh y + \sinh x \sinh y & \cosh x \sinh y + \sinh x \cosh y \\ \cosh x \sinh y + \sinh x \cosh y & \cosh x \cosh y + \sinh x \sinh y \end{bmatrix} \\ &= \begin{bmatrix} \cosh(x+y) & \sinh(x+y) \\ \sinh(x+y) & \cosh(x+y) \end{bmatrix} \\ &= A(x+y), \end{aligned} \tag{1.77}$$

where we have used the known identities

$$\cosh x \cosh y + \sinh x \sinh y = \cosh(x+y)$$

and

$$\cosh x \sinh y + \sinh x \cosh y = \sinh(x + y).$$

2. It is clear that $A^0 = I_2 = A(0)$. Now, let $n > 0$; then by (1) above we have

$$[A(x)]^2 = A(x)A(x) = A(2x).$$

We show by induction that for all $n > 0$,

$$[A(x)]^n = A(nx). \tag{1.78}$$

First, it is clear that (1.78) holds for $n = 0, 1$, and 2.

Second, assume that (1.78) holds for n and prove it for $n + 1$. Thus, we have

$$[A(x)]^n = [A(x)]^n A(x) = A(nx)A(x) = A((n + 1)x).$$

Thus, (1.78) holds for $n + 1$ and therefore, it is true for all $n \geq 0$.

Now, for $n < 0$, we see first, by using (1.77), that

$$A(x)A(-x) = A(-x)A(x) = A(0) = I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Therefore, the uniqueness of the inverse shows that $[A(x)]^{-1} = A(-x)$, and by definition $A^{-p} = [A^{-1}]^p, \quad p > 0$. Hence, we have for $n = -p < 0$,

$$[A(x)]^n = [A(x)]^{-p} = [A^{-1}(x)]^p = A(-px) = A(nx).$$

Consequently, for any integer n, we have

$$[A(x)]^n = A(nx).$$

◀

Exercise 1.2 (Multiplication by a Diagonal Matrix)
Let D be the diagonal matrix in $\mathscr{M}_n(\mathbb{K})$ with diagonal entries $d_1, \ldots, d_n$ and A be a square matrix in $\mathscr{M}_n(\mathbb{K})$. Compute the products DA and AD. ■

Solution
First, let us examine the simple case $n = 2$; then we will generalize it for all n. So, let

$$D = \begin{bmatrix} d_1 & 0 \\ 0 & d_2 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}.$$

We can easily verify that

$$DA = \begin{bmatrix} d_1a_{11} & d_1a_{12} \\ d_2a_{21} & d_2a_{22} \end{bmatrix} \quad \text{and} \quad AD = \begin{bmatrix} d_1a_{11} & d_2a_{12} \\ d_1a_{21} & d_2a_{22} \end{bmatrix}.$$

So, we see that the multiplication of the matrix A from the left by D is effected by multiplying the successive rows of A by the successive diagonal entries of D, and the multiplication of A from the right by D is effected by multiplying the successive columns of A by the successive diagonal entries of D.

Now, we want to show that this property holds for any $n \geq 2$. So, let $A = (a_{jk})$, $1 \leq j \leq n$, $1 \leq k \leq n$ and $D = d_{ij}$, $1 \leq i \leq n$, $1 \leq j \leq n$ with $d_{ij} = 0$ for $i \neq j$ and $d_{ii} = d_i$. Using Definition 1.1.11, we get

$$DA = (c_{ik}), \qquad 1 \leq i \leq n, \quad 1 \leq k \leq n,$$

with the entries of the ith row being

$$c_{ik} = \sum_{j=1}^{n} d_{ij}a_{jk} = d_{ii}a_{ik} = d_ia_{ik}, \quad 1 \leq k \leq n.$$

Thus,

$$DA = \begin{bmatrix} d_1a_{11} & d_1a_{12} & d_1a_{13} & \dots & d_1a_{1n} \\ d_2a_{21} & d_2a_{22} & d_2a_{23} & \dots & d_2a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ d_na_{n1} & d_na_{n2} & d_na_{n3} & \dots & d_na_{nn} \end{bmatrix}.$$

The same argument shows

$$AD = \begin{bmatrix} d_1a_{11} & d_2a_{12} & d_3a_{13} & \dots & d_na_{1n} \\ d_1a_{21} & d_2a_{22} & d_3a_{23} & \dots & d_na_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ d_1a_{n1} & d_2a_{n2} & d_3a_{n3} & \dots & d_na_{nn} \end{bmatrix}.$$

◀

Exercise 1.3 (Nilpotent Matrices)

A square matrix in $\mathscr{M}_n(\mathbb{K})$ is *nilpotent* of order k if $A^k = \mathbf{0}$.

1. Show that if A is nilpotent, then $I + A$ is invertible.
2. Calculate the inverse of the matrix

$$L = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

■

Solution

1. Assume that A is nilpotent of order k. We want to show that $(I + A)^{-1}$ exists. In the case of real 1×1 matrices, we have (under some assumptions) the Taylor series expansion

$$(1 + a)^{-1} = \frac{1}{a + 1} = 1 - a + a^2 - a^3 + \cdots = \sum_{n=0}^{\infty} (-1)^n a^n.$$

Analogously, we may look for the inverse of $I + A$ as the matrix B defined by

$$B = I - A + A^2 - A^3 + \cdots = \sum_{n=0}^{k-1} (-1)^n A^n. \tag{1.79}$$

The sum in the above equation will be finite since $A^k = 0$. It remains to verify that B is the inverse of $(I + A)$, that is we have to prove that

$$(I + A)B = B(I + A) = I.$$

Indeed, since $A^k = 0$, we have

$$\begin{aligned}
(I + A)B &= (I + A) \sum_{n=0}^{k-1} (-1)^n A^n \\
&= \sum_{n=0}^{k-1} (-1)^n A^n + \sum_{n=0}^{k-1} (-1)^n A^{n+1} \\
&= \sum_{n=0}^{k-1} (-1)^n A^n + \sum_{n=0}^{k-2} (-1)^n A^{n+1} \\
&= \sum_{n=0}^{k-1} (-1)^n A^n + \sum_{n=1}^{k-1} (-1)^{n-1} A^n \\
&= I.
\end{aligned}$$

The same argument easily shows that $B(I + A) = I$. Consequently,

$$B = (I + A)^{-1}.$$

2. It is clear that the matrix L can be written as $L = I_3 + A$, with

$$A = \begin{bmatrix} 0 & 2 & 3 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{bmatrix}.$$

Then, we can easily calculate

$$A^2 = \begin{bmatrix} 0 & 0 & 4 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad A^3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Hence, A is nilpotent of order 3. Using (1.79), we find that

$$L^{-1} = I_3 - A + A^2 = \begin{bmatrix} 1 & -2 & 1 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}.$$

◀

Exercise 1.4

Let A be a matrix in $\mathscr{M}_2(\mathbb{K})$ of the general form

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Show that

$$p(A) = A^2 - (a+d)A + (ad - bc)I_2 = \mathbf{0}.^2$$

■

Solution

We compute first A^2:

$$A^2 = AA = \begin{bmatrix} a^2 + bc & b(a+d) \\ c(a+d) & cb + d^2 \end{bmatrix}$$

and then

$$(a+d)A = \begin{bmatrix} a(a+d) & b(a+d) \\ c(a+d) & d(a+d) \end{bmatrix}, \qquad (ad-bc)I_2 = \begin{bmatrix} ad-bc & 0 \\ 0 & ad-bc \end{bmatrix}.$$

This clearly yields,

$$A^2 - (a+d)A + (ad-bc)I_2 = \mathbf{0}.$$

◀

[2] As we will see later, the number $a + d$ is called the *trace* of A, the number $ad - bc$ is called the *determinant* of A, and the polynomial $p(\lambda) = \lambda^2 - (a+d)\lambda + (ad-bc)$ is called the *characteristic polynomial* of A. See Definition 7.3.2.

1

Exercise 1.5 (Idempotent Matrices)

Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$. Then A is said to be *idempotent* if $A^2 = A$.

1. Show that if A is idempotent, then so is $I - A$.
2. Show that if A is idempotent, then $2A - I$ is invertible and is its own inverse.
3. Find all the idempotent matrices in $\mathscr{M}_2(\mathbb{R})$.
4. Show that if A is idempotent, and if p is a positive integer, then $A^p = A$.

■

Solution

1. Since $AI = IA = A$, we can easily show that

$$\begin{aligned}(I-A)^2 &= I - 2A + A^2\\ &= I - 2A + A\\ &= I - A,\end{aligned}$$

where we have used the fact that $A^2 = A$. This shows that $I - A$ is idempotent since $(I-A)^2 = I - A$.

2. We have

$$\begin{aligned}(2A-I)(2A-I) &= (A + (A-I))(A + (A-I))\\ &= A^2 + (A-I)^2 + 2A(A-I)\\ &= A^2 + (I-A)^2 + 2A^2 - 2A\\ &= I,\end{aligned}$$

where we have used the fact that A and $I - A$ are idempotent matrices. Consequently,

$$(2A-I)^{-1} = 2A - I.$$

3. Let us consider the matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Then, we have

$$A^2 = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{bmatrix}.$$

Therefore, $A^2 = A$ leads to the system of equations

$$\begin{cases} a^2 + bc = a, \\ ab + bd = b, \\ ac + cd = c, \\ bc + d^2 = d. \end{cases} \tag{1.80}$$

From the first and the third equations, we deduce that if $b = 0$ and $a + d \neq 1$, then $a = 0$ or $a = 1$ and $c = 0$. If $a + d = 1$, then c can be any real number. Then from the fourth equation, we deduce that $a = 1$ or $d = 0$. Then in this case, the idempotent matrices are

$$\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 \\ c & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ c & 0 \end{bmatrix}.$$

If $b \neq 0$, then, from the second equation, we have $a + d = 1$ and from the first equation, we have $c = \frac{a^2 - a}{b} = \frac{d - d^2}{b}$. Thus, the idempotent matrices of $\mathscr{M}_2(\mathbb{R})$ are the matrices of the form

$$\begin{bmatrix} a & b \\ \dfrac{a - a^2}{b} & 1 - a \end{bmatrix}$$

where a is in $\mathbb{R}$ and b is in $\mathbb{R} - \{0\}$.

4. We can show by induction that if $A^2 = A$, then

$$A^p = A, \tag{1.81}$$

for any positive integer p. It is clear that (1.81) is satisfied for $p = 1$ and $p = 2$. Now, assume that (1.81) holds for p and show that it is still holds for $p + 1$. We have

$$A^{p+1} = A^p A = AA = A^2 = A.$$

Consequently, (1.81) holds for any positive integer p. ◄

Exercise 1.6 (Rotation Matrix)

We define the rotation matrix in $\mathscr{M}_2(\mathbb{R})$ as

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix},$$

where θ is the rotation angle.

1. Show that $R^{-1}(\theta) = R(-\theta)$ (rotate back by $-\theta$).
2. Show that $R(\theta_1)R(\theta_2) = R(\theta_1 + \theta_2)$.

■

Solution

1. Since $R(\theta)$ is a matrix in $\mathscr{M}_2(\mathbb{R})$, using (1.39) we deduce that

$$\begin{aligned} R^{-1}(\theta) &= \frac{1}{\cos^2\theta + \sin^2\theta}\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \\ &= \begin{bmatrix} \cos(-\theta) & -\sin(-\theta) \\ \sin(-\theta) & \cos(-\theta) \end{bmatrix} \\ &= R(-\theta). \end{aligned}$$

2. We have by a simple computation

$$\begin{aligned} R(\theta_1)R(\theta_2) &= \begin{bmatrix} \cos\theta_1 & -\sin\theta_1 \\ \sin\theta_1 & \cos\theta_2 \end{bmatrix}\begin{bmatrix} \cos\theta_2 & -\sin\theta_2 \\ \sin\theta_2 & \cos\theta_2 \end{bmatrix} \\ &= \begin{bmatrix} \cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2 & -\sin\theta_2\cos\theta_1 - \sin\theta_1\cos\theta_2 \\ \sin\theta_2\cos\theta_1 + \sin\theta_1\cos\theta_2 & \cos\theta_1\cos\theta_2 - \sin\theta_1\sin\theta_2 \end{bmatrix} \\ &= \begin{bmatrix} \cos(\theta_1+\theta_2) & -\sin(\theta_1+\theta_2) \\ \sin(\theta_1+\theta_2) & \cos(\theta_1+\theta_2) \end{bmatrix} \\ &= R(\theta_1+\theta_2). \end{aligned}$$

The above result means that rotating by θ_1 and then by θ_2, is the same as rotating by $\theta_1 + \theta_2$. ◀

Exercise 1.7 (Involutory Matrix)

Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$. We say that A is *involutory* matrix if

$$A^2 = I.$$

1. Check that for any real number θ, the matrix

$$A = \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}$$

   is an involutory matrix.
2. Find all the involutory matrices in $\mathscr{M}_2(\mathbb{R})$.
3. Show that a matrix A is involutory if and only if

$$(I - A)(I + A) = 0.$$

4. Show that if A is an involutory matrix, then, the matrix $B = \frac{1}{2}(I + A)$ is idempotent.

■

Solution

1. We need to verify that $A^2 = I$. By a simple computation,

$$\begin{aligned} A^2 = AA &= \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix} \\ &= \begin{bmatrix} \cos^2\theta + \sin^2\theta & 0 \\ 0 & \cos^2\theta + \sin^2\theta \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I. \end{aligned}$$

2. Let A be a matrix in $\mathscr{M}_2(\mathbb{R})$, thus,

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

with a, b, c and d real numbers. We compute A^2 to find

$$A^2 = AA = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2 + bc & ab + bd \\ ac + cd & bc + d^2 \end{bmatrix}.$$

Therefore, A is involutory if and only if $A^2 = I$, that is

$$\begin{cases} a^2 + bc = 1, \\ ab + bd = 0, \\ ac + cd = 0, \\ bc + d^2 = 1. \end{cases} \tag{1.82}$$

If $b = 0$, then $a = \pm 1$ and $d = \pm 1$. Thus, the third equation in the above system gives: if $a = 1$ and $d = 1$ or $a = -1$ and $d = -1$, then $c = 0$, in the other cases $a = 1$ and $d = -1$ or $a = -1$ and $d = 1$, then c can be any real number. Therefore, for $b = 0$, the involutory matrices are

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ c & -1 \end{bmatrix}, \quad \begin{bmatrix} -1 & 0 \\ c & 1 \end{bmatrix}.$$

Now, if $b \neq 0$, then the second equation in (1.82) yields $d = -a$ and $a^2 + bc = 1$. Therefore, the involutory matrices in this case are

$$\begin{bmatrix} a & b \\ c & -a \end{bmatrix}, \qquad \text{with} \qquad a^2 + bc = 1.$$

3. Let A be an involutory matrix in $\mathscr{M}_n(\mathbb{K})$. Then, since I commutes with any matrix in $\mathscr{M}_n(\mathbb{K})$, and since $A^2 = I$,

$$(I - A)(I + A) = I^2 - A^2 - AI + IA = I - I = \mathbf{0}.$$

4. The matrix B is idempotent (Exercise 1.5), if and only if $B^2 = B$. We have

$$\begin{aligned} B^2 = BB &= \frac{1}{2}(I + A)\frac{1}{2}(I + A) \\ &= \frac{1}{4}(A^2 + 2IA + I^2) \\ &= \frac{1}{4}(I + 2A + I) \\ &= \frac{1}{2}(I + A) = B, \end{aligned}$$

where we have used the fact that $A^2 = I$.

◀

Exercise 1.8

Let A and B be two matrices in $\mathscr{M}_n(\mathbb{K})$ and I be the identity matrix in $\mathscr{M}_n(\mathbb{K})$. Check that if $I + AB$ is invertible, then $I + BA$ is invertible, and find its inverse. ■

Solution

Assume that $I + AB$ is invertible, that is $(I + AB)^{-1}$ exists. Now, a matrix C in $\mathscr{M}_n(\mathbb{K})$ is the inverse of $(I + BA)$ if and only if

$$(I + BA)C = C(I + BA) = I. \tag{1.83}$$

The first equality, i.e.,

$$(I + BA)C = I,$$

leads to (since the of multiplication is associative and distributive over the addition)

$$C + B(AC) = I.$$

Or, equivalently,

$$B(AC) = I - C. \tag{1.84}$$

Multiply from the left by A, we get

$$(AB)(AC) = A - AC,$$

whence

$$AC + (AB)(AC) = A.$$

That is

$$(I + AB)(AC) = A.$$

So, multiplying from the left by $(I + AB)^{-1}$ we have

$$AC = (I + AB)^{-1}A. \tag{1.85}$$

Now, using the second identity in (1.83), we obtain

$$C(BA) = I - C. \tag{1.86}$$

Multiplying from the right by B we get

$$C(BA)B = B - CB.$$

That is

$$(CB)(I + AB) = B.$$

Multiplying here from the right by $(I + AB)^{-1}$, we get

$$CB = B(I + AB)^{-1}. \tag{1.87}$$

From (1.85) and (1.87) we deduce that

$$(CA)B = B(AC) = B(I + AB)^{-1}A.$$

On the other hand, (1.84) and (1.86) imply that

$$(CA)B = B(AC) = I - C.$$

Consequently,

$$I - C = B(I + AB)^{-1}A,$$

and so

$$(I + BA)^{-1} = C = I - B(I + AB)^{-1}A.$$

◀

Exercise 1.9
Solve in $\mathscr{M}_n(\mathbb{K})$ the equation

$$2A + 3A^T = I. \tag{1.88}$$

■

Solution
Using the properties of the transpose, we recast (1.88) as

$$(2A + 3A^T)^T = I^T.$$

That is,

$$2A^T + 3A = I. \tag{1.89}$$

Multiplying Eq. (1.88) by 2 and Eq. (1.89) by -3 and adding the results, we obtain

$$5A = I.$$

Therefore, $A = \frac{1}{5}I$. ◄

Exercise 1.10
Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

and B be two matrices in $\mathscr{M}_2(\mathbb{K})$ such that

$$A^2B = BA^2 \qquad \text{and} \qquad a + d \neq 0.$$

Show that

$$AB = BA.$$

■

Solution
We have seen in Exercise 1.4 that if A is a matrix in $\mathscr{M}_2(\mathbb{K})$, then

$$A^2 - (a + d)A + (ad - bc)I_2 = \mathbf{0}.$$

Consequently, we can write A as

$$A = \frac{1}{(a + d)}[A^2 + (ad - bc)I_2].$$

Since the two matrices A^2 and I_2 commute with B, we have

$$\begin{aligned} AB &= \frac{1}{(a+d)}[A^2 + (ad-bc)I_2]B \\ &= \frac{1}{(a+d)}[A^2B + (ad-bc)I_2B] \\ &= \frac{1}{(a+d)}[BA^2 + (ad-bc)BI_2] \\ &= B[\frac{1}{(a+d)}[A^2 + (ad-bc)I_2]] \\ &= BA. \end{aligned}$$

Thus, A and B commute. ◀

Exercise 1.11 (Subgroup of $GL_3(\mathbb{R})$)
Let $(G, *)$ be a group and let H be a nonempty subset of G. Then H is a subgroup of G if

(H1) For all x, y in H, $x * y$ is in H.
(H2) If x is in H, then the inverse x' is in H.

Let a be a real number, and define the matrix $M(a)$ in $\mathscr{M}_3(\mathbb{R})$ as

$$M(a) = \begin{bmatrix} 1 & a & a \\ a & 1+\frac{a^2}{2} & \frac{a^2}{2} \\ -a & -\frac{a^2}{2} & 1-\frac{a^2}{2} \end{bmatrix}.$$

Next introduce the set

$$H = \{M(a) : \text{ where } a \text{ is in } \mathbb{R}\}$$

and the matrix

$$U = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}.$$

1. Show that $M(a) = I_3 + aU + a^2U^2$.
2. Prove that, for all a and b in $\mathbb{R}$,

$$M(a+b) = M(a)M(b). \tag{1.90}$$

3. Show that H is a subgroup in $GL(3, \mathbb{R})$ with respect to multiplication of matrices.
4. Find $(M(a))^k$, where k is an integer.

■

1

Solution

1. We compute

$$U^2 = UU = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix}.$$

Now, we write the matrix $M(a)$ as

$$M(a) = \begin{bmatrix} 1 & a & a \\ a & 1 + \frac{a^2}{2} & \frac{a^2}{2} \\ -a & -\frac{a^2}{2} & 1 - \frac{a^2}{2} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + a \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} + \frac{a^2}{2} \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix}.$$

Thus,

$$M(a) = I_3 + aU + \frac{a^2}{2}U^2.$$

2. By assertion (1) above, and since I_3 commute with U and U^2,

$$M(a)M(b) = \left(I_3 + aU + \frac{a^2}{2}U^2\right)\left(I_3 + bU + \frac{b^2}{2}U^2\right)$$

$$= I_3 + (a+b)U + \frac{(a+b)^2}{2}U^2 + \left(\frac{a^2 b}{2} + \frac{b^2 a}{2}\right)U^3 + \frac{a^2 b^2}{4}U^4.$$

We can easily check that

$$U^3 = U^2 U = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Also,

$$U^4 = U^3 U = \mathbf{0}.$$

Consequently,

$$M(a)M(b) = I_3 + (a+b)U + \frac{(a+b)^2}{2}U^2$$

$$= M(a+b).$$

We may also prove the above identity by a direct computation, using the form of the matrices $M(a)$ and $M(b)$.

3. It is clear that H is nonempty, since $M(0) = I_3$ is in H. Also, by (1.90), if $M(a)$ and $M(b)$ are in H, then the product $M(a)M(b)$ is also in H. In addition, using (1.90) once again, we have

$$M(a)M(-a) = M(a-a) = M(0) = I_3.$$

Thus, if $M(a)$ is in H, then the inverse $M^{-1}(a) = M(-a)$ is in H and so H is a subgroup of $GL(3,\mathbb{R})$.

4. First, let us assume that $k \geq 0$. We claim that (1.90) yields

$$\begin{aligned}(M(a))^k &= \underbrace{M(a)M(a)\cdots M(a)}_{k \text{ times}} \\ &= M(\underbrace{a+a+\cdots+a}_{k \text{ times}}) = M(ka). \end{aligned} \tag{1.91}$$

This can be verified by induction. It is clear that (1.91) is true for $k = 0$, $k = 1$, and $k = 2$. Next, assume that (1.91) holds for k and show that it also holds for $k+1$. We have, by (1.90),

$$\begin{aligned}(M(a))^{k+1} &= (M(a))^k M(a) \\ &= M(ka)M(a) \\ &= M(ka+a) \\ &= M((k+1)a).\end{aligned}$$

Therefore, (1.91) also hold for $k+1$. Consequently, for any integer $k \geq 0$,

$$(M(a))^k = M(ka).$$

Now, if $k \leq 0$, we have $k' = -k \geq 0$, and so we can write

$$\begin{aligned}(M(a))^k &= [(M(a))^{-1}]^{-k} \\ &= (M(-a))^{k'} \\ &= M(-k'a) \\ &= M(ka).\end{aligned}$$

Consequently, for any integer k,

$$(M(a))^k = M(ka).$$

◀

Exercise 1.12

Show that any matrix A in $\mathscr{M}_2(\mathbb{R})$, with $A \neq I_2$, satisfying $A^3 = I_2$ has trace equal to -1. ■

Solution

As we have seen in Exercise 1.4, if A is a matrix in $\mathscr{M}_2(\mathbb{K})$, then we have

$$A^2 - \mathrm{tr}(A)A + \det(A)I_2 = \mathbf{0}. \tag{1.92}$$

Multiplying (1.92) by A, we get

$$A^3 - \mathrm{tr}(A)A^2 + \det(A)A = \mathbf{0}.$$

Since $A^3 = I_2$, this yields

$$I_2 - \mathrm{tr}(A)\Big(\mathrm{tr}(A)A - \det(A)I_2\Big) + \det(A)A = \mathbf{0}, \tag{1.93}$$

where we have used again (1.92). Rearranging (1.93), we get

$$\Big(\det(A) - (\mathrm{tr}(A))^2\Big)A + \Big(1 + \mathrm{tr}(A)\det(A)\Big)I_2 = \mathbf{0}.$$

This gives, since $A \neq I_2$,

$$\det(A) = (\mathrm{tr}(A))^2 \qquad \text{and} \qquad 1 + \mathrm{tr}(A)\det(A) = 0.$$

Consequently,

$$(\mathrm{tr}(A))^3 = -1.$$

Since A is a real matrix, $\mathrm{tr}(A)$ is a real number and so

$$\mathrm{tr}(A) = -1.$$

◄

Determinants

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2.1 Introduction

As, indicated before, one of the main goals in linear algebra is to be able to determine whether a given square matrix is invertible or not, and if invertible, to find its inverse. In this chapter, we give a general criterion for the invertibility of square matrices. So, let us first recall the equation

$$ax = b, \tag{2.1}$$

and its solution, given by

$$x = \frac{b}{a} = a^{-1}b. \tag{2.2}$$

So, the solution in (2.2) is defined if and only if

$$a \neq 0. \tag{2.3}$$

Now, for a system of two equations and two unknowns, we have seen that the system

$$\begin{cases} ax + by = p, \\ cx + dy = q, \end{cases} \tag{2.4}$$

has a unique solution if and only if

$$ad - bc \neq 0. \tag{2.5}$$

Using matrix notation, then system (2.4) can be rewritten as

$$AX = \mathbf{b}, \tag{2.6}$$

where

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \qquad X = \begin{bmatrix} x \\ y \end{bmatrix}, \qquad \mathbf{b} = \begin{bmatrix} p \\ q \end{bmatrix}.$$

The number $ad-bc$ constructed from the entries of the matrix A is called the *determinant* of the 2×2 matrix A and is denoted by

$$\det(A) = ad - bc.$$

In analogy with this, and if we regard the constant a in (1.1) as a square matrix in $\mathscr{M}_1(\mathbb{K})$, then the number a in (2.3) is the determinant of the matrix $[a]$. As (2.3) and (2.5) show, the Eq. (2.1) and the system (2.4) have unique solutions, that is to say, the associated matrices are invertible, if and only if their determinants are not zero. So, the natural question is the following: *can we extend this condition for any square matrix A in* $\mathscr{M}_n(\mathbb{K})$*?* That is, can we show that *the matrix A in* $\mathscr{M}_n(\mathbb{K})$ *is invertible if and only if its determinant is not zero?* Before answering this question, we need to explain *how to find the determinant of a square matrix A in* $\mathscr{M}_n(\mathbb{K})$. A main goal in this chapter is to give answers to the above two questions.

2.2 Determinants by Cofactor Expansion

Since according to the above definitions, the determinant of the 1×1 matrix $[a]$ is a, we may write the determinant of the 2×2 matrix as

$$\det(A) = ad - bc = \det[a]\det[d] - \det[b]\det[c]. \tag{2.7}$$

Thus, we expressed the determinant of the 2×2 matrix A using determinants of 1×1 matrices. Now, we can define the determinant recursively, meaning that the definition of the determinant of an $n \times n$ matrix makes use of the determinant of $(n-1) \times (n-1)$ matrices.

So, we want to obtain an analogous formula of (2.7) for a matrix in $\mathscr{M}_n(\mathbb{K})$ for any $n \geq 1$. For this purpose, it is more convenient to rewrite the matrix A in $\mathscr{M}_2(\mathbb{K})$ as

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

and rewrite (2.7) as

$$\det(A) = \det[a_{11}]\det[a_{22}] - \det[a_{12}]\det[a_{21}]. \tag{2.8}$$

Now in order to find and expression for the determinant of a matrix A in $\mathscr{M}_3(\mathbb{K})$, we consider the system of linear equations

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 = b_1, \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 = b_2, \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 = b_3. \end{cases} \tag{2.9}$$

We leave it to the reader to check that the above system has a unique solution if and only if

$$a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{21}(a_{12}a_{33} - a_{13}a_{32}) + a_{31}(a_{12}a_{23} - a_{13}a_{22}) \neq 0. \tag{2.10}$$

We call the number in (2.10) the determinant of the matrix

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

The left-hand side in (2.10) can be rewritten as

$$\begin{aligned} &a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{13}a_{32}) + a_{31}(a_{12}a_{23} - a_{13}a_{22}) \\ &= a_{11} \det \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix} - a_{12} \det \begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix} + a_{31} \det \begin{bmatrix} a_{12} & a_{22} \\ a_{13} & a_{23} \end{bmatrix}. \end{aligned}$$

So, we have written the determinant of a 3×3 matrix in terms of determinants of 2×2 matrices.

If, we denote

$$M_{11} = \det \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}, \quad M_{21} = \det \begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}, \quad M_{31} = \det \begin{bmatrix} a_{12} & a_{22} \\ a_{13} & a_{23} \end{bmatrix},$$

we can rewrite the determinant of the 3×3 matrix as

$$\det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}. \tag{2.11}$$

We observe that M_{11} is obtained by removing the first row and the first column from the matrix A and computing the determinant of the resulting 2×2 matrix. Similarly, we can find M_{12} by removing the first row and the second column and computing the determinant of the remaining matrix, and so on. These M_{ij} are called the *minors* of the matrix A.

2

Definition 2.2.1 (Minor)

Let $A = (a_{ij})$, $1 \leq i, j \leq n$, be a matrix in $\mathscr{M}_n(\mathbb{K})$, $n \geq 2$. For any $1 \leq i, j \leq n$, the determinant of the matrix in $\mathscr{M}_{n-1}(\mathbb{K})$ obtained by deleting row i and column j from A is called a *minor* and is denoted by M_{ij}, and we write

$$M_{ij} = \det \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1(j-1)} & a_{1(j+1)} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2(j-1)} & a_{2(j+1)} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{(i-1)1} & a_{(i-1)2} & \dots & a_{(i-1)(j-1)} & a_{(i-1)(j+1)} & \dots & a_{(i-1)n} \\ a_{(i+1)1} & a_{(i+1)2} & \dots & a_{(i+1)(j-1)} & a_{(i+1)(j+1)} & \dots & a_{(i+1)n} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{n(j-1)} & a_{n(j+1)} & \dots & a_{nn} \end{bmatrix}. \tag{2.12}$$

Example 2.1

Consider the matrix

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & 2 \\ 0 & 5 & 1 \end{bmatrix}.$$

Then,

$$M_{11} = \det \begin{bmatrix} 1 & 2 \\ 5 & 1 \end{bmatrix} = 1 \cdot 1 - 2 \cdot 5 = -9, \quad M_{12} = \det \begin{bmatrix} 2 & 2 \\ 0 & 1 \end{bmatrix} = 2,$$

$$M_{13} = \det \begin{bmatrix} 2 & 1 \\ 0 & 5 \end{bmatrix} = 10, \quad M_{21} = \det \begin{bmatrix} 0 & 3 \\ 5 & 1 \end{bmatrix} = -15,$$

$$M_{22} = \det \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} = 1, \quad M_{23} = \det \begin{bmatrix} 1 & 0 \\ 0 & 5 \end{bmatrix} = 5,$$

$$M_{31} = \det \begin{bmatrix} 0 & 3 \\ 1 & 2 \end{bmatrix} = -3, \quad M_{32} = \det \begin{bmatrix} 1 & 3 \\ 2 & 2 \end{bmatrix} = -4,$$

$$M_{33} = \det \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} = 1.$$

■

In (2.11), we saw that the second term has a negative sign, while the first and the last terms have a positive sign. So, to avoid the negative signs and to be able to write an easy formula for the determinant, we define what we call the *cofactor*.

Definition 2.2.2 (Cofactor)

Let M_{ij}, $1 \leq i,j \leq n$ be the minors associated to the square matrix $A = (a_{ij})$, $1 \leq i,j \leq n$. Then, we define the *cofactor* C_{ij} as

$$C_{ij} = (-1)^{i+j} M_{ij}. \tag{2.13}$$

Example 2.2

In Example 2.1, we have, for instance,

$$C_{11} = (-1)^{1+1} M_{11} = M_{11} = -9$$

and

$$C_{23} = (-1)^{2+3} M_{23} = -M_{23} = -5.$$

■

Using the above definition, we can rewrite (2.11) as

$$\begin{aligned} \det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \\ &= a_{21}C_{21} + a_{22}C_{22} + a_{23}C_{2}3 \\ &= a_{31}C_{31} + a_{32}C_{32} + a_{33}C_{33}. \end{aligned}$$

We can also write the above determinant using the columns, as follows:

$$\begin{aligned} \det \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} &= a_{11}C_{11} + a_{21}C_{21} + a_{31}C_{31} \\ &= a_{12}C_{12} + a_{22}C_{22} + a_{32}C_{32} \\ &= a_{13}C_{13} + a_{23}C_{23} + a_{33}C_{33}. \end{aligned}$$

Now, the above formulas for the determinant can be generalized to any square matrix in $\mathscr{M}_n(\mathbb{K})$ as follows.

2

Definition 2.2.3 (Determinant)

Let $A = (a_{ij})$, $1 \leq i, j \leq n$ be a matrix in $\mathscr{M}_n(\mathbb{K})$, $n \geq 2$. Then, we define the *determinant* of A, using the rows of the matrix A, as

$$\det(A) = a_{i1}C_{i1} + a_{i2}C_{i2} + \cdots + a_{in}C_{in}, \tag{2.14}$$

for any fixed i. Or, using the columns of the matrix A, as

$$\det(A) = a_{1j}C_{1j} + a_{2j}C_{2j} + \cdots + a_{nj}C_{nj}, \tag{2.15}$$

for any fixed j. The above two formulas are called the *cofactor expansion* of the determinant.

Example 2.3

Find the determinant of the matrix A given by

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & 2 \\ 0 & 5 & 1 \end{bmatrix}.$$

■

Solution

To calculate the determinant of A, we need first to choose one row or one column and make use of Definition 2.2.3 accordingly. The smart choice is to take the row or the column that contain the largest number of zeros. In this case, we may choose the first row and use it to do the cofactor expansion. So, we use (2.15) with $i = 1$ and write

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}.$$

Since $a_{12} = 0$, this becomes

$$\begin{aligned} \det(A) &= a_{11}C_{11} + a_{13}C_{13} \\ &= C_{11} + 3C_{13}. \end{aligned} \tag{2.16}$$

We have computed the minors of the above matrix A in Example 2.1, so we have

$$C_{11} = (-1)^{1+1}M_{11} = M_{11} = -9, \qquad C_{13} = M_{13} = 10.$$

Inserting these values in (2.16), we get

$$\det(A) = -9 + 3 \cdot 10 = 21.$$

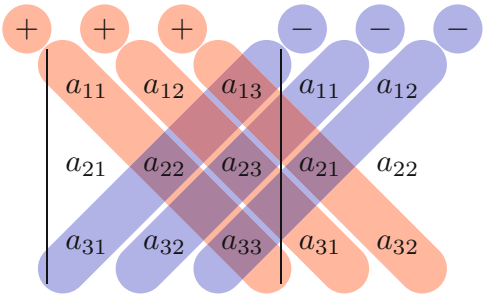


Fig. 2.1 To evaluate the 3×3 determinant, we take the products along the main diagonal and the lines parallel to it with a $(+)$ sign, and the products along the second diagonal and the lines parallel to it wit a $(-)$ sing

We can obtain the above result using the trick in Fig. 2.1 as follows:

$$\det(A) = 1 \cdot 1 \cdot 1 + 0 \cdot 2 \cdot 0 + 3 \cdot 2 \cdot 5 - 3 \cdot 1 \cdot 0 - 1 \cdot 2 \cdot 5 - 0 \cdot 2 \cdot 1 = 31 - 10 = 21.$$

◀

Example 2.4

Calculate the determinant by a cofactor expansion for

$$A = \begin{bmatrix} 2 & -1 & 5 \\ 1 & -4 & 3 \\ 4 & 2 & 0 \end{bmatrix}.$$

■

Solution

We may calculate the determinant of A using the third row:

$$\begin{aligned} \det(A) &= a_{13}C_{13} + a_{23}C_{23} + a_{33}C_{33} \\ &= 4C_{13} + 2C_{23}. \end{aligned}$$

Now, we have

$$C_{13} = (-1)^{1+3}M_{13} = \det \begin{bmatrix} -1 & 5 \\ -4 & 3 \end{bmatrix} = 17$$

and

$$C_{23} = (-1)^{2+3}M_{23} = -\det \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix} = -1.$$

Consequently,

$$\det(A) = 4 \cdot 17 + 2 \cdot (-1) = 66.$$

◄

2.3 Properties of the Determinants

In this section, we give the determinant of some particular matrices and establish some properties of the determinant.

It is clear from Definition 2.2.3 that if we use the cofactor expansion along one of the rows or along one of the columns of the matrix A, then we obtain the same value for the determinant. This implies that, A and A^T have the same determinant and we state this in the following theorem.

Theorem 2.3.1 (Determinant of the Transpose)
Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$. Then

$$\det(A) = \det(A^T). \tag{2.17}$$

In the following theorems, we calculate the determinant of diagonal and triangular matrices.

Theorem 2.3.2 (Determinant of a Diagonal Matrix)
Let $D = (d_{ij})$, $1 \leq i \leq n$, $1 \leq j \leq n$ be a diagonal matrix in $\mathscr{M}_n(\mathbb{K})$. Then the determinant of D is the product of the entries of the main diagonal. That is,

$$\det(D) = d_{11}d_{22}\cdots d_{nn}. \tag{2.18}$$

Proof
The proof of (2.18) can be done by induction. We first take $n = 2$ and let D_2 be the matrix

$$D_2 = \begin{bmatrix} d_{11} & 0 \\ 0 & d_{22} \end{bmatrix}.$$

Clearly,

$$\det(D_2) = d_{11}d_{22}.$$

Therefore, (2.18) holds for $n = 2$. Now, assume that (2.18) holds for $n - 1$, that is

$$\det(D_{n-1}) = \det \begin{bmatrix} d_{11} & 0 & 0 & \cdots & 0 \\ 0 & d_{22} & 0 & \cdots & 0 \\ 0 & 0 & d_{33} & \cdots & 0 \\ \vdots & & & \ddots & \\ 0 & 0 & 0 & \cdots & d_{n-1} \end{bmatrix} = d_{11}d_{22}\cdots d_{(n-1)(n-1)},$$

and let us show that (2.18) holds for n. Choosing $i = n$ and applying formula (2.14), we get

$$\begin{aligned} \det(D) &= d_{n1}C_{n1} + d_{n2}C_{n2} + \cdots + d_{nn}C_{nn} \\ &= d_{nn}C_{nn} \\ &= d_{nn}(-1)^{n+n}M_{nn} \\ &= d_{nn}\det(D_{n-1}) \\ &= d_{nn}d_{11}d_{22}\cdots d_{(n-1)(n-1)}. \end{aligned} \tag{2.19}$$

Thus, (2.18) holds for n as claimed. □

Example 2.5

Let

$$A = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

Then

$$\det(A) = (-1)\cdot 3 \cdot 5 = -15.$$

■

Remark 2.3.3 We deduce immediately from Theorem 2.3.2 that if I_n is the identity matrix in $\mathscr{M}_n(\mathbb{K})$, then

$$\det(I_n) = 1.$$

Theorem 2.3.4 (Determinant of a Triangular Matrix)
Let $A = (a_{ij})$, $1 \leq i,j \leq n$ be a triangular matrix in $\mathscr{M}_n(\mathbb{K})$. Then the determinant of A is the product of the entries of the main diagonal. That is

$$\det(A) = a_{11}a_{22}\cdots a_{nn}. \tag{2.20}$$

2

Proof

We prove this statement for upper triangular matrices; the same argument works for lower triangular matrices. So, let

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ 0 & a_{22} & a_{23} & \cdots & a_{2n} \\ 0 & 0 & a_{33} & \cdots & a_{3n} \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix}$$

As in the case of diagonal matrices, we proceed by induction. For $n = 2$, we take

$$A_2 = \begin{bmatrix} a_{11} & a_{21} \\ 0 & a_{22} \end{bmatrix},$$

and then

$$\det(A_2) = a_{11} \cdot a_{22}.$$

Now, assume that the determinant of the matrix

$$A_{n-1} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1(n-1)} \\ 0 & a_{22} & a_{23} & \cdots & a_{2(n-1)} \\ 0 & 0 & a_{33} & \cdots & a_{3(n-1)} \\ \vdots & & & \ddots & \\ 0 & 0 & 0 & \cdots & a_{(n_1)(n-1)} \end{bmatrix}$$

is equal to

$$\det(A_{n-1}) = a_{11}a_{22} \cdots a_{(n-1)(n-1)}.$$

Letting $i = n$ in (2.14), we get

$$\begin{aligned} \det A = a_{nn}C_{nn} &= a_{nn}(-1)^{n+n}M_{nn} \\ &= a_{nn} \det(A_{n-1}) \\ &= a_{nn}a_{11}a_{22} \cdots a_{(n-1)(n-1)}. \end{aligned}$$

This completes the proof of Theorem 2.3.4. □

Example 2.6
Find the determinant of the matrices

$$A = \begin{bmatrix} 1 & 3 & -1 \\ 0 & 2 & 0 \\ 0 & 0 & -4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 3 & -1 & 0 & 0 \\ 0 & 5 & 3 & 0 \\ 3 & 1 & 6 & -3 \end{bmatrix}.$$

■

Solution
Since A is upper triangular and B is lower triangular using (2.20) we get

$$\det(A) = 1 \cdot 2 \cdot (-4) = -8$$

and

$$\det(B) = 1 \cdot (-1) \cdot 3 \cdot (-3) = 9.$$

◀

Remark 2.3.5 We have seen in Theorems 1.2.11 and 1.2.13 that diagonal and triangular matrices are invertible if and only if all the entries of the main diagonal are not zero. That is, if the product of those entries is not zero, which is equivalent to the fact that the determinant is not zero.

2.4 Evaluating Determinants by Row Reduction

As we have seen above, it is easy to compute the determinant of diagonal and triangular matrices. So, if we can apply the row operations to transform a square matrix into a triangular matrix (which is easier than transforming it to a diagonal one), then the determinant of the new matrix can be calculated by just taking the product of the entries of the main diagonal. Therefore, the question is: *how do the row operations affect the determinant?* In this section, we answer this question and compute some determinants using the row reduction method.

We begin by a fundamental theorem that will lead us to an efficient procedure for evaluating the determinant of square matrices.

Theorem 2.4.1
Let $A = (a_{ij})$, $1 \leq i, j \leq n$, be a square matrix in $\mathscr{M}_n(\mathbb{K})$. If A has a row of zeros or a column of zeros, then

$$\det(A) = 0.$$

2

Proof

Suppose that there exists $1 \leq i_0 \leq n$, such that $a_{i_0 j} = 0$ for all $1 \leq j \leq n$. Then, using (2.14) for $i = i_0$, we deduce that

$$\det(A) = a_{i_0 1} C_{i_0 1} + a_{i_0 2} C_{i_0 2} + \cdots + a_{i_0 n} C_{i_0 n} = 0.$$

Similarly, if there exists $1 \leq j_0 \leq n$, such that $a_{i j_0} = 0$ for all $1 \leq i \leq n$, then using (2.15) for $j = j_0$, we get

$$\det(A) = a_{1 j_0} C_{1 j_0} + a_{2 j_0} C_{2 j_0} + \cdots + a_{n j_0} C_{n j_0} = 0.$$

This finishes the proof of Theorem 2.4.1. □

Now let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

be a matrix in $\mathscr{M}_2(\mathbb{K})$ and let B_1 be the matrix that results by interchanging the two rows and B_2 be the matrix that results by interchanging the two columns; that is,

$$B_1 = \begin{bmatrix} c & d \\ a & b \end{bmatrix} \quad \text{and} \quad B_2 = \begin{bmatrix} b & a \\ d & c \end{bmatrix}.$$

Then,

$$\det(B_1) = \det(B_2) = cb - ad = -(ad - bc) = -\det(A).$$

Next, let B_3 the matrix that results by multiplying one row (the first row for instance) by a scalar k and B_4 be the matrix that results from multiplying the first column by a scalar k; that is

$$B_3 = \begin{bmatrix} ka & kb \\ c & d \end{bmatrix} \quad \text{and} \quad B_4 = \begin{bmatrix} ka & b \\ kc & d \end{bmatrix}.$$

Then,

$$\det(B_3) = \det(B_4) = k(ad - bc) = k\det(A).$$

Finally for this case, let B_5 be the matrix that results by adding a multiple of one row of the matrix A to another row and B_6 be the matrix that results by adding a multiple of one column of A to another column, that is, for instance,

$$B_5 = \begin{bmatrix} a + kc & b + kd \\ c & d \end{bmatrix} \quad \text{and} \quad B_6 = \begin{bmatrix} a + kb & b \\ c + kd & d \end{bmatrix}.$$

Then,

$$\det(B_5) = \det(B_6) = ad - bc = \det(A).$$

The ways the above row operations affect the value of the determinant remain valid for any square matrix in $\mathscr{M}_n(\mathbb{K})$, $n \geq 1$.

Theorem 2.4.2

Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$.

1. *If B is the matrix that obtained by interchanging two rows or two columns of A, then*

$$\det(B) = -\det(A). \tag{2.21}$$

2. *If B is the matrix that obtained by multiplying a single row or a single column of A by a scalar k, then*

$$\det(B) = k\det(A). \tag{2.22}$$

3. *If B is the matrix that obtained by multiplying one row of A by a scalar k and adding it to another row, or from multiplying one column of A by a scalar k and adding it to another column, then*

$$\det(B) = \det(A). \tag{2.23}$$

Proof

Let $A = (a_{ij})$ and $B = (b_{ij})$, $1 \leq i, j \leq n$.

1. Without loss of generality, we can consider for instance the case where B is the matrix obtained from A by interchanging the first two rows. Let M_{ij} and M'_{ij}, $1 \leq i, j \leq n$, denote the minors of A and B and C_{ij} and C'_{ij} denote the cofactors of A and B, respectively. Then, by using the cofactor expansion through the first row of A, we have

$$\begin{aligned}\det(A) &= a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n} \\ &= a_{11}M_{11} - a_{12}M_{12} + \cdots \pm a_{1n}M_{1n}.\end{aligned}$$

On the other hand, we have, by using the second row of B,

$$\begin{aligned}\det(B) &= b_{21}C'_{21} + b_{21}C'_{21} + \cdots + b_{2n}C'_{2n} \\ &= a_{11}C'_{21} + a_{12}C'_{22} + \cdots + a_{1n}C'_{2n}, \qquad \text{since} \quad b_{2j} = a_{1j}, \quad 1 \leq j \leq n, \\ &= -a_{11}M'_{21} + a_{12}M'_{22} - \cdots \pm a_{1n}M'_{2n}\end{aligned}$$

$$= -a_{11}M_{11} + a_{12}M_{12} - \cdots \pm a_{1n}M_{1n}, \qquad \text{since} \quad M'_{2j} = M_{1j}, \quad 1 \le j \le n,$$
$$= -\det(A).$$

2. Assume that the matrix B is obtained by multiplying a row i_0 of A by a nonzero constant k. Then, using the cofactor expansion along the row i_0, we may calculate $\det(A)$ and $\det(B)$ using C_{i_0j} and C'_{i_0j}, $1 \le j \le n$, to denote the cofactors of a_{i_0j} and b_{i_0j}, respectively. We have

$$\det(A) = a_{i_0 1}C_{i_0 1} + a_{i_0 2}C_{i_0 2} + \cdots + a_{i_0 n}C_{i_0 n}$$

and

$$\det(B) = b_{i_0 1}C'_{i_0 1} + b_{i_0 2}C'_{i_0 2} + \cdots + b_{i_0 n}C'_{i_0 n}.$$

Since

$$a_{ij} = b_{ij}, \quad \text{for} \quad i \neq i_0,$$

we have

$$C_{i_0j} = C'_{i_0j}, \quad 1 \le j \le n.$$

We also have $b_{i_0j} = ka_{i_0j}$. Therefore,

$$\begin{aligned}\det(B) &= b_{i_0 1}C'_{i_0 1} + b_{i_0 2}C'_{i_0 2} + \cdots + b_{i_0 n}C'_{i_0 n} \\ &= (ka)_{i_0 1}C_{i_0 1} + (ka)_{i_0 2}C_{i_0 2} + \cdots + (ka)_{i_0 n}C_{i_0 n} \\ &= k(a_{i_0 1}C_{i_0 1} + a_{i_0 2}C_{i_0 2} + \cdots + a_{i_0 n}C_{i_0 n}) \\ &= k\det(A).\end{aligned}$$

Similarly, we can show (3). We leave this as an exercise to the reader. Also, the same argument can be applied if we use columns instead of rows. □

Elementary Matrices

If the matrix A in Theorem 2.4.2 is the identity matrix, then the matrix B is called an *elementary matrix*.

Definition 2.4.1 (Elementary Matrix)

A matrix E is called an *elementary matrix* if it can be obtained from the identity matrix by performing a single elementary row (or column) operation.

Example 2.7
The following matrices are elementary matrices:

$$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}}_{\text{Interchange the second and the third row of } I_3}, \quad \underbrace{\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_{\text{Multiply the first row of } I_4 \text{ by 2}}, \quad \underbrace{\begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{\text{Add 3 times the third row of } I_3 \text{ to the first row}}.$$

■

Our goal now is to show that the row operations on a matrix A are equivalent of multiplying the matrix A from the left by a finite sequence of elementary matrices, and similarly the column operations on A are equivalent of multiplying the matrix A from the right by a finite sequence of elementary matrices. To see this first on an example, consider the matrix

$$A = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 6 & 0 \\ 5 & -1 & 7 \end{bmatrix}$$

and let B be the matrix obtained from A by interchanging the first and the second rows:

$$B = \begin{bmatrix} 4 & 6 & 0 \\ 1 & -2 & 3 \\ 5 & -1 & 7 \end{bmatrix}.$$

Passing from A to B is equivalent to multiply A by the elementary matrix

$$E_1 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

that is

$$B = E_1 A.$$

Next, let C be the matrix obtained by multiplying the second column of A by 2:

$$C = \begin{bmatrix} 1 & -4 & 3 \\ 4 & 12 & 0 \\ 5 & -2 & 7 \end{bmatrix}.$$

2

Then,

$$C = AE_2,$$

with

$$E_2 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Remark 2.4.3 It is not hard to see that every elementary matrix is invertible and its inverse is also an elementary matrix.

Now, from the definition of elementary matrices, we can easily deduce the following:

- If E_1 is the elementary matrix obtained by interchanging two rows or two columns of the identity matrix, then $\det(E_1) = -1$.
- If E_2 is the elementary matrix obtained by multiplying a single row or a single column of the identity matrix by a scalar k, then $\det(E_2) = k$.
- If E_3 is the elementary matrix obtained by adding a multiple of one row (respectively, one column) of the identity matrix to another row (respectively, another column), then $\det(E_3) = 1$.

The following theorem is also very useful.

Theorem 2.4.4
Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$. If A contains two proportional rows or two proportional columns, then

$$\det(A) = 0.$$

Proof
Let $A = (a_{ij})$, $1 \leq i,j \leq n$, be a matrix in $\mathscr{M}_n(\mathbb{K})$. Assume that there exist $1 \leq i_0, i_1 \leq n$ such that the row r_{i_0} and the row r_{i_1} satisfy $r_{i_1} = kr_{i_0}$.

Let B be the matrix that obtained by adding r_{i_1} to $-kr_{i_0}$ in A. Then, by using (2.23), we have

$$\det(B) = \det(A).$$

But all the entries of the resulting row in B are zero. Thus, Theorem 2.4.1 implies that $\det(B) = 0$ and therefore, $\det(A) = 0$. The same method can be applied if two columns of A are proportional. □

Example 2.8
Consider the matrices

$$A = \begin{bmatrix} 1 & 3 & -1 \\ 0 & 2 & 0 \\ 2 & 6 & -2 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 3 & 9 \\ 0 & 2 & 6 \\ 0 & -1 & -3 \end{bmatrix}.$$

Since the first row and the third row of A are proportional ($r_3 = 2r_1$), $\det(A) = 0$. Similarly, since the second column and the third column of B are proportional, $\det(B) = 0$. ■

Example 2.9
Use the row reduction method to calculate the determinant of the matrix

$$A = \begin{bmatrix} 0 & 1 & 5 \\ 3 & -6 & 9 \\ 2 & 6 & 1 \end{bmatrix}.$$

■

Solution
Since the determinant of a triangular matrix is the product of the entries of the main diagonal, we apply the necessary row operations in order to get a triangular matrix. First, let A_1 be the matrix that obtained by interchanging r_1 (the first row) and r_2 in A, that is

$$A_1 = \begin{bmatrix} 3 & -6 & 9 \\ 0 & 1 & 5 \\ 2 & 6 & 1 \end{bmatrix}.$$

Theorem 2.4.2 leads to

$$\det(A_1) = -\det(A).$$

Next, let A_2 be the matrix obtained by multiplying the first row in A_1 by $k = 1/3$, i.e.,

$$A_2 = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 2 & 6 & 1 \end{bmatrix}.$$

By Theorem 2.4.2,

$$\det(A_2) = \frac{1}{3}\det(A_1).$$

2

Let A_3 be the matrix obtained by replacing r_3 in A_2 by $r_3 - 2r_1$, i.e.,

$$A_3 = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 10 & -5 \end{bmatrix}.$$

Then again by Theorem 2.4.2,

$$\det(A_3) = \det(A_2).$$

Finally, let A_4 be the matrix obtained by replacing r_3 in A_3 by $r_3 - 10r_2$, i.e.,

$$A_4 = \begin{bmatrix} 1 & -2 & 3 \\ 0 & 1 & 5 \\ 0 & 0 & -55 \end{bmatrix}.$$

Then

$$\det(A_4) = \det(A_3) = -55,$$

since A_4 is a triangular matrix.

Now, using the above formulas, we have

$$\det(A) = -\det(A_1) = -3\det(A_2) = -3\det(A_3) = -3\det(A_4) = -3 \cdot (-55) = 165.$$

◀

Theorem 2.4.5

Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$ and k be a scalar. Then, for $B = kA$, we have

$$\det(B) = k^n \det(A). \tag{2.24}$$

Proof

Let $A = (a_{ij})$, $1 \leq i,j \leq n$. Then the matrix B is given by $B = (ka_{ij})$, $1 \leq i,j \leq n$. So, to get B we need to do n row operations. Let $A_0 = A$, $A_n = B$ and A_i, $1 \leq i \leq n$ be the matrix obtained by multiplying the row r_i of the matrix A_{i-1} by k. Then, applying Theorem 2.4.2, we get

$$\det(A_i) = k\det(A_{i-1})$$

and therefore

$$\det(B) = \det(A_n) = k^n \det(A_0) = k^n \det(A).$$

This finishes the proof of Theorem 2.4.5. □

Theorem 2.4.6 (The Determinant is Not Linear)
Let A and B be two matrices in $\mathscr{M}_n(\mathbb{K})$, $n \geq 2$. Then in general

$$\det(A + B) \neq \det(A) + \det(B). \tag{2.25}$$

Proof
To show (2.25), we provide a counterexample. Thus, consider the two matrices

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 3 & 0 \\ 1 & -2 \end{bmatrix}.$$

Then,

$$A + B = \begin{bmatrix} 4 & 2 \\ 1 & 1 \end{bmatrix}.$$

We have

$$\det(A) = 3, \quad \det(B) = -6, \quad \text{and} \quad \det(A + B) = 2.$$

so,

$$2 = \det(A + B) \neq \det(A) + \det(B) = -3.$$

□

A very important property of the determinant is the following multiplicative property.

Theorem 2.4.7 (Multiplicativity)
Let A and B be two matrices in $\mathscr{M}_n(\mathbb{K})$. Then

$$\det(AB) = \det(A) \det(B). \tag{2.26}$$

2

Proof

If A is a singular matrix, then $\det(A)$ and $\det(AB)$ are both zero (Theorem 2.4.8). Hence, (2.26) holds. So, we can assume that A is invertible. Then, A can be row reduced (using the Gauss–Jordan elimination method in ▶ Sect. 1.3.1) to the identity matrix. That is, we can find a finite sequence of elementary matrices $E_1, E_2, \ldots, E_\ell$ such that

$$E_1 E_2 \cdots E_\ell A = I_n. \tag{2.27}$$

Hence,

$$A = E_\ell^{-1} \cdots E_2^{-1} E_1^{-1} I_n$$

and E_i^{-1}, $1 \leq i \leq \ell$ are elementary matrices.

Since

$$AB = E_\ell^{-1} \cdots E_2^{-1} E_1^{-1} B,$$

the proof of (2.26) is reduced to show that for any elementary matrix E and any square matrix M in $\mathscr{M}_n(\mathbb{K})$, we have

$$\det(EM) = \det(E)\det(M). \tag{2.28}$$

It is clear that (2.28) is satisfied, since (see Theorem 2.4.2):

- If E represents a row exchange, the $\det(E) = -1$ and then $\det(EM) = -\det(M)$, since the product EM is equivalent to exchange two rows in M.
- If E represents the multiplication of the ith row by a nonzero constant k, then $\det(E) = k$ and $\det(EM) = k\det(M)$.
- If E represents adding k times row j to row i, then $\det(E) = 1$ and $\det(EM) = \det(M)$.

Consequently,

$$\det(AB) = \det(E_\ell^{-1} \cdots E_2^{-1} E_1^{-1} B) = \det(E_\ell^{-1}) \det(E_{\ell-1}^{-1} \cdots E_2^{-1} E_1^{-1} B)$$

since $E = E_\ell^{-1}$ is an elementary matrix and we may take $M = E_{\ell-1}^{-1} \cdots E_2^{-1} E_1^{-1} B$. We can continue the process for $E = E_{\ell-1}^{-1}$ and $M = E_{\ell-2}^{-1} \cdots E_2^{-1} E_1^{-1} B$, eventually obtaining

$$\begin{aligned} \det(AB) &= \det(E_\ell^{-1}) \det(E_{\ell-1}^{-1}) \cdots \det(E_1^{-1}) \det(B) \\ &= \det\left(E_\ell^{-1} E_{\ell-1}^{-1}\right) \det(E_{\ell-2}^{-1}) \cdots \det(E_1^{-1}) \det(I_n) \det(B), \end{aligned}$$

by using (2.28) with $E = E_\ell^{-1}$ and $M = E_{\ell-1}^{-1}$ and the fact that $\det(I_n) = 1$. Finally, we arrive at

$$\begin{aligned} \det(AB) &= \det\left(E_\ell^{-1} E_{\ell-1}^{-1} \cdots E_1^{-1} I_n\right) \det(B) \\ &= \det(A)\det(B). \end{aligned}$$

This ends the proof of Theorem 2.4.7. □

Example 2.10
Consider the matrices

$$A = \begin{bmatrix} 3 & 0 \\ 1 & -1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 2 & 0 \\ 4 & -3 \end{bmatrix}.$$

Then,

$$AB = \begin{bmatrix} 6 & 0 \\ -2 & 3 \end{bmatrix}.$$

Hence,

$$\det(A) = -3, \quad \det(B) = -6,$$

and

$$\det(AB) = 18 = \det(A)\det(B).$$

■

2.4.1 Determinant Test for Invertibility

We have seen that Eq. (2.1) has a unique solution if and only if

$$a \neq 0, \qquad \text{or} \qquad \det[a] = a \neq 0.$$

Similarly, we have shown that system (2.4) has unique solution if and only if

$$ad - bc \neq 0, \qquad \text{or} \qquad \det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc \neq 0.$$

This is equivalent to say that the inverse of the above matrix exists if and only its determinant is not zero. In fact, this is the case for any matrix A in $\mathscr{M}_n(\mathbb{K})$.

Theorem 2.4.8
Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$. Then, A is invertible if and only if

$$\det(A) \neq 0.$$

2

Proof

First assume that A is invertible, and A^{-1} is its inverse. Then we have

$$AA^{-1} = A^{-1}A = I_n.$$

Hence, applying Theorem 2.4.7, we get

$$\det(AA^{-1}) = \det(A)\det(A^{-1}) = \det(I_n) = 1.$$

This shows that (since $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$) $\det(A) \neq 0$.

Conversely, assume that $\det(A) \neq 0$. Then using the row operation method (or the Gauss–Jordan elimination method), we write A as in (2.27):

$$E_1E_2\cdots E_\ell A = I_n, \tag{2.29}$$

and so as before, that

$$A = E_\ell^{-1}\cdots E_2^{-1}E_1^{-1}$$

and E_i^{-1}, $1 \leq i \leq \ell$, is the elementary matrix corresponding to the ith row operation applied by the Gauss–Jordan elimination algorithm.

Now, denoting

$$B = E_1E_2\cdots E_\ell,$$

we get

$$AB = BA = I_n.$$

Hence, A is invertible and $A^{-1} = B$. □

Now, if A is an invertible matrix, then *what is the relationship between the determinant of A and the determinant of A^{-1}?* This is answered in the following theorem.

Theorem 2.4.9 (Determinant of the Inverse)
Let A be an invertible matrix in $\mathscr{M}_n(\mathbb{K})$. Then,

$$\det(A^{-1}) = \frac{1}{\det(A)}. \tag{2.30}$$

Proof

Since A is invertible, Theorem 2.4.8 implies that $\det(A) \neq 0$. Writing the invertibility relation

$$AA^{-1} = A^{-1}A = I,$$

taking the determinants, and using Theorem 2.4.7, we obtain

$$\det(A^{-1}A) = \det(A^{-1})\det(A) = \det(I) = 1.$$

Hence,

$$\det(A^{-1}) = \frac{1}{\det(A)},$$

as claimed. □

Example 2.11

Consider the matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

We have seen in Theorem 1.2.4 that if A is invertible, then

$$A^{-1} = \frac{1}{ad - bc}\begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Now, we have by using (2.24),

$$\det(A^{-1}) = \frac{1}{(ad-bc)^2}\det\begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \frac{ad-bc}{(ad-bc)^2} = \frac{1}{ad-bc} = \frac{1}{\det(A)}.$$

■

2.5 The Adjoint of a Square Matrix

We have seen in Theorem 2.4.8 that the inverse of A exists if and only $\det(A) \neq 0$. Since our ultimate goal is to compute A^{-1}, we may ask *whether there is a way to compute* A^{-1} *by using the determinant?* To answer this question, let us consider a matrix A in $\mathscr{M}_2(\mathbb{K})$,

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

Recall again that the inverse A^{-1} is given by

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}. \tag{2.31}$$

We want now to explain the relationship between A and the matrix

$$B = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

It can be easily seen that the cofactors of A are

$$C_{11} = d, \quad C_{12} = -c, \quad C_{21} = -b, \quad C_{22} = a.$$

If we form the matrix

$$C = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix} = \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}.$$

Therefore, we can easily seen that

$$B = C^T.$$

Consequently, formula (2.31) can be rewritten as

$$A^{-1} = \frac{1}{\det(A)} C^T. \tag{2.32}$$

The matrix C is called the *cofactor matrix* of A or *matrix of cofactors* of A and C^T is called the *adjoint* of A and we may generalize this idea and give the following definition.

Definition 2.5.1 (Adjoint of a Matrix)

Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$ and C_{ij}, $1 \leq i, j \leq n$, be the cofactors of A, the matrix C defined by

$$C = \begin{bmatrix} C_{11} & C_{12} & \dots & C_{1n} \\ C_{21} & C_{22} & \dots & C_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ C_{n1} & C_{n2} & \dots & C_{nn} \end{bmatrix}$$

is called the *cofactor matrix* or *matrix of cofactors* of A and the transpose of C is called the *adjoint* of A, and we write

$$\operatorname{adj}(A) = C^T.$$

Example 2.12
Find the adjoint of the matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 3 & 0 \\ 1 & 0 & -2 \end{bmatrix}.$$

■

Solution
We compute the cofactors of A as

$$C_{11} = \det \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix} = -6, \quad C_{12} = -\det \begin{bmatrix} -1 & 0 \\ 1 & -2 \end{bmatrix} = -2,$$

and similarly,

$$C_{13} = -3, \quad C_{21} = 0, \quad C_{22} = -4, \quad C_{23} = 0,$$
$$C_{31} = -6, \quad C_{32} = -2, \quad C_{33} = 3.$$

Consequently, the cofactor matrix of A is

$$C = \begin{bmatrix} -6 & -2 & -3 \\ 0 & -4 & 0 \\ -6 & -2 & 3 \end{bmatrix},$$

and then

$$\text{adj}(A) = C^T = \begin{bmatrix} -6 & 0 & -6 \\ -2 & -4 & -2 \\ -3 & 0 & 3 \end{bmatrix}.$$

◀

Now, as in the case of the inverse of a 2×2 matrix considered in (2.32), we have the following theorem.

Theorem 2.5.1 (The Adjoint Formula for the Inverse)
Let A be an invertible matrix in $\mathscr{M}_n(\mathbb{K})$. Then its inverse is given by

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A). \tag{2.33}$$

2

Proof

We need to show that the matrix B defined by

$$B = \frac{1}{\det(A)} \operatorname{adj}(A)$$

satisfies

$$AB = BA = I. \tag{2.34}$$

Then, B is an inverse of A and the uniqueness of the inverse (Theorem 1.2.3) leads to $B = A^{-1}$. To check (2.34), let $A = (a_{ij})$, $1 \leq i,j \leq n$, and $\operatorname{adj}(A) = (d_{ji})$, $1 \leq j,i \leq n$, with $d_{ji} = C_{ij}$. By Definition 1.1.11,

$$A \operatorname{adj}(A) = (b_{ij}), \qquad 1 \leq i,j \leq n,$$

with the entries of the above product satisfying

$$\begin{aligned} b_{ij} &= a_{i1}d_{1j} + a_{i2}d_{2j} + \cdots + a_{in}d_{nj} \\ &= a_{i1}C_{j1} + a_{i2}C_{j2} + \cdots + a_{in}C_{jn}. \end{aligned}$$

Now, if $i = j$, then the above formula is the cofactor expansion of the determinant of the matrix A along the ith row.

On the other hand, if $i \neq j$, then

$$a_{i1}C_{j1} + a_{i2}C_{j2} + \cdots + a_{in}C_{jn} = 0.$$

The above equation is just the determinant of the matrix A, where we replace the ith row by the jth row. Then, in this case the matrix contains two identical rows and so its determinant is zero (Theorem 2.4.4).

Therefore, we obtain

$$A \operatorname{adj}(A) = \begin{bmatrix} \det(A) & 0 & \dots & 0 \\ 0 & \det(A) & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \det(A) \end{bmatrix} = \det(A) I.$$

Since A is invertible, $\det(A) \neq 0$, and we have

$$A\left(\frac{1}{\det(A)} \operatorname{adj}(A)\right) = AB = I.$$

By the same method, we can show that $BA = I$, and therefore, $B = A^{-1}$. This completes the proof of Theorem 2.5.1. □

Example 2.13

Use the adjoint formula to find the inverse of the matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ -1 & 3 & 0 \\ 1 & 0 & -2 \end{bmatrix}.$$

■

Solution

We have computed the cofactors of A in Example 2.12. Let us now find the determinant of A. Using the cofactor expansion along the second column, we have

$$\begin{aligned} \det(A) &= a_{12}C_{12} + a_{22}C_{22} + a_{32}C_{32} \\ &= 3C_{22} \\ &= -12. \end{aligned}$$

Thus, since $\det(A) \neq 0$, A^{-1} exists and

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A).$$

From Example 2.12, we have

$$\operatorname{adj}(A) = \begin{bmatrix} -6 & 0 & -6 \\ -2 & -4 & -2 \\ -3 & 0 & 3 \end{bmatrix}.$$

Therefore

$$A^{-1} = -\frac{1}{12} \begin{bmatrix} -6 & 0 & -6 \\ -2 & -4 & -2 \\ -3 & 0 & 3 \end{bmatrix} = \begin{bmatrix} 1/2 & 0 & 1/2 \\ 1/6 & 1/3 & 1/6 \\ 1/4 & 0 & -1/4 \end{bmatrix}.$$

◀

Example 2.14

Use formula (2.33) to find the inverse of the matrix

$$A = \begin{bmatrix} -3 & 2 & -1 \\ 2 & 0 & 1 \\ -1 & 2 & 1 \end{bmatrix}.$$

■

2

Solution

We need first to compute the cofactors of A as in Example 2.12. We find

$$C_{11} = -2, \quad C_{12} = -3, \quad C_{13} = 4, \quad C_{21} = -4, \quad C_{22} = -4, \quad C_{23} = 4,$$
$$C_{31} = 2, \quad C_{32} = 1, \quad C_{33} = -4$$

Consequently, the cofactor matrix is

$$C = \begin{bmatrix} -2 & -3 & 4 \\ -4 & -4 & 4 \\ 2 & 1 & -4 \end{bmatrix}$$

and so

$$\text{adj}(A) = \begin{bmatrix} -2 & -4 & 2 \\ -3 & -4 & 1 \\ 4 & 4 & -4 \end{bmatrix}.$$

Now, we use the cofactor expansion along the second row to find the determinant of A as

$$\begin{aligned} \det(A) &= a_{21}C_{21} + a_{22}C_{22} + a_{23}C_{23} \\ &= 2C_{21} + C_{23} \\ &= -4. \end{aligned}$$

Since $\det(A) \neq 0$, A^{-1} exists and is given by

$$A^{-1} = \frac{1}{\det(A)}\text{adj}(A) = -\frac{1}{4}\begin{bmatrix} -2 & -4 & 2 \\ -3 & -4 & 1 \\ 4 & 4 & -4 \end{bmatrix} = \begin{bmatrix} 1/2 & 1 & -1/2 \\ 3/4 & 1 & -1/4 \\ -1 & -1 & 1 \end{bmatrix}.$$

◀

Example 2.15

Use the adjoint matrix to find the inverse of the matrix

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & -1 & 3 \\ 0 & 0 & -2 \end{bmatrix}.$$

■

Solution

First, it is clear that since A is a triangular matrix,

$$\det(A) = 2 \cdot (-1) \cdot (-2) = 4 \neq 0.$$

This means that A is invertible. Now, we need to find the adjoint of A. We compute first the cofactor matrix C of A. We have

$$C_{11} = \det \begin{bmatrix} -1 & 3 \\ 0 & -2 \end{bmatrix} = 2, \quad C_{12} = 0, \quad C_{13} = 0, \quad C_{21} = 2,$$
$$C_{22} = -4, \quad C_{23} = 0, \quad C_{31} = 2, \quad C_{32} = -6, \quad C_{33} = -2.$$

Consequently,

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} = \begin{bmatrix} 2 & 0 & 0 \\ 2 & -4 & 0 \\ 2 & -6 & -2 \end{bmatrix}.$$

Thus,

$$\operatorname{adj}(A) = C^T = \begin{bmatrix} 2 & 2 & 2 \\ 0 & -4 & -6 \\ 0 & 0 & -2 \end{bmatrix},$$

and so

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A) = \frac{1}{4} \begin{bmatrix} 2 & 2 & 2 \\ 0 & -4 & -6 \\ 0 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 1/2 & 1/2 & 1/2 \\ 0 & -1 & -3/2 \\ 0 & 0 & -1/2 \end{bmatrix}.$$

◄

Example 2.16

1. Use the row reduction method to find the determinant of the matrix

$$A = \begin{bmatrix} 2 & 4 & 6 \\ 0 & 0 & -1 \\ 2 & -1 & 5 \end{bmatrix}.$$

2. Use the adjoint matrix to find A^{-1}.
3. Use the above results to solve the system of equations

$$\begin{cases} 2x_1 + 4x_2 + 6x_3 = 1, \\ -x_3 = 2, \\ 2x_1 - x_2 + 5x_3 = -1. \end{cases} \tag{2.35}$$

■

2

Solution

1. Denote by r_1, r_2, and r_3 the rows of A and of all the matrices obtained by means of row operations. Our goal is to apply the row operation method to get a triangular matrix from A. First, we exchange r_2 and r_3 and get

$$A_1 = \begin{bmatrix} 2 & 4 & 6 \\ 2 & -1 & 5 \\ 0 & 0 & -1 \end{bmatrix},$$

and $\det(A_1) = -\det(A)$. Next, we replace r_2 by $r_2 - r_1$ and obtain

$$A_2 = \begin{bmatrix} 2 & 4 & 6 \\ 0 & -5 & -1 \\ 0 & 0 & -1 \end{bmatrix}$$

and $\det(A_2) = \det(A_1)$. Now, since A_2 is a triangular matrix,

$$\det(A_2) = 2 \cdot (-5) \cdot (-1) = 10.$$

Consequently,

$$\det(A) = -\det(A_1) = -\det(A_2) = -10.$$

2. Since $\det(A) \neq 0$, A is invertible. We need to find the adjoint of A. We compute first the cofactor matrix C of A. We have

$$C_{11} = \det \begin{bmatrix} 0 & -1 \\ -1 & 5 \end{bmatrix} = -1, \quad C_{12} = -2, \quad C_{13} = 0, \quad C_{21} = -26,$$
$$C_{22} = -2, \quad C_{23} = 10, \quad C_{31} = -4, \quad C_{32} = 2, \quad C_{33} = 0.$$

Consequently,

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} = \begin{bmatrix} -1 & -2 & 0 \\ -26 & -1 & 10 \\ -4 & 2 & 0 \end{bmatrix},$$

and so

$$\operatorname{adj}(A) = C^T = \begin{bmatrix} -1 & -26 & -4 \\ -2 & -2 & 2 \\ 0 & 10 & 0 \end{bmatrix}.$$

This gives

$$A^{-1} = \frac{1}{\det(A)}\operatorname{adj}(A) = -\frac{1}{10}\begin{bmatrix} -1 & -26 & -4 \\ -2 & -2 & 2 \\ 0 & 10 & 0 \end{bmatrix} = \begin{bmatrix} 1/10 & 13/5 & 2/5 \\ 1/5 & 1/5 & -1/5 \\ 0 & -1 & 0 \end{bmatrix}.$$

3. We may write the system (2.35) as

$$AX = b, \quad \text{with} \quad A = \begin{bmatrix} 2 & 4 & 6 \\ 0 & 0 & -1 \\ 2 & -1 & 5 \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \text{and} \quad b = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}.$$

Since A is invertible, the solution of (2.35) is given by

$$X = A^{-1}b = \begin{bmatrix} 1/10 & 13/5 & 2/5 \\ 1/5 & 1/5 & -1/5 \\ 0 & -1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 49/10 \\ 4/5 \\ -2 \end{bmatrix}.$$

◀

2.5.1 Cramer's Rule

In this subsection, we will use the adjoint formula to find the solution of the system

$$AX = b \tag{2.36}$$

where A is an invertible matrix in $\mathscr{M}_n(\mathbb{K})$ and X and b are vectors in $\mathscr{M}_{n\times 1}(\mathbb{K})$. That is

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} b_1 \\ \vdots \\ b_n \end{bmatrix}. \tag{2.37}$$

To clarify the idea, take $n = 2$. Then we have

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}.$$

Since A is invertible, the solution of (2.36) is given by

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = A^{-1}b = \frac{1}{\det(A)} \operatorname{adj}(A)b$$
$$= \frac{1}{\det(A)} \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$$
$$= \frac{1}{\det(A)} \begin{bmatrix} b_1a_{22} - b_2a_{12} \\ -b_1a_{21} + b_2a_{11} \end{bmatrix}.$$

Now, if we consider the two matrices

$$A_1 = \begin{bmatrix} b_1 & a_{12} \\ b_2 & a_{22} \end{bmatrix} \quad \text{and} \quad A_2 = \begin{bmatrix} a_{11} & b_1 \\ a_{21} & b_2 \end{bmatrix},$$

then we have

$$b_1a_{22} - b_2a_{12} = \det(A_1) \quad \text{and} \quad -b_1a_{21} + b_2a_{11} = \det(A_2).$$

It is clear that the matrix A_1 is obtained by replacing the first column of A by the vector b and the matrix A_2 is obtained by replacing the second column of A by the vector b. This shows that the solution of (2.36) is given by

$$x_1 = \frac{\det(A_1)}{\det(A)} \quad \text{and} \quad x_2 = \frac{\det(A_2)}{\det(A)}.$$

This method of finding x_1 and x_2 is called the *Cramer rule* and is generalized in the following theorem.

Theorem 2.5.2 (Cramer's Rule)
Consider the system of linear equations (2.36). *Assume that* $\det(A) \neq 0$. *Then, the components of the unique solution of* (2.36) *are given by*

$$x_1 = \frac{\det(A_1)}{\det(A)}, \quad x_2 = \frac{\det(A_2)}{\det(A)}, \ldots, \quad x_n = \frac{\det(A_n)}{\det(A)}, \tag{2.38}$$

where A_j, $1 \leq j \leq n$, *is the matrix obtained by replacing the entries in the jth column of the matrix A by the entries of the column*

$$b = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}.$$

Proof

First method. It is clear that if $\det(A) \neq 0$, then A is invertible and the unique solution of (2.36) is given by $X = A^{-1}b$. Now using formula (2.33), we have

$$X = \frac{1}{\det(A)} \operatorname{adj}(A)b$$
$$= \frac{1}{\det(A)} \begin{bmatrix} C_{11} & C_{21} & \dots & C_{n1} \\ C_{12} & C_{22} & \dots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \dots & C_{nn} \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix},$$

whence

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \frac{1}{\det(A)} \begin{bmatrix} b_1C_{11} + b_2C_{21} + \cdots + b_nC_{n1} \\ b_1C_{12} + b_2C_{22} + \cdots + b_nC_{n2} \\ \vdots \\ b_1C_{1n} + b_2C_{2n} + \cdots + b_nC_{nn} \end{bmatrix}.$$

Thus, the jth component of X is given by

$$x_j = \frac{b_1C_{1j} + b_2C_{2j} + \cdots + b_nC_{nj}}{\det(A)}.$$

But it is clear that if we replace a_{1j}, $1 \leq j \leq n$, by b_j, then

$$b_1C_{1j} + b_2C_{2j} + \cdots + b_nC_{nj}$$

is the cofactor expansion of the determinant of the resulting matrix

$$A_j = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1(j-1)} & b_1 & a_{1(j+1)} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2(j-1)} & b_2 & a_{2(j+1)} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{n(j-1)} & b_n & a_{n(j+1)} & \dots & a_{nn} \end{bmatrix}$$

since $C_{1j}, C_{2j}, \dots, C_{nj}$ are simultaneously cofactors of A and of A_j. Consequently,

$$x_j = \frac{\det(A_j)}{\det(A)}.$$

Second method. We denote by $A_j(b)$ the matrix obtained by replacing the jth column of A by the vector b. Let $\mathbf{a_1}, \mathbf{a_2}, \dots, \mathbf{a_n}$ be the column vectors of A and let $e_1, e_2, \dots, e_n$ be the column vectors of the identity matrix I. Then, we have, for $1 \leq j \leq n$,

$$I_j(X) = [e_1, \dots, X, \dots, e_n],$$

where we replaced e_j by X. Clearly,

$$\begin{aligned} AI_j(X) &= [Ae_1, \ldots, AX, \ldots, Ae_n] \\ &= [\mathbf{a_1}, \ldots, b, \ldots, \mathbf{a_n}] = A_j(b). \end{aligned}$$

Now, applying Theorem 2.4.7, we have

$$\det(AI_j(X)) = \det(A)\det(I_j(X)) = \det(A_j(b)).$$

Since $\det(I_j(X)) = x_j$, we conclude that

$$x_j = \frac{\det(A_j(b))}{\det(A)} = \frac{\det(A_j}{\det(A)}.$$

This finishes the proof of Theorem 2.5.2. □

Example 2.17

Use Cramer's rule to find the solution of the linear system

$$\begin{cases} x_1 - x_2 = 1, \\ x_1 + 2x_2 = 3. \end{cases} \tag{2.39}$$

■

Solution

System (2.39) can be written in matrix form as

$$AX = b,$$

with

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \quad \text{and} \quad b = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

Following Cramer's rule, we introduce the matrices

$$A_1 = \begin{bmatrix} 1 & -1 \\ 3 & 2 \end{bmatrix} \quad \text{and} \quad A_2 = \begin{bmatrix} 1 & 1 \\ 1 & 3 \end{bmatrix}.$$

Since $\det(A) = 3 \neq 0$, A is invertible and the components of the unique solution of (2.39) are given by

$$x_1 = \frac{\det(A_1)}{\det(A)} = \frac{5}{3} \quad \text{and} \quad x_2 = \frac{\det(A_2)}{\det(A)} = \frac{2}{3}.$$

◄

Example 2.18

Use Cramer's rule to find the solution of the system

$$\begin{cases} x_1 + 2x_2 + 3x_3 = 1, \\ 2x_1 + 5x_2 + 3x_3 = 6, \\ x_1 + 8x_3 = -6. \end{cases} \tag{2.40}$$

■

Solution

The system (2.40) can be written in the form

$$AX = b,$$

with

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}, \quad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}, \quad \text{and} \quad b = \begin{bmatrix} 1 \\ 6 \\ -6 \end{bmatrix}.$$

We can use the method of cofactor expansion to show that

$$\det(A) = -1 \neq 0.$$

Therefore, A is invertible, and following Cramer's rule we introduce

$$A_1 = \begin{bmatrix} 1 & 2 & 3 \\ 6 & 5 & 3 \\ -6 & 0 & 8 \end{bmatrix}, \qquad A_2 = \begin{bmatrix} 1 & 1 & 3 \\ 2 & 6 & 3 \\ 1 & -6 & 8 \end{bmatrix}, \qquad A_3 = \begin{bmatrix} 1 & 2 & 1 \\ 2 & 5 & 6 \\ 1 & 0 & -6 \end{bmatrix}.$$

By the cofactor expansion, we show that

$$\det(A_1) = -2, \qquad \det(A_2) = -1, \qquad \det(A_3) = 1.$$

Hence, the components of the solution of (2.40) are given by

$$x_1 = \frac{\det(A_1)}{\det(A)} = \frac{-2}{-1} = 2, \quad x_2 = \frac{\det(A_2)}{\det(A)} = 1, \quad x_3 = \frac{\det(A_3)}{\det(A)} = -1.$$

◄

2

2.6 Exercises

Exercise 2.1

1. Consider the matrix

$$A = \begin{bmatrix} c & 2 & 0 \\ 1 & c & 2 \\ 0 & 1 & c \end{bmatrix},$$

where c is a real number. Find all values of c, if any, for which A is invertible.

2. Put $c = 1$ and use the adjoint matrix to find A^{-1}.

■

Solution

1. The matrix A is invertible if and only if $\det(A) \neq 0$. Now, using the cofactor expansion, we have

$$\begin{aligned} \det(A) &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \\ &= c \det \begin{bmatrix} c & 2 \\ 1 & c \end{bmatrix} - 2 \det \begin{bmatrix} 1 & 2 \\ 0 & c \end{bmatrix} \\ &= c(c^2 - 2) - 2c \\ &= c(c^2 - 4) = c(c-2)(c+2). \end{aligned}$$

So, the matrix A is invertible if and only if $c \neq 0$, $c \neq 2$ and $c \neq -2$.

2. Since for $c = 1$, $\det(A) \neq 0$, then A is invertible. We need to find the adjoint of A. We compute first the cofactor matrix C of A. A simple computation (see ▶ Sect. 2.5) shows that

$$C = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix} = \begin{bmatrix} -1 & -1 & 1 \\ -2 & 1 & -1 \\ 4 & -2 & -1 \end{bmatrix}.$$

Thus,

$$\operatorname{adj}(A) = C^T = \begin{bmatrix} -1 & -2 & 4 \\ -1 & 1 & -2 \\ 1 & -1 & -1 \end{bmatrix},$$

and so

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A) = -\frac{1}{3} \begin{bmatrix} -1 & -2 & 4 \\ -1 & 1 & -2 \\ 1 & -1 & -1 \end{bmatrix} = \begin{bmatrix} 1/3 & 2/3 & -4/3 \\ 1/3 & -1/3 & 2/3 \\ -1/3 & 1/3 & 1/3 \end{bmatrix}.$$

◀

Exercise 2.2
Let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}.$$

1. Find A^2 and $\mathrm{tr}(A^2)$.
2. Show that

$$\det(A) = \frac{1}{2}\det\begin{bmatrix} \mathrm{tr}(A) & 1 \\ \mathrm{tr}(A^2) & \mathrm{tr}(A) \end{bmatrix}.$$

■

Solution
1. We have

$$A^2 = AA = \begin{bmatrix} a & b \\ c & d \end{bmatrix}\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} a^2 + bc & ab + db \\ ac + dc & d^2 + bc \end{bmatrix}.$$

Thus, the trace of A^2 is

$$\mathrm{tr}(A^2) = a^2 + bc + d^2 + bc = a^2 + d^2 + 2bc.$$

2. We have

$$\begin{aligned} \det\begin{bmatrix} \mathrm{tr}(A) & 1 \\ \mathrm{tr}(A^2) & \mathrm{tr}(A) \end{bmatrix} &= (\mathrm{tr}(A))^2 - \mathrm{tr}(A^2) \\ &= (a+d)^2 - (a^2d^2 + 2bc) \\ &= 2ad - 2bc \\ &= 2\det(A), \end{aligned}$$

which gives the desired result. ◄

Exercise 2.3
Let A and B be two invertible matrices in $\mathscr{M}_n(\mathbb{R})$. Show that if

$$AB = -BA,$$

then n is even. ■

2

Solution

Since $AB = -BA$, we have

$$\det(AB) = \det(-BA).$$

Using the properties of the determinant (Theorem 2.4.5), we get

$$\det(AB) = (-1)^n \det(BA).$$

Using the product rule (Theorem 2.4.7) and the fact that $\det(A) \neq 0$ and $\det(B) \neq 0$ (Theorem 2.4.8), we obtain $(-1)^n = 1$. This shows that n is even. ◀

Exercise 2.4

1. Find the determinant of the matrix A in $\mathscr{M}_2(\mathbb{C})$ given by

$$A = \begin{bmatrix} \omega & \omega \\ -1 & \omega \end{bmatrix}$$

and the determinant of the matrix B in $\mathscr{M}_3(\mathbb{C})$ given by

$$B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{bmatrix},$$

where $\omega = \cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}$.

2. Find B^{-1}. ■

Solution

1. By direct computation,

$$\det(A) = \omega^2 + \omega.$$

Using Euler's formula, we have

$$\omega^3 = \cos(2\pi) + i\sin(2\pi) = 1.$$

Thus, ω is a cubic root of 1, so

$$\omega^3 - 1 = (\omega - 1)(\omega^2 + \omega + 1) = 0,$$

which gives

$$\omega^2 + \omega + 1 = 0,$$

since $\omega \neq 1$. Hence,

$$\det(A) = \omega^2 + \omega = -1.$$

Similarly, to find the determinant of B, we use the cofactor expansion along the first row, to get

$$\begin{aligned}\det(B) &= a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13} \\ &= C_{11} + C_{12} + C_{13} \\ &= \det\begin{bmatrix} \omega & \omega^2 \\ \omega^2 & \omega \end{bmatrix} - \det\begin{bmatrix} 1 & \omega^2 \\ 1 & \omega \end{bmatrix} + \det\begin{bmatrix} 1 & \omega \\ 1 & \omega^2 \end{bmatrix} \\ &= -\omega^4 + 3\omega^2 - 2\omega.\end{aligned}$$

Since $\omega^3 = 1$, we have $\omega^4 = \omega$. Therefore,

$$\det(B) = 3(\omega^2 - \omega).$$

On the other hand, since ω satisfies the equation

$$a^2 + a + 1 = 0 \tag{2.41}$$

and since $\omega^3 = 1$, it follows that ω^2 satisfies the same equation, that is

$$\omega^4 + \omega^2 + 1 = 0.$$

Since the coefficients of the Eq. (2.41) are real, one necessarily has that $\bar{\omega}$ is also a solution to (2.41), therefore, $\omega^2 = \bar{\omega}$. Consequently,

$$\begin{aligned}\det(B) &= 3(\omega^2 - \omega) \\ &= 3(\bar{\omega} - \omega) \\ &= -6i\sin(2\pi/3).\end{aligned}$$

2. To find the inverse of B, we need first to find the adjoint matrix $\operatorname{adj}(B)$. We compute the cofactors of B as follows:

$$C_{11} = \det\begin{bmatrix} \omega & \omega^2 \\ \omega^2 & \omega \end{bmatrix} = \omega^2 - \omega^4 = \omega^2 - \omega, \quad C_{12} = -\det\begin{bmatrix} 1 & \omega^2 \\ 1 & \omega \end{bmatrix} = \omega^2 - \omega,$$

$$C_{13} = \det\begin{bmatrix} 1 & \omega \\ 1 & \omega^2, \end{bmatrix} = \omega^2 - \omega, \quad C_{21} = -\det\begin{bmatrix} 1 & 1 \\ \omega^2 & \omega \end{bmatrix} = \omega^2 - \omega,$$

$$C_{22} = \det\begin{bmatrix} 1 & 1 \\ 1 & \omega \end{bmatrix} = \omega - 1, \quad C_{23} = -\det\begin{bmatrix} 1 & 1 \\ 1 & \omega^2 \end{bmatrix} = 1 - \omega^2,$$

$$C_{31} = \det\begin{bmatrix} 1 & 1 \\ \omega & \omega^2 \end{bmatrix} = \omega^2 - \omega, \quad C_{23} = -\det\begin{bmatrix} 1 & 1 \\ 1 & \omega^2 \end{bmatrix} = 1 - \omega^2,$$

$$C_{33} = \det\begin{bmatrix} 1 & 1 \\ 1 & \omega \end{bmatrix} = \omega - 1.$$

Consequently, the cofactor matrix is

$$C = \begin{bmatrix} \omega^2 - \omega & \omega^2 - \omega & \omega^2 - \omega \\ \omega^2 - \omega & \omega - 1 & 1 - \omega^2 \\ \omega^2 - \omega & 1 - \omega^2 & \omega - 1 \end{bmatrix},$$

and thus

$$\text{adj}(B) = C^T = \begin{bmatrix} \omega^2 - \omega & \omega^2 - \omega & \omega^2 - \omega \\ \omega^2 - \omega & \omega - 1 & 1 - \omega^2 \\ \omega^2 - \omega & 1 - \omega^2 & \omega - 1 \end{bmatrix}.$$

Consequently, applying (2.33), we get

$$B^{-1} = \frac{1}{\det(B)}\text{adj}(B) = \frac{1}{3(\omega^2 - \omega)}\begin{bmatrix} \omega^2 - \omega & \omega^2 - \omega & \omega^2 - \omega \\ \omega^2 - \omega & \omega - 1 & 1 - \omega^2 \\ \omega^2 - \omega & 1 - \omega^2 & \omega - 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3\omega} & \frac{\omega}{3} \\ \frac{1}{3} & \frac{\omega}{3} & \frac{1}{3\omega} \end{bmatrix},$$

where we have used the fact that $\omega^2 + \omega + 1 = 0$. ◀

Exercise 2.5 (Vandermonde Determinant)
Let $(a_1, a_2, \ldots, a_n)$ be in $\mathbb{K}^n$. We define the Vandermonde determinant $V(a_1, a_2, \ldots, a_n)$ to be the determinant of the Vandermonde matrix

$$V_n = \begin{bmatrix} 1 & a_1 & a_1^2 & \ldots & a_1^{n-1} \\ 1 & a_2 & a_2^2 & \ldots & a_2^{n-1} \\ \vdots & \vdots & & & \vdots \\ 1 & a_n & a_n^2 & \ldots & a_n^{n-1} \end{bmatrix}. \tag{2.42}$$

1. Find $V(a_1, a_2, \ldots, a_n)$ for all $n \geq 1$.
2. Deduce that the Vandermonde determinant is zero if and only if at least two of the a_i's coincide.

■

Solution

1. We find the determinant of Vandermonde by induction. First, for $n = 1$, we have $V(a_1) = 1$. Next, for $n = 2$, we have

$$V(a_1, a_2) = \det \begin{bmatrix} 1 & a_1 \\ 1 & a_2 \end{bmatrix} = a_2 - a_1.$$

For $n = 3$, we have

$$V(a_1, a_2, a_3) = \det \begin{bmatrix} 1 & a_1 & a_1^2 \\ 1 & a_2 & a_2^2 \\ 1 & a_3 & a_3^2 \end{bmatrix}.$$

To find the above determinant, we use Theorem 2.4.2 and replace c_2 in the above matrix by $c_2 - a_1 c_1$ and c_3 by $c_3 - a_1 c_2$, where c_1, c_2 and c_3 are the first, the second, and the third columns of the above matrix to get

$$\begin{aligned} V(a_1, a_2, a_3) &= \det \begin{bmatrix} 1 & 0 & 0 \\ 1 & a_2 - a_1 & a_2(a_2 - a_1) \\ 1 & a_3 - a_1 & a_3(a_3 - a_1) \end{bmatrix} \\ &= \det \begin{bmatrix} a_2 - a_1 & a_2(a_2 - a_1) \\ a_3 - a_1 & a_3(a_3 - a_1) \end{bmatrix} \\ &= (a_2 - a_1)a_3(a_3 - a_1) - (a_3 - a_1)a_2(a_2 - a_1) \\ &= (a_2 - a_1)(a_3 - a_1)(a_3 - a_2) \\ &= \prod_{1 \leq j < i \leq 3} (a_i - a_j). \end{aligned}$$

Now, we assume that

$$V(a_1, a_2, \ldots, a_{n-1}) = \prod_{1 \leq j < i \leq n-1} (a_i - a_j) \tag{2.43}$$

and show that

$$V(a_1, a_2, \ldots, a_n) = \prod_{1 \leq j < i \leq n} (a_i - a_j). \tag{2.44}$$

2

Indeed, using the same method as before and replacing c_2 by $c_2 - a_1c_1$, c_3 by $c_3 - a_1c_2$ up to c_n, which we replace by $c_n - a_1c_{n-1}$, we get

$$\begin{aligned}
V(a_1, a_2, \ldots, a_n) &= \det \begin{bmatrix} 1 & 0 & 0 & \ldots & 0 \\ 1 & (a_2 - a_1) & (a_2 - a_1)a_2 & \ldots & (a_2 - a_1)a_2^{n-2} \\ \vdots & \vdots & & & \vdots \\ 1 & (a_n - a_1) & (a_n - a_1)a_n & \ldots & (a_n - a_1)a_n^{n-2} \end{bmatrix} \\
&= \det \begin{bmatrix} (a_2 - a_1) & (a_2 - a_1)a_2 & \ldots & (a_2 - a_1)a_2^{n-2} \\ \vdots & & & \vdots \\ (a_n - a_1) & (a_n - a_1)a_n & \ldots & (a_n - a_1)a_n^{n-2} \end{bmatrix} \\
&= (a_2 - a_1)(a_3 - a_1) \cdot \cdots \cdot (a_n - a_1) \det \begin{bmatrix} 1 & a_2 & \ldots & a_2^{n-2} \\ \vdots & & & \vdots \\ 1 & a_n & \ldots & a_n^{n-2} \end{bmatrix} \\
&= (a_2 - a_1)(a_3 - a_1) \cdot \cdots \cdot (a_n - a_1) V(a_1, a_2, \ldots, a_{n-1}) \\
&= \prod_{1<i\leq n} (a_i - a_1) \cdot \prod_{1\leq j<i\leq n-1} (a_i - a_j).
\end{aligned}$$

Now, by using (2.43), we conclude that (2.44) holds.

2. It is clear that from (2.44) if there exists $1 \leq i_0, j_0 \leq n$ and $a_{i_0} = a_{j_0}$ with $i_0 \neq j_0$, then $V(a_1, a_2, \ldots, a_n) = 0$. ◀

Exercise 2.6 (Hankel Matrix)

We define the Hankel matrix by

$$H_n = \begin{bmatrix} s_0 & s_1 & s_2 & \ldots & s_{n-1} \\ s_1 & s_2 & s_3 & \ldots & s_n \\ \vdots & \vdots & & & \vdots \\ s_{n-1} & s_n & s_{n+1} & \ldots & s_{2n-2} \end{bmatrix} = (s_{i+j-2}), \quad 1 \leq i, j \leq n, \tag{2.45}$$

where

$$s_k = \sum_{i=1}^{n} a_i^k, \qquad 0 \leq k \leq 2n-2,$$

for some $a_1, a_2, \ldots, a_n$ in $\mathbb{K}$.

1. Show that $H_n = V_n^T V_n$, where V_n is the Vandermonde matrix (2.42).
2. Find $\det(H_n)$.

■

Solution

We have

$$V_n^T V_n = \begin{bmatrix} 1 & 1 & \dots & 1 \\ a_1 & a_2 & \dots & a_n \\ \vdots & \vdots & & \vdots \\ a_1^{n-1} & a_2^{n-1} & \dots & a_n^{n-1} \end{bmatrix} \begin{bmatrix} 1 & a_1 & a_1^2 & \dots & a_1^{n-1} \\ 1 & a_2 & a_2^2 & \dots & a_2^{n-1} \\ \vdots & \vdots & & & \vdots \\ 1 & a_n & a_n^2 & \dots & a_n^{n-1} \end{bmatrix}$$

To get s_{i+j-2}, we need to multiply row r_{i-1} of V_n^T with column c_{j-1} of V_n. For instance

$$s_0 = s_{1+1-2} = r_1 \cdot c_1, \quad i = j = 1,$$

$$s_1 = s_{1+2-2} = s_{2+1-2} = r_1 \cdot c_2 = r_2 \cdot c_1, \quad i = 1, j = 2, \quad \text{or} \quad i = 2, j = 1.$$

$$\vdots$$

$$s_k = r_i \cdot c_j, \quad \text{with} \quad i + j = k + 2.$$

Thus, for example

$$s_k = [a_1^{i-1}, a_2^{i-1}, \dots, a_n^{i-1}] \cdot \begin{bmatrix} a_1^{j-1} \\ a_2^{j-1} \\ \vdots \\ a_n^{j-1} \end{bmatrix} = \sum_{i=1}^{n} a_i^{i+j-2} = \sum_{i=1}^{n} a_i^k, \qquad (0 \le k \le 2n-2).$$

Consequently, it is clear that $V_n^T V_n = H_n$.

2. Using Theorem 2.4.7 and Theorem 2.3.1, we have

$$\begin{aligned} \det(H_n) = \det(V_n^T V_n) &= \det(V_n^T) \det(V_n) \\ &= [\det(V)]^2 \\ &= \prod_{1 \le j < i \le n} (a_i - a_j)^2, \end{aligned}$$

where we have used (2.44). ◄

Exercise 2.7 (Skew–Symmetric Matrix)

Let A be a square matrix in $\mathscr{M}_n(\mathbb{K})$. A is called *skew symmetric* if $A^T = -A$.

1. Show that for any matrix A in $\mathscr{M}_n(\mathbb{K})$, then $A + A^T$ is symmetric and $A - A^T$ is skew symmetric.
2. Show that any matrix A in $\mathscr{M}_n(\mathbb{K})$ can be written as the sum of a symmetric matrix and a skew symmetric one.
3. Let A be a skew symmetric matrix in $\mathscr{M}_n(\mathbb{K})$. Show that if n is odd, then

$$\det(A) = 0.$$

■

2

Solution

1. Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$. Then,

$$(A + A^T)^T = A^T + (A^T)^T = A^T + A.$$

Consequently, $A + A^T$ is symmetric.

Similarly,

$$(A - A^T)^T = A^T - (A^T)^T = A^T - A = -(A - A^T).$$

So, $A - A^T$ is skew symmetric.

2. Now, for any matrix A in $\mathscr{M}_n(\mathbb{K})$, we have

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T).$$

Since $A + A^T$ is symmetric, $\frac{1}{2}(A + A^T)$ is also symmetric; moreover, since $A - A^T$ is skew symmetric, so is $\frac{1}{2}(A - A^T)$.

3. Since A is skew symmetric, we have $A = -A^T$. This gives

$$\det(A) = \det(-A^T) = (-1)^n \det(A^T),$$

by making use of (2.24). Now, since $\det(A) = \det(A^T)$ (Theorem 2.3.1), then by the above reasoning if n is odd, then

$$2\det(A) = 0.$$

This yields

$$\det(A) = 0.$$

◀

Exercise 2.8 (Companion Matrix)

The *companion matrix* associated with the polynomial

$$p(x) = a_0 + a_1 x + \cdots + a_{n-1}x^{n-1} + x^n$$

is defined to be the matrix

$$C_p = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \vdots & 0 \\ \vdots & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-1} \end{bmatrix},$$

where $a_0, a_1, \dots a_{n-1}$ are in $\mathbb{K}$.

1. Show that $\det(C_p) = (-1)^n a_0$.
2. Prove that $\det(\lambda I - C_p) = p(\lambda)$.
3. Deduce that $\det(\lambda_0 I - C_p) = 0$ if and only if λ_0 is a root of $p(\lambda)$.

■

Solution

1. We use the row reduction method (Theorem 2.4.2) to compute the determinant of C_p. We can apply two methods:

For the first method, we denote the columns of the matrix C_p and the columns of the obtained matrices after any column operation by $C_1, C_2, \ldots, C_n$. Permuting the first and the second columns, we get

$$C_{p,1} = \begin{bmatrix} 1 & 0 & 0 & \ldots & 0 \\ 0 & 0 & 1 & \vdots & 0 \\ \vdots & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & 1 \\ -a_1 & -a_0 & -a_2 & \ldots & -a_{n-1} \end{bmatrix}$$

and $\det(C_{p,1}) = -\det(C_p)$. Similarly, permuting C_2 and C_3, we obtain

$$C_{p,2} = \begin{bmatrix} 1 & 0 & 0 & \ldots & 0 \\ 0 & 1 & 0 & \vdots & 0 \\ \vdots & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & \ldots & 1 \\ -a_1 & -a_2 & -a_0 & \ldots & -a_{n-1} \end{bmatrix}$$

and $\det(C_{p,2}) = -\det(C_{p,1}) = (-1)^2 \det(C_p)$. We continue in this way and in each operation, we permute C_i and C_{i+1}, $i = 1, \ldots, n-1$, and finally, after $(n-1)$ operations, we get

$$C_{p,(n-1)} = \begin{bmatrix} 1 & 0 & \ldots & 0 & 0 \\ 0 & 1 & \vdots & 0 & 0 \\ \vdots & \vdots & & \vdots & 0 \\ 0 & 0 & \ldots & 1 & 0 \\ -a_1 & -a_2 & \ldots & -a_{n-1} & -a_0 \end{bmatrix}$$

and $\det(C_{p,(n-1)}) = (-1)^{n-1} \det(C_p)$. Since $C_{p,(n-1)}$ is a triangular matrix, its determinant is the product of the entries of the main diagonal, that is

$$\det(C_{p,(n-1)}) = -a_0.$$

2

Consequently,

$$\det(C_p) = (-1)^n a_0.$$

For the second method, we just replace the nth row r_n in C_p by $r_n + a_1 r_1 + a_2 r_2 + \cdots + a_{n-1} r_{n-1}$, and obtain

$$A = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \vdots & 0 \\ \vdots & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_0 & 0 & 0 & \dots & 0 \end{bmatrix},$$

with $\det(A) = \det(C_p)$. Now, we use the cofactor expansion along the first column to compute the determinant of A

$$\det(A) = -a_0 C_{n1} = -a_0(-1)^{n+1} M_{n1},$$

where

$$M_{n1} = \det \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \vdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix} = 1.$$

Thus,

$$\det(C_p) = \det(A) = (-1)^n a_0.$$

2. We have

$$\lambda I - C_p = \begin{bmatrix} \lambda & -1 & 0 & \dots & 0 \\ 0 & \lambda & -1 & \vdots & 0 \\ \vdots & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & \dots & -1 \\ a_0 & a_1 & a_2 & \dots & \lambda + a_{n-1} \end{bmatrix}.$$

Now, to compute the determinant of $\lambda I - C_p$, we use the cofactor expansion along the first column, obtaining

$$\det(\lambda I - C_p) = \lambda C_{11} + a_0 C_{n1} = \lambda M_{11}^{(n-1)} + a_0(-1)^{n+1} M_{n1}^{(n-1)},$$

where

$$M_{11}^{(n-1)} = \det \begin{bmatrix} \lambda & -1 & \vdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & -1 \\ a_1 & a_2 & \dots & \lambda + a_{n-1} \end{bmatrix}$$

and

$$M_{n1}^{(n-1)} = \det \begin{bmatrix} -1 & 0 & \dots & 0 \\ \lambda & -1 & \vdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & -1 \end{bmatrix} = (-1)^{n-1}, \tag{2.46}$$

since the above matrix is triangular. Thus,

$$\det(\lambda I - C_p) = \lambda M_{11}^{(n-1)} + a_0(-1)^{n+1}(-1)^{n-1} = \lambda M_{11}^{(n-1)} + a_0.$$

Now, observe that the matrix in (2.46) is the same as the matrix $\lambda I - C_p$, but of size $(n-1) \times (n-1)$. So, the same method as before yields

$$M_{11}^{(n-1)} = \lambda M_{11}^{(n-2)} + a_1,$$

where $M_{11}^{(n-2)}$ is the determinant of the $(n-2) \times (n-2)$ matrix

$$\begin{bmatrix} \lambda & -1 & \vdots & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \dots & -1 \\ a_2 & a_3 & \dots & \lambda + a_{n-1} \end{bmatrix}.$$

Consequently, we obtain

$$\det(\lambda I - C_p) = \lambda^2 M_{11}^{(n-2)} + a_1\lambda + a_0.$$

Repeating this process until the end, we obtain

$$\det(\lambda I - C_p) = \lambda^{n-2} M_{11}^{(2)} + a_{n-3}\lambda^{n-3} + \cdots + a_1\lambda + a_0, \tag{2.47}$$

where

$$M_{11}^{(2)} = \det \begin{bmatrix} \lambda & -1 \\ a_{n-2} & \lambda + a_{n-1} \end{bmatrix} = \lambda^2 + \lambda a_{n-1} + a_{n-2}.$$

2

Plugging this into (2.47), we get

$$\det(\lambda I - C_p) = \lambda^n + \lambda^{n-1}a_{n-1} + a_{n-2}\lambda^{n-2} + a_{n-3}\lambda^{n-3} + \cdots + a_1\lambda + a_0 = p(\lambda).$$

3. It is clear, since $\det(\lambda I - C_p) = p(\lambda)$, that if $p(\lambda_0) = 0$, then $\det(\lambda_0 I - C_p) = 0$ and if $\det(\lambda_0 I - C_p) = 0$, then $p(\lambda_0) = 0$. ◀

Exercise 2.9 (Tridiagonal Matrix)
The matrix A_n in $\mathscr{M}_n(\mathbb{K})$ is called *tridiagonal* if it has the form

$$A_n = \begin{bmatrix} a_1 & b_1 & 0 & \dots & \dots & 0 \\ c_1 & a_2 & b_2 & & & \vdots \\ 0 & c_2 & a_3 & b_3 & \dots & 0 \\ \vdots & & \ddots & \ddots & \ddots & \vdots \\ \vdots & & \ddots & \ddots & \ddots & b_{n-1} \\ 0 & \dots & & 0 & c_{n-1} & a_n \end{bmatrix}.$$

1. Show that

$$\det(A_n) = a_n \det(A_{n-1}) - b_{n-1}c_{n-1}\det(A_{n-2}), \qquad n \geq 3. \tag{2.48}$$

2. The Fibonacci sequence is the sequence of numbers $1, 2, 3, 5, 8, 13, \dots$ generated by the relation $F_n = F_{n-1} + F_{n-2}$, $n \geq 3$, with $F_1 = 1$ and $F_2 = 2$. Show that the determinant of the Fibonacci matrix

$$B_n = \begin{bmatrix} 1 & 1 & 0 & \dots & \dots & 0 \\ -1 & 1 & 1 & & & \vdots \\ 0 & -1 & 1 & 1 & \dots & 0 \\ \vdots & & \ddots & \ddots & \ddots & \\ \vdots & & \ddots & \ddots & \ddots & 1 \\ 0 & \dots & & 0 & -1 & 1 \end{bmatrix}$$

is the nth Fibonacci number F_n.

■

Solution
1. The cofactor expansion along the last column, yields

$$\det(A_n) = a_n C_{nn} + b_{n-1}C_{(n-1)n}.$$

Consequently,

$$\det(A_n) = a_n(-1)^{n+n}\det\begin{bmatrix} a_1 & b_1 & 0 & \dots & \dots & 0 \\ c_1 & a_2 & b_2 & & & \vdots \\ 0 & c_2 & a_3 & b_3 & \dots & 0 \\ \vdots & \ddots & \ddots & \ddots & & \\ \vdots & \ddots & \ddots & \ddots & b_{n-2} & \\ 0 & \dots & 0 & c_{n-2} & a_{n-1} & \end{bmatrix}$$

$$+ b_{n-1}(-1)^{n-1+n}\det\begin{bmatrix} c_1 & a_1 & b_1 & \dots & \dots & 0 \\ 0 & c_2 & a_2 & b_2 & & \vdots \\ 0 & 0 & c_3 & a_3 & \ddots & 0 \\ \vdots & & \ddots & \ddots & \ddots & b_{n-3} \\ \vdots & & \ddots & \ddots & c_{n-3} & a_{n-2} \\ 0 & \dots & & 0 & 0 & c_{n-1} \end{bmatrix}. \tag{2.49}$$

On the other hand, we have

$$\det\begin{bmatrix} c_1 & a_1 & b_1 & \dots & \dots & 0 \\ 0 & c_2 & a_2 & b_2 & & \vdots \\ 0 & 0 & c_3 & a_3 & \ddots & 0 \\ \vdots & & \ddots & \ddots & \ddots & b_{n-3} \\ \vdots & & \ddots & \ddots & c_{n-3} & a_{n-2} \\ 0 & \dots & & 0 & 0 & c_{n-1} \end{bmatrix}$$

$$= c_{n-1}(-1)^{2n-2}\det\begin{bmatrix} a_1 & b_1 & 0 & \dots & \dots & 0 \\ c_1 & a_2 & b_2 & & & \vdots \\ 0 & c_2 & a_3 & b_3 & \dots & 0 \\ \vdots & & \ddots & \ddots & \ddots & \\ \vdots & & \ddots & \ddots & \ddots & b_{n-3} \\ 0 & \dots & & 0 & c_{n-3} & a_{n-2} \end{bmatrix}$$

$$= c_{n-2}\det(A_{n-2}).$$

Plugging the last expression into (2.49), we get

$$\det(A_n) = a_n\det(A_{n-1}) - b_{n-1}c_{n-1}\det(A_{n-2}),$$

which is exactly (2.48).

2

2. We may apply (2.48) for $a_n = 1$, $b_n = 1$ and $c_n = -1$ for all n, we get

$$\det(B_n) = \det(B_{n-1}) + \det(B_{n-2}), \qquad n \geq 3.$$

Thus, for $n = 1$, we have $\det(B_1) = \det[1] = 1$. For $n = 2$, we have

$$\det(B_2) = \det \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = 2.$$

Therefore, $\det(B_n) = F_n$ for all $n \geq 1$. ◀

Exercise 2.10 (Determinant of Block Matrices and Schur's Formula)
Let M be a matrix in $\mathscr{M}_n(\mathbb{K})$ and let A, B, C and D be matrices in $\mathscr{M}_p(\mathbb{K})$ with $2p = n$.

1. Show that if

$$M = \begin{bmatrix} A & \mathbf{0}_{p\times p} \\ \mathbf{0}_{p\times p} & D \end{bmatrix},$$

then

$$\det(M) = \det(A)\det(D) = \det(AD). \tag{2.50}$$

2. Show that (2.50) remains true if M has one of the forms

$$M = \begin{bmatrix} A & \mathbf{0}_{p\times p} \\ C & D \end{bmatrix}, \qquad \text{or} \qquad M = \begin{bmatrix} A & B \\ \mathbf{0}_{p\times p} & D \end{bmatrix}.$$

3. Show that if M is the matrix written blockwise as

$$M = \begin{bmatrix} A & B \\ C & D \end{bmatrix},$$

then the formula

$$\det M = \det(AD - BC) \tag{2.51}$$

is not true in general.

4. Show that if D is invertible, then

$$\det(M) = \det(D)\det(A - BD^{-1}C) \qquad \text{(Schur's formula).}$$

■

Solution

1. Assume that $A = (a_{ij})$ is an $r \times r$ matrix ($r \leq p$) and D is a $p \times p$ matrix. We use induction on r to show (2.50). If $r = 1$, then by expanding along the first row, we obtain

$$\det \left[\begin{array}{c|c} a_{11} & 0 \ldots 0 \\ \hline \begin{matrix} 0 \\ \vdots \\ 0 \end{matrix} & D \end{array}\right] = a_{11} \det(D) = \det(A) \det(D).$$

Now, we assume that (2.50) is true for $r = p - 1$ and show that it remains true for $r = p$. We have

$$\det(M) = \det \begin{bmatrix} A & \mathbf{0}_{p\times p} \\ \mathbf{0}_{p\times p} & D \end{bmatrix},$$

where the matrix A can be written as

$$A = \left[\begin{array}{cc|c} & & a_{1p} \\ & A_{(p-1)\times(p-1)} & \vdots \\ & & \\ \hline a_{p1} & \ldots \quad a_{p(p-1)} & a_{pp} \end{array}\right],$$

with $A_{(p-1)\times(p-1)} = (a_{ij})$, $1 \leq i, j \leq p-1$. Obviously,

$$\det(A) = a_{pp} \det(A_{(p-1)\times(p-1)})$$

and

$$\begin{aligned} \det(M) = a_{pp} \det \begin{bmatrix} A_{(p-1)\times(p-1)} & \mathbf{0}_{(p-1)\times p} \\ \mathbf{0}_{p\times(p-1)} & D \end{bmatrix} &= a_{pp} \det(A_{(p-1)\times(p-1)}) \det(D) \\ &= \det(A) \det(D), \end{aligned}$$

where we have used the induction hypothesis.

2. The first formula can be proved by the same method as in the first question. The second one can be proved by using the determinant of the transpose (Theorem 2.3.1).

3. It is clear that (2.51) is true for $p = 1$. We provide a counterexample for $p \geq 2$. Assume, for instance, that $p = 2$ and we consider the matrices

$$A = \begin{bmatrix} 1 & 2 \\ 1 & -1 \end{bmatrix}, \quad B = \begin{bmatrix} 2 & 1 \\ -1 & 3 \end{bmatrix}, \quad C = \begin{bmatrix} 1 & 0 \\ -1 & 4 \end{bmatrix}, \quad D = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}.$$

Then,

$$\det(AD - BC) = \det \begin{bmatrix} 7 & 4 \\ 6 & -13 \end{bmatrix} = -115.$$

On the other hand, we have

$$\det(M) = \det \begin{bmatrix} 1 & 2 & 2 & 1 \\ 1 & -1 & -1 & 3 \\ 1 & 0 & 4 & 2 \\ -1 & 4 & 2 & 3 \end{bmatrix} = -94.$$

4. To prove Schur's formula, we observe that

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} I_p & \mathbf{0}_{p\times p} \\ -D^{-1}C & I_p \end{bmatrix} = \begin{bmatrix} A - BD^{-1}C & B \\ \mathbf{0}_{p\times p} & D \end{bmatrix}.$$

Then, we apply (2.50), to get

$$\det \begin{bmatrix} A & B \\ C & D \end{bmatrix} \det \begin{bmatrix} I_p & \mathbf{0}_{p\times p} \\ -D^{-1}C & I_p \end{bmatrix} = \det \begin{bmatrix} A - BD^{-1}C & B \\ \mathbf{0}_{p\times p} & D \end{bmatrix}.$$

That is,

$$\det(M) = \det(D) \det(A - BD^{-1}C),$$

since $\det I_p = 1$. ◄

Euclidean Vector Spaces

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3.1 Introduction

The main objectives in this chapter are to generalize the basic geometric ideas in $\mathbb{R}^2$ or $\mathbb{R}^3$ to nontrivial higher-dimensional spaces $\mathbb{R}^n$. Our approach is to start from geometric concepts in $\mathbb{R}^2$ or $\mathbb{R}^3$ and then extend them to $\mathbb{R}^n$ in a purely algebraic manner.

In engineering and physics, many quantities, such as force and velocity, are defined by their direction and magnitude. Mathematically, this is what we call a *vector.* Thus, we can simply define a vector in 2- or 3-dimensional space as a *line segment* with a definite direction, or graphically as an arrow connecting a point A (called *initial point*) and a point B (called *terminal point*), as shown in Fig. 3.1

In engineering and physics, some quantities like weight or height can be represented by a scalar. On the other hand, a force, for example, can be described by its magnitude and direction, therefore, a vector is characterized by its magnitude (length) and its direction. Accordingly, two vectors are *equivalent* (or *equal*) if they have the same length and the same direction. See Fig. 3.2.

3.2 Vector Addition and Multiplication by a Scalar

3.2.1 Vector Addition

Consider the vector v with initial point A and terminal point B and let w be the vector with initial point B and terminal point C. Then it is not hard to see that $v + w$ is the vector with initial point A and terminal point C, as in Fig. 3.3.

Now, the question is that: *how can we add v to w if the terminal point of v is not the initial point of w?* In this case, as we stated before, two vectors with the same length and the same direction are equal. Thus, as in Fig. 3.4, the dashed vector is equal to the vector v and its initial point is in the same time the terminal point of v and thus we can

Fig. 3.1 The vector $v = \vec{AB}$

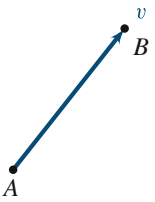


Fig. 3.2 The vectors v and w are equal since they have the same length and the same direction

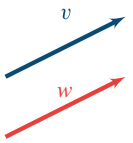


Fig. 3.3 The sum of two vectors $v + w$

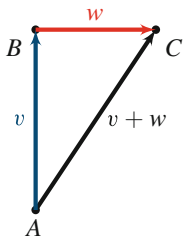
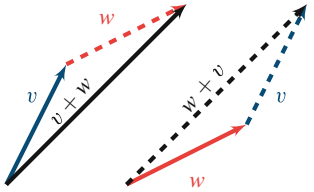


Fig. 3.4 The sum of two vectors $v + w$



add the two vectors as above. Now, from this simple remark, we may deduce directly that:

1. The sum of vectors is commutative. That is

$$v + w = w + v, \tag{3.1}$$

as illustrated in Fig. 3.4.

2. The sum of vectors is associative, that is

$$u + (v + w) = (u + v) + w, \tag{3.2}$$

as illustrated in Fig. 3.5.

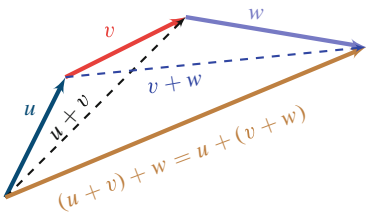


Fig. 3.5 The sum of vectors is associative

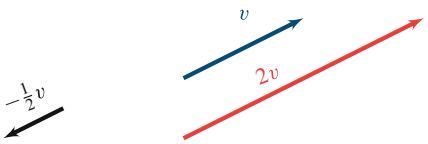


Fig. 3.6 Multiplication of a vector by a scalar

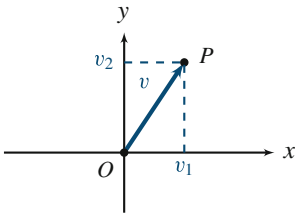


Fig. 3.7 Components of a vector if the initial point is the origin

3.2.2 Multiplication of a Vector by a Scalar

Let v be a nonzero vector in 2- or 3-dimensional space and let k be a scalar. Then, we define the product kv to be the vector whose length is $|k|$ times the length of v and whose direction is the same direction of v if $k > 0$ and the opposite one if $k < 0$. See Fig. 3.6.

3.2.3 Vectors in Coordinate Systems

Let v be a vector in 2-dimensional space with its initial point at the origin of a rectangular coordinate system. Then the vector is completely determined by the coordinates of its terminal point; see Fig. 3.7.

Thus, $v = \overrightarrow{OP} = (v_1, v_2)$, where v_1 and v_2 are called the *components* of v. Now, in the coordinate system the vector is determined by its components rather than by its direction and its length.

Now, if the initial point P_1 is not the origin, then we just need to translate the coordinate system in such a way that the point P_1 will be the origin in the new coordinate system as in Fig. 3.8.

Let $P(x_1, y_1)$ be the initial point of the vector v and $P_2(x_2, y_2)$ be the terminal point of v. Then in the $x'y'$-plane the components of the vector v are $v = (x_2', y_2')$, since P_1 is the origin in the $x'y'$-plane. Since in the xy-plane we have $x_2' = x_2 - x_1$ and $y_2' = y_2 - y_1$, the components of v in the xy-plane are $v = (x_2 - x_1, y_2 - y_1)$.

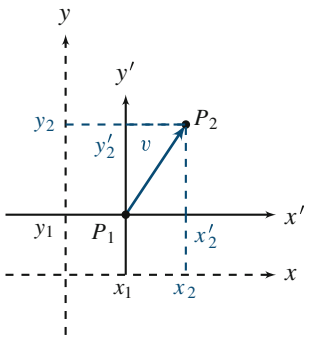


Fig. 3.8 Components of a vector if the initial point is not the origin

Now, we are going to generalize the above notions to any n-dimensional space.

We have seen in (1.15) how to add two vectors in 2-dimensional space ($\mathbb{R}^2$) and 3-dimensional space ($\mathbb{R}^3$). To generalize (1.15), let us first define the space $\mathbb{R}^n$.

Definition 3.2.1 (Euclidean Space $\mathbb{R}^n$)

Let n be a positive integer. An *ordered n-tuple* is a finite sequence of n real numbers $(v_1, v_2, \ldots, v_n)$. The space $\mathbb{R}^n$ is the set of all ordered n-tuples. This space $\mathbb{R}^n$ can be also regarded as the space of all vectors with n components.

Now, similarly to the geometric approach in (1.15), we may define the addition in $\mathbb{R}^n$. Thus, if $v = (v_1, v_2, \ldots, v_n)$ and $w = (w_1, w_2, \ldots, w_n)$ are two vectors in $\mathbb{R}^n$, we define the sum $v + w$ to be the vector

$$v + w = (v_1 + w_1, v_2 + w_2, \ldots, v_n + w_n). \tag{3.3}$$

Example 3.1

Let $v = (1, 3, -1)$ and $w = (-1, 0, 2)$ be two vectors of $\mathbb{R}^3$. Then,

$$v + w = (0, 3, 1).$$

■

As depicted in Fig. 3.9, if v is the vector with the initial point A and terminal point B and w is the vector with initial point B and terminal point C, then the sum $v + w$ is the vector with the initial point A and terminal point C.

The real numbers $v_1, v_2, \ldots, v_n$ are called the *components* of the vector v. Form (3.3), we may easily show that

$$v + w = w + v.$$

Thus, the addition $(+)$ of vectors is *commutative*. Moreover, and as we have seen for the 2-dimensional space, it is clear that

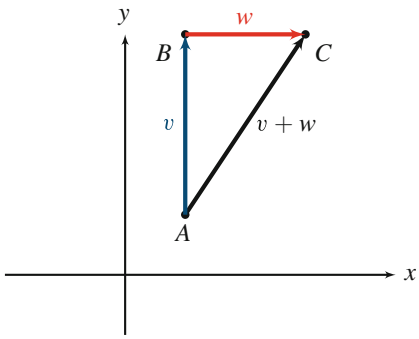


Fig. 3.9 The sum of two vectors $v + w$ in a coordinate system

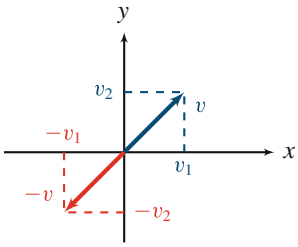


Fig. 3.10 The vector $-v$

$$u + (v + w) = (u + v) + w,$$

for any three vectors u, v and w in $\mathbb{R}^n$. Thus, the addition of vectors is also *associative*.

In particular, if $w = v$, then

$$v + v = 2v = (v_1 + v_1, v_2 + v_2, \ldots, v_n + v_n) = (2v_1, 2v_2, \ldots, 2v_n).$$

We may also define the vector $(-1)v = -v$ (Fig. 3.10) to be the vector with the same length as v and its direction the opposite of that of v. Using the geometric representation in $\mathbb{R}^2$, for example, we find that if $v = (v_1, v_2)$, then $-v = (-v_1, -v_2)$.

Analogously, if k is a scalar, then we define kv to be the vector

$$kv = (kv_1, kv_2, \ldots, kv_n). \tag{3.4}$$

Also, we define the zero vector in $\mathbb{R}^n$ to be the vector whose components are all zero. That is, $\mathbf{0} = 0_{\mathbb{R}^n} = (0, 0, \ldots, 0)$, and this vector has the property

$$\mathbf{0} + v = v + \mathbf{0} = v$$

for any vector v in $\mathbb{R}^n$. Thus, the zero vector is an *identity element* with respect to the addition of vectors. Furthermore, for any vector v in $\mathbb{R}^n$,

$$v + (-v) = \mathbf{0}.$$

Thus, $-v$ is the *inverse* of v with respect to the addition of vectors.

Consequently, we may collect all the above properties in the following theorem (see Definition 1.1.10).

3

Theorem 3.2.1 (Group Structure of $\mathbb{R}^n$)
The space $(\mathbb{R}^n, +)$ *is an* Abelian group *with respect to the addition* $(+)$.

Let us list a number of further properties:
Using (3.4), we have for any vector v in $\mathbb{R}^n$,

$$1v = 1(v_1, v_2, \ldots, v_n) = (1v_1, 1v_2, \ldots, 1v_n) = (v_1, v_2, \ldots, v_n) = v.$$

Thus, 1 is the *identity element* for multiplication by scalars.

Next, it is clear that for any two scalars k_1 and k_2, we have

$$(k_1k_2)v = (k_1k_2v_1, k_1k_2v_2, \ldots, k_1k_2v_n) = k_1(k_2v_1, k_2v_2, \ldots, k_2v_n) = k_1(k_2v).$$

This property represents the *compatibility of scalar multiplication* with the multiplication in $\mathbb{R}$.

Next, (3.3) and (3.4) imply that

$$\begin{aligned} k(v + w) &= k(v_1 + w_1, v_2 + w_2, \ldots, v_n + w_n) \\ &= (k(v_1 + w_1), k(v_2 + w_2), \ldots, k(v_n + w_n)) \\ &= (kv_1 + kw_1, kv_2 + kw_2, \ldots, kv_n + kw_n) \\ &= k(v_1, v_2, \ldots, v_n) + k(w_1, w_2, \ldots, w_n) \\ &= kv + kw. \end{aligned}$$

Thus, *distributivity of scalar multiplication* with respect to vector addition holds.

Finally, *distributivity of scalar multiplication* with respect to the addition in $\mathbb{R}$ also holds. That is, for any two scalar k_1 and k_1, we have

$$\begin{aligned} (k_1 + k_2)v &= ((k_1 + k_2)v_1, (k_1 + k_2)v_2, \ldots, (k_1 + k_2)v_n) \\ &= (k_1v_1 + k_2v_1, k_1v_2 + k_2v_2, \ldots, k_1v_n + k_2v_n) \\ &= (k_1v_1, k_1v_2, \ldots, k_1v_n) + (k_2v_1, k_2v_2, \ldots, k_2v_n) \\ &= k_1v + k_2v. \end{aligned}$$

The above four properties together with Theorem 3.2.1 endow $\mathbb{R}^n$ with an algebraic structure called *vector space*.

Let us give a general definition of a vector space E over a field $\mathbb{K}$.

Definition 3.2.2 (Vector Space)

Let E be a nonempty set and $\mathbb{K}$ be a field. Let two operations be given on E: one internal denoted by $(+)$ and defined as:

$$E \times E \to E : \qquad (v, w) \mapsto v + w$$

and one external denoted by $(\cdot)$ and defined as:

$$\mathbb{K} \times E \to E : \qquad (k, v) \mapsto k \cdot v = kv.$$

Then, E is called a *vector space* over the field $\mathbb{K}$ if for all x, y in E and for all λ, μ in $\mathbb{K}$ the following properties hold:

1. $(E, +)$ is an Abelian group.
2. $1_{\mathbb{K}} x = x$.
3. $\lambda(\mu x) = (\lambda\mu)x$.
4. $\lambda(x + y) = \lambda x + \lambda y$.
5. $(\lambda + \mu)x = \lambda x + \mu x$.

Example 3.2

As we have seen above, $\mathbb{R}^n$ is a vector space over $\mathbb{R}$. ■

Notation. We ca also write a vector $v = (v_1, v_2, \ldots, v_n)$ in $\mathbb{R}^n$ as a column vector

$$v = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}.$$

In this case and as we have seen in ▶ Chap. 1, the vector v can be seen as an $n \times 1$ matrix.

3.2.4 Linear Combination

As shown in Fig. 3.11, the vector v can be written in the coordinate system as

$$(v_1, v_2) = (v_1, 0) + (0, v_2) = v_1(1, 0) + v_2(0, 1).$$

Now, if we denote $e_1 = (1, 0)$ and $e_2 = (0, 1)$, then v can be written as

$$v = v_1 e_1 + v_2 e_2, \tag{3.5}$$

where v_1 and v_2 are scalars. Relation (3.5) says that the vector v is a *linear combination* of the two vectors e_1 and e_2.

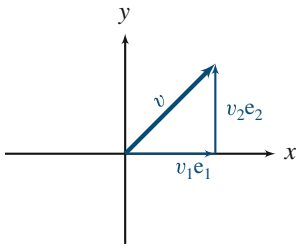


Fig. 3.11 The vector v is a linear combination of e_1 and e_2

Now, we generalize the above notion to the space $\mathbb{R}^n$ as follows.

Definition 3.2.3 (Linear Combination)

Let $v, v_1, v_2, \ldots, v_p$ be vectors in $\mathbb{R}^n$. We say that v is a *linear combination* of $v_1, v_2, \ldots, v_p$ if there exist scalars $k_1, k_2, \ldots, k_p$ such that

$$v = k_1v_1 + k_2v_2 + \cdots + k_pv_p. \tag{3.6}$$

Example 3.3

Let $v = (v_1, v_2, v_3)$ be a vector in $\mathbb{R}^3$; then v can be written as

$$v = v_1(1, 0, 0) + v_2(0, 1, 0) + v_3(0, 0, 1).$$

Thus, v is a linear combination of the vectors $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$, and $e_3 = (0, 0, 1)$. ■

3.3 Norm, Dot Product, and Distance in $\mathbb{R}^n$

In this section, we define some numbers associated to vectors, called norm, dot product, and distance.

3.3.1 The Norm of a Vector in $\mathbb{R}^n$

As, we have said earlier, a vector is determined by its length (or magnitude or *norm*) and its direction. Now, to define the norm of a vector in $\mathbb{R}^n$, let us first consider the vector $v = (v_1, v_2)$ in the coordinate system $\mathbb{R}^2$ as in Fig. 3.11. Then, using Pythagoras' theorem, we have

$$\|v\|^2 = v_1^2 + v_2^2, \qquad \text{or} \qquad \|v\| = \sqrt{v_1^2 + v_2^2},$$

where $\|v\|$ denotes the length of v. This indicates how to generalize the above notion to vectors in $\mathbb{R}^n$ for any $n \geq 2$.

Definition 3.3.1 (Norm of a Vector)

Let $v = (v_1, v_2, \ldots, v_n)$ be a vector in $\mathbb{R}^n$. Then the *norm* of v is denoted by $\|v\|$ and is defined as

$$\|v\| = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}. \tag{3.7}$$

The norm in (3.7) is called the *Euclidean norm* since it is associated with the Euclidean geometry.

Example 3.4

Consider the vector $v = (1, 0, -2)$. Then,

$$\|v\| = \sqrt{1^2 + 0^2 + (-2)^2} = \sqrt{5}.$$

■

Let us list some properties of the norm of a vector.

Theorem 3.3.1 (Properties of the Norm of a Vector)

Let $v = (v_1, v_2, \ldots, v_n)$ be a vector in $\mathbb{R}^n$ and k be a scalar. Then, we have

1. $\|v\| \geq 0$.
2. $\|v\| = 0$ *if and only if* $v = \mathbf{0}$.
3. $\|kv\| = |k|\|v\|$.

Proof

1. The first property is a direct consequence of (3.7).

2. Assume first that $\|v\| = 0$. Then by (3.7), we deduce that

$$v_1^2 + v_2^2 + \cdots + v_n^2 = 0,$$

whence, $v_1 = v_2 = \cdots = v_n = 0$. Thus, $v = (0, 0, \ldots, 0) = \mathbf{0}$. Conversely, by definition,

$$\mathbf{0} = \sqrt{0^2 + 0^2 + \cdots + 0^2} = 0.$$

3. Since $kv = (kv_1, kv_2, \ldots, kv_n)$, then,

$$\begin{aligned}
\|kv\| &= \sqrt{(kv_1)^2 + (kv_2)^2 + \cdots + (kv_n)^2} \\
&= \sqrt{k^2(v_1^2 + v_2^2 + \cdots + v_n^2)} \\
&= |k| \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2} \\
&= |k|\|v\|.
\end{aligned}$$

Remark 3.3.2
It is clear from above that the norm of a vector in $\mathbb{R}^n$ can be seen as a mapping $\mathcal{N}$ defined on the spaces $\mathbb{R}^n$ and with values into $\mathbb{R}^+$ as:

$$\mathcal{N} : \mathbb{R}^n \to \mathbb{R}^+, \quad v \mapsto \mathcal{N}(v) = \|v\|,$$

such that the properties in Theorem 3.3.1 are satisfied. □

Unit Vectors

Definition 3.3.2 (Unit Vector)
A vector of norm 1 is called a *unit vector*.

Example 3.5
The vectors $e_1 = (1, 0)$ and $e_2 = (0, 1)$ are unit vectors in $\mathbb{R}^2$, since $\|e_1\| = \|e_2\| = 1$. ■

Example 3.6 (Normalizing a Nonzero Vector)
Let v be a nonzero vector in $\mathbb{R}^n$. Let

$$u = \frac{v}{\|v\|}.$$

Then, u is a unit vector since

$$\|u\| = \left\| \frac{v}{\|v\|} \right\| = \frac{1}{\|v\|}\|v\| = 1,$$

where we have applied the last property in Theorem 3.3.1, with $k = 1/\|v\|$. The above process of obtaining the unit vector u is called the *normalization of* v. We can write v as

$$v = \|v\|u.$$

Since $\|v\| \geq 0$, we see that the vectors u and v have the same direction. ■

Example 3.7 (Standard Unit Vectors in $\mathbb{R}^n$)
The vectors

$$e_1 = (1, 0, 0, \ldots, 0) \quad e_2 = (0, 1, 0, \ldots, 0), \ldots, \quad e_n = (0, \ldots, 0, 1)$$

are unit vectors in $\mathbb{R}^n$, called the *standard* unit vectors in $\mathbb{R}^n$. ■

Remark 3.3.3 It is clear that any vectors $v = (v_1, v_2, \ldots, v_n)$ can be written as a linear combination of the standard unit vectors $e_1, e_2, \ldots, e_n$ as

$$v = v_1e_1 + v_2e_2 + \ldots v_ne_n.$$

3.3.2 Distance in $\mathbb{R}^n$

Definition 3.3.3 (Distance)

Let $P_1(x_1, x_2, \ldots, x_n)$ and $P_2(y_1, y_2, \ldots, y_n)$ be two points in $\mathbb{R}^n$. We define the *distance* between P_1 and P_2 to be the norm of the vector $\overrightarrow{P_1P_2}$ and we write

$$d(P_1, P_2) = \|\overrightarrow{P_1P_2}\| = \sqrt{(y_1 - x_1)^2 + (y_2 - x_2)^2 + \cdots + (y_n - x_n)^2}. \tag{3.8}$$

Example 3.8

Let $P_1(0, 1, 2)$ and $P_2(-1, 3, 1)$ be two points in $\mathbb{R}^3$. Then

$$d(P_1, P_2) = \|\overrightarrow{P_1P_2}\| = \sqrt{(-1 - 0)^2 + (3 - 1)^2 + (1 - 2)^2} = \sqrt{6}.$$

■

In the following theorem, we list some properties of the distance.

Theorem 3.3.4 (Properties of the Distance)

Let $P_1(x_1, x_2, \ldots, x_n)$, $P_2(y_1, y_2, \ldots, y_n)$ and $P_3(z_1, z_2, \ldots, z_n)$ be three points in $\mathbb{R}^n$. Then, we have

1. $d(P_1, P_2) \geq 0$.
2. $d(P_1, P_2) = 0$ *if and only if* $P_1 = P_2$.
3. $d(P_1, P_2) = d(P_2, P_1)$.
4. $d(P_1, P_2) \leq d(P_1, P_2) + d(P_2, P_3)$ *(triangle inequality).*

Proof

The first and the second properties are obvious. To prove the third property, we have

$$\begin{aligned} d(P_1, P_2) &= \sqrt{(y_1 - x_1)^2 + (y_2 - x_2)^2 + \cdots + (y_n - x_n)^2} \\ &= \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2 + \cdots + (x_n - y_n)^2} \\ &= d(P_2, P_1). \end{aligned}$$

For the fourth property, since the distance is positive, it is enough to show that

$$\big(d(P_1, P_2)\big)^2 \leq \big(d(P_1, P_2) + d(P_2, P_3)\big)^2. \tag{3.9}$$

We have

$$\begin{aligned}\big(d(P_1,P_2)\big)^2 &= (x_1-y_1)^2+(x_2-y_2)^2+\cdots+(x_n-y_n)^2\\ &= ((x_1-z_1)+(z_1-y_1))^2+((x_2-z_2)+(z_2-y_2))^2+\ldots\\ &\quad+((x_n-z_n)+(z_n-y_n))^2\\ &= (x_1-z_1)^2+(x_2-z_2)^2+\cdots+(x_n-z_n)^2\\ &\quad+(z_1-y_1)^2+(z_2-y_2)^2+\cdots+(z_n-y_n)^2\\ &\quad+2(x_1-z_1)(z_1-y_1)+2(x_2-z_2)(z_2-y_2)+\cdots+2(x_n-z_n)(z_n-y_n).\end{aligned} \tag{3.10}$$

Clearly,

$$\begin{aligned}&(x_1-z_1)(z_1-y_1)+(x_2-z_2)(z_2-y_2)+\cdots+(x_n-z_n)(z_n-y_n)\\ &\le\sqrt{\Big[(x_1-z_1)^2+(x_2-z_2)^2+\cdots+(x_n-z_n)^2\Big]\Big[(z_1-y_1)^2+(z_2-y_2)^2+\cdots+(z_n-y_n)^2\Big]}\\ &= d(P_1,P_2)d(P_2,P_3)\end{aligned}$$

Plugging this last inequality into (3.10), then, we obtain (3.9). □

3.3.3 The Dot Product of Two Vectors in $\mathbb{R}^n$

Since vectors in $\mathbb{R}^n$ can be seen as column matrices, we can not multiply two vectors using matrix multiplication (Definition 1.1.11). Instead, we can define the *dot product* of two vectors. To examine a simple case first consider two vectors $u=(u_1,u_2)$ and $v=(v_1,v_2)$ in $\mathbb{R}^2$ and let θ be the acute angle between them, as in Fig. 3.12. We define the dot product of these two vectors as the product of their norms and the cosine of the angle θ:

$$u\cdot v=\|u\|\|v\|\cos\theta. \tag{3.11}$$

Formula (3.11) is based on 2-dimensional geometry and it cannot be applied for vectors in $\mathbb{R}^n$ with $n>3$. For this reason we will provide an algebraic formula to define the

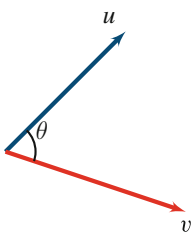


Fig. 3.12 The dot product of u and v

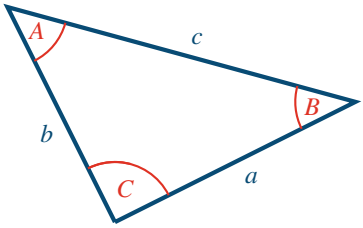


Fig. 3.13 The cosine law

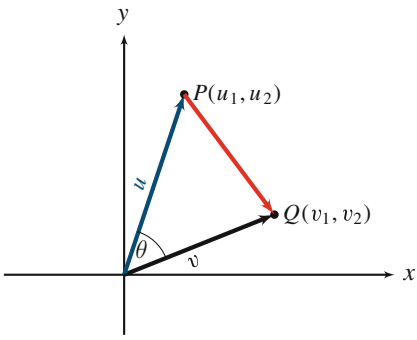


Fig. 3.14 The sum of two vectors $v + w$ in a coordinate system

dot product of two vectors in $\mathbb{R}^n$ for any $n \geq 1$. To do this, let us first review the cosine formula for any triangle with A, B and C as its angles and a, b and c the lengths of its sides, as shown in Fig. 3.13.

We have

$$\begin{aligned} c^2 &= a^2 + b^2 - 2ab\cos C, \\ a^2 &= b^2 + c^2 - 2bc\cos A, \\ b^2 &= a^2 + c^2 - 2ac\cos B. \end{aligned} \tag{3.12}$$

Consider two vectors $u = (u_1, u_2)$ and $v = (v_1, v_2)$ in the standard coordinate system of $\mathbb{R}^2$ as in Fig. 3.14.

Using the cosine rule (3.12), we have

$$\|\overrightarrow{PQ}\|^2 = \|u\|^2 + \|v\|^2 - 2\|u\|\|v\|\cos\theta. \tag{3.13}$$

As we have seen before,

$$\|\overrightarrow{PQ}\|^2 = (v_1 - u_1)^2 + (v_2 - u_2)^2, \quad \|v\|^2 = v_1^2 + v_2^2, \quad \|u\|^2 = u_1^2 + u_2^2.$$

Plugging these last expressions into (3.13), we have

$$u \cdot v = \|u\|\|v\|\cos\theta = u_1v_1 + u_2v_2.$$

In this last formula, we got a rid of the cosine of the angle θ and so, by knowing the components of the vectors u and v, we can easily compute the dot product between the two vectors using the algebraic expression

$$u \cdot v = u_1 v_1 + u_2 v_2.$$

This indicates how to generalize the definition of the dot product for any two vectors $u = (u_1, u_2, \ldots, u_n)$ and $v = (v_1, v_2, \ldots, v_n)$ in $\mathbb{R}^n$, for all $n \geq 2$.

Definition 3.3.4 (Dot Product)

Let $u = (u_1, u_2, \ldots, u_n)$ and $v = (v_1, v_2, \ldots, v_n)$ be two vectors in $\mathbb{R}^n$. Then, the *dot product* (also called the *inner product* or *scalar product*) of u and v is defined as

$$u \cdot v = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n. \tag{3.14}$$

Example 3.9

Consider the vectors $u = (1, 2, -1)$ and $v = (3, 0, -1)$ in $\mathbb{R}^3$. Then

$$u \cdot v = 1 \cdot 3 + 2 \cdot 0 + (-1) \cdot (-1) = 2.$$

■

It is clear that formula (3.14) allows us to deduce several properties of the dot product. We list some of them.

Theorem 3.3.5 (Properties of the Dot Product)

Let $u = (u_1, u_2, \ldots, u_n)$, $v = (v_1, v_2, \ldots, v_n)$, and $w = (w_1, w_2, \ldots, w_n)$ be vectors in $\mathbb{R}^n$ and k be a scalar. Then, we have

1. $u \cdot \mathbf{0} = 0$.
2. $u \cdot u = \|u\|^2$.
3. $u \cdot v = v \cdot u$.
4. $u \cdot (v + w) = u \cdot v + u \cdot w$.
5. $k(u \cdot v) = (ku) \cdot v = u \cdot (kv)$.

Proof

1. It is clear that, since $\mathbf{0} = (0, 0, \ldots, 0)$, then $u \cdot \mathbf{0} = 0$.
2. By Definition 3.3.1,

$$u \cdot u = u_1^2 + u_2^2 + \cdots + u_n^2 = \|u\|^2.$$

3. It is obvious that

$$\begin{aligned} u \cdot v &= u_1v_1 + u_2v_2 + \cdots + u_nv_n \\ &= v_1u_1 + v_2u_2 + \cdots + v_nu_n \\ &= v \cdot u. \end{aligned}$$

4. By (3.3),

$$v + w = (v_1 + w_1, v_2 + w_2, \ldots, v_n + w_n).$$

Hence,

$$\begin{aligned} u \cdot (v + w) &= u_1(v_1 + w_1) + u_2(v_2 + w_2) + \cdots + u_n(v_n + w_n) \\ &= (u_1v_1 + u_2v_2 + \cdots + u_nv_n) + (u_1w_1 + u_2w_2 + \cdots + u_nw_n) \\ &= u \cdot v + u \cdot w. \end{aligned}$$

5. Similarly, using (3.4), we have

$$\begin{aligned} k(u \cdot v) &= (ku_1)v_1 + (ku_2)v_2 + \cdots + (ku_n)v_n \\ &= (ku) \cdot v \\ &= u_1(kv_1) + u_2(kv_2) + \cdots + u_n(kv_n) \\ &= u \cdot (kv). \end{aligned}$$

□

It is clear from above that the dot product has very similar properties to those of the usual multiplication in $\mathbb{R}$. Here is a simple example.

Example 3.10
For any two vectors u and v in $\mathbb{R}^n$, we have

$$\begin{aligned} (u - 2v) \cdot (u + 3v) &= u \cdot (u + 3v) + (-2v) \cdot (u + 3v) \\ &= (u \cdot u) + (u \cdot (3v)) + (-2v) \cdot u + (-2v) \cdot (3v) \\ &= \|u\|^2 + 3(u \cdot v) - 2(v \cdot u) - 6\|v\|^2 \\ &= \|u\|^2 + (u \cdot v) - 6\|v\|^2. \end{aligned}$$

■

Next, we prove a very important inequality, called the *Cauchy–Schwarz inequality*. This inequality evaluates the absolute value of the dot product of two vectors in terms

of their norms. Its idea is very simple and for vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$ it is a direct consequence of the definition of the dot product (formula (3.11)) and the properties of the cosine function. So, let u and v be two nonzero vectors in $\mathbb{R}^2$ or $\mathbb{R}^3$. Then, from (3.11), we have

$$\cos\theta = \frac{u \cdot v}{\|u\|\|v\|},$$

and since

$$|\cos\theta| \leq 1,$$

it follows that

$$\left| \frac{u \cdot v}{\|u\|\|v\|} \right| \leq 1.$$

This yields

$$|u \cdot v| \leq \|u\|\|v\|. \tag{3.15}$$

Inequality (3.15) is called the *Cauchy–Schwarz inequality* (in $\mathbb{R}^2$ or $\mathbb{R}^3$) and trivially holds if u or v is the zero vector. Now, we can state and prove this inequality for any two vectors u and v in $\mathbb{R}^n$.

Theorem 3.3.6 (Cauchy–Schwarz Inequality)
Let $u = (u_1, u_2, \ldots, u_n)$ and $v = (v_1, v_2, \ldots, v_n)$ be two vectors in $\mathbb{R}^n$. Then,

$$|u \cdot v| \leq \|u\|\|v\|. \tag{3.16}$$

That is,

$$|u_1v_1 + u_2v_2 + \cdots + u_nv_n| \leq \left(\sqrt{u_1^2 + u_2^2 + \cdots + u_n^2} \right) \left(\sqrt{v_1^2 + v_2^2 + \cdots + v_n^2} \right).$$

The equality holds if and only if $u = kv$ for some a scalar $k = \frac{u \cdot v}{\|v\|^2}$.

Proof

In fact there are several proofs of the Cauchy–Schwarz inequality; here we present one that uses the above properties of the dot product.

As, we have seen above, it is clear that for any two vectors u and v in $\mathbb{R}^n$ and for any λ in $\mathbb{R}$, we have

$$(u + \lambda v) \cdot (u + \lambda v) = \|u + \lambda v\|^2 \geq 0.$$

On the other hand, by using the properties of the dot product, we have

$$\begin{aligned}(u+\lambda v)\cdot(u+\lambda v) &= (u\cdot u)+\lambda^2(v\cdot v)+2\lambda(u\cdot v)\\ &= \lambda^2\|v\|^2+2\lambda(u\cdot v)+\|u\|^2\\ &= a\lambda^2+2b\lambda+c, \end{aligned} \tag{3.17}$$

with

$$a=\|v\|^2, \qquad b=(u\cdot v), \qquad c=\|u\|^2.$$

Since $a>0$, the polynomial of λ in (3.17) is positive for any λ in $\mathbb{R}$ if and only if the discriminant

$$\Delta = b^2-ac<0,$$

that is

$$|b|\leq \sqrt{a}\,\sqrt{c},$$

or equivalently

$$|(u\cdot v)|\leq \|u\|\|v\|,$$

which is exactly the Cauchy–Schwarz inequality.

Now, it is clear that if $u=kv$, then,

$$\begin{aligned}|u\cdot v| = k\|v\|^2 &= k\|v\|\|v\|\\ &= \|u\|\|v\|.\end{aligned}$$

On the other hand, if $|u\cdot v|=\|u\|\|v\|$, then, $u=\frac{u\cdot v}{\|v\|^2}v$. This finishes the proof of Theorem 3.3.6. □

Example 3.11

Consider the vectors $u=(1,0,-1)$ and $v=(-1,2,0)$. Thus, we have

$$\|u\|=\sqrt{2}, \qquad \|u\|=\sqrt{5}, \quad u\cdot v=-1,$$

It is clear that

$$1<\sqrt{2}\,\sqrt{5},$$

which means that the Cauchy–Schwarz inequality holds. ■

Example 3.12 (Young's Inequality)
Consider the two vectors $u = (a, b)$ and $v = (b, a)$ in $\mathbb{R}^2$. Then, by applying the Cauchy–Schwarz inequality, we have

$$|u \cdot v| = 2|ab| \leq \|u\|\|v\| = a^2 + b^2.$$

Thus, for any two real numbers a and b,

$$|ab| \leq \frac{a^2}{2} + \frac{b^2}{2}. \tag{3.18}$$

Now, let $\eta > 0$ be any real number. Applying (3.18) for $a' = \sqrt{\eta}a$ and $b' = \dfrac{b}{\sqrt{\eta}}$, we obtain

$$|a'b'| \leq \frac{a'^2}{2} + \frac{b'^2}{2},$$

or equivalently,

$$|ab| \leq \frac{\eta}{2}a^2 + \frac{b^2}{2\eta}. \tag{3.19}$$

Inequality (3.19) is known as *Young's inequality*. If we put $\eta/2 = \gamma$, then inequality (3.19) becomes

$$|ab| \leq \gamma a^2 + \frac{b^2}{4\gamma}. \tag{3.20}$$

■

Example 3.13
Show that for any real numbers $a_1, a_2, \ldots, a_n$, we have the inequality

$$(a_1 + a_2 + \cdots + a_n)^2 \leq n(a_1^2 + a_2^2 + \cdots + a_n^2). \tag{3.21}$$

■

Solution
Consider in $\mathbb{R}^n$ the two vectors

$$u = (1, 1, \ldots, 1) \qquad \text{and} \qquad v = (a_1, a_2, \ldots, a_n).$$

Then, we have the Cauchy–Schwarz inequality,

$$|u \cdot v| \leq \|u\|\|v\|.$$

That is,

$$|a_1 + a_2 + \cdots + a_n| \leq \sqrt{n}\sqrt{a_1^2 + a_2^2 + \cdots + a_n^2},$$

or equivalently

$$(a_1 + a_2 + \cdots + a_n)^2 \leq n(a_1^2 + a_2^2 + \cdots + a_n^2).$$

Which is exactly (3.21). ◄

Now, having the Cauchy–Schwarz inequality, we can prove some important inequalities and identities.

Theorem 3.3.7 (Triangle Inequality)
Let u and v be two vectors in $\mathbb{R}^n$. Then,

$$\|u + v\| \leq \|u\| + \|v\|. \tag{3.22}$$

Proof
We know that

$$(u + v) \cdot (u + v) = \|u + v\|^2. \tag{3.23}$$

On the other hand, by using the properties of the dot product,

$$\begin{aligned}(u + v) \cdot (u + v) &= u \cdot u + v \cdot v + 2(u \cdot v)\\ &= \|u\|^2 + \|v\|^2 + 2(u \cdot v).\end{aligned} \tag{3.24}$$

By the Cauchy–Schwarz inequality, we have

$$2(u \cdot v) \leq 2|(u \cdot v)| \leq 2\|u\|\|v\|.$$

Plugging this last inequality into (3.24), we obtain

$$\begin{aligned}(u + v) \cdot (u + v) &\leq \|u\|^2 + \|v\|^2 + 2\|u\|\|v\|\\ &= (\|u\| + \|v\|)^2.\end{aligned} \tag{3.25}$$

Comparing (3.23) and (3.25), we get

$$\|u + v\|^2 \leq (\|u\| + \|v\|)^2,$$

which eventually leads to (3.22). □

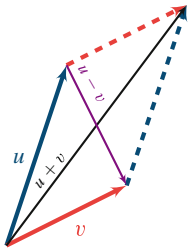


Fig. 3.15 The parallelogram identity

Theorem 3.3.8 (Parallelogram Identity)
Let u and v be two vectors in $\mathbb{R}^n$. Then,

$$\|u+v\|^2 + \|u-v\|^2 = 2(\|u\|^2 + \|v\|^2). \tag{3.26}$$

Proof
The parallelogram identity in $\mathbb{R}^2$ is illustrated in Fig. 3.15. To prove it, we have, on one hand

$$\begin{aligned}\|u+v\|^2 = (u+v)\cdot(u+v) &= (u\cdot u) + (v\cdot v) + 2(u\cdot v)\\ &= \|u\|^2 + \|v\|^2 + 2(u\cdot v).\end{aligned} \tag{3.27}$$

On the other hand,

$$\begin{aligned}\|u-v\|^2 = (u-v)\cdot(u-v) &= u\cdot u + v\cdot v - 2(u\cdot v)\\ &= \|u\|^2 + \|v\|^2 - 2(u\cdot v).\end{aligned} \tag{3.28}$$

Combining (3.27) and (3.28), we obtain (3.26). □

Above, we wrote the norm of a vector u in terms of the dot product. Now, we want to do the opposite. So, *can we write the dot product of two vectors using their norms?* The answer of this question is given in the following theorem.

Theorem 3.3.9 (Polarization Identity)
Let u and v be two vectors in $\mathbb{R}^n$. Then, we have

$$u\cdot v = \frac{1}{4}\|u+v\|^2 - \frac{1}{4}\|u-v\|^2. \tag{3.29}$$

Proof

To prove (3.29), we subtract (3.28) from (3.27), obtaining

$$\|u+v\|^2 - \|u-v\|^2 = 4(u \cdot v),$$

which is exactly (3.29). □

3.4 Orthogonality in $\mathbb{R}^n$

From Euclidean geometry we know that two vectors u and v in $\mathbb{R}^2$ are *orthogonal* or *perpendicular* if and only if the acute angle between them is $\theta = \pi/2$. This is a geometric definition. To define the orthogonality of vectors in $\mathbb{R}^n$ for any $n \geq 2$, we need to find instead an algebraic formulation. We have seen from formula (3.11) that if we know the angle between two vectors, then we can define the algebraic quantity that we called the dot product. Thus, if $\theta = \pi/2$, then we have from (3.11) that

$$u \cdot v = 0. \tag{3.30}$$

We see immediately that the two vectors are orthogonal if and only if their dot product is zero. Since (3.30) is an algebraic definition of the orthogonality, then we can take it as a definition of the orthogonality of vectors in $\mathbb{R}^n$ for any $n \geq 2$.

Definition 3.4.1 (Orthogonality in $\mathbb{R}^n$)

Let u and v be two vectors in $\mathbb{R}^n$. Then u and v are said to be *orthogonal* if

$$u \cdot v = 0. \tag{3.31}$$

Example 3.14

1. The vectors $e_1 = (1, 0)$, and $e_2 = (0, 1)$ are orthogonal in $\mathbb{R}^2$ since $e_1 \cdot e_2 = 0$.
2. In $\mathbb{R}^3$, the vectors $e_1 = (1, 0, 0)$, $e_2 = (0, 1, 0)$ and $e_3 = (0, 0, 1)$ are pairwise orthogonal since

$$e_1 \cdot e_2 = e_2 \cdot e_3 = e_3 \cdot e_1 = 0.$$

3. The vectors $u = (-1, 0, 1, 0)$ and $v = (1, 0, 1, 2)$ are orthogonal in $\mathbb{R}^4$.

■

Example 3.15

1. Show that the vectors $u = (a, b)$ and $v = (b, -a)$ are orthogonal in $\mathbb{R}^2$.
2. Use the above result to find two vectors that are orthogonal to $w = (-2, 3)$.
3. Find a unit vector that is orthogonal to $y = (5, 12)$.

■

Solution

1. We have

$$u \cdot v = ab - ba = 0,$$

so u and v are orthogonal.

2. Let $w = (-2, 3)$. Then, we have $w = (a, b)$ with $a = -2$ and $b = 3$. From the first question, we deduce that w is orthogonal to the vector $(b, -a) = (3, 2)$. Also, the vector (a, b) is orthogonal to the vector $(-b, a)$. Thus, w is also orthogonal to $(-3, -2)$.

3. Let $y = (5, 12)$. Then from above, we deduce that y is orthogonal to $(12, -5)$. Denote $z = (12, -5)$. We form a unit vector from this vector by putting (see Example 3.6)

$$u = \frac{1}{\|z\|} z = \frac{1}{\sqrt{(12)^2 + (-5)^2}} (12, -5) = \frac{1}{13} (12, -5) = (12/13, -5/13).$$

It is clear that u is orthogonal to y since

$$y \cdot u = 5(12/13) + 12(-5/13) = 0.$$

◀

3.4.1 Orthogonal Projections in $\mathbb{R}^n$

Let $\vec{F}$ be a force applied to a particle moving along a path. The magnitude of this force in the direction of a nonzero vector v is given by

$$\|\vec{F_v}\| = \|\vec{F}\| \cos\theta \frac{v}{\|v\|}$$

where $v/\|v\|$ is the unit vector in the direction of v. The vector $\vec{F_v}$ is called the *orthogonal projection* of $\vec{F}$ on v.

Before we define the projection of a vector u on a nonzero vector v, we prove the following assertion.

Theorem 3.4.1

Let u and v be two vectors in $\mathbb{R}^n$. Assume that v is a nonzero vector. Then, u can be written in exactly one way in the form

$$u = w_1 + w_2, \tag{3.32}$$

where $w_1 = kv$ and w_2 is orthogonal to v (as shown in Fig. 3.16).

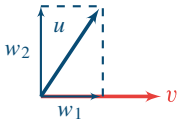


Fig. 3.16 The projection of u on v

Proof

Since $w_1 = kv$, where k is a scalar, and since w_2 is orthogonal to v, we have $w_2 \cdot v = 0$. Thus, we obtain

$$\begin{aligned} u \cdot v &= (w_1 + w_2) \cdot v \\ &= w_1 \cdot v + w_2 \cdot v \\ &= (kv) \cdot v \\ &= k\|v\|^2. \end{aligned}$$

Consequently,

$$k = \frac{u \cdot v}{\|v\|^2},$$

whence

$$w_1 = \frac{u \cdot v}{\|v\|^2} v \qquad \text{and} \qquad w_2 = u - \frac{u \cdot v}{\|v\|^2} v.$$

□

Definition 3.4.2 (Orthogonal Projection)

The above vector w_1 is called the *orthogonal projection* of u on v and we denote

$$\text{proj}_v u = \frac{u \cdot v}{\|v\|^2} v. \tag{3.33}$$

Remark 3.4.2 It is clear that the vector w_1 can be written as

$$w_1 = \frac{u \cdot v}{\|v\|} \frac{v}{\|v\|},$$

and since $v/\|v\|$ is a unit vector, we call the number

$$\frac{u \cdot v}{\|v\|}$$

the *component of u in the direction of v*.

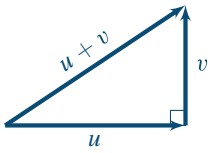


Fig. 3.17 Pythagoras' theorem in a right triangle

Example 3.16
Let $u = (1, -1, 0)$ and $v = (2, 0, 1)$. Then, we have

$$\begin{aligned} \text{proj}_v\, u &= \frac{u \cdot v}{\|v\|^2} v \\ &= \frac{2}{\sqrt{5}}(2, 0, 1) \\ &= \left(\frac{4}{\sqrt{5}}, 0, \frac{2}{\sqrt{5}}\right). \end{aligned}$$

■

Pythagoras' Theorem in $\mathbb{R}^n$

In Euclidean geometry, Pythagoras' theorem says that in a right triangle as in Fig. 3.17, we have

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2. \tag{3.34}$$

Since the angle between the vectors u and v is $\theta = \pi/2$, we have by making use of (3.11) that $u \cdot v = 0$. Consequently, instead of the geometric property discussed above, we formulated an algebraic property that the two vectors should satisfy in order for the Pythagoras theorem to hold. That is to say, if the dot product of two vectors is zero, then the identity (3.34) is satisfied. Thus, using this algebraic assumption, we can generalize the Pythagoras theorem for any two orthogonal vectors in $\mathbb{R}^n$ and we have the following statement.

Theorem 3.4.3 (Pythagoras' Theorem in $\mathbb{R}^n$)
Let $u = (u_1, u_2, \ldots, u_n)$ and $v = (v_1, v_2, \ldots, v_n)$ be two orthogonal vectors in $\mathbb{R}^n$. Then,

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2. \tag{3.35}$$

Proof

Since u and v are orthogonal, $u \cdot v = 0$. On the other hand,

$$\begin{aligned}\|u+v\|^2 = (u+v)\cdot(u+v) &= (u\cdot u) + (v\cdot v) + 2(u\cdot v)\\ &= \|u\|^2 + \|v\|^2 + 2(u\cdot v)\\ &= \|u\|^2 + \|v\|^2,\end{aligned}$$

which is exactly (3.35). □

3.5 Exercises

Exercise 3.1 (Orthogonal Matrix)

Consider the matrix

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

Show that:

1. Each row of A is a unit vector and each column of A is a unit vector.
2. The row vectors of A are pairwise orthogonal.
3. The column vectors of A are pairwise orthogonal.

A matrix with the above properties is called an *orthogonal matrix*. See ▶ Sect. 8.1. ■

Solution

We denote by

$$r_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0\right), \quad r_2 = \left(-\frac{1}{2}, \frac{1}{2}, \frac{1}{\sqrt{2}}\right), \quad r_3 = \left(\frac{1}{2}, -\frac{1}{2}, \frac{1}{\sqrt{2}}\right)$$

the row vectors of A, and by

$$c_1 = \left(\frac{1}{\sqrt{2}}, -\frac{1}{2}, \frac{1}{2}\right), \quad c_2 = \left(\frac{1}{\sqrt{2}}, \frac{1}{2}, -\frac{1}{2}\right), \quad c_3 = \left(0, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$$

the column vectors of A.

1. It is clear that

$$\|r_1\| = \|c_3\| = \sqrt{\frac{1}{2} + \frac{1}{2}} = 1.$$

Similarly, we have

$$\|r_2\| = \|r_3\| = \|c_1\| = \|c_2\| = \sqrt{\frac{1}{4} + \frac{1}{4} + \frac{1}{2}} = 1.$$

2. We have

$$r_1 \cdot r_2 = \frac{1}{2} \cdot \frac{1}{\sqrt{2}} - \frac{1}{2} \cdot \frac{1}{\sqrt{2}} = 0.$$

Similarly,

$$r_1 \cdot r_3 = r_2 \cdot r_3 = 0.$$

Thus, the row vectors r_1, r_2, and r_3 are pairwise orthogonal.

3. We can easily check that

$$c_1 \cdot c_2 = c_2 \cdot c_3 = c_3 \cdot c_1 = 0,$$

i.e., the column vectors c_1, c_2, and c_3 are pairwise orthogonal. ◀

Exercise 3.2

Let u and v be two vectors in $\mathbb{R}^n$.

1. Show that

$$\Big|\|u\| - \|v\|\Big| \leq \|u - v\|. \tag{3.36}$$

2. Prove that

$$\|u + v\|\|u - v\| \leq \|u\|^2 + \|v\|^2. \tag{3.37}$$

When does equality hold? Give a geometric interpretation of this inequality in $\mathbb{R}^2$.

■

Solution

1. To establish (3.36), we write $u = v + (u - v)$. Then, by the triangle inequality (3.22),

$$\begin{aligned} \|u\| &= \|v + (u - v)\| \\ &\leq \|v\| + \|u - v\|. \end{aligned}$$

That is

$$\|u\| - \|v\| \leq \|u - v\|. \tag{3.38}$$

On the other hand, we also have $v = u + (v - u)$, so, as above

$$\begin{aligned} \|v\| &= \|u + (v - u)\| \\ &\leq \|u\| + \|v - u\|. \end{aligned}$$

Therefore,

$$\|v\| - \|u\| \leq \|v - u\| = \|u - v\|. \tag{3.39}$$

Now (3.38) and (3.39) yield (3.36).

2. To prove (3.37), we have, as in (3.27) and (3.28),

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2 + 2(u \cdot v) \tag{3.40}$$

and

$$\|u - v\|^2 = \|u\|^2 + \|v\|^2 - 2(u \cdot v), \tag{3.41}$$

which in turn yield

$$\begin{aligned} \|u + v\|^2 \|u - v\|^2 &= \left(\|u\|^2 + \|v\|^2 + 2(u \cdot v)\right)\left(\|u\|^2 + \|v\|^2 - 2(u \cdot v)\right) \\ &= (\|u\|^2 + \|v\|^2)^2 - 4(u \cdot v)^2. \end{aligned}$$

Since $(u \cdot v)^2 \geq 0$, then we have

$$\|u + v\|^2 \|u - v\|^2 \leq (\|u\|^2 + \|v\|^2)^2,$$

which is exactly (3.37). Of course equality holds if and only if $u \cdot v = 0$, which means if and only if the vectors are orthogonal. Geometrically, this means that the product of the lengths of the diagonals in a parallelogram as in Fig. 3.15 is less than or equal to the sum $\|u\|^2 + \|v\|^2$, and we have equality if and only if the parallelogram is a square, that is, if and only if the vectors u and v are orthogonal. ◀

Exercise 3.3

1. Suppose that u, v, w, and z are vectors in $\mathbb{R}^n$ such that

$$u + v + w + z = \mathbf{0}.$$

Prove that $u + v$ is orthogonal to $u + z$ if and only if

$$\|u\|^2 + \|w\|^2 = \|v\|^2 + \|z\|^2. \tag{3.42}$$

2. Prove that

$$16 \leq (a+b+c+d)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}\right) \tag{3.43}$$

for all positive numbers a, b, c, d. ■

Solution

1. We first compute $\|w\|^2$, obtaining

$$\begin{aligned}\|w\|^2 &= w \cdot w \\ &= (-(u+v+z)) \cdot (-(u+v+z)) \\ &= (u \cdot u) + (v \cdot v) + (z \cdot z) + 2\Big((u \cdot v) + (v \cdot z) + (u \cdot z)\Big) \\ &= \|u\|^2 + \|v\|^2 + \|z\|^2 + 2\Big((u \cdot v) + (u \cdot u) - (u \cdot u) + (v \cdot z) + (u \cdot z)\Big). \end{aligned} \tag{3.44}$$

We have

$$(u \cdot v) + (u \cdot u) = (u \cdot (u+v)) \qquad \text{and} \qquad (v \cdot z) + (u \cdot z) = (u+v) \cdot z.$$

Consequently,

$$\begin{aligned}(u \cdot v) + (u \cdot u) + (v \cdot z) + (u \cdot z) &= u \cdot (u+v) + (v+u) \cdot z \\ &= (u+v) \cdot (u+z).\end{aligned}$$

Plugging this last identity into (3.44), we get

$$\begin{aligned}\|w\|^2 &= \|u\|^2 + \|v\|^2 + \|z\|^2 + 2\Big((u+v) \cdot (u+z) - u \cdot u\Big) \\ &= \|v\|^2 + \|z\|^2 - \|u\|^2 + 2(u+v) \cdot (u+z),\end{aligned}$$

since $u \cdot u = \|u\|^2$. Now, from this last formula, we see that if $u+v$ and $u+z$ are orthogonal, then

$$\|w\|^2 + \|u\|^2 = \|v\|^2 + \|z\|^2. \tag{3.45}$$

Conversely, if (3.45) holds, then $(u+v) \cdot (u+z) = 0$, i.e., $u+v$ and $u+z$ are orthogonal.

2. We consider the two vectors in $\mathbb{R}^4$

$$u = (\sqrt{a}, \sqrt{b}, \sqrt{c}, \sqrt{d}) \qquad \text{and} \qquad v = \left(\frac{1}{\sqrt{a}}, \frac{1}{\sqrt{b}}, \frac{1}{\sqrt{c}}, \frac{1}{\sqrt{d}}\right).$$

Then

$$\|u\| = \sqrt{a+b+c+d} \qquad \text{and} \qquad \|v\| = \sqrt{\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}}.$$

Also, we have

$$u \cdot v = \sqrt{a}\frac{1}{\sqrt{a}} + \sqrt{b}\frac{1}{\sqrt{b}} + \sqrt{c}\frac{1}{\sqrt{c}} + \sqrt{d}\frac{1}{\sqrt{d}} = 4.$$

Applying the Cauchy–Schwarz inequality, we get

$$(u \cdot v)^2 \leq \|u\|^2 \|v\|^2,$$

that is

$$16 \leq (a+b+c+d)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}+\frac{1}{d}\right),$$

which is exactly inequality (3.43). In fact, the above inequality can be generalized to any positive numbers a_i, $1 \leq i \leq n$, as follows:

$$n^2 \leq \left(\sum_{i=1}^{n} a_i\right)\left(\sum_{i=1}^{n} \frac{1}{a_i}\right).$$

◀

Exercise 3.4 (Apollonius' Identity)
Let u, v, and w be vectors in $\mathbb{R}^n$. Show that

$$\|u-v\|^2 = 2\|w-u\|^2 + 2\|w-v\|^2 - 4\|w - \frac{1}{2}(u+v)\|^2. \tag{3.46}$$

■

Solution
Recall the parallelogram identity:

$$\|U+V\|^2 + \|U-V\|^2 = 2(\|U\|^2 + \|V\|^2), \tag{3.47}$$

for any two vectors U and V in $\mathbb{R}^n$. Applying (3.47) to the vectors

$$U = \frac{w-u}{2} \qquad \text{and} \qquad V = \frac{w-v}{2},$$

we get

$$\|w - \frac{1}{2}(u+v)\|^2 + \|\frac{1}{2}(u-v)\|^2 = 2\|\frac{1}{2}(w-u)\|^2 + 2\|\frac{1}{2}(w-v)\|^2,$$

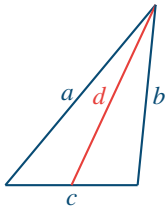


Fig. 3.18 Apollonius' identity

or

$$\|w - \frac{1}{2}(u+v)\|^2 + \frac{1}{4}\|u-v\|^2 = \frac{1}{2}\|w-u\|^2 + \frac{1}{2}\|(w-v)\|^2.$$

This gives (3.46).

In $\mathbb{R}^2$, Apollonius' identity has the following geometric meaning: in a triangle with sides of lengths a, b and c, let d be the length of the line segment from the midpoint of the side of length c to the opposite vertex, see Fig. 3.18. Then,

$$a^2 + b^2 = \frac{1}{2}c^2 + 2d^2.$$

◀

Exercise 3.5

Prove that the following statements are equivalent:

1. $\|u - v\| = \|u + v\|$.
2. $\|u + v\|^2 = \|u\|^2 + \|v\|^2$.
3. The vectors u and v are orthogonal.

■

Solution

In order to show that the above statements are equivalent, we need to show that (1) ⇒ (2) ⇒ (3) ⇒ (1).

First, assume that $\|u - v\| = \|u + v\|$. Then,

$$\|u - v\|^2 = \|u + v\|^2.$$

On the other hand,

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2 + 2(u \cdot v) \tag{3.48}$$

and

$$\|u - v\|^2 = \|u\|^2 + \|v\|^2 - 2(u \cdot v). \tag{3.49}$$

Hence, the right-hand sides of (3.48) and (3.49) must coincide, i.e.,

$$u \cdot v = -u \cdot v,$$

or equivalently, $u \cdot v = 0$. Using this last equality in (3.48), we obtain

$$\|u + v\|^2 = \|u\|^2 + \|v\|^2.$$

Thus, we have proved that (1) $\Rightarrow$ (2).

Next, assume that (2) is satisfied. Then (3.48) implies that $u \cdot v = 0$. This means that u and v are orthogonal. This shows that (2) $\Rightarrow$ (3).

Finally, we assume that u and v are orthogonal, that is $u \cdot v = 0$. Using (3.48) and (3.49), we deduce that $\|u + v\|^2 = \|u - v\|^2$, which is exactly (1). Thus, we have proved that (3) $\Rightarrow$ (1). ◀

Exercise 3.6 (Dunkl–Williams Inequality)
Let u and v be two nonzero vectors in $\mathbb{R}^n$. Show that

$$\frac{1}{2}(\|u\| + \|v\|) \left\| \frac{u}{\|u\|} - \frac{v}{\|v\|} \right\| \leq \|u - v\|, \tag{3.50}$$

where equality holds if and only if $\|u\| = \|v\|$ or $\|u\| + \|v\| = \|u - v\|$. ■

Solution
We have

$$\begin{aligned} \left\| \frac{u}{\|u\|} - \frac{v}{\|v\|} \right\|^2 &= \left(\frac{u}{\|u\|} - \frac{v}{\|v\|} \right) \cdot \left(\frac{u}{\|u\|} - \frac{v}{\|v\|} \right) \\ &= \frac{1}{\|u\|^2}(u \cdot u) + \frac{1}{\|v\|^2}(v \cdot v) - \frac{2}{\|u\|\|v\|}(u \cdot v) \\ &= 2 - \frac{2}{\|u\|\|v\|}(u \cdot v). \end{aligned} \tag{3.51}$$

Using (3.41), we recast then, (3.51) as

$$\left\| \frac{u}{\|u\|} - \frac{v}{\|v\|} \right\|^2 = \frac{1}{\|u\|\|v\|} \left(2\|u\|\|v\| - (\|u\|^2 + \|v\|^2 - \|u - v\|^2) \right).$$

Hence

$$\left\| \frac{u}{\|u\|} - \frac{v}{\|v\|} \right\|^2 = \frac{1}{\|u\|\|v\|} \left(\|u - v\|^2 - (\|u\| - \|v\|)^2 \right).$$

Multiplying both side in the last identity by $-(\|u\| + \|v\|)^2/4$ and adding $\|u - v\|^2$ to both sides, we get

$$\begin{aligned}
&\|u - v\|^2 - \left\| \frac{u}{\|u\|} - \frac{v}{\|v\|} \right\|^2 \left(\frac{\|u\| + \|v\|}{2} \right)^2 \\
&= \|u - v\|^2 - \frac{(\|u\| + \|v\|)^2}{4\|u\|\|v\|} \left(\|u - v\|^2 - (\|u\| - \|v\|)^2 \right) \\
&= \frac{(\|u\| - \|v\|)^2}{4\|u\|\|v\|} \left((\|u\| + \|v\|)^2 - \|u - v\|^2 \right). \qquad (3.52)
\end{aligned}$$

Using the triangle inequality (3.22), we deduce that the right-hand side in (3.52) is positive. Therefore, we obtain

$$\|u - v\|^2 - \left\| \frac{u}{\|u\|} - \frac{v}{\|v\|} \right\|^2 \left(\frac{\|u\| + \|v\|}{2} \right)^2 \geq 0,$$

which in turn yields (3.50).

It is clear from (3.52) that its right-hand side is zero if and only if $\|u\| = \|v\|$ or $\|u\| + \|v\| = \|u - v\|$. These last conditions lead to the identity

$$\frac{1}{2}(\|u\| + \|v\|) \left\| \frac{u}{\|u\|} - \frac{v}{\|v\|} \right\| = \|u - v\|.$$

◀

Exercise 3.7 (Hölder's Inequality)
Let $u = (u_1, u_2, \ldots, u_n)$ and $v = (v_1, v_2, \ldots, v_n)$ be two vectors in $\mathbb{R}^n$. Show that for any $p, q > 1$ with $1/p + 1/q = 1$,

$$|u \cdot v| \leq \sum_{i=1}^{n} |u_i v_i| \leq \left(\sum_{i=1}^{n} |u_i|^p \right)^{1/p} \left(\sum_{i=1}^{n} |v_i|^q \right)^{1/q} = \|u\|_p \|v\|_q,$$

with

$$\|u\|_p = \left(\sum_{i=1}^{n} |u_i|^p \right)^{1/p} \quad \text{and} \quad \|v\|_q = \left(\sum_{i=1}^{n} |v_i|^q \right)^{1/q}$$

For $p = 2$, the Hölder inequality reduces to the Cauchy–Schwarz inequality. ■

Solution
Without loss of generality, we may assume that

$$\|u\|_p \neq 0 \quad \text{and} \quad \|v\|_q \neq 0.$$

We introduce the new vectors

$$\mathbf{u} = \frac{u}{\|u\|_p} = (\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_n) \quad \text{and} \quad \mathbf{v} = \frac{v}{\|v\|_q} = (\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n),$$

where

$$\mathbf{u}_i = \frac{u_i}{\|u\|_p} \quad \text{and} \quad \mathbf{v}_i = \frac{v_i}{\|v\|_q}, \quad i = 1, \ldots, n.$$

We can easily check that

$$\|\mathbf{u}\|_p = \|\mathbf{v}\|_q = 1.$$

Now, using the more general Young inequality[1]

$$|ab| \leq \frac{|a|^p}{p} + \frac{|b|^q}{q}, \qquad \frac{1}{p} + \frac{1}{q} = 1,$$

for $a = \mathbf{u}_i$ and $b = \mathbf{v}_i$, we get

$$|\mathbf{u}_i \mathbf{v}_i| \leq \frac{|\mathbf{u}_i|^p}{p} + \frac{|\mathbf{v}_i|^q}{q}.$$

Summing over i we have

$$\begin{aligned} |u \cdot v| &= \left| \sum_{i=1}^{n} u_i v_i \right| \leq \sum_{i=1}^{n} |u_i v_i| \\ &= \|u\|_p \|v\|_q \sum_{i=1}^{n} |\mathbf{u}_i \mathbf{v}_i| \\ &\leq \|u\|_p \|v\|_q \left(\sum_{i=1}^{n} \frac{|\mathbf{u}_i|^p}{p} + \sum_{i=1}^{n} \frac{|\mathbf{v}_i|^q}{q} \right) \\ &= \|u\|_p \|v\|_q \left(\frac{\|\mathbf{u}\|^p}{p} + \frac{\|\mathbf{v}\|^q}{q} \right) = \|u\|_p \|v\|_q, \end{aligned}$$

since $\|\mathbf{u}\|_q = \|\mathbf{v}\|_q = 1$ and $1/p + 1/q = 1$. This yields the desired result. ◄

Exercise 3.8 (Minkowski's Inequality)
Let $x_i, y_i \geq 0$ for $1 \leq i \leq n$, and let $p > 1$. Show that

$$\left(\sum_{i=1}^{n} (x_i + y_i)^p \right)^{1/p} \leq \left(\sum_{i=1}^{n} x_i^p \right)^{1/p} + \left(\sum_{i=1}^{n} y_i^p \right)^{1/p}. \tag{3.53}$$

■

[1] This Young inequality can be easily shown by using the concavity of the logarithm function.

Solution

The inequality is trivial if $x_i = y_i = 0$ for all $1 \leq i \leq n$.

Now, without loss of generality, assume that

$$\sum_{i=1}^{n}(x_i + y_i)^p \neq 0.$$

We have

$$\sum_{i=1}^{n}(x_i + y_i)^p = \sum_{i=1}^{n} x_i(x_i + y_i)^{p-1} + \sum_{i=1}^{n} y_i(x_i + y_i)^{p-1}. \tag{3.54}$$

Applying the Hölder inequality to the two terms in the right-hand side of (3.54), with $q = p/(p-1)$, we get

$$\sum_{i=1}^{n} x_i(x_i + y_i)^{p-1} \leq \left(\sum_{i=1}^{n} x_i^p\right)^{1/p} \left(\sum_{i=1}^{n}(x_i + y_i)^p\right)^{1/q}$$

and

$$\sum_{i=1}^{n} y_i(x_i + y_i)^{p-1} \leq \left(\sum_{i=1}^{n} y_i^p\right)^{1/p} \left(\sum_{i=1}^{n}(x_i + y_i)^p\right)^{1/q}.$$

Using these two estimates and (3.54), we get

$$\sum_{i=1}^{n}(x_i + y_i)^p \leq \left(\sum_{i=1}^{n} x_i^p\right)^{1/p} \left(\sum_{i=1}^{n}(x_i + y_i)^p\right)^{1/q} + \left(\sum_{i=1}^{n} y_i^p\right)^{1/p} \left(\sum_{i=1}^{n}(x_i + y_i)^p\right)^{1/q},$$

which is equivalent to (3.53) since $1/p + 1/q = 1$.

We leave it to reader to show that the inequality (3.53) is reversed for $0 < p < 1$, that is

$$\left(\sum_{i=1}^{n}(x_i + y_i)^p\right)^{1/p} \geq \left(\sum_{i=1}^{n} x_i^p\right)^{1/p} + \left(\sum_{i=1}^{n} y_i^p\right)^{1/p}. \tag{3.55}$$

◀

Exercise 3.9

Let u and v be two nonzero vectors in $\mathbb{R}^n$. Prove that

$$\left\| \frac{1}{\|u\|}u - \|u\|v \right\| = \left\| \frac{1}{\|v\|}v - \|v\|u \right\|. \tag{3.56}$$

■

Solution

We have

$$\begin{aligned}\left\| \frac{1}{\|u\|}u - \|u\|v \right\|^2 &= \left(\frac{1}{\|u\|}u - \|u\|v \right) \cdot \left(\frac{1}{\|u\|}u - \|u\|v \right) \\ &= \frac{1}{\|u\|^2}(u \cdot u) - 2(u \cdot v) + \|u\|^2(v \cdot v) \\ &= 1 - 2(u \cdot v) + \|u\|^2\|v\|^2.\end{aligned}$$

On the other hand, we have

$$\begin{aligned}\left\| \frac{1}{\|v\|}v - \|v\|u \right\|^2 &= \left(\frac{1}{\|v\|}v - \|v\|u \right) \cdot \left(\frac{1}{\|v\|}v - \|v\|u \right) \\ &= \frac{1}{\|v\|^2}(v \cdot v) - 2(u \cdot v) + \|v\|^2(u \cdot u) \\ &= 1 - 2(u \cdot v) + \|u\|^2\|v\|^2.\end{aligned}$$

Comparing the two identities above, we obtain

$$\left\| \frac{1}{\|u\|}u - \|u\|v \right\|^2 = \left\| \frac{1}{\|v\|}v - \|v\|u \right\|^2.$$

Taking the square root of both sides in the above identity, then we obtain (3.56). ◀

Exercise 3.10

Let u and v be two nonzero vectors in $\mathbb{R}^n$.

1. Show that if $\|u\| \neq \|v\|$, then for any $\nu > 0$, we have

$$\|(u\|u\|^\nu - v\|v\|^\nu)\| \leq \frac{\|u\|^{\nu+1} - \|v\|^{\nu+1}}{\|u\| - \|v\|}\|u - v\|. \tag{3.57}$$

2. Show that if $u \neq \mathbf{0}$ and $v \neq \mathbf{0}$, then

$$\left\| \left(\frac{u}{\|u\|^{\nu+2}} - \frac{v}{\|v\|^{\nu+2}} \right) \right\| \leq \frac{\|u\|^{\nu+1} - \|v\|^{\nu+1}}{\|u\| - \|v\|} \frac{\|u - v\|}{\|u\|^{\nu+1}\|v\|^{\nu+1}}. \tag{3.58}$$

■

Solution

1. To show (3.57), it is enough to prove that (since the right-hand side of (3.57) is positive for $\nu > -1$)

$$\|(u\|u\|^\nu - v\|v\|^\nu)\|^2 (\|u\| - \|v\|)^2 \leq (\|u\|^{\nu+1} - \|v\|^{\nu+1})^2\|u - v\|^2.$$

By expanding both sides in the above inequality, and using the fact that

$$\begin{aligned}\|(u\|u\|^\nu - v\|v\|^\nu)\|^2 &= \left(u\|u\|^\nu - v\|v\|^\nu\right)\cdot\left(u\|u\|^\nu - v\|v\|^\nu\right)\\ &= \|u\|^{2\nu+2} + \|v\|^{2\nu+2} - 2\|u\|^\nu\|v\|^\nu(u\cdot v)\end{aligned}$$

and

$$\|u-v\|^2 = \|u\|^2 + \|v\|^2 - 2(u\cdot v),$$

we obtain

$$\begin{aligned}&\left[\|u\|^{2\nu+2} + \|v\|^{2\nu+2} - 2\|u\|^\nu\|v\|^\nu(u\cdot v)\right]\left[\|u\|^2 + \|v\|^2 - 2\|u\|\|v\|\right]\\ &\le \left[\|u\|^{2\nu+2} + \|v\|^{2\nu+2} - 2\|u\|^{\nu+1}\|v\|^{\nu+1}\right]\left[\|u\|^2 + \|v\|^2 - 2(u\cdot v)\right].\end{aligned}$$

After cancelling like terms and moving all terms to the right side, we get the equivalent inequality

$$0 \le (2\|u\|\|v\| - 2(u\cdot v))(\|u\|^{\nu+2} - \|v\|^{\nu+2})(\|u\|^\nu - \|v\|^\nu). \tag{3.59}$$

By the Cauchy–Schwarz inequality, we deduce that

$$2\|u\|\|v\| - 2(u\cdot v) \ge 0.$$

In addition, the two factors

$$(\|u\|^{\nu+2} - \|v\|^{\nu+2})(\|u\|^\nu - \|v\|^\nu)$$

have the same sign for $\nu > 0$. Therefore, (3.59) is satisfied and consequently, (3.57) holds true.

2. To show (3.58), we have first (by squaring each side and expanding)

$$\begin{aligned}\left\|\left(\frac{u}{\|u\|^{\nu+2}} - \frac{v}{\|v\|^{\nu+2}}\right)\right\| &= \left\|\left(\frac{u\|v\|^{\nu+2} - v\|u\|^{\nu+2}}{\|u\|^{\nu+2}\|v\|^{\nu+2}}\right)\right\|\\ &= \left\|\left(\frac{u\|u\|^\nu - v\|v\|^\nu}{\|u\|^{\nu+1}\|v\|^{\nu+1}}\right)\right\|\\ &= \frac{1}{\|u\|^{\nu+1}\|v\|^{\nu+1}}\left\|(u\|u\|^\nu - v\|v\|^\nu)\right\|.\end{aligned}$$

Applying (3.57) to the last term on the right-hand side in the above identity, then (3.58) holds. ◀

Exercise 3.11

Let u and v be two nonzero vectors in $\mathbb{R}^n$. Show that

$$u \cdot v \leq \frac{1}{4}\Big(\|u+v\|^2 - (\|u\| - \|v\|)^2\Big) \tag{3.60}$$

and

$$u \cdot v \geq \frac{1}{4}\Big((\|u\| - \|v\|)^2 - \|u-v\|^2\Big). \tag{3.61}$$

■

Solution

To show (3.60), we have, by the polarization identity (identity (3.29))

$$u \cdot v = \frac{1}{4}\|u+v\|^2 - \frac{1}{4}\|u-v\|^2. \tag{3.62}$$

Now, using (3.38), we obtain

$$\|u\| - \|v\| \leq \|u-v\|,$$

which gives

$$-\|u-v\|^2 \leq -(\|u\| - \|v\|)^2. \tag{3.63}$$

Now (3.60) follows by using (3.63) in (3.62).

To show (3.62) we have first

$$\begin{aligned}\|u+v\|^2 - (\|u\| - \|v\|)^2 &= \|u\|^2 + \|v\|^2 + 2(u \cdot v) - (\|u\|^2 + \|v\|^2 - 2\|u\|\|v\|) \\ &= 2\|u\|\|v\| + 2(u \cdot v).\end{aligned}$$

Using the Cauchy–Schwarz inequality, we have

$$2\|u\|\|v\| + 2(u \cdot v) \geq 0.$$

Therefore, we deduce that

$$\|u+v\|^2 \geq (\|u\| - \|v\|)^2.$$

Plugging this last inequality into (3.62), then, (3.61) is satisfied. ◄

Exercise 3.12 (Lagrange's Identity)

Let $u = (u_1, u_2, u_3)$ and $v = (u_1, v_2, v_3)$ be two vectors in $\mathbb{R}^3$. We define the *cross product* $u \times v$ of u and v in $\mathbb{R}^3$ to be the vector

$$u \times v = (u_2v_3 - u_3v_2, u_3v_1 - u_1v_3, u_1v_2 - u_2v_1).$$

Show that

$$\|u\|^2\|v\|^2 = (u \cdot v)^2 + \|u \times v\|^2. \tag{3.64}$$

This identity relates norms, the dot product, and the cross product. ■

Solution

To show (3.64), we compute all the terms in this identity. So, we have

$$\begin{aligned}\|u \times v\|^2 &= (u_2v_3 - u_3v_2)^2 + (u_3v_1 - u_1v_3)^2 + (u_1v_2 - u_2v_1)^2 \\ &= u_2^2v_3^2 + u_3^2v_2^2 + u_3^2v_1^2 + u_1^2v_3^2 + u_1^2v_2^2 + u_2^2v_1^2 \\ &\quad -2(u_2v_3u_3v_2 + u_3v_1u_1v_3 + u_1v_2u_2v_1).\end{aligned} \tag{3.65}$$

On the other hand,

$$\begin{aligned}(u \cdot v)^2 &= (u_1v_1 + u_2v_2 + u_3v_3)^2 \\ &= u_1^2v_1^2 + u_2^2v_2^2 + u_3^2v_3^2 + 2(u_1v_1u_2v_2) + 2(u_1v_1u_3v_3) + 2(u_2v_2u_3v_3).\end{aligned} \tag{3.66}$$

Adding (3.65) to (3.66), we get

$$\begin{aligned}&\|u \times v\|^2 + (u \cdot v)^2 \\ &= u_2^2v_3^2 + u_3^2v_2^2 + u_3^2v_1^2 + u_1^2v_3^2 + u_1^2v_2^2 + u_2^2v_1^2 + u_1^2v_1^2 + u_2^2v_2^2 + u_3^2v_3^2 \\ &= (u_1^2 + u_2^2 + u_3^2)(v_1^2 + v_2^2 + v_3^2) \\ &= \|u\|^2\|v\|^2.\end{aligned}$$

◄

General Vector Spaces

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4.1 Definition and Examples

In ▶ Chap. 3 we studied the properties of the space $\mathbb{R}^n$ and extended the definition of its algebraic structure to a general vector space (Definition 3.2.2). In this chapter, we study some properties of general vector spaces. We first recall Definition 3.2.2.

Definition 4.1.1 (Vector Space)

Let E be a nonempty set and $\mathbb{K}$[a] be a commutative field. We define two operations. An internal operation on E denoted by $(+)$ and defined as:

$$E \times E \to E : \qquad (v, w) \mapsto v + w,$$

and an external one denoted by $(\cdot)$ and defined as:

$$\mathbb{K} \times E \to E : \qquad (\lambda, v) \mapsto \lambda \cdot v = \lambda v.$$

So, E is a *vector space* over the field $\mathbb{K}$ if $(E, +)$ is an Abelian group and for all x, y in E and for all λ, μ in $\mathbb{K}$ the following properties hold:

1. $1 \cdot x = x$.[b]
2. $\lambda(\mu x) = (\lambda\mu)x$.
3. $\lambda(x + y) = \lambda x + \lambda y$.
4. $(\lambda + \mu)x = \lambda x + \mu x$.

[a] A commutative field is a commutative unitary ring (Definition 1.2.1) such that $e \neq 0$ and all nonzero elements are invertible with respect to multiplication. For example, $(\mathbb{R}, +, \cdot)$ and $(\mathbb{C}, +, \cdot)$ are fields.
[b] Here 1 is the identity element with respect to the multiplication in $\mathbb{K}$.

Example 4.1

As we have seen in Example 3.2, $\mathbb{R}^n$ is a vector space over the field $\mathbb{R}$. ■

Example 4.2 ($\mathbb{C}$ Is a Vector Space over $\mathbb{R}$)
We consider the set of complex numbers $z = x + iy$ with $i^2 = -1$ and x, y real numbers. We define the addition and multiplication by scalars as follows: if $z = x + iy$ and $w = a + ib$ are two complex numbers and λ is a real number, then

$$z + w = (x + a) + i(y + b) \qquad \text{and} \qquad \lambda z = (\lambda x) + i(\lambda y).$$

Then, it is easily checked that $(\mathbb{C}, +, \cdot)$ is a vector space over $\mathbb{R}$. In fact this vector space has the same algebraic property as $\mathbb{R}^2$. See Example 5.8 for more details.

■

Example 4.3 (The Vector Space of $m \times n$ Matrices)
The set of matrices $(\mathscr{M}_{m\times n}(\mathbb{K}), +, \cdot)$ with the usual matrix operations of addition and multiplication by scalars is a vector space over the field $\mathbb{K} = \mathbb{R}$ or $\mathbb{K} = \mathbb{C}$. Indeed, we have proved in Theorem 1.1.5 that $(\mathscr{M}_{m\times n}(\mathbb{K}), +)$ is an Abelian group. Also, it is clear that for any λ and μ in $\mathbb{K}$ and for any two matrices A and B in $\mathscr{M}_{m\times n}(\mathbb{K})$, we have

1. $1 \cdot A = A$.
2. $\lambda(\mu A) = (\lambda\mu)A$.
3. $\lambda(A + B) = \lambda A + \lambda B$.
4. $(\lambda + \mu)A = \lambda A + \mu A$.

■

Example 4.4 (The Vector Space of Real-Valued Functions)
Let X be an arbitrary nonempty set and let $E = \mathscr{F}(X, \mathbb{R})$ be the set of all functions from X to $\mathbb{R}$. We define the operation of addition $(+)$ and multiplication by scalars $(\cdot)$, respectively, as

$$E \times E \to E : \qquad (f, g) \mapsto f + g$$

where for all x in X,

$$(f + g)(x) = f(x) + g(x)$$

and

$$\mathbb{R} \times E \to E : \qquad (\lambda, f) \mapsto \lambda \cdot f = \lambda f$$

where

$$(\lambda f)(x) = \lambda f(x),$$

for all x in X. Then one can easily check that $E = (\mathscr{F}(X, \mathbb{R}), +, \cdot)$ is a vector space over the field $\mathbb{R}$. ■

We have seen in Example 3.2 that $\mathbb{R}^n$ is a vector space over the field $\mathbb{R}$. In fact, this is true for any field $\mathbb{K}$.

Example 4.5 ($\mathbb{K}^n$ Is a Vector Space over $\mathbb{K}$)
Define the addition in $\mathbb{K}^n$ by

$$(a_1, a_2, \ldots, a_n) + (b_1, b_2, \ldots, b_n) = (a_1 + b_1, a_2 + b_2, \ldots, a_n + b_n)$$

and multiplication by scalar by

$$\lambda(a_1, a_2, \ldots, a_n) = (\lambda a_1, \lambda a_2, \ldots, \lambda_n a_n)$$

for a_i, b_i, $1 \leq i \leq n$, and λ in $\mathbb{K}$. Then, we can easily show that $\mathbb{K}^n$ is a vector space over $\mathbb{K}$. ■

Notation We denote the vector space $(E, +, \cdot)$ simply by E.

4.2 Properties of Vector Spaces

Let us list some important algebraic properties of vector spaces. These properties will be used later on in the book. We start with the following theorem.

Theorem 4.2.1 (Properties of Vector Spaces)
Let E be a vector space over a field $\mathbb{K}$. Let u be a vector in E and λ be a scalar in $\mathbb{K}$. Then, we have

1. $0_{\mathbb{K}} u = 0_E$,
2. $\lambda 0_E = 0_E$,
3. $(-\lambda)u = -\lambda u$,
4. *if* $\lambda u = 0_E$, *the* $\lambda = 0_{\mathbb{K}}$ *or* $u = 0_E$,

where $0_{\mathbb{K}}$ is the zero of the field $\mathbb{K}$ (the identity element with respect to the addition in $\mathbb{K}$) and 0_E is the zero of E (the identity element of the Abelian group $(E, +)$). If $\mathbb{K} = \mathbb{R}$, we denote $0_{\mathbb{R}}$ simply by 0.

Proof
To prove the first property, we have by using axiom (4) in Definition 4.1.1 that

$$(\lambda - \mu)u + \mu u = ((\lambda - \mu) + \mu)u = \lambda u.$$

Since $(E, +)$ is an Abelian group, it follows that

$$(\lambda - \mu)u = \lambda u - \mu u, \tag{4.1}$$

which holds for any λ and μ in $\mathbb{K}$ and for any u in E. Recall that if μ is in $\mathbb{K}$, then $-\mu$ is also in $\mathbb{K}$, since $(\mathbb{K}, +)$ is an Abelian group with respect to addition in $\mathbb{K}$ and $-\mu$ is the inverse of μ with respect the addition in $\mathbb{K}$. Since (4.1) holds for any λ and μ in $\mathbb{K}$, setting $\lambda = \mu$ we have

$$0_{\mathbb{K}} u = 0_E.$$

Next, to prove the second property, we use axiom (3) in Definition 4.1.1 and the fact that $(E, +)$ is an Abelian group ($-v$ is in E for any v in E), to get

$$\lambda(u - v) + \lambda v = \lambda((u - v) + v) = \lambda v$$

for any u and v in E and any λ in $\mathbb{K}$. This gives, for the same reason as before,

$$\lambda(u - v) = \lambda u - \lambda v, \tag{4.2}$$

which for $u = v$ yields

$$\lambda 0_E = 0_E.$$

For the third property, we have

$$(-\lambda)u = (0_{\mathbb{K}} - \lambda)u = 0_{\mathbb{K}} u - \lambda u = 0_E - \lambda u = -\lambda u.$$

Finally, for the fourth property, we deduce from the first and the second properties that

$$0_{\mathbb{K}} u = \lambda 0_E = 0_E.$$

Conversely, the identity

$$\lambda u = 0_E,$$

implies either $\lambda = 0_{\mathbb{K}}$ (the second property), or $\lambda \neq 0_{\mathbb{K}}$ and in this case λ is invertible (since $(\mathbb{K}, +, \cdot)$ is a commutative field) and we have

$$\lambda^{-1}(\lambda u) = (\lambda^{-1}\lambda)u = 1 \cdot u = u = 0_E,$$

where we have used axiom (1) in Definition 4.1.1 and the fact that $\lambda u = 0_E$. Thus, the fourth property holds and the proof of Theorem 4.2.1 is complete. □

4.3 Subspaces

Verifying all the axioms in Definition 3.2.2 seems rather tedious, but as we will see in many situations, we do not need to verify all of them since in many cases, the vector spaces under consideration are contained in a larger vector space. For example we can easily show that the set of 2×2 matrices with real entries is a vector space over $\mathbb{R}$ with the usual matrix operations of addition and multiplication by scalars. But this space is contained in the larger vector space $(\mathscr{M}_{m\times n}(\mathbb{R}), +, \cdot)$. So, in this section, we will discuss how to recognize such vector spaces. We start with the following definition.

Definition 4.3.1 (Subspace)
A nonempty subset F of a vector space E is called a *subspace* of E if F is itself a vector space under the addition and multiplication by scalar defined on E.

Now, we should gain something from the fact that F is a subset of the larger space E. Indeed, to show that F is itself a vector space under the addition and the scalar multiplication defined on E, it is not necessary to verify all the axioms in Definition 4.1.1 since many of them are "inherited" from E. For example, there is no need to prove that $u + v = v + u$ for all elements of F since this property is already satisfied for all the elements of E (recall that $(E, +)$ is an Abelian group) and hence for all elements of F, since F is a subset of E. However, other properties are not inheritable. For example, for two elements u and v in F, $u + v$ is of course in E, but it may not lie in the subset F; similarly, λu is in E for any λ in $\mathbb{K}$, but it might not be in F.

Thus, we can easily prove the following theorem.

Theorem 4.3.1 (Characterization of Subspaces)
Let E be a vector space and let F be a nonempty subset of E. Then, F is a subspace of E if and only if for all u and v in F and for all λ in $\mathbb{K}$, we have
1. $u - v \in F$, *and*
2. $\lambda u \in F$.

Remark 4.3.2 The first property in Theorem 4.3.1 means that F is a subgroup of the group $(E, +)$ (see Exercise 1.11). In addition, this property can be replaced by

$$u + v \in F. \tag{4.3}$$

Of course, this property alone does not imply that F is a subgroup of $(E, +)$.

The proof of Theorem 4.3.1 is very simple and we omit it. We simply need to verify that the axioms of Definition 4.1.1 are satisfied if and only if the two properties of Theorem 4.3.1 hold.

Now, we may combine the two properties in Theorem 4.3.1 and state the following result.

Theorem 4.3.3

Let E be a vector space over a field $\mathbb{K}$ and F be a nonempty subset of E. Then, F is subspace of E if and only if for all u and v in F and all λ and μ in $\mathbb{K}$, we have

$$\lambda u + \mu v \in F. \tag{4.4}$$

Proof

First, assume that F is a subspace of E. Then, by the second property in Theorem 4.3.1, λu and μv belong to F; therefore, (4.3) yields (4.4). Conversely, assume that (4.4) is satisfied, then we get for $\lambda = \mu = 1$ the property (4.3). In addition, for $\mu = 0_{\mathbb{K}}$, we get the second property in Theorem 4.3.1. This completes the proof of Theorem 4.3.3. □

Remark 4.3.4 From now on, we consider Theorem 4.3.1 or Theorem 4.3.3 as the definition of a subspace.

Example 4.6

Let E be a vector space over a field $\mathbb{K}$ and F be a subspace of E. Then F always contains 0_E. In addition, the set $\{0_E\}$ is itself a subspace of E which is the smallest subspace of E, called the *zero subspace* of E. Also, E is a subspace of itself and it is the largest subspace of E. ■

Example 4.7

Let E be a vector space over a field $\mathbb{K}$ and u be an element of E. Then, the set

$$F = \{\lambda u : \quad \lambda \in \mathbb{K}\}$$

is a subspace of E. ■

Solution

Let v and w be two elements of F. Then there exist λ and μ in $\mathbb{K}$ such that

$$v = \lambda u \quad \text{and} \quad w = \mu u.$$

Then

$$v + w = \lambda u + \mu u = (\lambda + \mu)u,$$

which means that $v + w \in F$, since $\lambda + \mu \in \mathbb{K}$.

Next, for any $\alpha \in \mathbb{K}$, we have

$$\alpha v = \alpha(\lambda u) = (\alpha\lambda)u.$$

This shows that $\alpha v \in F$, since $\alpha\lambda \in \mathbb{K}$. Consequently, F is a subspace of E. ◀

Example 4.8 (Orthogonal Subspace)
Consider the vector space $\mathbb{R}^n$ over the field $\mathbb{R}$ and let u be a vector of $\mathbb{R}^n$. Denote by F the set which consists of all the vectors of $\mathbb{R}^n$ that are orthogonal to u. Show that F is a subspace of $\mathbb{R}^n$. ■

Solution
We may define F to be the set of all vectors whose dot product with u is zero (Definition 3.4.1): if $v \in F$, then $u \cdot v = 0$. It is clear that F is not empty, since $u \cdot 0_{\mathbb{R}^n} = 0$. This implies that $0_{\mathbb{R}^n} \in F$. Now, let u and w be two elements of F and α and β be two real numbers. Then we have, by using the properties of the dot product,

$$u \cdot (\alpha v + \beta w) = \alpha(u \cdot v) + \beta(u \cdot w) = 0,$$

and so $\alpha v + \beta w \in F$. Hence, F is a subspace of $\mathbb{R}^n$. ◀

Example 4.9 (Subspace of Linear Combinations)
Let E be a vector space over a field $\mathbb{K}$ and $u_1, u_2, \ldots, u_n$ be n elements of E. Then the set

$$F = \left\{ u = \lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_n u_n = \sum_{i=1}^{n} \lambda_i u_i \right\},$$

where $\lambda_1, \lambda_2, \ldots, \lambda_n$ are elements of $\mathbb{K}$ is a subspace of E. F it is called the *subspace of linear combination* of $u_1, u_2, \ldots, u_n$, or the subspace *spanned* by $u_1, u_2, \ldots, u_n$. ■

Solution
First of all, it is clear that the set F is not empty since it does contain 0_E because

$$0_E = 0_{\mathbb{K}} u_1 + 0_{\mathbb{K}} u_2 + \cdots + 0_{\mathbb{K}} u_n.$$

Second, let u and v be two elements of F and α and β be two elements of $\mathbb{K}$. Then there exist λ_i, $1 \leq i \leq n$, and μ_i, $1 \leq i \leq n$, in $\mathbb{K}$ such that

$$u = \lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_n u_n \qquad \text{and} \qquad v = \mu_1 u_1 + \mu_2 u_2 + \cdots + \mu_n u_n.$$

Consequently,

$$\begin{aligned} \alpha u + \beta v &= \alpha(\lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_n u_n) + \beta(\mu_1 u_1 + \mu_2 u_2 + \cdots + \mu_n u_n) \\ &= (\alpha\lambda_1)u_1 + (\alpha\lambda_2)u_2 + \cdots + (\alpha\lambda_n)u_n + (\beta\mu_1)u_1 + (\beta\mu_2)u_2 + \cdots + (\beta\mu_n)u_n \\ &= (\alpha\lambda_1 + \beta\mu_1)u_1 + (\alpha\lambda_2 + \beta\mu_2)u_2 + \cdots + (\alpha\lambda_n + \beta\mu_n)u_n, \end{aligned}$$

which is an element of F, since $\alpha\lambda_i + \beta\mu_i$, $i \leq 1 \leq n$ are elements of $\mathbb{K}$. Thus, F is a subspace of E. ◀

Example 4.10 (Subspaces of Even and Odd Functions)
A real-valued function defined on the real line is said to be an even function if $f(-x) = f(x)$ for all x in $\mathbb{R}$ and it is called an odd function if $f(-x) = -f(x)$ for all x in $\mathbb{R}$. Let F_1 be the set of all even functions and F_2 the set of all odd functions. Show that F_1 and F_2 are subspaces of the space $\mathscr{F}(\mathbb{R}, \mathbb{R})$ with the operations of addition and multiplication by scalars defined in Example 4.4. ■

Solution
We will show that F_1 is a subspace of $\mathscr{F}(\mathbb{R}, \mathbb{R})$, leaving it to the reader to show that F_2 is also a subspace of $\mathscr{F}(\mathbb{R}, \mathbb{R})$.

First, it is clear that F_1 is not empty since the zero function $0_{\mathscr{F}(\mathbb{R},\mathbb{R})}$, defined as

$$0_{\mathscr{F}(\mathbb{R},\mathbb{R})} : \mathbb{R} \to \mathbb{R} : \qquad x \mapsto 0_{\mathscr{F}(\mathbb{R},\mathbb{R})}(x) = 0_{\mathbb{R}},$$

is an element of F_1 since $0_{\mathscr{F}(\mathbb{R},\mathbb{R})}(x) = 0_{\mathscr{F}(\mathbb{R},\mathbb{R})}(-x) = 0_{\mathbb{R}}$ for all x in $\mathbb{R}$.

Now, consider two functions f and g in F_1 and let α and β be two real numbers. Then,

$$f(-x) = f(x), \qquad \text{and} \qquad g(-x) = g(x)$$

for all x in $\mathbb{R}$, and so

$$\begin{aligned}(\alpha f + \beta g)(-x) &= \alpha f(-x) + \beta g(-x)\\ &= \alpha f(x) + \beta g(x)\\ &= (\alpha f + \beta g)(x)\end{aligned}$$

for all x in $\mathbb{R}$, This means that $\alpha f + \beta g$ is an even function and thus it is an element of F_1. Therefore, F_1 is a subspace of $\mathscr{F}(\mathbb{R}, \mathbb{R})$. ◀

Example 4.11 (Subspaces of Diagonal and Triangular Matrices)
Consider the space of 2×2 square matrices $\mathscr{M}_2(\mathbb{R})$ over the field $\mathbb{R}$.

We define F_1 to be the set of upper triangular matrices of the form

$$\begin{bmatrix} a & b \\ 0 & d \end{bmatrix}$$

and F_2 to be the set of diagonal matrices of the form

$$\begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix},$$

with a, b and d are real numbers. Show that F_1 and F_2 are subspaces of $\mathscr{M}_2(\mathbb{R})$. ■

Solution

First, it is clear that

$$0_{\mathscr{M}_2(\mathbb{R})} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$

belongs to both F_1 and F_2. Second, it is clear that the sum of two upper triangular matrices is an upper triangular matrix and also the multiplication of an upper triangular matrix by a scalar yields an upper triangular matrix. This means that F_1 is a subspace of $\mathscr{M}_2(\mathbb{R})$. We can show similarly that F_2 is a subspace of $\mathscr{M}_2(\mathbb{R})$. ◄

Example 4.12 (Null Space)

In the vector space $\mathbb{R}^n$, consider the set

$$\mathscr{N} = \mathscr{N}(A) = \{\, u \text{ is a column vector in } \mathbb{R}^n \text{ such that } Au = 0_{\mathbb{R}^m}\},^{1}$$

where A is a matrix in $\mathscr{M}_{m\times n}(\mathbb{R})$. Show that $\mathscr{N}$ is a subspace of $\mathbb{R}^n$. This subspace is called the *null* space of the matrix A. ■

Solution

First of all $\mathscr{N}$ is not empty, since $A0_{\mathbb{R}^n} = 0_{\mathbb{R}^m}$, which means that $0_{\mathbb{R}^n}$ belongs to $\mathscr{N}$. Now, let u and v be two elements in $\mathscr{N}$ and α and β be two elements of $\mathbb{R}$. Then, we have

$$Au = 0_{\mathbb{R}^m} \qquad \text{and} \qquad Av = 0_{\mathbb{R}^m}.$$

Now, using the properties of matrices, we have

$$\begin{aligned} A(\alpha u + \beta v) &= \alpha(Au) + \beta(Av) \\ &= \alpha 0_{\mathbb{R}^m} + \beta 0_{\mathbb{R}^m} = 0_{\mathbb{R}^m}. \end{aligned}$$

Hence, $\mathscr{N}$ is a subspace of $\mathbb{R}^n$. ◄

In the following theorem we will show that if we have a family of subspaces of a vector space E, then we can always construct other subspaces from this family.

Theorem 4.3.5 (Intersection of Subspaces)

Let E be a vector space over a field $\mathbb{K}$ and let F_1 and F_2 be two subspaces of E. Then

$$F = F_1 \cap F_2$$

(Continued)

[1] We may also denote $\mathscr{N}(A) = \text{Ker}(A)$.

Theorem 4.3.5 (continued)
is also a subspace of E. More generally, if F_i is a family of subspaces of E indexed by a nonempty set I, then

$$F = \bigcap_{i \in I} F_i,$$

is a subspace of E.

Proof
To simplify things, we prove the above theorem for two vector spaces F_1 and F_2. The same prove can be adapted for the general case.

It is clear that

$$F = F_1 \cap F_2$$

is not empty, since 0_E belongs to both subspaces F_1 and F_2 due to the fact that both are subspaces of E.

Next, consider two elements u and v in F and two elements α and β in $\mathbb{K}$. Then, u and v are also elements of F_1 and of F_2. Then, we deduce that $\alpha u + \beta v$ is an element of F_1 and F_2, again because F_1 and of F_2 are subspaces of E, and therefore, $\alpha u + \beta v$ is an element of F. This shows that F is a subspace of E. □

Remark 4.3.6 The subspace F spanned by $u_1, u_2, \ldots, u_n$, defined in Example 4.9, is the smallest subspace containing $u_1, u_2, \ldots, u_n$. In fact, F is the intersection of all subspaces of E containing $u_1, u_2, \ldots, u_n$.

Remark 4.3.7 If F_1 and F_2 are two subspaces of a vector space E, then the union

$$F = F_1 \cup F_2$$

is not a vector space in general, unless F_1 contains F_2 or F_2 contains F_1. See Exercise 4.9 for more details.

As we have seen above, the union of two subspaces is not necessarily a subspace. However, for any subspaces F_1 and F_2 of a vector space E, there is at least one subspace containing F_1 and F_2, called their sum and denoted $F_1 + F_2$.

Theorem 4.3.8 (Sum of Two Subspaces)
Let E be a vector space over a field $\mathbb{K}$ and F_1 and F_2 be two subspaces of E. Then, the set

$$F = F_1 + F_2 = \{u \in E \text{ such that } u = u_1 + u_2 \text{ with } u_1 \in F_1 \text{ and } u_2 \in F_2\}$$

is a subspace of E. In fact, F is the smallest subspace containing F_1 and F_2.

Proof
It is clear that 0_E belongs to both F_1 and F_2 sine both are subspaces of E. In addition, since we can write

$$0_E = 0_E + 0_E,$$

0_E belongs to F, and so F is not empty.

Next, let u and v be two elements of F and α and β be two elements of $\mathbb{K}$. Thus,

$$u = u_1 + u_2 \qquad \text{and} \qquad v = v_1 + v_2,$$

with $u_1, v_1 \in F_1$ and $u_2, v_2 \in F_2$. Then,

$$\begin{aligned} \alpha u + \beta v &= \alpha(u_1 + u_2) + \beta(v_1 + v_2) \\ &= (\alpha u_1 + \beta v_1) + (\alpha u_2 + \beta v_2). \end{aligned}$$

Now, since F_1 and F_2 are subspaces of E, $\alpha u_1 + \beta v_1 \in F_1$ and $\alpha u_2 + \beta v_2 \in F_2$, whence $\alpha u + \beta v \in F$. Consequently, F is a subspace of E.

Now, if U is another subspace of E containing F_1 and F_2, then U contains all the elements u_1 of F_1 as well as all the elements u_2 of F_2, hence, it contains all the elements $u = u_1 + u_2$, since U is a subspace. Thus, U contains all the elements u of F. Hence, $F_1 + F_2$ is the smallest subspace containing F_1 and F_2. □

Remark 4.3.9 Let E be a vector space over a field $\mathbb{K}$ and F be a subspace of E. Define

$$C_E(F) = \{u \in E \quad \text{such that } u \notin F\}.$$

Then we have

$$E = F + C_E(F),$$

but $C_E(F)$ is not a subspace of E since it does not contain 0_E.

4.3.1 Direct Sum Decomposition

We have seen in Theorem 3.4.1, that if we fix a vector v in $\mathbb{R}^n$, then each vector u in $\mathbb{R}^n$ can be written in exactly one way in the form

$$u = w_1 + w_2, \tag{4.5}$$

where $w_1 = \lambda v$ and w_2 is orthogonal to v. In other words, we consider $\mathbb{R}^n$ as a vector space over the field $\mathbb{R}$ and introduce the sets F_1 consisting of all vectors parallel to v,

$$F_1 = \{w_1 = \lambda v, \quad \text{with} \quad \lambda \in \mathbb{R}\}$$

and the set F_2 of all vectors orthogonal to v,

$$F_1 = \{w_2 \quad \text{such that} \quad v \cdot w_2 = 0\}.$$

We have seen in Example 4.7 that F_1 is a subspace of $\mathbb{R}^n$ and in Example 4.8 that F_2 is also a subspace of $\mathbb{R}^n$.

Form (4.5), we can say that

$$\mathbb{R}^n = F_1 + F_2,$$

with the additional property, that each element of $\mathbb{R}^n$ can be written in a unique way as the sum of an element of F_1 and an element of F_2. In this case, we say that $\mathbb{R}^n$ is the *direct sum* of F_1 and F_2. More generally, we have

Definition 4.3.2 (Direct Sum)

Let F_1 and F_2 be two subspaces of a vector space E. Then E is said to be the *direct sum* of F_1 and F_2, and one writes

$$E = F_1 \oplus F_2,$$

if for each u in E there exist unique vectors v in F_1 and w in F_2 such that

$$u = v + w.$$

Theorem 4.3.10

Let F_1 and F_2 be two subspaces of a vector space E. Then E is the direct sum of F_1 and F_2 if and only if $E = F_1 + F_2$ and $F_1 \cap F_2 = \{0_E\}$.

Proof

First, assume that $E = F_1 \oplus F_2$. Let u be an element in $F_1 \cap F_2$. Then we can write u as

$$u = u + 0_E,$$

with $u \in F_1$ and $0_E \in F_2$, or

$$u = 0_E + u,$$

with $0_E \in F_1$ and $u \in F_2$. Since u can be written in exactly one way as the sum of an element of F_1 and an element of F_2, we deduce that $u = 0_E$. Therefore, $F_1 \cap F_2 = \{0_E\}$.

Conversely, assume that $E = F_1 + F_2$ and $F_1 \cap F_2 = \{0_E\}$ and let u be an element of E such that

$$u = v_1 + w_1 = v_2 + w_2$$

with $v_1, v_2 \in F_1$ and $w_1, w_2 \in F_2$. Then,

$$v_1 - v_2 = w_1 - w_2.$$

Thus, we deduce that $v_1 - v_2$ belongs to F_2 and thus belongs to $F_1 \cap F_2$, and $w_1 - w_2$ belongs to F_1 and therefore belongs to $F_1 \cap F_2$. Since $F_1 \cap F_2 = \{0_E\}$, we deduce that

$$v_1 - v_2 = w_1 - w_2 = 0_E.$$

This gives $v_1 = v_2$ and $w_1 = w_2$. Therefore, $E = F_1 \oplus F_2$. □

Example 4.13

Consider the two subspaces F_1 of even functions and F_2 of odd functions defined in Example 4.10. Show that

$$\mathscr{F}(\mathbb{R}, \mathbb{R}) = F_1 \oplus F_2.$$

■

Solution

Let f be a function in $\mathscr{F}(\mathbb{R}, \mathbb{R})$. Then for all x in $\mathbb{R}$, f can be written as

$$\begin{aligned} f(x) &= \frac{f(x) + f(-x)}{2} + \frac{f(x) - f(-x)}{2} \\ &= g(x) + h(x), \end{aligned}$$

with

$$g(x) = \frac{f(x) + f(-x)}{2} \quad \text{and} \quad h(x) = \frac{f(x) - f(-x)}{2}.$$

It is clear that g is an even function, since

$$g(-x) = \frac{f(-x) + f(x)}{2} = g(x)$$

and h is an odd function since,

$$h(-x) = \frac{f(-x) - f(x)}{2} = -h(x).$$

Thus, we have shown that $\mathscr{F}(\mathbb{R}, \mathbb{R}) = F_1 + F_2$. It remains to prove that $F_1 \cap F_2 = \{0_{\mathscr{F}(\mathbb{R},\mathbb{R})}\}$. Let f be an element in $F_1 \cap F_2$; then f is odd and even at the same time. That is, for all x in $\mathbb{R}$, we have

$$f(-x) = f(x) \qquad \text{and} \qquad f(-x) = -f(x).$$

This gives $f(x) = 0$ for all x in $\mathbb{R}$. Hence $f = 0_{\mathscr{F}(\mathbb{R},\mathbb{R})}$. Therefore, $F_1 \cap F_2 = \{0_{\mathscr{F}(\mathbb{R},\mathbb{R})}\}$ and consequently, Theorem 4.3.10 implies that $\mathscr{F}(\mathbb{R}, \mathbb{R}) = F_1 \oplus F_2$. ◀

4.4 Linear Independence

In this section, we investigate whether the elements in a given set of a vector space are interrelated in the sense that one element can be written as a linear combination of the others, as we have seen in Example 4.9. As we will see later, it is really important to know if such a relation exists. Now, we start with the following definition.

Definition 4.4.1 (Linear Dependence and Independence)

Let E be a vector space over a field $\mathbb{K}$ and $u_1, u_2, \ldots, u_n$ be elements of E. The elements $u_1, u_2, \ldots, u_n$ are *linearly dependent* (or the family of elements $u_1, u_2, \ldots, u_n$ is *linearly dependent*) if there exist $\lambda_1, \lambda_2, \ldots, \lambda_n \in \mathbb{K}$, not all $0_{\mathbb{K}}$, such that

$$\lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_n u_n = 0_E. \tag{4.6}$$

Otherwise, they are *linearly independent.* In other words, $u_1, u_2, \ldots, u_n$ are linearly independent if

$$\lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_n u_n = 0_E$$

implies that $\lambda_1 = \lambda_2 = \cdots = \lambda_n = 0_{\mathbb{K}}$.

Example 4.14
In the vector space $\mathbb{R}^2$, the two vectors $e_1 = (1, 0)$ and $e_2 = (0, 1)$ are linearly independent, since the relation

$$\lambda_1 e_1 + \lambda_2 = 0_{\mathbb{R}^2},$$

implies that $\lambda_1 = \lambda_2 = 0$. But the vectors $u_1 = (1, 1)$ and $u_2 = (2, 2)$ are linearly dependent, since

$$-2u_1 + u_2 = 0_{\mathbb{R}^2}.$$

■

Example 4.15
Show that in the vector space $\mathscr{F}(\mathbb{R}, \mathbb{R})$, the two functions e^{st} and e^{rt} are linearly independent if and only if $s \neq r$. ■

Solution
First, assume that $s \neq r$ and let α and β be two real numbers such that

$$\alpha e^{st} + \beta e^{rt} = 0.$$

Differentiating both sides in the above equation, we get

$$\alpha s e^{st} + \beta r e^{rt} = 0.$$

Multiplying the first equation by s and subtracting the result from the second equation, we obtain $\beta(r - s)e^t = 0$. Since $s \neq r$, we have $\beta = 0$. Plugging this into the first equation, we obtain $\alpha = 0$. Hence the two functions e^{st} and e^{rt} are linearly independent.

Conversely, assume that the functions e^{st} and e^{rt} are linearly independent and $r = s$. Then, we have

$$e^{st} - e^{rt} = 0,$$

which contradicts the linear independence of e^{st} and e^{rt}. Thus, necessarily $r \neq s$. ◄

We shall often use the following linear dependence criterion.

Theorem 4.4.1
Let E be a vector space over a field $\mathbb{K}$ and $u_1, u_2, \ldots, u_n$ be elements of E. The elements $u_1, u_2, \ldots, u_n$ are linearly dependent if and only if (at least) one of the elements is a linear combination of the rest.

Proof

First, assume for instance that u_1 is a linear combination of the other elements. Then, there exist $\alpha_2, \alpha_3, \ldots, \alpha_n \in \mathbb{K}$ such that

$$u_1 = \alpha_2 u_2 + \alpha_3 u_3 + \cdots + \alpha_n u_n.$$

That is

$$u_1 - \alpha_2 u_2 - \alpha_3 u_3 - \cdots - \alpha_n u_n = 0_E.$$

Hence, (4.6) is satisfied with $\lambda_1 = 1, \lambda_i = -\alpha_i$, $i = 2, \ldots, n$.

Conversely, assume that $u_1, u_2, \ldots, u_n$ are linearly dependent. Then, there exists at least one p, $1 \leq p \leq n$, such that $\lambda_p \neq 0_{\mathbb{K}}$ and

$$\lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_p u_p + \cdots + \lambda_n u_n = 0_E.$$

This gives

$$u_p = -\frac{\lambda_1}{\lambda_p} u_1 - \frac{\lambda_2}{\lambda_p} u_2 - \cdots - \frac{\lambda_n}{\lambda_p} u_n,$$

i.e., u_p is a linear combination of the remaining elements. □

Theorem 4.4.2

Let E be a vector space over a field $\mathbb{K}$ and $u_1, u_2, \ldots, u_n$ be elements of E. If $u_1, u_1, \ldots, u_n$ are linearly independent, then $u_i \neq 0_E$ for all $1 \leq i \leq n$. In particular, if u is an element of E, then the family $\{u\}$ is linearly independent if and only if $u \neq 0_E$.

Proof

The proof is straightforward since if there is $1 \leq p \leq n$ with $u_p = 0_E$, then, we have for some $\lambda \neq 0_{\mathbb{K}}$,

$$0_{\mathbb{K}} u_1 + 0_{\mathbb{K}} u_2 + \cdots + \lambda u_p + \cdots + 0_{\mathbb{K}} u_n = 0_E,$$

which contradicts the fact that $u_1, u_2, \ldots, u_n$ are linearly independent. □

Theorem 4.4.3
Let E be a vector space over a field $\mathbb{K}$ and $u_1, u_2, \ldots, u_n$ be linearly independent elements of E. Let $\lambda_1, \lambda_2, \ldots, \lambda_n$ and $\mu_1, \mu_2, \ldots, \mu_n$ be elements of $\mathbb{K}$ such that

$$\lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_n u_n = \mu_1 u_1 + \mu_2 u_2 + \cdots + \mu_n u_n. \tag{4.7}$$

Then $\lambda_i = \mu_i$, $i = 1, \ldots, n$.

Proof
We can write (4.7) as

$$(\lambda_1 - \mu_1)u_1 + (\lambda_2 - \mu_2)u_2 + \cdots + (\lambda_n - \mu_n)u_n = 0_E.$$

Since $u_1, u_2, \ldots, u_n$ are linearly independent, it follows that

$$\lambda_i = \mu_i, \qquad i = 1, \ldots, n.$$

□

4.5 Bases of Vector Spaces

In this section, our main goal is to show how a small finite set of vectors or elements of a vectors space (called a *basis*) can be used to describe all the other vectors in that vector space. We introduce this fundamental property of *basis* in linear algebra. To explain the idea, consider the vector space $\mathbb{R}^2$ and take the two vectors $e_1 = (1, 0)$ and $e_2 = (0, 1)$ of $\mathbb{R}^2$. Then any vector $u = (x, y)$ of $\mathbb{R}^2$ can be written as a linear combination of the vectors e_1 and e_2 as

$$u = xe_1 + ye_2.$$

In this case, we say that the set $\{e_1, e_2\}$ *spans* the vector space $\mathbb{R}^2$. In addition, since the vectors e_1 and e_2 are linearly independent (Example 4.14), the above linear combination is unique (Theorem 4.4.3). That is to say, there is one and only one way to write u as a combination of the two vectors e_1 and e_2. So, in order to be able to write u in a unique way as a linear combination of the two vectors e_1 and e_2, two properties should be satisfied:

- The set $\{e_1, e_2\}$ spans $\mathbb{R}^2$, and
- the two vectors e_1 and e_2 are linearly independent.

Any set of two vectors in $\mathbb{R}^2$ satisfying the above two properties is called *basis*. Now, we extend the above idea to general vector spaces.

Definition 4.5.1 (Span of a Vector Space)

Let E be a vector space over a field $\mathbb{K}$. We say that the set of elements $\{u_1, u_2, \ldots, u_n\}$ of E *spans* E if their linear combinations fill E. That is, for each u in E, there exist $\lambda_1, \lambda_2, \ldots, \lambda_n$ in $\mathbb{K}$ such that

$$u = \lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_n u_n.$$

We may also say that the elements $u_1, u_2, \ldots, u_n$ *generate* the vector space E and write

$$E = \text{span}(u_1, u_2, \ldots, u_n).$$

Example 4.16

It is trivial to see that in the vector space $\mathbb{R}^n$, the set of vectors $\{e_1, e_2, \ldots, e_n\}$ with

$$e_i = (0, 0, \ldots, 0, 1, 0, \ldots, 0) \qquad (0 \text{ everywhere except the } i\text{th component})$$

spans $\mathbb{R}^n$. ■

Example 4.17

Let $\mathscr{P}_n$ be the set of all polynomials with real coefficients and of degree less than or equal to n. One can easily show that this set is a vector space over the field $\mathbb{R}$. Now, the family $\{1, x, x^2, \ldots, x^n\}$ spans this vector space, since any polynomial $p(x)$ in $\mathscr{P}_n$ can be written as

$$p(x) = a_0 + a_1 x + \cdots + a_n x^n,$$

where $a_0, a_1, \ldots, a_n$ are real numbers. ■

Definition 4.5.2 (Basis)

Let E be a vector space over a field $\mathbb{K}$. A *basis* $\mathbf{B}$ of the vector space E is a set of elements of E, $\mathbf{B} = \{u_1, u_2, \ldots, u_n\}$, which is linearly independent and spans E. In this case each element u of E can be written in exactly one way as

$$u = \lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_n u_n$$

for some $\lambda_1, \lambda_2, \ldots, \lambda_n$ in $\mathbb{K}$. The elements $\lambda_1, \lambda_2, \ldots, \lambda_n$ are called the *components* of u in the basis $\mathbf{B}$.

Example 4.18

The set of vectors $\{e_1, e_2, \ldots, e_n\}$ defined in Example 4.16 is a basis of the vector space $\mathbb{R}^n$ since it spans $\mathbb{R}^n$ and is linearly independent. ■

Example 4.19

Show that the family $\{1, x, x^2, \ldots, x^n\}$ is a basis of the vector space $\mathscr{P}_n$ defined in Example 4.17. ■

Solution

We have seen in Example 4.17 that the family $\{1, x, x^2, \ldots, x^n\}$ spans $\mathscr{P}_n$. It remains to show that this family is linearly independent. So, let $\lambda_0, \lambda_1, \ldots, \lambda_n$ be elements in $\mathbb{R}$ such that for all x in $\mathbb{R}$, we have

$$\lambda_0 + \lambda_1 x + \cdots + \lambda_n x^n = 0. \tag{4.8}$$

Since (4.8) holds for all x in $\mathbb{R}$, it also holds for $x = 0$. Thus, putting $x = 0$ in (4.8), we get $\lambda_0 = 0$. Now, by taking the derivative of (4.8) with respect to x, we obtain

$$\lambda_1 + 2\lambda_2 x + \cdots + n\lambda_n x^{n-1} = 0. \tag{4.9}$$

Similarly, taking $x = 0$ in (4.9), we get $\lambda_1 = 0$. By repeating this process up to the nth derivative of (4.8), we obtain

$$\lambda_0 = \lambda_1 = \cdots = \lambda_n = 0.$$

Thus, the family $\{1, x, x^2, \ldots, x^n\}$ is linearly independent and hence forms a basis in $\mathscr{P}_n$. ◄

Example 4.20

We consider the space $\mathscr{M}_2(\mathbb{R})$ of 2×2 square matrices over the field $\mathbb{R}$.

We define in this space the elements

$$M_1 = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \quad M_2 = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad M_3 = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}, \quad M_4 = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

Show that the set $\{M_1, M_2, M_2, M_4\}$ is a basis of $\mathscr{M}_2(\mathbb{R})$. ■

Solution

First, we need to show that the family $\{M_1, M_2, M_2, M_4\}$ spans $\mathscr{M}_2(\mathbb{R})$. So, consider a matrix

$$M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

in $\mathscr{M}_2(\mathbb{R})$. Then, it is clear that

$$M = aM_1 + bM_2 + cM_3 + dM_4.$$

Thus, $\{M_1, M_2, M_2, M_4\}$ spans $\mathscr{M}_2(\mathbb{R})$.

Second, we need to show that $\{M_1, M_2, M_2, M_4\}$ is linearly independent. So, let $\lambda_1, \ldots, \lambda_4 \in \mathbb{R}$ be such that

$$\lambda_1 M_1 + \lambda_2 M_2 + \lambda_3 M_3 + \lambda_4 M_4 = 0_{\mathscr{M}_2(\mathbb{R})}.$$

That is

$$\lambda_1 \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} + \lambda_2 \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} + \lambda_3 \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} + \lambda_4 \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix},$$

or equivalently,

$$\begin{bmatrix} \lambda_1 & \lambda_2 \\ \lambda_3 & \lambda_4 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

This yields $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = 0$. Thus, the family $\{M_1, M_2, M_2, M_4\}$ is linearly independent and consequently form a basis in $\mathscr{M}_2(\mathbb{R})$. ◀

Remark 4.5.1 (Non-uniqueness of Bases) A basis in a vector space is not unique, meaning that we may find more than one basis in the same vector space. As an example, we can consider the vector space $\mathbb{R}^2$ and the two vectors $u_1 = (1, 2)$ and $u_2 = (1, 3)$. Then, we easily show that $\{u_1, u_2\}$ is a basis in $\mathbb{R}^2$. But we have also shown in Example 4.16 that the set $\{e_1, e_2\} = \{(1, 0), (0, 1)\}$ is a basis in $\mathbb{R}^2$.

4.6 Dimension of a Vector Space

In this section, we will associate a number to a vector space. Recall that the *cardinality* of a set **B** is the number of elements of **B**.

Definition 4.6.1 (Dimension of a Vector Space)
Let E be a vector space over a field $\mathbb{K}$. The *dimension* of E, denoted by $\dim_{\mathbb{K}} E$ is defined to be the cardinality of a basis **B** for E. In addition, the zero vector space is defined to have dimension zero.

Example 4.21
Let us list the dimensions of some spaces that we have encountered before.

- The dimension of $\mathbb{R}^n$ is n since $\{e_1, e_2, \ldots, e_n\}$ is a basis of $\mathbb{R}^n$ (Example 4.18).
- The dimension of $\mathscr{M}_2(\mathbb{R})$ is 4 since $\{M_1, M_2, M_3, M_4\}$ is a basis of $\mathscr{M}_2(\mathbb{R})$ (Example 4.20). In general, we can easily prove that $\dim_{\mathbb{K}} \mathscr{M}_{m\times n}(\mathbb{K}) = m \times n$.
- We have proved in Example 4.19 that the set $\{1, x, x^2, \ldots, x^n\}$ is a basis of the space $\mathscr{P}_n$. Consequently, $\dim_{\mathbb{R}} \mathscr{P}_n = n + 1$.

■

Now, we state one of the main theorems in linear algebra.

Theorem 4.6.1 (Existence of a Basis)
Every non-zero vector space E of finite dimension has at least one basis.

The proof of this theorem is based on Zorn's lemma and is beyond the scope of this book.

Definition 4.6.2 (Maximal Linearly Independent Set)
Let E be a vector space over a field $\mathbb{K}$ and let S be a set of elements of E. Then the set S is called a *maximal linearly independent set* if the set $S \cup \{v\}$ is linearly dependent for every element v in the vector space E, that does not belong to S.

Example 4.22
In $\mathbb{R}^2$, the set $S = \{e_1, e_2\} = \{(1,0),(0,1)\}$ is maximal linearly independent subset, since for any vector $u = (x, y)$ in $\mathbb{R}^2$, the set $S \cup \{u\} = \{e_1, e_2, u\}$ is linearly independent because $u = xe_1 + ye_2$. ■

Theorem 4.6.2
Let E be a vector space over a field $\mathbb{K}$ and let $S = \{u_1, u_1, \ldots, u_n\}$ be a maximal set of linearly independent elements of E. Then, S is a basis of E.

Proof
We just need to show that S spans E. That is, each element of E can be expressed as a linear combination of the elements of S. Let w be an element in E. Since S is a maximal linearly independent set, the elements $w, u_1, u_2, \ldots, u_n$ are linearly dependent. Hence there exists $\lambda_0, \lambda_1, \ldots, \lambda_n$ in $\mathbb{K}$, not all $0_{\mathbb{K}}$, such that

$$\lambda_0 w + \lambda_1 u_1 + \cdots + \lambda_n u_n = 0_E. \tag{4.10}$$

It is clear that $\lambda_0 \neq 0_{\mathbb{K}}$: otherwise the elements $u_1, u_2, \ldots, u_n$ would be linearly dependent. Thus, we obtain from (4.10)

$$w = -\frac{\lambda_1}{\lambda_0}u_1 - \frac{\lambda_2}{\lambda_0}u_2 + \cdots + \frac{\lambda_n}{\lambda_0}u_n.$$

This shows that w is a linear combination of the elements of S. Thus, S is a basis of E. □

Theorem 4.6.3
Different bases of the same vector space E have the same number of elements (the same cardinality).

In order to prove this theorem, we first establish the following lemma.

Lemma 4.6.4 *Let E be a vector space over a field $\mathbb{K}$ and $\mathbf{B} = \{u_1, u_2, \ldots, u_n\}$ be a basis of E. Let $v_1, v_2, \ldots, v_m$ be elements in E. If $m > n$, then $v_1, v_2, \ldots, v_m$ are linearly dependent. That is, any linearly independent set S of n elements is maximal.*

Proof
To prove this lemma, we assume that $v_1, v_2, \ldots, v_m$ are linearly independent. Since $\mathbf{B} = \{u_1, u_1, \ldots, u_n\}$ is a basis of E, each element from the set $\{v_1, v_2, \ldots, v_m\}$ can be written in a unique way as a linear combination of the elements of $\mathbf{B}$. For instance, we have

$$v_1 = \lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_n u_n$$

for some $\lambda_1, \lambda_2, \ldots, \lambda_n$ in $\mathbb{K}$. By assumption, we know that $v_1 \neq 0_E$, therefore, there exists at least $1 \leq p \leq n$, such that $\lambda_p \neq 0_{\mathbb{K}}$ (without loss of generality, we may assume that $p = 1$). Hence, we get

$$u_1 = \frac{1}{\lambda_1} v_1 - \frac{\lambda_2}{\lambda_1} u_2 - \cdots - \frac{\lambda_n}{\lambda_1} u_n.$$

This implies that the set

$$G_1 = \{v_1, u_2, \ldots, u_n\}$$

spans E. Indeed, for w in E, we have, since $\mathbf{B}$ is a basis,

$$\begin{aligned} w &= a_1 u_1 + a_2 u_2 + \cdots + a_n u_n \\ &= a_1 \left(\frac{1}{\lambda_1} v_1 - \frac{\lambda_2}{\lambda_1} u_2 - \cdots - \frac{\lambda_n}{\lambda_1} u_n \right) + a_2 u_2 + \cdots + a_n u_n \\ &= \frac{a_1}{\lambda_1} v_1 + \left(a_2 - \frac{a_1 \lambda_2}{\lambda_1} \right) u_2 + \cdots + \left(a_n - \frac{a_1 \lambda_n}{\lambda_1} \right) u_n \end{aligned}$$

for some $a_1, a_2, \ldots, a_n$ in $\mathbb{K}$. Next, since G_1 spans E, we have, for some $\mu_1, \mu_2, \ldots, \mu_n$ in $\mathbb{K}$

$$v_2 = \mu_1 v_1 + \mu_2 u_2 + \cdots + \mu_n u_n.$$

Now, it is clear that there exists at least $2 \leq j \leq n$, such that $\mu_j \neq 0_{\mathbb{K}}$: otherwise, if $\mu_2 = \mu_3 = \cdots = \mu_n = 0_{\mathbb{K}}$, then we have $v_2 = \mu_1 v_1$, which contradicts the assumed linear independence of the family $\{v_1, v_2, \ldots, v_m\}$. Thus, as before, without loss of generality, we

may assume that $\mu_2 \neq 0_{\mathbb{K}}$. Then, we have as before,

$$u_2 = \frac{1}{\mu_2} v_2 - \frac{\mu_1}{\mu_2} v_1 - \frac{\mu_3}{\mu_2} u_3 - \cdots - \frac{\mu_n}{\mu_2} u_n.$$

Hence, the set

$$G_2 = \{v_1, v_2, u_3, \ldots, u_n\}$$

also spans E. The idea now is to continue our procedure and to replace $u_1, u_2, \ldots$ by $v_1, v_2, \ldots$ to conclude at the end (by induction) that

$$G_n = \{v_1, v_2, \ldots, v_n\}$$

spans E, and since $m > n$, the elements $v_{n+1}, v_{n+2}, \ldots, v_m$ are linear combinations of $v_1, v_2, \ldots, v_n$, which contradicts our assumption on the linear independence of $v_1, v_2, \ldots, v_m$. This concludes our proof. □

Proof of Theorem 4.6.3

Let $\mathbf{B}_1 = \{u_1, u_2, \ldots, u_n\}$ and $\mathbf{B}_2 = \{v_1, v_2, \ldots, v_m\}$ be two bases of the same vector space E. Lemma 4.6.4 implies that it is not possible to have $m > n$ or $n > m$. Then necessarily $m = n$. □

Theorem 4.6.5

Let E be a vector space over a field $\mathbb{K}$ with $\dim_{\mathbb{K}} E = n$ and let $u_1, u_2, \ldots, u_n$ be linearly independent elements of E. Then the set $\{u_1, u_2, \ldots, u_n\}$ constitutes a basis of E.

Proof

According to Lemma 4.6.4, the set $\{u_1, u_2, \ldots, u_n\}$ is a maximal set of linearly independent elements of E, thus, Theorem 4.6.2, implies that this set is a basis of E. □

Remark 4.6.6 Let E be a vector space over a field $\mathbb{K}$ with $\dim_{\mathbb{K}} E = n$. Then we deduce from above that:

- Any set of linearly independent elements of E has at most n elements.
- Any set that has at least $n + 1$ elements is linearly dependent.

4.6.1 Dimension of a Subspace

As we have said before, any subspace F of a vector space E is itself a vector space. So, we need to find the dimension of this subspace and compare it with the dimension of E. Thus, we have the following theorem.

Theorem 4.6.7 (Dimension of a Subspace)
Let E be a vector space over a field $\mathbb{K}$ with $\dim_{\mathbb{K}} E = n$ ($n > 0$, that is $E \neq \{0_E\}$). Let F be a subspace of E with $F \neq \{0_E\}$. Then,

$$\dim_{\mathbb{K}} F \leq \dim_{\mathbb{K}} E.$$

In particular, if

$$\dim_{\mathbb{K}} F = \dim_{\mathbb{K}} E, \qquad \text{then} \qquad F = E.$$

Proof
Suppose that $\dim_{\mathbb{K}} F > \dim_{\mathbb{K}} E$. Then there exists at least one basis in F with at least $n + 1$[2] elements, that is, there exists at least one linearly independent set of F with at least $n + 1$ elements. But each linearly independent set of F is also linearly independent in E. Hence, we obtain a linearly independent set in E with at least $n + 1$ elements, which contradicts Remark 4.6.6. Thus, $\dim_{\mathbb{K}} F \leq \dim_{\mathbb{K}} E$.

Now, if $\dim_{\mathbb{K}} F = \dim_{\mathbb{K}} E$, then there exists a basis $\mathbf{B} = \{u_1, u_2, \ldots, u_n\}$ in F. Then, $\mathbf{B}$ is also a basis of E (Theorem 4.6.5). Therefore, any $v \in E$ can be written as a linear combination of elements of F of the form

$$v = \lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_n u_n$$

with $\lambda_1, \lambda_2, \ldots, \lambda_n \in \mathbb{K}$. Consequently, v also belongs to F. This yields $E \subset F$ and since $F \subset E$ (by definition), we have $F = E$. □

4.6.2 Construction of a Basis

Suppose that we are given r linearly independent vectors $u_1, u_2, \ldots, u_r$ of a vector space E of dimension n, with $r < n$. Then, we may ask the following question: *is it possible to extend the set of vectors $\{u_1, u_2, \ldots, u_r\}$ so as to obtain a basis of E?* The answer turns

[2] This basis has at least one element $u_0 \neq 0_E$ since $F \neq \{0_E\}$.

out to be positive:

Theorem 4.6.8 (Construction of a Basis)
Let E be a vector space over a field $\mathbb{K}$ with $\dim_{\mathbb{K}} E = n$. Let r be a positive integer with $r < n$ and let $v_1, v_2, \ldots, v_r$ be linearly independent elements of E. Then, there exist elements $v_{r+1}, \ldots, v_n$ of E such that the set

$$\{v_1, v_2, \ldots, v_n\}$$

is a basis of E.

Proof
Since $\dim_{\mathbb{K}} E = n$ and $r < n$, the set $S = \{v_1, v_2, \ldots, v_r\}$ cannot be a basis of E, and then by Theorem 4.6.2, S cannot be a maximal set of linearly independent elements of E. Hence, by Definition 4.6.2, there exists $v_{r+1} \in E$ such that the set $S \cup \{v_{r+1}\}$ is linearly independent. Now, if $r + 1 = n$, then according to Theorem 4.6.5, S is a basis of E. If $r + 1 < n$, then we repeat the same procedure until we construct (by induction) a set of n linearly independent elements $\{u_1, u_2, \ldots, u_n\}$ of E and then, this should be a basis of E due to the same reason as before (Theorem 4.6.5). □

Example 4.23
We know that $\dim_{\mathbb{R}} \mathbb{R}^3 = 3$. Consider the two vectors $u = (1, 0, 1)$ and $v = (1, 3, 2)$. It is clear that u and v are linearly independent. If we now take the vector $w = (-1, 2, 4)$, then we may easily show that u, v, and w are linearly independent and thus form a basis of $\mathbb{R}^3$. ■

4.7 Exercises

Exercise 4.1
Consider the vector space $\mathscr{F}(\mathbb{R}, \mathbb{R})$ introduced in Example 4.4, defined over the field $\mathbb{R}$. Show that the three functions in this vector space

$$f(x) = \sin x, \quad g(x) = \sin(x + p), \quad \text{and} \quad h(x) = \cos(x + q),$$

where p and q are two real numbers, are linearly dependent. ■

Solution
Using the sine and cosine laws, we have

$$\sin(x + p) = \sin x \cos p + \cos x \sin p \tag{4.11}$$

and

$$\cos(x+q) = \cos x \cos q - \sin x \sin q. \tag{4.12}$$

Multiplying (4.11) by $\cos q$ and (4.12) by $(-\sin p)$ and adding the results, we obtain

$$\begin{aligned}\cos q \sin(x+p) + \sin p \cos(x+q) &= \cos q \cos p \sin x + \sin q \sin p \sin x \\ &= (\cos q \cos p + \sin q \sin p) \sin x \\ &= \cos(p-q) \sin x.\end{aligned}$$

Denoting $\alpha = \cos q$, $\beta = \sin p$ and $\gamma = -\cos(p-q)$, we get

$$\alpha \sin(x+p) + \beta \cos(x+q) + \gamma \sin x = 0,$$

but α, β and γ are not all zero, for any real numbers p and q. Consequently, f, g, and h are linearly dependent. ◀

Exercise 4.2 (Components of a Vector in a Basis)
Consider the space $\mathbb{R}^4$ over the field $\mathbb{R}$. Let

$$u_1 = (1,2,-1,-2), \quad u_2 = (2,3,0,-1), \quad u_3 = (1,3,-1,0), \quad u_4 = (1,2,1,4)$$

be vectors in $\mathbb{R}^4$.

1. Show that the set $\mathbf{B} = \{u_1, u_2, u_3, u_4\}$ is a basis in $\mathbb{R}^4$.
2. Find the components of the vector

$$v = (7,14,-1,2)$$

in this basis.

■

Solution

1. We have seen in Example 4.21 that $\dim_{\mathbb{R}} \mathbb{R}^4 = 4$. Now, since $\mathbf{B}$ has also four elements, then according to Theorem 4.6.5, in order to show that $\mathbf{B}$ is a basis, it is enough to prove that $\mathbf{B}$ is a linearly independent set. So, let $\lambda_1, \lambda_2, \lambda_3$ and λ_4 be elements in $\mathbb{R}$ satisfying

$$\lambda_1 u_1 + \lambda_2 u_2 + \lambda_3 u_3 + \lambda_4 u_4 = 0_{\mathbb{R}^4}, \tag{4.13}$$

that is

$$\lambda_1(1,2,-1,-2) + \lambda_2(2,3,0,-1) + \lambda_3(1,3,-1,0) + \lambda_4(1,2,1,4) = (0,0,0,0).$$

This gives (see ▶ Chap. 3) the system of equations

$$\begin{cases} \lambda_1 + 2\lambda_2 + \lambda_3 + \lambda_4 = 0, \\ 2\lambda_1 + 3\lambda_2 + 3\lambda_3 + 2\lambda_4 = 0, \\ -\lambda_1 - \lambda_3 + \lambda_4 = 0, \\ -2\lambda_1 - \lambda_2 + 4\lambda_4 = 0. \end{cases}$$

Now, it is clear that the above system has $(\lambda_1, \lambda_2, \lambda_3, \lambda_4) = (0, 0, 0, 0)$ as a solution. This solution is unique, since the matrix

$$A = \begin{bmatrix} 1 & 2 & 1 & 1 \\ 2 & 3 & 3 & 2 \\ -1 & 0 & -1 & 1 \\ -2 & -1 & 0 & 4 \end{bmatrix}$$

is invertible (Theorem 1.2.9), because (Theorem 2.4.8)

$$\det(A) = -2 \neq 0.$$

Thus, (4.13) implies $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = 0$. Hence, the vectors u_1, u_2, u_3 and u_4 are linearly independent in $\mathbb{R}^4$ and thus $\mathbf{B}$ is a basis of $\mathbb{R}^4$.

2. Since $\mathbf{B}$ is a basis of $\mathbb{R}^4$, there exists $\alpha_1, \alpha_2, \alpha_3$, and α_4 in $\mathbb{R}$ such that

$$v = \alpha_1 u_1 + \alpha_2 u_2 + \alpha_3 u_3 + \alpha_4 u_4. \tag{4.14}$$

Hence, $\alpha_1, \alpha_2, \alpha_3$ and α_4 are the components of v in the basis $\mathbf{B}$. To find these components, we proceed as before and obtain the system of equations

$$\begin{cases} \alpha_1 + 2\alpha_2 + \alpha_3 + \alpha_4 = 7, \\ 2\alpha_1 + 3\alpha_2 + 3\alpha_3 + 2\alpha_4 = 14, \\ -\alpha_1 - \alpha_3 + \alpha_4 = -1, \\ -2\alpha_1 - \alpha_2 + 4\alpha_4 = 2. \end{cases}$$

Its solution is given by (see ▶ Sect. 1.2.3)

$$\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix} = A^{-1} \begin{bmatrix} 7 \\ 14 \\ -1 \\ -2 \end{bmatrix}. \tag{4.15}$$

We can easily use the method described in Theorem 2.5.1 to find

$$A^{-1} = \begin{bmatrix} 17/2 & -5 & -13/2 & 2 \\ -3 & 2 & 3 & -1 \\ -5 & 3 & 3 & -1 \\ 7/2 & -2 & -5/2 & 1 \end{bmatrix},$$

so (4.15) becomes

$$\begin{bmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \alpha_4 \end{bmatrix} = \begin{bmatrix} 17/2 & -5 & -13/2 & 2 \\ -3 & 2 & 3 & -1 \\ -5 & 3 & 3 & -1 \\ 7/2 & -2 & -5/2 & 1 \end{bmatrix} \begin{bmatrix} 7 \\ 14 \\ -1 \\ -2 \end{bmatrix} = \begin{bmatrix} 0 \\ 2 \\ 2 \\ 1 \end{bmatrix}.$$

Consequently, the components of the vector v in the basis $\mathbf{B}$ are $(0, 2, 2, 1)$. ◀

Exercise 4.3 (Dimension of a Direct Sum)
Let E be a finite-dimensional vector space over a field $\mathbb{K}$ and let F_1 and F_2 be two subspaces of E.

1. Show that

$$\dim_{\mathbb{K}}(F_1 + F_2) = \dim_{\mathbb{K}} F_1 + \dim_{\mathbb{K}} F_2 - \dim_{\mathbb{K}}(F_1 \cap F_2). \tag{4.16}$$

2. Deduce that if $E = F_1 \oplus F_2$, then

$$\dim_{\mathbb{K}} E = \dim_{\mathbb{K}} F_1 + \dim_{\mathbb{K}} F_2. \tag{4.17}$$

■

Solution
1. Denote

$$\dim_{\mathbb{K}} F_1 = n, \quad \dim_{\mathbb{K}} F_2 = m, \quad \text{and} \quad \dim_{\mathbb{K}}(F_1 \cap F_2) = r.$$

Of course, here $r \leq \min(n, m)$ since $F_1 \cap F_2 \subset F_1$ and $F_1 \cap F_2 \subset F_2$ (Theorem 4.6.7). To prove (4.16), we will construct a basis of the subspace $F_1 + F_2$ and find the dimension of $F_1 + F_2$ by proving that the number of elements (cardinality) of this basis is exactly $n+m-r$. Since $F_1 \cap F_2$ is a subspace of E (Theorem 4.3.5), then according to Theorem 4.6.1, this subspace has a basis. So, let $S = \{w_1, w_2, \ldots, w_r\}$ be a basis of $F_1 \cap F_2$. Thus, S is a linearly independent set in $F_1 \cap F_2$ and therefore, it is a linearly independent set in both F_1 and F_2. Theorem 4.6.8 implies that there exist elements $u_{r+1}, \ldots, u_n$ of F_1 and elements $v_{r+1}, \ldots, v_m$ of F_2, such that

$$\mathbf{B}_1 = \{w_1, w_2, \ldots, w_r, u_{r+1}, \ldots, u_n\}$$

is a basis of F_1 and

$$\mathbf{B}_2 = \{w_1, w_2, \ldots, w_r, v_{r+1}, \ldots, v_m\}$$

is a basis of F_2.

Now,

$$\mathbf{B} = \mathbf{B}_1 \cup \mathbf{B}_2 = \{w_1, w_2, \ldots, w_r, u_{r+1}, \ldots, u_n, v_{r+1}, \ldots, v_m\}$$

contains $n + m - r$ elements. Consequently, it is enough to show that $\mathbf{B}$ is a basis of $F_1 + F_2$. That is, $\mathbf{B}$ spans $F_1 + F_2$ and the elements of $\mathbf{B}$ are linearly independent elements in $F_1 + F_2$. First, it is clear that if u is an element of B, then u is either in F_1 or in F_2. If u is in F_1, then we may write it as

$$u = u + 0_E,$$

which is an element of $F_1 + F_2$ since 0_E belongs to F_2 (F_2 is a subspace). Similarly, if u belongs to F_2, we write

$$u = 0_E + u$$

which is an element of $F_1 + F_2$. Thus, it is clear that B is a set of $F_1 + F_2$.

Now, let z be an element in $F_1 + F_2$. Then there exist z_1 in F_1 and z_2 in F_2, such that

$$z = z_1 + z_2.$$

Since $\mathbf{B}_1$ is a basis of F_1 and $\mathbf{B}_2$ is a basis of F_2, we have

$$z_1 = \lambda_1 w_1 + \lambda_2 w_2 + \cdots + \lambda_r w_r + \lambda_{r+1} u_{r+1} + \cdots + \lambda_n u_n,$$

and

$$z_2 = \mu_1 w_1 + \mu_2 w_2 + \cdots + \mu_r w_r + \mu_{r+1} v_{r+1} + \cdots + \mu_m v_m,$$

where λ_i, $i = 1, \ldots, n$, and μ_j, $j = 1, \ldots, m$, are elements of $\mathbb{K}$. Hence, we may write z as

$$\begin{aligned} z = z_1 + z_2 = (\lambda_1 + \mu_1) w_1 + (\lambda_2 + \mu_2) w_2 + \cdots + \lambda_r w_r \\ + \lambda_{r+1} u_{r+1} + \cdots + \lambda_n u_n + \mu_{r+1} v_{r+1} + \cdots + \mu_m v_m. \end{aligned}$$

So, z is a linear combination of the elements of $\mathbf{B}$, which shows that $\mathbf{B}$ spans $F_1 + F_2$.

Next, we want to show that the elements of $\mathbf{B}$ are linearly independent. Let α_i, $i = 1, \ldots r$, λ_j, $j = r + 1, \ldots, n$, and μ_k, $k = r + 1, \ldots, m$, be elements of $\mathbb{K}$ satisfying

$$\alpha_1 w_1 + \alpha_2 w_2 + \cdots + \alpha_r w_r + \lambda_{r+1} u_{r+1} + \cdots + \lambda_n u_n + \mu_{r+1} v_{r+1} + \cdots + \mu_n v_n = 0_E,$$

or equivalently

$$\sum_{i=1}^{r} \alpha_i w_i + \sum_{j=r+1}^{n} \lambda_j u_j + \sum_{k=r+1}^{m} \mu_k v_k = 0_E. \tag{4.18}$$

4

This implies that

$$\sum_{k=r+1}^{m} \mu_k v_k = -\sum_{i=1}^{r} \alpha_i w_i - \sum_{j=r+1}^{n} \lambda_j u_j,$$

which shows that $\sum_{k=r+1}^{m} \mu_k v_k$ belongs to F_1 and hence to $F_1 \cap F_2$. Thus, since $S = \{w_1, w_2, \ldots, w_r\}$ is a basis of $F_1 \cap F_2$, there exist $\beta_1, \beta_2, \ldots, \beta_r$ in $\mathbb{K}$ such that

$$\sum_{k=r+1}^{m} \mu_k v_k = \sum_{i=1}^{r} \beta_i w_i,$$

or equivalently,

$$\sum_{k=r+1}^{m} \mu_k v_k - \sum_{i=1}^{r} \beta_i w_i = 0_E.$$

Since $\mathbf{B}_2$ is a linearly independent set, this last relation yields

$$\beta_1 = \beta_2 = \cdots = \beta_r = \mu_{r+1} = \cdots = \mu_m = 0_{\mathbb{K}}. \tag{4.19}$$

By the same method, we may write, using (4.18),

$$\sum_{j=r+1}^{n} \lambda_j u_j = -\sum_{i=1}^{r} \alpha_i w_i - \sum_{k=r+1}^{m} \mu_k v_k.$$

This shows that $\sum_{j=r+1}^{n} \lambda_j u_j$ is an element in $F_1 \cap F_2$ and as before, we can show, by using the fact that $\mathbf{B}_1$ is a linearly independent set, that

$$\lambda_{r+1} = \cdots = \lambda_m = 0_{\mathbb{K}}. \tag{4.20}$$

Next, thanks to (4.19) and (4.20), relation (4.18) becomes

$$\sum_{i=1}^{r} \alpha_i w_i = 0_E,$$

which gives $\alpha_1 = \alpha_2 = \cdots = \alpha_r = 0_{\mathbb{K}}$. ◄

Consequently, the set $\mathbf{B}$ is linearly independent and therefore, is a basis of $F_1 + F_2$. Hence,

$$\dim_{\mathbb{K}}(F_1 + F_2) = n + m - r = \dim_{\mathbb{K}} F_1 + \dim_{\mathbb{K}} F_2 - \dim_{\mathbb{K}}(F_1 \cap F_2).$$

2. If $E = F_1 \oplus F_2$, then $F_1 \cap F_2 = \{0_E\}$ (Theorem 4.3.10) and then $\dim_K(F_1 \cap F_2) = 0$. Thus, by (4.16), the identity (4.17) holds.

Exercise 4.4 (Direct Sum of Symmetric and Skew-Symmetric Matrices)
We consider the space of square real matrices $\mathscr{M}_n(\mathbb{R})$. We define the set of symmetric matrices S to be the set of all matrices in $\mathscr{M}_n(\mathbb{R})$ satisfying $A^T = A$ and the set of skew-symmetric matrices W to be the set of all matrices in $\mathscr{M}_n(\mathbb{R})$ satisfying $A^T = -A$.

1. Prove that S and W are subspaces of $\mathscr{M}_n(\mathbb{R})$.
2. Prove that

$$\mathscr{M}_n(\mathbb{R}) = S \oplus W. \tag{4.21}$$

■

Solution
1. It is clear that the zero matrix $\mathbf{0} = 0_{\mathscr{M}_n(\mathbb{R})}$ satisfies

$$\mathbf{0}^T = \mathbf{0} = -\mathbf{0}.$$

So, $0_{\mathscr{M}_n(\mathbb{R})}$ belongs to both S and W.

Next, we can easily deduce from Theorem 1.4.1 that the sum of two symmetric matrices is symmetric and the multiplication of a symmetric matrix by a scalar gives a symmetric matrix. Hence, S is a subspace of $\mathscr{M}_n(\mathbb{R})$. Using the same Theorem 1.4.1, we have for A and B two elements in W and λ in $\mathbb{R}$ that

$$(A + B)^T = A^T + B^T = -A - B = -(A + B)$$

and

$$(\lambda A)^T = \lambda A^T = \lambda(-A) = -(\lambda A).$$

Hence, W is a subspace of $\mathscr{M}_n(\mathbb{R})$.

2. Now, let A be a matrix in $\mathscr{M}_n(\mathbb{R})$. Then we can write A as

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T).$$

We have shown in Exercise 2.7 that $A + A^T$ is symmetric and $A - A^T$ is skew-symmetric. Thus, we have proved that

$$\mathscr{M}_n(\mathbb{R}) = S + W.$$

It remains to show that (see Theorem 4.3.10) $S \cap W = \{0_{\mathscr{M}_n(\mathbb{R})}\}$. So, let A be an element of the intersection $S \cap W$. Thus, A satisfies

$$A = A^T \qquad \text{and} \qquad A^T = -A.$$

This means $2A = 0_{\mathscr{M}_n(\mathbb{R})}$, and so $A = 0_{\mathscr{M}_n(\mathbb{R})}$. Hence, $S \cap W = \{0_{\mathscr{M}_n(\mathbb{R})}\}$ and therefore (4.21) holds. ◀

Exercise 4.5 (Linearly Independent Column Vectors of a Square Matrix)

1. Consider the vector space $\mathbb{R}^2$ over the field $\mathbb{R}$. Let $u_1 = (a, b)$ and $u_2 = (c, d)$ be two vectors in $\mathbb{R}^2$. Show that u_1 and u_2 are linearly independent if and only if

   $$ad - bc \neq 0.$$

2. Now, consider the vector space $\mathbb{R}^n$ $(n \geq 2)$ over $\mathbb{R}$ and let $u_1, u_2, \ldots, u_n$ be n vectors in $\mathbb{R}^n$. Show that $u_1, u_2, \ldots, u_n$ are linearly independent (form a basis of $\mathbb{R}^n$) if and only if

   $$\det(A) \neq 0,$$

   where A is the matrix in $\mathscr{M}_n(\mathbb{R})$ defined as

   $$A = [u_1, u_2, \ldots, u_n] \qquad (u_1, u_2, \ldots, u_n \text{ are the column vectors of } A).$$

3. Show that the vectors $v_1 = (1, 2, 3)$, $v_2 = (1, 1, 0)$, and $v_3 = (3, 4, 3)$ are linearly dependent in $\mathbb{R}^3$.

■

Solution

1. Assume that u_1 and u_2 are linearly independent. Then the equation

$$\lambda_1 u_1 + \lambda_2 u_2 = 0_{\mathbb{R}^2}, \tag{4.22}$$

with λ_1 and λ_2 in $\mathbb{R}$, has the trivial solution $\lambda_1 = \lambda_2 = 0$ as the uniques solution. Equation (4.22) is equivalent to the system

$$\begin{cases} a\lambda_1 + c\lambda_2 = 0, \\ b\lambda_1 + d\lambda_2 = 0. \end{cases}$$

This system has the trivial solution $\lambda_1 = \lambda_2 = 0$ as the unique solution if and only if (see Theorem 1.2.9) the matrix

$$A = \begin{bmatrix} a & c \\ b & d \end{bmatrix}$$

is invertible. That is, if and only if

$$\det(A) = ad - bc \neq 0.$$

2. By the same method, let $u_i = (u_{i1}, u_{i2}, \ldots, u_{in})$, $1 \leq i \leq n$, be vectors in $\mathbb{R}^n$. These vectors are linearly independent if and only if the equation

$$\lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_n u_n = 0_{\mathbb{R}^n}$$

has the unique trivial solution $\lambda_1 = \lambda_2 = \cdots = \lambda_n = 0$. Expanding this equation, we obtain

$$\begin{cases} u_{11}\lambda_1 + u_{21}\lambda_2 + \cdots + u_{n1}\lambda_n = 0, \\ u_{12}\lambda_1 + u_{22}\lambda_2 + \cdots + u_{n2}\lambda_n = 0, \\ \vdots \qquad\quad \vdots \qquad\quad \vdots \qquad\quad \vdots \\ u_{1n}\lambda_1 + u_{2n}\lambda_2 + \cdots + u_{nn}\lambda_n = 0. \end{cases}$$

This system has the unique solution $\lambda_1 = \lambda_2 = \cdots = \lambda_n = 0$ if and only if the matrix

$$A = \begin{bmatrix} u_{11} & u_{21} & u_{31} & \ldots & u_{n1} \\ u_{12} & u_{22} & u_{32} & \ldots & u_{n2} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ u_{1n} & u_{2n} & u_{3n} & \ldots & u_{nn} \end{bmatrix} = [u_1, u_2, \ldots, u_n]$$

is invertible. That is, if and only if $\det(A) \neq 0$.

3. We have

$$\det[v_1, v_2, v_3] = \det \begin{bmatrix} 1 & 1 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 3 \end{bmatrix} = 0.$$

Thus, according to our previous proof, v_1, v_2, and v_3 are linearly dependent in $\mathbb{R}^3$. ◀

Exercise 4.6

Let E be a vector space over a field $\mathbb{K}$. Let G, F_1, and F_2 be three subspaces of E. Show that if

$$E = G \oplus F_1 = G \oplus F_2 \qquad \text{and} \qquad F_1 \subset F_2, \tag{4.23}$$

then $F_1 = F_2$. ■

Solution

To show that $F_1 = F_2$, it is enough to prove that $F_2 \subset F_1$. Let w_2 be an element of F_2, then w_2 is an element of E. Hence, using (4.23), we write w_2 as

$$w_2 = v + w_1,$$

where $v \in G$ and $w_1 \in F_1$. Since $F_1 \subset F_2$, we deduce that $w_1 \in F_2$. This implies that

$$v = w_2 - w_1 \in F_2$$

and thus $v \in G \cap F_2 = \{0_E\}$. That is, $v = 0_E$. Therefore, $w_2 = w_1$, and so $w_2 \in F_1$. Hence $F_2 \subset F_1$. ◀

Exercise 4.7 (Direct Product of Two Vector Spaces)

Let E and F be two finite-dimensional vector spaces over the same field $\mathbb{K}$. We define the space (called direct product of E and F) $E \times F$ to be the set of all pairs (u, v) whose first component is an element u of E and whose second component is an element v of F. We define the addition in $E \times F$ by

$$(u_1, v_1) + (u_2, u_2) = (u_1 + u_2, v_1 + v_2),$$

where (u_1, v_1) and (u_2, v_2) are elements of $E \times F$. Also, for (u, v) in $E \times F$ and for λ in $\mathbb{K}$, we define the multiplication by scalars as

$$\lambda(u, v) = (\lambda u, \lambda v).$$

1. Show that $(E \times F, +, \cdot)$ is a vector space over $\mathbb{K}$ with $0_{E\times F} = (0_E, 0_F)$.
2. Show that

$$\dim_{\mathbb{K}}(E \times F) = \dim_{\mathbb{K}} E + \dim_{\mathbb{K}} F.$$

3. We define the sets E_1 and F_1 as follows:

$$E_1 = E \times \{0_F\} = \{(u, 0_F), \quad \text{where } u \in E\}$$

and

$$F_1 = \{0_E\} \times F = \{(0_E, v), \quad \text{where } v \in F\}.$$

Show that E_1 and F_1 are subspaces of $E \times F$.

4. Prove that

$$E \times F = E_1 \oplus F_1. \tag{4.24}$$

■

Solution

1. We leave it to the reader to verify that $E \times F$ satisfies the axioms in Definition 4.1.1.

2. Denote $\dim_{\mathbb{K}} E = r$ and $\dim_{\mathbb{K}} F = s$. Then, according to Theorem 4.6.1, there exist elements $u_1, u_2, \ldots, u_r$ of E and $v_1, v_2, \ldots, v_s$ of F such that $\{u_1, u_2, \ldots, u_r\}$ is a basis of E and $\{v_1, v_2, \ldots, v_s\}$ is a basis of F. Now define in $E \times F$ the sets

$$\mathbf{B}_1 = \{(u_1, 0_F), (u_2, 0_F), \ldots, (u_r, 0_F)\}$$

and

$$\mathbf{B}_2 = \{(0_E, v_1), (0_E, v_2), \ldots, (0_E, v_s)\}.$$

So, we want to show that the set $\mathbf{B} = \{\mathbf{B}_1, \mathbf{B}_2\}$ which consists of the elements of $\mathbf{B}_1 \cup \mathbf{B}_2$ is a basis of $E \times F$. First we prove that $\mathbf{B}$ spans $E \times F$. So, let (u, v) be an element of $E \times F$, with $u \in E$ and $v \in F$. There exist $\alpha_1, \alpha_2, \ldots, \alpha_r$ and $\beta_1, \beta_2, \ldots, \beta_s$ in $\mathbb{K}$ such that

$$u = \alpha_1 u_1 + \alpha_2 u_2 + \cdots + \alpha_s u_s \quad \text{and} \quad v = \beta_1 v_1 + \beta_2 v_2 + \cdots + \beta_s v_s.$$

Then

$$\begin{aligned}(u, v) = {} & \alpha_1(u_1, 0_F) + \alpha_2(u_2, 0_F) + \cdots + \alpha_r(u_r, 0_F) \\ & + \beta_1(0_E, v_1) + \beta_2(0_E, v_2) + \cdots + \beta_s(0_E, v_s).\end{aligned}$$

Thus, it is clear that the set $\mathbf{B}$ spans $E \times F$. Now, since $\mathbf{B}_1$ and $\mathbf{B}_2$ are linearly independent sets, one can easily show that $\mathbf{B}$ is a linearly independent set (we leave this to the reader) and thus conclude that $\mathbf{B}$ is a basis of $E \times F$. Hence

$$\dim_{\mathbb{K}}(E \times F) = r + s = \dim_{\mathbb{K}} E + \dim_K F.$$

3. We show that E_1 is a subspace of $E \times F$. First, it is clear that $0_{E\times F} = (0_E, 0_F) \in E_1$. Now, let w_1 and w_2 be two elements of E_1 and λ_1 and λ_2 be two elements of $\mathbb{K}$. Then, we have

$$w_1 = (u_1, 0_F) \qquad \text{and} \qquad w_2 = (u_2, 0_F)$$

for some u_1 and u_2 in E. We have

$$\lambda_1 w_1 + \lambda_2 w_2 = \lambda_1(u_1, 0_F) + \lambda_2(u_2, 0_F) = (\lambda_1 u_1 + \lambda_2 u_2, 0_F).$$

Since E is a vector space, $\lambda_1 u_1 + \lambda_2 u_2 \in E$, and thus $\lambda_1 w_1 + \lambda_2 w_2 \in E_1$. This shows that F_1 is a vector subspace of $E \times F$. By the same method, we can show that F_1 is a vector subspace of $E \times F$.

4. To prove (4.24), in view of Theorem 4.3.10, we need to show that

$$E \times F = E_1 + F_1 \qquad \text{and} \qquad F_1 \cap F_2 = \{0_{E\times F}\}.$$

So, let w be an element of $E \times F$. Then there exists $u \in E$ and $v \in F$ such that

$$w = (u, v) = (u, 0_F) + (0_E, v).$$

Since $(u, 0_F) \in E_1$ and $(0_E, v) \in F_1$, we deduce that $E \times F = E_1 + F_1$.

Now let w be an element of $E_1 \cap F_1$. Then there exists $u \in E$ and $v \in F$ such that

$$w = (u, 0_F) = (0_E, v).$$

This yields $u = 0_E$ and $v = 0_F$. Thus, $w = (0_E, 0_F) = 0_{E\times F}$. Hence $F_1 \cap F_2 = \{0_{E\times F}\}$, and so (4.24) holds. ◀

Exercise 4.8 (Complement of a Subspace)
Let E be a finite-dimensional vector space over a field $\mathbb{K}$, and U be a subspace of E. Show that there exists a subspace W of E such that

$$E = U \oplus W.$$

■

Solution
Since E has finite dimension, so has U, since U is a subspace of E (Theorem 4.6.7). Denote

$$\dim_{\mathbb{K}} E = n, \quad \dim_{\mathbb{K}} U = m, \quad \text{with} \quad m \leq n.$$

First, if $m = n$, then $F = E$ (Theorem 4.6.7) and thus in this case we can take $W = \{0_E\}$.

Second, if $m < n$, then, according to Theorem 4.6.1, U has a basis, say, $\mathbf{B}_1 = \{u_1, u_2, \ldots, u_m\}$. Thus, $\mathbf{B}$ is a linearly independent set in U and hence is a linearly independent set in E. By Theorem 4.6.8, there exist elements $w_{m+1}, \ldots, w_m$ of E such that the set

$$\mathbf{B} = \{u_1, u_2, \ldots, u_m, w_{m+1}, \ldots, w_n\}$$

is a basis of E. We define the space W as

$$W = \text{span}\{w_{m+1}, \ldots, w_n\}.$$

We need to show that $E = U \oplus W$, that is (Theorem 4.3.10),

$$E = U + W \qquad \text{and} \qquad U \cap W = \{0_E\}.$$

To prove the first equality, let v be an element of E. Since $\mathbf{B}$ is a basis in E, then it spans E and thus there exist $\lambda_1, \lambda_2, \ldots, \lambda_n$ in $\mathbb{K}$ such that

$$v = \lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_m u_m + \lambda_{m+1} w_{m+1} + \cdots + \lambda_n w_n.$$

Put

$$u = \lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_m u_m \quad \text{and} \quad w = \lambda_{m+1} w_{m+1} + \cdots + \lambda_n w_n.$$

Then $u \in U$, $w \in W$, and $v = u + w$. This shows that $E = U + W$.

Next, we need to prove that $U \cap W = \{0_E\}$. So, let v be an element in $U \cap W$. Then $v \in U$ and $v \in W$, that is

$$v = \alpha_1 u_1 + \alpha_2 u_2 + \cdots + \alpha_m u_m$$

and

$$v = \alpha_{m+1} w_{m+1} + \cdots + \alpha_n w_n.$$

Consequently,

$$\alpha_1 u_1 + \alpha_2 u_2 + \cdots + \alpha_m u_m - \alpha_{m+1} w_{m+1} - \cdots - \alpha_n w_n = 0_E.$$

Since **B** is a linearly independent set,

$$\alpha_i = 0, \qquad i = 1, \ldots, n.$$

Therefore $v = 0_E$. Thus our result holds. ◀

Exercise 4.9 (Union of Two Subspaces)
Let E be a vector space over a field $\mathbb{K}$. Let F_1 and F_2 be two subspaces of E. Show that $F_1 \cup F_2$ is a subspace of E if and only if $F_1 \subset F_2$ or $F_2 \subset F_1$. ■

Solution
First, assume that $F_1 \cup F_2$ is a subspace of E and let w_1 be an element of F_1 and w_2 be an element of F_2. Then, both w_1 and w_2 are elements of $F_1 \cup F_2$ and since $F_1 \cup F_2$ is a subspace,

$$w = w_1 + w_2$$

is also an element of $F_1 \cup F_2$. That is, $w \in F_1$ or $w \in F_2$. If $w \in F_1$, then

$$w_2 = w - w_1 \in F_1,$$

since F_1 is a subspace. This shows that $F_2 \subset F_1$. Similarly, if $w \in F_2$, then

$$w_1 = w - w_2 \in F_2,$$

since F_2 is a subspace. Thus, $F_1 \subset F_2$.

Conversely, assume for instance that $F_1 \subset F_2$. We need to show that $F_1 \cup F_2$ is a subspace. It is clear that 0_E belongs to $F_1 \cup F_2$ since it does belong to both subspaces. Now, let u and

v be two elements of $F_1 \cup F_2$ and λ and μ be two elements of $\mathbb{K}$. Then, u and v belong to F_2 since $F_1 \subset F_2$. Then $\lambda u + \mu v \in F_2$, since F_2 is a subspace. Thus, $\lambda u + \mu v \in F_1 \cup F_2$. Consequently, $F_1 \cup F_2$ is a subspace. ◄

4

Exercise 4.10 (Orthogonal Complement)
Consider the vector space $\mathbb{R}^n$ over the field $\mathbb{R}$. Let F be a subspace of $\mathbb{R}^n$. We define the orthogonal complement of F to be the set of all vectors in $\mathbb{R}^n$ that are orthogonal to all vectors of F, and denoted it $F^\perp$. That is

$$F^\perp = \{u \in \mathbb{R}^n, \text{ such that the dot product } u \cdot v = 0, \quad \text{for all } v \in F\}.$$

1. Show that $F^\perp$ is a subspace of $\mathbb{R}^n$ and $F \subset (F^\perp)^\perp$.[3]
2. Show that $\mathbb{R}^n = F \oplus F^\perp$.
3. Now, let G be another subspace of $\mathbb{R}^n$. Prove the following

$$(F + G)^\perp = F^\perp \cap G^\perp, \tag{4.25}$$

$$(F \cap G)^\perp = F^\perp + G^\perp. \tag{4.26}$$

4. Deduce that if $\mathbb{R}^n = F \oplus G$, then $\mathbb{R}^n = F^\perp \oplus G^\perp$.

■

Solution
1. It is clear that $0_{\mathbb{R}^n}$ is an element of $F^\perp$ since for any vector $u \in \mathbb{R}^n$, we have $u \cdot 0_{\mathbb{R}^n} = 0$. In particular $u \cdot 0_{\mathbb{R}^n} = 0$ for all vectors $u \in F$. Now, let u and v be two vectors in $F^\perp$ and λ and μ be two real numbers. By Theorem 3.3.5, for any w in F, we have

$$\begin{aligned}(\lambda u + \mu v) \cdot w &= \lambda(u \cdot w) + \mu(v \cdot w)\\ &= 0,\end{aligned}$$

since $u \cdot w = 0$ and $v \cdot w = 0$. Therefore, $F^\perp$ is a subspace of $\mathbb{R}^n$. To show that $(F^\perp)^\perp = F$, we need to prove that $F \subset (F^\perp)^\perp$ and $(F^\perp)^\perp \subset F$. Let u be an element of F. Then for any $w \in F^\perp$, we have $u \cdot w = 0$. That is $u \in (F^\perp)^\perp$. This yields $F \subset (F^\perp)^\perp$.

2. To show that $\mathbb{R}^n = F \oplus F^\perp$, let v be an element of F and u be an element of $\mathbb{R}^n$. Then, using Theorem 3.4.1, we deduce that u can be written in exactly one way as

$$u = w_1 + w_2$$

with $w_1 = kv \in F$ and w_2 is orthogonal to w_1, that is $w_2 \in F^\perp$. We deduce that $\mathbb{R}^n = F \oplus F^\perp$.

[3] In fact since $\dim_{\mathbb{K}} \mathbb{R}^n$ is finite, we even have $F = (F^\perp)^\perp$, but the proof of $(F^\perp)^\perp \subset F$ requires some knowledge of topology. We omit it here.

3. To prove (4.25), we need to show that

$$(F+G)^{\perp} \subset (F^{\perp} \cap G^{\perp}) \qquad \text{and} \qquad (F^{\perp} \cap G^{\perp}) \subset (F+G)^{\perp}.$$

Let v be an element of $(F+G)^{\perp}$. Then for any $u \in F+G$, we have

$$v \cdot u = 0. \tag{4.27}$$

It is clear that (4.27) is also satisfied for the elements of F and for the elements of G, since $F \subset F+G$ and $G \subset F+G$. Thus, $u \in F^{\perp}$ and $u \in G^{\perp}$, i.e., $u \in F^{\perp} \cap G^{\perp}$. This implies that $F+G \subset F^{\perp} \cap G^{\perp}$.

Next, let w be an element of $F^{\perp} \cap G^{\perp}$. Then

$$w \cdot u = 0, \qquad \text{and} \qquad w \cdot v = 0,$$

for all $u \in F$ and for all $v \in G$. This implies

$$w \cdot (u+v) = 0.$$

That is, $w \in (F+G)^{\perp}$, since any element of $F+G$ is written as $u+v$ with $u \in F$ and $v \in G$. Thus, $F^{\perp} \cap G^{\perp} \subset (F+G)^{\perp}$. Consequently, (4.25) holds.

Now to show (4.26), we have, since for any subspace H of $\mathbb{R}^n$, $(H^{\perp})^{\perp} = H$,

$$(F \cap G)^{\perp} = [(F^{\perp})^{\perp} \cap (G^{\perp})^{\perp}]^{\perp}.$$

Applying (4.25) to $F^{\perp}$ and $G^{\perp}$, we get

$$(F \cap G)^{\perp} = [(F^{\perp})^{\perp} \cap (G^{\perp})^{\perp}]^{\perp} = [(F^{\perp} + G^{\perp})^{\perp}]^{\perp} = F^{\perp} + G^{\perp}.$$

This ends the proof of (4.26).

4. Now, if $\mathbb{R}^n = F \oplus G$, then we have

$$\mathbb{R}^n = F+G \qquad \text{and} \qquad F \cap G = \{0_{\mathbb{R}^n}\}.$$

Hence, we have by using (4.26)

$$F^{\perp} + G^{\perp} = (F \cap G)^{\perp} = \{0_{\mathbb{R}^n}\}^{\perp} = \mathbb{R}^n,$$

since $\{0_{\mathbb{R}^n}\}$ is orthogonal to all the elements of $\mathbb{R}^n$. On the other hand, making use of (4.25), we get

$$F^{\perp} \cap G^{\perp} = (F+G)^{\perp} = \{\mathbb{R}^n\}^{\perp} = \{0_{\mathbb{R}^n}\},$$

since $\{0_{\mathbb{R}^n}\}$ is the only vector which is orthogonal to all vectors of $\mathbb{R}^n$. So, we have already proved that

$$\mathbb{R}^n = F^{\perp} + G^{\perp} \qquad \text{and} \qquad F^{\perp} \cap G^{\perp} = \{0_{\mathbb{R}^n}\}.$$

4

Consequently, $\mathbb{R}^n = F^{\perp} \oplus G^{\perp}$. ◄

Linear Transformations

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5.1 Definition and Examples

In ▶ Chap. 1, we defined a matrix as a rectangular array of numbers (Definition 1.1.5). In this chapter, we give the mathematical definition of matrices through *linear transformations*. We will see that the multiplication of two matrices is equivalent to the composition of two linear transformations. One of the important properties of linear transformations is that they carry some algebraic properties from one vector space to another. Sometimes, this will provide us with the necessary knowledge of some vector spaces, without even studying them in detail, but by rather seeing them as the result of a linear transformation of other well-known vector spaces.

We saw in ▶ Chap. 1 that if we multiply an $m \times n$ matrix A by a vector u in $\mathbb{R}^n$ we obtain a vector $v = Au$ in $\mathbb{R}^m$. This multiplication satisfies a very important property called *linearity*. That is, if u_1 and u_2 are two vectors in $\mathbb{R}^n$, $v_1 = Au_1$, and $v_2 = Au_2$, then

$$A(u_1 + u_2) = Au_1 + Au_1 = v_1 + v_2,$$

and for any scalar λ, we have

$$A(\lambda u_1) = \lambda A u_1 = \lambda v_1.$$

So, by doing the above matrix multiplication, we have *transformed linearly* the elements of the space $\mathbb{R}^n$ to elements of the space $\mathbb{R}^m$. This is what we call *linear transformation*, and we can generalize this notion to any two vector spaces as follows.

Definition 5.1.1 (Linear Transformation)

Let E and F be two vector spaces over the same field $\mathbb{K}$. A *linear transformation* (also called *homomorphism*) from E to F is defined to be a linear map f from E into F and satisfying, for all u and v in E and λ in $\mathbb{K}$:

$$f(u+v) = f(u) + f(v), \tag{5.1}$$

and

$$f(\lambda u) = \lambda f(u). \tag{5.2}$$

If, in addition, f is bijective (or one-to-one and onto), then f is called an *isomorphism*. If $E = F$, then f is called *a linear operator* or *endomorphism*. If in addition f is bijective, then f is called an *automorphism*.

Remark 5.1.1 The two properties (5.1) and (5.2) can be combined in a single property and one writes: f is a linear transformation from the vector space E to the vector space F if for all $u, v \in E$ and all $\lambda, \mu \in \mathbb{K}$, we have

$$f(\lambda u + \mu v) = \lambda f(u) + \mu f(v). \tag{5.3}$$

Notation The set of linear transformations from E to F is denoted by $\mathscr{L}(E,F)$. If $E = F$, then we denote $\mathscr{L}(E,F)$ simply by $\mathscr{L}(E)$.

Example 5.1 (The Zero Transformation)

Let E and F be two vector spaces over the same field $\mathbb{K}$. The zero transformation $0_{\mathscr{L}(E,F)}$ is the transformation that maps all elements u of E to the zero element 0_F of F. That is

$$0_{\mathscr{L}(E,F)}(u) = 0_F.$$

Clearly, $0_{\mathscr{L}(E,F)}$ is a linear transformation. Indeed, for all $u, v \in E$ and all $\lambda, \mu \in \mathbb{K}$, we have

$$\begin{aligned} 0_{\mathscr{L}(E,F)}(\lambda u + \mu v) &= 0_F \\ &= \lambda 0_F + \mu 0_F \\ &= \lambda 0_{\mathscr{L}(E,F)}(u) + \mu 0_{\mathscr{L}(E,F)}(u). \end{aligned}$$

■

Example 5.2 (The Identity Operator)

Let E be a vector space over a field $\mathbb{K}$. We define the identity Id_E to be the operator which maps each element u of E into itself. That is, for all u in E,

$$\mathrm{Id}_E(u) = u.$$

Then, Id_E is an endomorphism, because for all $u, v \in E$ and all $\lambda, \mu \in \mathbb{K}$, we have

$$\begin{aligned}\mathrm{Id}_E(\lambda u + \mu v) &= \lambda u + \mu v \\ &= \lambda \mathrm{Id}_E(u) + \mu \mathrm{Id}_E(v).\end{aligned}$$

■

Example 5.3 (Linear Transformation Associated to Matrix Multiplication)
Let A be a fixed matrix in $\mathscr{M}_{m\times n}(\mathbb{R})$. Consider the transformation f acting from the vector space $\mathbb{R}^n$ to the vector space $\mathbb{R}^m$ as follows:

$$f(u) = Au \tag{5.4}$$

for any column vector u in $\mathbb{R}^n$. This transformation f is linear. Indeed, let u, v be two vectors in $\mathbb{R}^n$, and λ, μ be real numbers. Then, using the properties of matrices, we have

$$\begin{aligned}f(\lambda u + \mu v) &= A(\lambda u + \mu v) \\ &= \lambda Au + \mu Av \\ &= \lambda f(u) + \mu f(v).\end{aligned}$$

■

Example 5.4 (The Derivative is a Linear Transformation)
Let $\mathscr{F}(\mathbb{R}, \mathbb{R})$ be the vector space of all real-valued functions defined on $\mathbb{R}$. Let $\mathscr{D}(\mathbb{R}, \mathbb{R})$ be the vector space of all differentiable real-valued functions defined on $\mathbb{R}$. Define the transformation D from $\mathscr{F}(\mathbb{R}, \mathbb{R})$ to $\mathscr{D}(\mathbb{R}, \mathbb{R})$ by

$$D(f) = \frac{df}{dx}.$$

Then, it is clear that D is linear, since

$$D(\lambda f + \mu g) = \frac{d}{dx}(\lambda f + \mu g) = \lambda D(f) + \mu D(g).$$

■

5.2 Fundamental Properties of Linear Transformations

In this section we list some fundamental properties of linear transformations. As usual, we start with algebraic properties. Let E and F be two vector spaces over the same field $\mathbb{K}$. For any f and g in $\mathscr{L}(E, F)$ and for any λ in $\mathbb{K}$, we define the addition and

multiplication by scalars as follows:

$$(f+g)(u) = f(u) + g(u) \qquad \text{and} \qquad (\lambda f)(u) = \lambda f(u).$$

One can easily verify that $\mathscr{L}(E,F)$ with the above addition and multiplication by scalars is a vector space over $\mathbb{K}$.

In addition to the above algebraic structure of the vector space $\mathscr{L}(E,F)$, we exhibit now more properties of linear transformations.

5

Theorem 5.2.1 (Composition of Linear Transformations)
Let E, F, and G be three vector spaces over the same field $\mathbb{K}$. Let f be an element of $\mathscr{L}(E,F)$ and g be an element of $\mathscr{L}(F,G)$. Then $g \circ f$ is an element of $\mathscr{L}(E,G)$.

Proof
We have

$$\begin{aligned} g \circ f : E &\to G \\ u &\mapsto (g \circ f)(u) = g[f(u)]. \end{aligned}$$

Now, let u, v be elements of E and λ, μ be elements of $\mathbb{K}$. Then by the linearity of f and g, we have

$$\begin{aligned} (g \circ f)(\lambda u + \mu v) &= g[f(\lambda u + \mu v)] \\ &= g[\lambda f(u) + \mu f(v)] \\ &= \lambda g[f(u)] + \mu g[f(v)] \\ &= \lambda (g \circ f)(u) + \mu (g \circ f)(v). \end{aligned}$$

This establishes the linearity of $g \circ f$. □

Linear transformations preserve some algebraic properties of vector spaces, as we will show in the following theorem. We have seen in ▶ Chap. 1 that the multiplication of any $m \times n$ matrix by the zero vector $0_{\mathbb{R}^n}$ of $\mathbb{R}^n$ results in the zero vector $0_{\mathbb{R}^m}$ in $\mathbb{R}^m$. In fact, this property is also true for any linear transformation as we will proof in the following theorem.

Theorem 5.2.2

Let E and F be two vector spaces over the same field $\mathbb{K}$ and f be a linear transformation from E into F. Then we have:

1. $f(0_E) = 0_F$.
2. $f(-u) = -f(u)$.
3. *If H is a subspace of E, then $f(H)$ is a subspace of F.*
4. *If M is a subspace of F, then $f^{-1}(M)$ is a subspace of E.*

Proof

1. To show that $f(0_E) = 0_F$, since $f(0_E)$ is an element of $f(E)$, we need to show that for all v in $f(E)$,

$$v + f(0_E) = f(0_E) + v = v; \tag{5.5}$$

this will imply that $f(0_E) = 0_F$. So, let v be an element of $f(E)$. Then there exists $u \in E$ such that $v = f(u)$. We have

$$u + 0_E = 0_E + u = u.$$

This implies, due to the linearity of f, that

$$f(u + 0_E) = f(u) + f(0_E) = f(u).$$

Similarly,

$$f(0_E + u) = f(0_E) + f(u) = f(u).$$

The last two relations give (5.5), since $v = f(u)$.

2. We have that for u in E,

$$u + (-u) = 0_E,$$

Thus, the linearity of f implies

$$f(u) + f(-u) = f(0_E) = 0_F.$$

This gives $f(-u) = -f(u)$, due to the uniqueness of the inverse in the Abelian group $(F, +)$.

3. If H is a subspace of E, then H contains 0_E. Consequently, $f(H)$ contains $f(0_E) = 0_F$. This implies that $f(H)$ is not empty. Now, let v_1, v_2 be two elements of $f(H)$ and λ, μ be two elements of $\mathbb{K}$. Then there exist u_1 and u_2 in H such that

$$v_1 = f(u_1) \qquad \text{and} \qquad v_2 = f(u_2).$$

It is clear that $\lambda u_1 + \mu u_2 \in H$, since H is a subspace of E. By using the linearity of f,

$$\begin{aligned}\lambda v_1 + \mu v_2 &= \lambda f(u_1) + \mu f(u_2)\\ &= f(\lambda u_1 + \mu u_2)\end{aligned}$$

which is an element of $f(H)$. Hence, $f(H)$ is a subspace of F.

4. Let M be a subspace of F. Clearly, $f^{-1}(M)$ contains 0_E. Let u_1 and u_2 be two elements of $f^{-1}(M)$. Then, $f(u_1)$ and $f(u_2)$ are elements of M. Hence, for any α and β in $\mathbb{K}$, $\alpha f(u_1) + \beta f(u_2) \in M$, since M is a subspace of F. The linearity of f implies

$$\alpha f(u_1) + \beta f(u_2) = f(\alpha u_1 + \beta u_2).$$

This shows that $\alpha u_1 + \beta u_2$ is an element of $f^{-1}(M)$. Consequently, $f^{-1}(M)$ is a subspace of E. □

5.2.1 The Kernel and the Image of a Linear Transformation

We saw in Example 4.12 that for a fixed matrix A in $\mathscr{M}_{m\times n}(\mathbb{R})$ the null space

$$\mathscr{N} = \{\text{ column vectors } u \text{ in } \mathbb{R}^n, \text{ such that } Au = 0_{\mathbb{R}^m}\}$$

is a subspace of $\mathbb{R}^n$. In view of (5.4), the subspace $\mathscr{N}$ can be written as

$$\mathscr{N} = \{\text{ column vectors } u \text{ in } \mathbb{R}^n, \text{ such that } f(u) = 0_{\mathbb{R}^m}\} = f^{-1}\{0_{\mathbb{R}^m}\},$$

where f is defined in (5.4). The space $\mathscr{N}$ is called the null space of the matrix A, or the *kernel* of the linear transformation f. We can generalize this to any linear transformation as follows.

Definition 5.2.1 (The Kernel of a Linear Transformation)
Let E and F be two vector spaces over the same field $\mathbb{K}$ and f be a linear transformation from E into F. The *kernel* of f is the set of all elements $u \in E$ such that $f(u) = 0_F$. We denote the kernel of f by $\text{Ker}(f)$ and write

$$\text{Ker}(f) = \{u \in E, \text{ such that } f(u) = 0_F\} = f^{-1}\{0_F\}.$$

As usual, when introducing a new set in algebra, a natural question is to check if this set has an algebraic structure. Here we show that the kernel of a linear transformation is a subspace.

Theorem 5.2.3 (The Subspace $\mathrm{Ker}(f)$)
Let E and F be two vector spaces over the same field $\mathbb{K}$ and f be a linear transformation from E into F. Then $\mathrm{Ker}(f)$ is a subspace of E.

Proof
We may prove Theorem 5.2.3 directly, by showing that $\mathrm{Ker}(f)$ satisfies the properties of a subspace. Or, we may easily see that since $\{0_F\}$ is a subspace of F and since $\mathrm{Ker}(f) = f^{-1}\{0_F\}$, Theorem 5.2.2 gives that $\mathrm{Ker}(f)$ is a subspace of E. □

Example 5.5
Consider the linear transformation $f : \mathbb{R}^2 \to \mathbb{R}$ defined as

$$f(u, v) = u - 2v.$$

Then

$$\begin{aligned}\mathrm{Ker}(f) &= \{(u, v) \in \mathbb{R}^2, \text{ such that } f(u, v) = 0\} \\ &= \{(u, v) \in \mathbb{R}^2, \text{ such that } u = 2v\}\end{aligned}$$

Thus, $\mathrm{Ker}(f)$ is the subspace of $\mathbb{R}^2$ spanned by the vector $(2, 1)$. ■

The kernel of a linear transformation is useful in determining when the transformation is injective (one-to-one). Indeed, we have:

Theorem 5.2.4 (Injective Linear Transformation)
Let E and F be two vector spaces over the same field $\mathbb{K}$ and f be a linear transformation from E into F. Then, the following two statements are equivalent:
1. $\mathrm{Ker}(f) = \{0_E\}$.
2. *The linear transformation f is injective. That is, if u and v are elements in E such that $f(u) = f(v)$, then $u = v$.*

Proof
First, we show that (1) implies (2). That is, we assume that $\mathrm{Ker}(f) = \{0_E\}$ and show that f is injective. So, let u, v be elements in E such that $f(u) = f(v)$, That is

$$f(u) - f(v) = f(u - v) = 0_F, \tag{5.6}$$

where we have used the linearity of f. The identity (5.6) implies that $u - v \in \text{Ker}(f)$, and since $\text{Ker}(f) = \{0_E\}$, we deduce that $u - v = 0_E$, that is $u = v$. This implies that f is injective.

Conversely, assume that f is injective; we need to show that $\text{Ker}(f) = \{0_E\}$. First, it is clear that $\{0_E\} \subset \text{Ker}(f)$ since $\text{Ker}(f)$ is a subspace of E (or since $f(0_E) = 0_F$). Now, let u be an element of $\text{Ker}(f)$. Then, by definition, we have

$$f(u) = 0_F = f(0_E).$$

Since f is injective, it follows that $u = 0_E$. Thus, we proved that $\text{Ker}(f) \subset \{0_E\}$ and therefore $\text{Ker}(f) = \{0_E\}$. Hence, we have shown that (2) implies (1). This completes the proof of Theorem 5.2.4. □

As mentioned earlier, one of the main properties of linear transformations is that they allow to transfer some algebraic properties from one vector space to another. As we saw in ▶ Chap. 4, it is very important to know the dimension of a vector space. In other words, it is important to know, or to be able to construct a basis in a vector space. More precisely, let E and F be two vector spaces over the same field $\mathbb{K}$ and let f be a linear transformation from E into F. Assume that $\dim_{\mathbb{K}} E = n$ and let $\mathbf{B}_E = \{u_1, u_2, \ldots, u_n\}$ be a basis of E (which exists according to Theorem 4.6.1). A natural question is: *under what conditions on f we have $\dim_{\mathbb{K}} E = \dim_{\mathbb{K}} F$? And if $\dim_{\mathbb{K}} E = \dim_{\mathbb{K}} F$, then could $B_F = \{f(u_1), f(u_2), \ldots, f(u_n)\}$ be a basis of F?* To answer these questions, we start with the following statement.

Theorem 5.2.5

Let E and F be two vector spaces over the same field $\mathbb{K}$ and f be an injective linear transformation from E into F. Let $u_1, u_2, \ldots, u_n$ be linearly independent elements of E. Then $f(u_1), f(u_2), \ldots, f(u_n)$ are linearly independent elements of F.

Proof

Let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be elements of $\mathbb{K}$ such that

$$\lambda_1 f(u_1) + \lambda_2 f(u_2) + \cdots + \lambda_n f(u_n) = 0_F. \tag{5.7}$$

The linearity of f implies that

$$f(\lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_n u_n) = 0_F.$$

This means that $\lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_n u_n \in \text{Ker}(f)$. Since f is injective, $\text{Ker}(f) = \{0_E\}$ (Theorem 5.2.4). Hence,

$$\lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_n u_n = 0_E.$$

Since $u_1, u_2, \ldots, u_n$ are linearly independent elements of E, it follows

$$\lambda_1 = \lambda_2 = \cdots = \lambda_n = 0_{\mathbb{K}}. \tag{5.8}$$

Thus, we have proved that (5.7) yields (5.8), which proves that $f(u_1), f(u_2), \ldots, f(u_n)$ are linearly independent. □

From the above result, we deduce the following result.

Theorem 5.2.6
Let E and F be two vector spaces over the same field $\mathbb{K}$, with $\dim_{\mathbb{K}} F = n$, and f be an injective linear transformation from E into F. Let $u_1, u_2, \ldots, u_n$ be linearly independent elements of E. Then the set $\mathbf{B}_F = \{f(u_1), f(u_2), \ldots, f(u_n)\}$ is a basis of F.

Proof
By Theorem 5.2.5, $\mathbf{B}_F$ is a linearly independent set of elements in F. Since the cardinality of $\mathbf{B}_F$ is equal to n, Lemma 4.6.4 implies that $\mathbf{B}_F$ is a maximal linearly independent set of elements in F. Hence, Theorem 4.6.2 implies that $\mathbf{B}_F$ is a basis in F. □

Remark 5.2.7 Theorem 5.2.6 can be used to construct a basis of a vector space as follows: suppose that we have a vector space F of finite dimension n and we want to construct a basis for F. Then it is enough to find an injective linear transformation f from another space E to F and n linearly independent elements of E. The basis of F will then be given by the images of the n linearly independent elements of E under f.

Definition 5.2.2 (The Image of a Linear Transformation)
Let E and F be two vector spaces over the same field $\mathbb{K}$ and f be a linear transformation from E into F. We define the *image* of f to be the set of elements w in F for which there exists an element u in E such that $w = f(u)$. We denote the image of f by $\mathrm{Im}(f)$ and write

$$\mathrm{Im}(f) = \{f(u) \in F, \text{ such that } u \in E\} = f(E).$$

Example 5.6
In Example 5.3, the image of the linear transformation defined by $f(u) = Au$ is the set of all vectors w in $\mathbb{R}^m$ that can be written as the product of the matrix A and a vector u in $\mathbb{R}^n$. ■

Next, we endow the image of a linear transformation with an algebraic structure.

Theorem 5.2.8 (The Subspace $\mathrm{Im}(f)$)
Let E and F be two vector spaces over the same field $\mathbb{K}$ and f be a linear transformation from E into F. Then, $\mathrm{Im}(f)$ is a subspace of F.

Proof
First, it is clear that 0_F is an element of F since $0_F = f(0_E)$ (Theorem 5.2.2). Second, let w_1, w_2 be elements of $\mathrm{Im}(f)$ and λ, μ be elements of $\mathbb{K}$. Then there exist u_1 and u_2 in E such that

$$w_1 = f(u_1) \qquad \text{and} \qquad w_2 = f(u_2).$$

Since f is linear,

$$\begin{aligned} \lambda w_1 + \mu w_2 &= \lambda f(u_1) + \mu f(u_2) \\ &= f(\lambda u_1 + \mu u_2). \end{aligned} \tag{5.9}$$

Since E is a vector space, $\lambda u_1 + \mu u_2 \in E$. Thus, in (5.9) we have expressed $\lambda w_1 + \mu w_2$ as the image of an element of E. Hence, $\lambda w_1 + \mu w_2 \in \mathrm{Im}(f)$. Consequently, $\mathrm{Im}(f)$ is a subspace of F. □

The image of f can be used to determine if the linear transformation is surjective (or onto), as in the following theorem.

Theorem 5.2.9 (Surjective Linear Transformation)
Let E and F be two vector spaces over the same field $\mathbb{K}$ and f be a linear transformation from E into F. Then, the following two statements are equivalent:

1. $\mathrm{Im}(f) = F$.
2. *f is surjective. That is, every element w in F has a corresponding element u in E such that $w = f(u)$.*

Proof
We prove first that (1) implies (2). Let w be an element of F. Since $\mathrm{Im}(f) = F$, then $w \in \mathrm{Im}(f)$. Thus, by definition there exists $u \in E$ such that $w = f(u)$. This means that f is surjective. Conversely, assume that f is surjective and let z be an element of F. Then there exists v in E such that $z = f(v)$. This implies that $z \in \mathrm{Im}(f)$, which shows that $F \subset \mathrm{Im}(f)$. Since by definition $\mathrm{Im}(f) \subset F$, we deduce that $\mathrm{Im}(f) = F$, and so (2) implies (1). □

Next, we introduce one of the fundamental theorems of linear algebra that relates the dimension of the kernel and the dimension of the image of a linear transformation to the dimension of the vector space on which the transformation is defined.

Theorem 5.2.10 (Rank-Nullity Theorem)
Let E and F be two vector spaces over the same field $\mathbb{K}$ and f be a linear transformation from E into F. Assume that the dimension of E is finite. Then, we have

$$\dim_{\mathbb{K}} E = \dim_{\mathbb{K}} \operatorname{Ker}(f) + \dim_{\mathbb{K}} \operatorname{Im}(f). \tag{5.10}$$

The dimension of the image of f, $\dim_{\mathbb{K}} \operatorname{Im}(f)$, is also called the rank *of f and is denoted by* $\operatorname{rank}(f)$. *Also, the dimension of the kernel of f $\dim_{\mathbb{K}} \operatorname{Ker}(f)$ is called the* nullity *of f and is denoted by* $\operatorname{null}(f)$. *Thus, (5.10) can be also recast as*

$$\dim_{\mathbb{K}} E = \operatorname{null}(f) + \operatorname{rank}(f).$$

Proof
Denote

$$\dim_K E = n, \qquad \dim_{\mathbb{K}} \operatorname{Ker}(f) = q, \qquad \text{and} \qquad \operatorname{rank}(f) = s.$$

First, if $\operatorname{Im}(f) = \{0_F\}$, then $f(u) = 0_F$ for any u in E. This means that $u \in \operatorname{Ker}(f)$. Hence, $E = \operatorname{Ker}(f)$ and thus

$$\dim_{\mathbb{K}} E = \dim_{\mathbb{K}} \operatorname{Ker}(f),$$

which is exactly (5.10). Now, if $\operatorname{Im}(f) \neq \{0_F\}$, then we have $s > 0$, and so by Theorem 4.6.1 the space $\operatorname{Im}(f)$ has a basis. Let $\{w_1, w_2, \ldots, w_s\}$ be a basis of $\operatorname{Im}(f)$. Thus, by definition there exist $u_1, \ldots, u_s \in E$ such that

$$w_i = f(u_i), \qquad i = 1, \ldots, s. \tag{5.11}$$

The elements $u_1, u_2, \ldots, u_s$ are linearly independent in E. Indeed, let $\alpha_1, \alpha_2, \ldots, \alpha_s$ be elements of $\mathbb{K}$ such that

$$\alpha_1 u_1 + \alpha_2 u_2 + \cdots + \alpha_s u_s = 0_E.$$

Then, by the linearity of f and (5.11), we have

$$f(\alpha_1 u_1 + \alpha_2 u_2 + \cdots + \alpha_s u_s) = f(0_E) = 0_F,$$

or equivalently

$$\alpha_1 w_1 + \alpha_2 w_2 + \cdots + \alpha_s w_s = 0_F.$$

Since $\{w_1, w_2, \ldots, w_s\}$ is a basis of $\mathrm{Im}(f)$, it follows that

$$\alpha_1 = \alpha_2 = \cdots = \alpha_s = 0_{\mathbb{K}}.$$

Now, if $\mathrm{Ker}(f) = \{0_E\}$, then we can show that the set $\{u_1, u_2, \ldots, u_s\}$ spans E. Indeed, let v be an element of E. Then $f(v) \in \mathrm{Im}(f)$. Since $\{w_1, w_2, \ldots, w_s\}$ spans $\mathrm{Im}(f)$, there exist $\lambda_1, \lambda_2, \ldots, \lambda_s$ in $\mathbb{K}$ such that

$$\begin{aligned} f(v) &= \lambda_1 w_1 + \lambda_2 w_2 + \cdots + \lambda_s w_s \\ &= \lambda_1 f(u_1) + \lambda_2 f(u_2) + \cdots + \lambda_s f(u_s) \\ &= f(\lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_s u_s). \end{aligned}$$

Since f is injective (Theorem 5.2.4), then we have from above

$$v = \lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_s u_s.$$

This shows that $\{u_1, u_2, \ldots, u_s\}$ spans E and thus is a basis of E. Consequently,

$$s = \dim_{\mathbb{K}} E = \dim_{\mathbb{K}} \mathrm{Ker}(f) + \mathrm{rank}(f) = 0 + s,$$

so (5.10) holds.

Next, if $\mathrm{Ker}(f) \neq \{0_E\}$, then $q > 0$, and hence there exists a basis $\{v_1, v_2, \ldots, v_q\}$ of $\mathrm{Ker}(f)$. Our goal is to show that the set

$$\mathbf{B} = \{u_1, u_2, \ldots, u_s, v_1, v_2, \ldots, v_q\}$$

is a basis for E. This will suffice to prove (5.10). First, we show that $\mathbf{B}$ spans E. Let v be an element in E. Then, as above, there exist $\lambda_1, \lambda_2, \ldots, \lambda_s$ in $\mathbb{K}$ such that

$$f(v) = f(\lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_s u_s).$$

By the linearity of f, we have

$$f(v - \lambda_1 u_1 - \lambda_2 u_2 - \cdots - \lambda_s u_s) = 0_F.$$

Hence $v - \lambda_1 u_1 - \lambda_2 u_2 - \cdots - \lambda_s u_s \in \mathrm{Ker}(f)$. Thus, there exist $\beta_1, \beta_2, \ldots, \beta_q \in \mathbb{K}$ such that

$$v - \lambda_1 u_1 - \lambda_2 u_2 - \cdots - \lambda_s u_s = \beta_1 v_1 + \beta_2 v_2 + \cdots + \beta_q v_q.$$

This gives

$$v = \lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_s u_s + \beta_1 v_1 + \beta_2 v_2 + \cdots + \beta_q v_q,$$

which shows that **B** spans E. To prove that **B** is a linearly independent set, let $\gamma_1, \gamma_2 \ldots, \gamma_s, \delta_1, \delta_2 \ldots, \delta_q$ be elements of $\mathbb{K}$ satisfying

$$\sum_{i=1}^{s} \gamma_i u_i + \sum_{j=1}^{q} \delta_j v_j = 0_E. \tag{5.12}$$

This implies, by the linearity of f,

$$f\left(\sum_{i=1}^{s} \gamma_i u_i\right) + f\left(\sum_{j=1}^{q} \delta_j v_j\right) = f(0_E) = 0_F,$$

whence

$$f\left(\sum_{i=1}^{s} \gamma_i u_i\right) = \sum_{i=1}^{s} \gamma_i w_i = 0_F,$$

since $\sum_{j=1}^{q} \delta_j v_j \in \mathrm{Ker}(f)$. Therefore,

$$\gamma_1 = \gamma_2 = \cdots = \gamma_s = 0_{\mathbb{K}},$$

since $\{w_1, w_2, \ldots, w_s\}$ is a linearly independent set. Plugging this into (5.12) yields (for the same reason)

$$\delta_1 = \delta_2 = \cdots = \delta_q = 0_{\mathbb{K}}.$$

Therefore, **B** is a linearly independent set and hence a basis of E. This completes the proof of Theorem 5.2.10. □

5.3 Isomorphism of Vector Spaces

In an attempt to carry over some algebraic properties of a vector space to another, we now define what we call *isomorphism* between two vector spaces.

Definition 5.3.1 (Isomorphism of Vector Spaces)
Let E and F be two vector spaces over the same field $\mathbb{K}$ and f be a linear transformation from E into F. Then f is said to be an *isomorphism* if f is bijective (surjective and injective). If $F = E$, then this isomorphism is called an *automorphism*.

(Continued)

Definition 5.3.1 (continued)

If such isomorphism exists, then the two spaces E and F have the same algebraic properties (are algebraically equivalent) and we say that they are *isomorphic*, and write $E \cong F$.

Example 5.7

The identity Id_E transformation defined in Example 5.2 is an automorphism. ■

Example 5.8 ($\mathbb{C}$ Is Isomorphic to $\mathbb{R}^2$)

We consider the vector space $\mathbb{C}$ over the field $\mathbb{R}$. Then $\mathbb{C}$ is isomorphic to $\mathbb{R}^2$. This can be easily seen by considering the linear transformation f from $\mathbb{R}^2$ to $\mathbb{C}$ given by

$$f : \mathbb{R}^2 \to \mathbb{C}$$
$$(x, y) \mapsto x + iy.$$

We can easily show that f is an isomorphism. Thus, we deduce directly that $\dim_{\mathbb{R}} \mathbb{C} = 2$, since $\dim_{\mathbb{R}} \mathbb{R}^2 = 2$.

We may also easily prove that the vector space $\mathbb{C}^n$ over the field $\mathbb{R}$ is isomorphic to $\mathbb{R}^{2n}$. ■

Theorem 5.3.1 (Inverse of an Isomorphism)

Let E and F be two vector spaces over the same field $\mathbb{K}$ and f be an isomorphism from E into F. Then f^{-1} is an isomorphism from F into E.

Proof

It is clear that if f is bijective, then f^{-1} is bijective. We need to show only that if f is linear, then f^{-1} is also linear. So, let w_1, w_2 be elements in F and λ, μ be elements of $\mathbb{K}$. Then, since f is bijective, then there exist a unique u_1 and a unique u_2 in E such that

$$w_1 = f(u_1) \quad \text{and} \quad w_2 = f(u_2),$$

and so

$$u_1 = f^{-1}(w_1) \quad \text{and} \quad u_2 = f^{-1}(w_2).$$

Now, we have

$$\begin{aligned} \lambda w_1 + \mu w_2 &= \lambda f(u_1) + \mu f(u_2) \\ &= f(\lambda u_1 + \mu u_2), \end{aligned}$$

whence

$$f^{-1}(\lambda w_1 + \mu w_2) = \lambda u_1 + \mu u_2$$
$$= \lambda f^{-1}(w_1) + \mu f^{-1}(w_2).$$

Therefore, f^{-1} is linear, as needed. □

In the following theorem we show that the composition of two isomorphisms is also an isomorphism.

Theorem 5.3.2 (Composition of Isomorphism)
Let E, F and G be three vector spaces over the same field $\mathbb{K}$, f be an isomorphism from E into F, and g be an isomorphism from F to G. Then the composition $g \circ f$ is an isomorphism from E to G.

The proof of Theorem 5.3.2 is obvious, since the composition of two bijective transformations is bijective and in Theorem 5.2.1 we have seen that the composition of two linear transformations is a linear transformation.

We have observed above that in order to prove that a transformation is an isomorphism, we need to show that this transformation is linear and bijective. This last requirement can be relaxed under some algebraic assumptions as we will see in the following theorem.

Theorem 5.3.3
Let E and F be two vector spaces over the same field $\mathbb{K}$ and f be a linear transformation from E into F. Assume that

$$\dim_{\mathbb{K}} E = \dim_{\mathbb{K}} F. \tag{5.13}$$

If f is injective or surjective, then f is an isomorphism.

Proof
First, assume that f is injective. Then according to Theorem 5.2.4, $\mathrm{Ker}(f) = \{0_E\}$. Then using (5.10) and (5.13), we deduce that

$$\dim_{\mathbb{K}} F = \mathrm{rank}(f). \tag{5.14}$$

Since $\mathrm{Im}(f)$ is a subspace of F, Theorem 4.6.7 together with (5.14) implies that $F = \mathrm{Im}(f)$. Hence, by Theorem 5.2.9, f is surjective and since we assumed that f is injective, then f is bijective and therefore, an isomorphism.

Now, if f is surjective, then we have as above $F = \mathrm{Im}(f)$. This implies that

$$\dim_{\mathbb{K}} E = \dim_{\mathbb{K}} F = \mathrm{rank}(f).$$

Therefore, (5.10) leads to $\dim_{\mathbb{K}} \mathrm{Ker}(f) = 0$. Then, $\mathrm{Ker}(f) = \{0_E\}$, and hence by Theorem 5.2.4, f is injective and consequently f is bijective. □

5

Remark 5.3.4 Theorem 5.3.3 implies that if f is an element in $\mathscr{L}(E)$ (with $\dim_{\mathbb{K}} E$ finite), then to show that f is an automorphism, we need just to show that $\mathrm{Ker}(f) = \{0_E\}$ or $\mathrm{Im}(f) = E$.

As we have seen in ▶ Chap. 4, it is very useful to find a basis of a vector space, since this could allow one to infer many properties of the space. Now suppose that we have two vector spaces E and F over the same field $\mathbb{K}$, f is an isomorphism from E to F, and B is a basis of E. So the natural question is whether $f(B)$ is basis of F. This turns out to be true as we will see in the following theorem.

Theorem 5.3.5
Let E and F be two vector spaces over the same field $\mathbb{K}$ and f be an isomorphism from E into F. Let $\mathbf{B}$ be a basis of E. Then $f(\mathbf{B})$ is a basis of F.

Proof
Let $\mathbf{B}$ be a basis of E. Since f is injective and $\mathbf{B}$ is a linearly independent set in E, then according to Theorem 5.2.5, $f(\mathbf{B})$ is a linearly independent set in F. Clearly, since $\mathbf{B}$ spans E, then $f(\mathbf{B})$ spans $f(E)$. Since f is surjective, then we have (see Theorem 5.2.9) $f(E) = F$ and thus $f(\mathbf{B})$ spans F. Consequently, $f(\mathbf{B})$ is a basis of F. □

Corollary 5.3.6 *Two finite-dimensional vector spaces over the same field are isomorphic if and only if they have the same dimension.*

Proof
First, assume that $E \cong F$. Then, by Theorem 5.3.5, we deduce that

$$\dim_{\mathbb{K}} E = \dim_{\mathbb{K}} F.$$

Conversely, assume that

$$\dim_{\mathbb{K}} E = \dim_{\mathbb{K}} F = n.$$

Then, Theorem 4.6.1 implies that there exists $\mathbf{B}_E = \{u_1, u_2, \ldots, u_n\}$, a basis of E, and $\mathbf{B}_F = \{w_1, w_2, \ldots, w_n\}$, a basis of F. We define the transformation f from E to F as follows:

$$f : E \to F$$
$$u = \lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_n u_n \mapsto f(u) = w = \lambda_1 w_1 + \lambda_2 w_2 + \cdots + \lambda_n w_n,$$

for $\lambda_1, \lambda_2, \ldots, \lambda_n \in \mathbb{K}$. It is clear that f is linear. Now, let u and v be two elements in E, such that

$$f(u) = f(v).$$

Since $\mathbf{B}_E$ is a basis of E, there exists λ_i and μ_i, $i = 1, 2, \ldots, n$ elements in $\mathbb{K}$ such that

$$u = \lambda_1 u_1 + \lambda_2 u_2 + \cdots + \lambda_n u_n \quad \text{and} \quad v = \mu_1 u_1 + \mu_2 u_2 + \cdots + \mu_n u_n.$$

Thus, $f(u) = f(v)$ which implies that

$$\lambda_1 w_1 + \lambda_2 w_2 + \cdots + \lambda_n w_n = \mu_1 w_1 + \mu_2 w_2 + \cdots + \mu_n w_n.$$

Theorem 4.4.3 shows that

$$\lambda_i = \mu_i, \quad i = 1, 2, \ldots, n.$$

Hence, $u = v$. Therefore, f is injective. Hence, since $\dim_{\mathbb{K}} E = \dim_{\mathbb{K}} F$, Theorem 5.3.3 shows that f is bijective. □

5.4 Exercises

Exercise 5.1 (The Kernel and the Image of a Projection)
Let E be a vector space over a field $\mathbb{K}$ and let $f : E \to E$ be an endomorphism of E such that $f^2 = f \circ f = f$.[1]

Show that

1. $\mathrm{Ker}(f) = \mathrm{Ker}(f^2)$ if and only if $\mathrm{Im}(f) \cap \mathrm{Ker}(f) = \{0_E\}$.
2. $E = \mathrm{Ker}(f) \oplus \mathrm{Im}(f)$.
3. If g is an endomorphism of E, then $g \circ f = f \circ g$ if and only if

$$g(\mathrm{Ker}(f)) \subset \mathrm{Ker}(f) \quad \text{and} \quad g(\mathrm{Im}(f)) \subset \mathrm{Im}(f).$$

■

[1] A linear transformation satisfying this property is called a projection.

Solution

1. We need to show that $\text{Ker}(f) \subset \text{Ker}(f^2)$ and $\text{Ker}(f^2) \subset \text{Ker}(f)$. Let u be an element in $\text{Ker}(f)$. i.e., $f(u) = 0_E$. This gives

$$f^2(u) = f(f(u)) = (f \circ f)(u) = f(0_E) = 0_E,$$

so $u \in \text{Ker}(f^2)$. Thus, $\text{Ker}(f) \subset \text{Ker}(f^2)$. Conversely, let w be an element of $\text{Ker}(f^2)$. Then, since $f^2 = f$,

$$f^2(w) = f(w) = 0_E,$$

so $w \in \text{Ker}(f)$. Hence, $\text{Ker}(f^2) \subset \text{Ker}(f)$.

2. In view of Theorem 4.3.10, we need to prove that

$$E = \text{Ker}(f) + \text{Im}(f), \qquad \text{and} \qquad \text{Ker}(f) \cap \text{Im}(f) = \{0_E\}. \tag{5.15}$$

So, let u be an element in E. Then we can write u as

$$u = (u - f(u)) + f(u).$$

It is clear that $f(u) \in \text{Im}(f)$. On the other hand, we have

$$f(u - f(u)) = f(u) - f^2(u) = 0.$$

Thus, $u - f(u) \in \text{Ker}(f)$. Hence, we have proved the first assertion in (5.15).

Next, let v be an element in $\text{Ker}(f) \cap \text{Im}(f)$. Then there exists u in E such that

$$v = f(u) \quad \text{and} \quad f(v) = 0_E.$$

This gives $f^2(u) = 0_E$. Thus, $u \in \text{Ker}(f^2)$. Since $\text{Ker}(f) = \text{Ker}(f^2)$, then $u \in \text{Ker}(f)$ and we have $v = f(u) = 0_E$. Hence, $\text{Ker}(f) \cap \text{Im}(f) = \{0_E\}$ and consequently, $E = \text{Ker}(f) \oplus \text{Im}(f)$.

3. Assume that $g \circ f = f \circ g$ and let w be an element in $g(\text{Ker}(f))$. Then there exists u in $\text{Ker}(f)$ such that $w = g(u)$. Now, since $g \circ f = f \circ g$, we have

$$f(w) = f(g(u)) = g(f(u)) = g(0_E) = 0_E,$$

because $u \in \text{Ker}(f)$ and g is a linear transformation. Thus, we have shown that $w \in \text{Ker}(f)$. Therefore $g(\text{Ker}(f)) \subset \text{Ker}(f)$.

Now, let z be an element of $g(\text{Im}(f))$. Then there exists w in $\text{Im}(f)$ such that $z = g(w)$. Since $w \in \text{Im}(f)$, there exists u in E such that $w = f(u)$. Thus,

$$z = g(f(u)) = f(g(u)) = f(y), \qquad y = g(u).$$

Hence, $z \in \text{Im}(f)$, which proves that $g(\text{Im}(f)) \subset \text{Im}(f)$.

Conversely, assume that $g(\mathrm{Ker}(f)) \subset \mathrm{Ker}(f)$ and $g(\mathrm{Im}(f)) \subset \mathrm{Im}(f)$. Let u be an element of E. We claim that

$$f(g(u)) = g(f(u)). \tag{5.16}$$

Indeed, using (2), we deduce that

$$u = u_1 + u_2,$$

with $u_1 \in \mathrm{Ker}(f)$ and $u_2 \in \mathrm{Im}(f)$. Thus, there exists u_3 in E such that

$$f(u_1) = 0 \quad \text{and} \quad u_2 = f(u_3).$$

Now, we have

$$\begin{aligned} g(f(u)) &= g(f(u_1 + u_2)) = g(f(u_1)) + g(f(u_2)) \\ &= g(0_E) + g(f(u_2)) \\ &= g(f(u_2)) \end{aligned} \tag{5.17}$$

and

$$f(g(u)) = f(g(u_1 + u_2)) = f(g(u_1)) + f(g(u_2)). \tag{5.18}$$

Next, it is clear that $g(u_1) \in g(\mathrm{Ker}(f))$, and since $g(\mathrm{Ker}(f)) \subset \mathrm{Ker}(f)$, we deduce that $g(u_1) \in \mathrm{Ker}(f)$ and hence $f(g(u_1)) = 0$. Also, since $u_2 \in \mathrm{Im}(f)$, we have $g(u_2) \in g(\mathrm{Im}(f)) \subset \mathrm{Im}(f)$, and so there exists u_4 in E such that

$$g(u_2) = f(u_4).$$

Consequently, taking all these into account, we can rewrite (5.17) and (5.18) as

$$g(f(u)) = g(f(u_2)) = g(f^2(u_3)) = g(f(u_3)) = g(u_2) \tag{5.19}$$

and respectively

$$f(g(u)) = f(g(u_2)) = f(f(u_4)) = f(u_4) = g(u_2). \tag{5.20}$$

Now, (5.19) and (5.20) yield (5.16), and so $f \circ g = g \circ f$. ◀

Exercise 5.2

Consider the endomorphism $f : \mathbb{R}^n \to \mathbb{R}^n$ defined as

$$f(u) = (u_1 + u_n, u_2 + u_{n-1}, \ldots, u_n + u_1), \quad \text{for} \quad u = (u_1, u_2, \ldots, u_n).$$

Find $\dim_{\mathbb{R}} \mathrm{Ker}(f)$ and $\dim_{\mathbb{R}} \mathrm{Im}(f) = \mathrm{rank}(f)$. ■

Solution

First, we need to find the subspace $\mathrm{Ker}(f)$. So, let $u = (u_1, u_2, \ldots, u_n)$ be a vector in $\mathbb{R}^n$. Then u is a vector in $\mathrm{Ker}(f)$ if and only $f(u) = 0_{\mathbb{R}^n}$. This implies that

$$\begin{cases} u_1 + u_n = 0, \\ u_2 + u_{n-1} = 0, \\ \vdots \\ u_n + u_1 = 0. \end{cases}$$

Solving this system, we find

$$\begin{cases} u_1 = -u_n, \\ u_2 = -u_{n-1}, \\ \vdots \\ u_n = -u_1. \end{cases}$$

First, if n is even, then we have $n = 2p$ then u can be written as

$$u = (u_1, u_2, \ldots, u_p, -u_p, -u_{p-1}, \ldots, -u_1).$$

This vector can be written as

$$u = u_1(1, 0, \ldots, 0, \ldots, -1) + u_2(0, 1, \ldots, 0, \ldots, -1, 0) + \cdots, +u_p(0, \ldots, 1, -1, \ldots, 0).$$

Thus, the set of vectors $\mathbf{B} = \{a_1, a_2, \ldots, a_p\}$ such that a_j, $1 \leq j \leq p$ has all components zero except for the jth component which is 1 and the $n - j + 1$ component which is -1, spans $\mathrm{Ker}(f)$. We may easily show that $\mathbf{B}$ is a linearly independent set, so a basis of $\mathrm{Ker}(f)$. Thus, if n is even, then

$$\dim_{\mathbb{R}} \mathrm{Ker}(f) = p = \frac{n}{2}.$$

Now, applying Theorem 5.2.10, we deduce that

$$\mathrm{rank}(f) = \dim_{\mathbb{R}} \mathbb{R}^n - \dim_{\mathbb{R}} \mathrm{Ker}(f) = n - \frac{n}{2} = \frac{n}{2}.$$

Second, if n is odd, i.e., $n = 2q + 1$, each element u of $\mathrm{Ker}(f)$ can be written as

$$u = (u_1, u_2, \ldots, u_q, 0, -u_q, \ldots, -u_1),$$

and then

$$u = u_1 b_1 + u_2 b_2 + \cdots + u_q b_q,$$

where b_k, $1 \leq k \leq q$, is the vector with all components zero, except for the kth component which is equal to 1 and the $(n-(k+1)+2)$-nd component, which is equal to -1. As above, we can easily show that the set of vectors $S = \{b_1, b_2, \ldots, b_q\}$ is a basis of $\mathrm{Ker}(f)$ and we have

$$\dim_{\mathbb{R}} \mathrm{Ker}(f) = q = \frac{n-1}{2}.$$

Consequently, we obtain as above

$$\mathrm{rank}(f) = n - \frac{n-1}{2} = \frac{n+1}{2}.$$

◀

Exercise 5.3 (Rank of the Composition)
Let E, F and G be three vector spaces over the same field $\mathbb{K}$. Assume that $\dim_{\mathbb{K}} E$ is finite. Let f be an element of $\mathscr{L}(E,F)$ and g be an element in $\mathscr{L}(F,G)$.

1. Show that $\mathrm{Im}(g \circ f) \subset \mathrm{Im}(g)$.
2. Prove that $\mathrm{Ker}(f) \subset \mathrm{Ker}(g \circ f)$.
3. Deduce that

$$\mathrm{rank}(g \circ f) \leq \min(\mathrm{rank}(f), \mathrm{rank}(g)). \tag{5.21}$$

■

Solution
First, it is clear that $g \circ f$ is an element of $\mathscr{L}(E,G)$, (see Theorem 5.2.1).

1. Let w be an element in $\mathrm{Im}(g \circ f)$. Then there exists u in E such that

$$w = (g \circ f)(u) = g(f(u)).$$

Put $v = f(u)$; then $v \in F$ and $w = g(v)$, so $w \in \mathrm{Im}(g)$.

2. Let u be an element in $\mathrm{Ker}(f)$, then we have $f(u) = 0_F$. Since g is linear, then it holds that

$$g(f(u)) = g(0_F) = 0_G$$

(see Theorem 5.2.2). This means that $u \in \mathrm{Ker}(g \circ f)$.

3. Since $\mathrm{Im}(g \circ f) \subset \mathrm{Im}(g)$, then we have, according to Theorem 4.6.7,

$$\mathrm{rank}(g \circ f) = \dim_{\mathbb{K}} \mathrm{Im}(g \circ f) \leq \dim_{\mathbb{K}} \mathrm{Im}(g) = \mathrm{rank}(g). \tag{5.22}$$

On the other hand, from (2) we have, for the same reason,

$$\dim_{\mathbb{K}} \mathrm{Ker}(f) \leq \dim_{\mathbb{K}} \mathrm{Ker}(g \circ f).$$

Thus, applying Theorem 5.2.10 twice, we have

$$\operatorname{rank}(g \circ f) = \dim_{\mathbb{K}} E - \dim_{\mathbb{K}} \operatorname{Ker}(g \circ f) \leq \dim_{\mathbb{K}} E - \dim_{\mathbb{K}} \operatorname{Ker}(f) = \operatorname{rank}(f). \quad (5.23)$$

Finally (5.22) and (5.23) yield (5.21).

◀

5

Exercise 5.4

Let E be a vector space over a field $\mathbb{K}$ such that $\dim_{\mathbb{K}} E$ is finite and f be an element of $\mathscr{L}(E)$. Show that the following statements are equivalent:

1. There exist two projections P and Q in $\mathscr{L}(E)$ such that

$$f = P - Q \qquad \text{and} \qquad \operatorname{Im}(P) = \operatorname{Im}(Q).$$

2. $f^2 = 0_{\mathscr{L}(E)}$.

■

Solution

We have seen in Exercise 5.1 that P and Q satisfy $P^2 = P$ and $Q^2 = Q$. Now, we need to show that (1) $\Rightarrow$ (2) and (2) $\Rightarrow$ (1).

First, assume that (1) is satisfied and let u be an element of F. Then, we have

$$\begin{aligned} f^2(u) = f(f(u)) &= (P - Q)\left((P - Q)(u)\right) \\ &= P^2(u) - Q(P(u)) - P(Q(u)) + Q^2(u) \\ &= P(u) - Q(P(u)) - P(Q(u)) + Q(u). \end{aligned}$$

Now, it is clear that $P(u) \in \operatorname{Im}(P) = \operatorname{Im}(Q)$. Then, we have $Q(P(u)) = P(u)$.[2] By the same argument $P(Q(u)) = Q(u)$. Consequently, taking these into account, we have from above that $f(f(u)) = 0_E$ for all u in E. This means that $f^2 = 0_{\mathscr{L}(E)}$. Thus, we have proved that (1) $\Rightarrow$ (2).

Conversely, since $\dim_{\mathbb{K}} E$ is finite, then (see Exercise 4.8), $\operatorname{Ker}(f)$ has a complement in E. Hence, there exists a projection P such that $\operatorname{Im}(P) = \operatorname{Ker}(f)$.

We define Q as $Q = P - f$ so $f = P - Q$. To show that Q is a projection, we need to prove that $Q^2 = Q$. We have

$$Q^2 = (P - f) \circ (P - f) = P^2 - P \circ f - f \circ P + f^2.$$

Since $P^2 = P$ and $f^2 = 0_{\mathscr{L}(E)}$ (by assumption), we deduce that

$$Q^2 = P - P \circ f - f \circ P. \quad (5.24)$$

[2] Since Q is a projection, we have $Q(y) = y$ for all y in $\operatorname{Im}(Q)$.

Now, since $\mathrm{Im}(P) \subset \mathrm{Ker}(f)$, then we deduce that $f \circ P = 0_{\mathscr{L}(E)}$. Also, since $f^2 = 0_{\mathscr{L}(E)}$, we have

$$\mathrm{Im}(f) \subset \mathrm{Ker}(f) = \mathrm{Im}(P) = \mathrm{Ker}(\mathrm{Id}_{\mathscr{L}(E)} - P).$$

This gives

$$(\mathrm{Id}_{\mathscr{L}(E)} - P) \circ f = 0_{\mathscr{L}(E)},$$

whence $P \circ f = f$. ◀

Inserting these relations into (5.24), we get

$$Q^2 = P - P \circ f - f \circ P = P - f = Q.$$

Thus, Q is a projection.

Now, we need to show that $\mathrm{Im}(P) = \mathrm{Im}(Q)$. That is $\mathrm{Im}(P) \subset \mathrm{Im}(Q)$ and $\mathrm{Im}(Q) \subset \mathrm{Im}(P)$. So, first, let w be an element of $\mathrm{Im}(P) = \mathrm{Ker}(f)$. Since $w \in \mathrm{Im}(P)$, then $P(w) = w$. We also have $f(w) = 0_E$. On the other hand, we have

$$Q(w) = (P - f)(w) = P(w) = w.$$

This means that $w \in \mathrm{Im}(Q)$ and so $\mathrm{Im}(P) \subset \mathrm{Im}(Q)$.

On the other hand, let u be an element of $\mathrm{Im}(Q)$. Then

$$u = Q(u) = P(u) - f(u),$$

whence

$$P(u) = P^2(u) - (P \circ f)(u) = P(u) - f(u).$$

This yields $f(u) = 0_E$ and consequently,

$$u = Q(u) = P(u).$$

This means that $u \in \mathrm{Im}(P)$. Thus $\mathrm{Im}(Q) \subset \mathrm{Im}(P)$, which concludes our proof.

Exercise 5.5

Let E be a vector space over a field $\mathbb{K}$ such that $\dim_{\mathbb{K}} E = n$ and f be an element of $\mathscr{L}(E)$. Show that the following statements are equivalent:

1. $\mathrm{Ker}(f) = \mathrm{Im}(f)$.
2. $f^2 = 0_{\mathscr{L}(E)}$ and $\mathrm{rank}(f) = \dfrac{n}{2}$.

■

Solution

As usual, we show that (1) $\Rightarrow$ (2) and (2) $\Rightarrow$ (1). First, assume that $\text{Ker}(f) = \text{Im}(f)$ and let u be an element of E. Then $f(u) \in \text{Im}(f)$, and since $\text{Ker}(f) = \text{Im}(f)$, then $f(u) \in \text{Ker}(f)$ and so

$$f(f(u)) = f^2(u) = 0_E.$$

Since this holds for all elements of E, we deduce that $f^2 = 0_{\mathscr{L}(E)}$.

Now, using Theorem 5.2.10, we have

$$\dim_{\mathbb{K}} E = \dim_{\mathbb{K}} \text{Ker}(f) + \text{rank}(f) = \dim_{\mathbb{K}} \text{Im}(f) + \text{rank}(f) = 2\,\text{rank}(f).$$

This gives $\text{rank}(f) = n/2$. Thus, we have proved that (1) $\Rightarrow$ (2).

Conversely, assume that (2) holds; we need to show that $\text{Ker}(f) = \text{Im}(f)$. Let w be an element in $\text{Im}(f)$, then there exists u in E such that $w = f(u)$. Now, we have

$$f(w) = f(f(u)) = f^2(0) = 0_E,$$

so, $w \in \text{Ker}(f)$. This shows that $\text{Im}(f) \subset \text{Ker}(f)$.

Now, since $\text{rank}(f) = n/2$, Theorem 5.2.10 implies that $\dim_{\mathbb{K}} \text{Ker}(f) = n/2$. Since, $\text{Im}(f) \subset \text{Ker}(f)$ and $\text{rank}(f) = \dim_{\mathbb{K}} \text{Ker}(f)$, Theorem 4.6.7 shows that $\text{Im}(f) = \text{Ker}(f)$. ◄

Exercise 5.6 (Nilpotent Linear Transformation)

Let E be a vector space over a field $\mathbb{K}$ such that $\dim_{\mathbb{K}} E = n$ and f be an element of $\mathscr{L}(E)$. Let k be a positive integer. f is called *nilpotent* of index k[3] if

$$f^k = \underbrace{f \circ f \circ \cdots \circ f}_{k \text{ times}} = 0_{\mathscr{L}(E)}.$$

Show that if f is nilpotent of index n, then for any u in E satisfying $f^{n-1}(u) \neq 0_E$, the set $\mathbf{B} = \{u, f(u), f^2(u), \ldots, f^{n-1}(u)\}$ is a basis of E. ■

Solution

Since the cardinality of $\mathbf{B}$ equals n, Theorem 4.6.5 shows that it is enough to show that $\mathbf{B}$ is a linearly independent set. First, it is clear that since $f^{n-1}(u) \neq 0_E$, the linearity of f shows that for any $0 \leq k \leq (n-1)$, we have $f^k(u) \neq 0_E$. Now, let $\lambda_0, \lambda_1, \ldots, \lambda_{n-1}$ be elements in $\mathbb{K}$ satisfying

$$\lambda_0 u + \lambda_1 f(u) + \lambda_2 f^2(u) + \cdots + \lambda_{n-1} f^{n-1}(u) = 0_E.$$

[3] See Exercise 1.2 for the definition of a nilpotent matrix.

Applying f^{n-1} to this identity, using the linearity of f^{n-1} and the fact that $f^n(u) = 0_E$, we get

$$f^{n-1}(\lambda_0 u + \lambda_1 f(u) + \lambda_2 f^2(u) + \cdots + \lambda_{n-1} f^{n-1}(u)) = f^{n-1}(0_E) = 0_E. \tag{5.25}$$

Since $f^m(u) = 0_E$, this gives for all $m \geq n$,

$$\lambda_0 f^{n-1}(u) = 0_E.$$

Since $f^{n-1}(u) \neq 0_E$, then $\lambda_0 = 0_{\mathbb{K}}$. Next, arguing in the same way, we apply f^{n-2} to (5.25) and using the fact that $\lambda_0 = 0_{\mathbb{K}}$, we obtain $\lambda_1 = 0_{\mathbb{K}}$. By continuing the process and applying each time $f^{n-\ell}$, $1 \leq \ell \leq n$ to (5.25), we can show that $\lambda_{\ell-1} = 0_{\mathbb{K}}$. Hence,

$$\lambda_0 = \lambda_1 = \cdots = \lambda_{n-1} = 0_{\mathbb{K}}.$$

Consequently, **B** is a linearly independent set, and thus a basis of E. ◄

Exercise 5.7 (Triangle and Frobenius Inequalities for the Rank)
Let E and F be two finite-dimensional vector spaces over the same field $\mathbb{K}$. Let f and g be elements of $\mathscr{L}(E, F)$.

1. Prove that $\mathrm{Im}(f + g) \subset \mathrm{Im}(f) + \mathrm{Im}(g)$.
2. Show that $\mathrm{rank}(f + g) \leq \mathrm{rank}(f) + \mathrm{rank}(g)$.
3. Deduce that $|\mathrm{rank}(f) - \mathrm{rank}(g)| \leq \mathrm{rank}(f + g)$.
4. Show that if f_1, f_2, and f_3 are elements of $\mathscr{L}(F_1, F_2), \mathscr{L}(F_2, F_3)$, and $\mathscr{L}(F_3, F_4)$, respectively, where F_1, F_2, F_3, and F_4 are four finite-dimensional vector spaces over the same field $\mathbb{K}$, then

$$\mathrm{rank}(f_2) - \mathrm{rank}(f_3 \circ f_2) = \dim_{\mathbb{K}}(\mathrm{Im}(f_2) \cap \mathrm{Ker}(f_3)). \tag{5.26}$$

Deduce that

$$\mathrm{rank}(f_2 \circ f_1) + \mathrm{rank}(f_3 \circ f_2) \leq \mathrm{rank}\, f_2 + \mathrm{rank}(f_3 \circ f_2 \circ f_1) \quad \text{(Frobenius inequality)}. \tag{5.27}$$

■

Solution
1. First, it is clear that $f + g \in \mathscr{L}(E, F)$ since $\mathscr{L}(E, F)$ is a vector space. It is also obvious that $\mathrm{Im}(f)$, $\mathrm{Im}(g)$ and $\mathrm{Im}(f + g)$ are subspaces of F (see Theorem 5.2.8). Now, let w be an element of $\mathrm{Im}(f + g)$. Then there exists u in E such that

$$w = (f + g)(u) = f(u) + g(u).$$

Since $f(u) \in \mathrm{Im}(f)$ and $g(u) \in \mathrm{Im}(g)$, we see that $w \in \mathrm{Im}(f+g)$. This shows that $\mathrm{Im}(f+g) \subset \mathrm{Im}(f) + \mathrm{Im}(g)$.

2. Applying formula (4.16) for $F_1 = \mathrm{Im}(f)$ and $F_2 = \mathrm{Im}(g)$, we get

$$\dim_{\mathbb{K}}(\mathrm{Im}(f) + \mathrm{Im}(g)) \leq \dim_{\mathbb{K}} \mathrm{Im}(f) + \dim_{\mathbb{K}} \mathrm{Im}(g) = \mathrm{rank}(f) + \mathrm{rank}(g).$$

On the other hand, since $\mathrm{Im}(f+g) \subset \mathrm{Im}(f) + \mathrm{Im}(g)$, Theorem 4.6.7 shows that

$$\mathrm{rank}(f+g) = \dim_{\mathbb{K}} \mathrm{Im}(f+g) \leq \dim_{\mathbb{K}}(\mathrm{Im}(f) + \mathrm{Im}(g)) \leq \mathrm{rank}(f) + \mathrm{rank}(g).$$

3. We can apply (2) for $f+g$ and $-g$, and get

$$\mathrm{rank}(f+g+(-g)) \leq \mathrm{rank}(f+g) + \mathrm{rank}(-g).$$

Since $\mathrm{rank}(-g) = \mathrm{rank}(g)$, it follows that

$$\mathrm{rank}(f) \leq \mathrm{rank}(f+g) + \mathrm{rank}(g).$$

Equivalently,

$$\mathrm{rank}(f) - \mathrm{rank}(g) \leq \mathrm{rank}(f+g).$$

By applying the same method to $f+g$ and $(-f)$, we get

$$\mathrm{rank}(g) - \mathrm{rank}(f) \leq \mathrm{rank}(f+g).$$

Combining the above two relations, then we obtain the desired result.

4. First, we need to prove (5.26). Since

$$\mathrm{rank}(f_3 \circ f_2) = \dim_{\mathbb{K}} f_3(\mathrm{Im}(f_2)),$$

applying Theorem 5.2.10 with $H = \mathrm{Im}(f_2)$ and $f = f_3$, we have

$$\dim_{\mathbb{K}} f_3(H) = \dim_{\mathbb{K}} \mathrm{Im}(f_3) = \dim_{\mathbb{K}} H - \dim_{\mathbb{K}} \mathrm{Ker}(f_3).$$

It is clear that the kernel of the restriction of f_2 to H is $H \cap \mathrm{Ker}(f_3) = \mathrm{Im}(f_2) \cap \mathrm{Ker}(f_3)$. It follows that

$$\mathrm{rank}(f_2) - \mathrm{rank}(f_3 \circ f_2) = \dim_{\mathbb{K}}(\mathrm{Im}(f_2) \cap \mathrm{Ker}(f_3)).$$

Now to prove (5.27) we apply (5.26). We have

$$\mathrm{rank}(f_2) - \mathrm{rank}(f_3 \circ f_2) = \dim_{\mathbb{K}}(\mathrm{Im}(f_2) \cap \mathrm{Ker}(f_3))$$

and

$$\operatorname{rank}(f_2 \circ f_1) - \operatorname{rank}(f_3 \circ f_2 \circ f_1) = \dim_{\mathbb{K}}(\operatorname{Im}(f_2 \circ f_1) \cap \operatorname{Ker}(f_3)).$$

Consequently, to show (5.27) it is enough to prove that

$$\dim_{\mathbb{K}}(\operatorname{Im}(f_2 \circ f_1) \cap \operatorname{Ker}(f_3)) \leq \dim_{\mathbb{K}}(\operatorname{Im}(f_2) \cap \operatorname{Ker}(f_3)),$$

which is true since $\operatorname{Im}(f_2 \circ f_1) \subset \operatorname{Im}(f_2)$ (see the first question in Exercise 5.3). ◄

Linear Transformations and Matrices

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6.1 Definition and Examples

The goal of this chapter is to make a connection between matrices and linear transformations. So, let E and F be two finite-dimensional vector spaces over the same field $\mathbb{K}$ such that $\dim_{\mathbb{K}} E = n$ and $\dim_{\mathbb{K}} F = m$. Then, according to Theorem 4.6.1, both spaces have bases. So, let $\mathbf{B}_E = \{u_1, u_2, \ldots, u_n\}$ be a basis of E and $\mathbf{B}_F = \{w_1, w_2, \ldots, w_m\}$ be a basis of F. Let f be an element of $\mathscr{L}(E, F)$. Since $\mathbf{B}_F$ is a basis of F, for any u_j, $1 \leq j \leq n$ in $\mathbf{B}_E$, $f(u_j)$ is uniquely written as a linear combination of the elements of $\mathbf{B}_F$:

$$f(u_j) = a_{1j}w_1 + a_{2j}w_2 + \cdots + a_{mj}w_m, \quad 1 \leq j \leq n,$$

where

$$a_{ij}, \qquad 1 \leq i \leq m, \quad 1 \leq j \leq n$$

are elements of $\mathbb{K}$. It is clear that knowledge of the a_{ij}, completely determines the linear transformation f and allows us to define the matrix associate to f as follows.

Definition 6.1.1 (Matrix Associated to a Linear Transformation)
The matrix associated to the linear transformation f given above is defined as the map defined from $I \times J$ to $\mathbb{K}$ as

$$(i, j) \mapsto a_{ij},$$

where $I = \{1, 2, \ldots, m\}$ and $J = \{1, 2, \ldots, n\}$, i is an element of the set I and j is an element of the set J and a_{ij} are as above. We denote this matrix as $M(f, \mathbf{B}_E, \mathbf{B}_F)$ or sometimes just $M(f)$ (when there is no confusion) and write it as

(Continued)

Definition 6.1.1 (continued)

$$M(f) = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}.$$

It is clear that the entries of the jth column ($1 \leq j \leq n$) of this matrix $M(f)$ are the components of $f(u_j)$ in the basis $\mathbf{B}_F$. We call $M(f)$ *the matrix of* f *in the bases* $\mathbf{B}_E$ *and* $\mathbf{B}_F$.

Example 6.1

Let f be the linear transformation defined as

$$f : \mathbb{R}^2 \to \mathbb{R}^3,$$
$$(u, v) \mapsto (u - v, u + 2v, v).$$

Find the matrix for f in the standard bases of $\mathbb{R}^2$ and $\mathbb{R}^3$. ■

Solution

The standard bases of $\mathbb{R}^2$ and $\mathbb{R}^3$ are respectively

$$\mathbf{B}_{\mathbb{R}^2} = \{(1,0),(0,1)\}, \quad \text{and} \quad \mathbf{B}_{\mathbb{R}^3} = \{(1,0,0),(0,1,0),(0,0,1)\}.$$

To find the matrix $M(f)$ associated to the linear transformation f, we need to find the components of $f(1,0)$ and $f(0,1)$ in the basis $\mathbf{B}_{\mathbb{R}^3}$; then $f(1,0)$ will be the first column of $M(f)$ and $f(1,0)$ will be the second column of $M(f)$. We have

$$f(1,0) = (1,1,0) = 1(1,0,0) + 1(0,1,0) + 0(0,0,1).$$

Thus,

$$f(1,0) = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

Similarly,

$$f(0,1) = (-1,2,1) = -1(1,0,0) + 2(0,1,0) + 1(0,0,1),$$

and so

$$f(0,1) = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}.$$

Consequently, the matrix $M(f)$ is

$$M(f) = \begin{bmatrix} 1 & -1 \\ 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

It is clear that

$$f(1,0) = M(f) \cdot \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

and

$$f(1,0) = M(f) \cdot \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}.$$

◀

Example 6.2 (Identity Matrix Associated to the Identity Transformation)
Let E be a vector space over a field $\mathbb{K}$ with $\dim_{\mathbb{K}} E = n$ and f be the identity operator defined in Example 5.2. Then, for any basis $\mathbf{B} = \{u_1, u_2, \ldots, u_n\}$ of E, we can easily see that

$$\begin{cases} f(u_1) = u_1 = 1u_1 + 0u_1 + \cdots + 0u_n, \\ f(u_2) = u_2 = 0u_1 + 1u_1 + \cdots + 0u_n, \\ \vdots \\ f(u_n) = u_n = 0u_1 + 0u_2 + \cdots + 1u_n. \end{cases}$$

Consequently, we easily obtain

$$M(f) = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & & & \ddots & \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix} = I,$$

where I is identity matrix given in Definition 1.2.2. ■

Example 6.3 (Zero Matrix Associated to the Zero Transformation)
Let E and F be two vector spaces over the same field $\mathbb{K}$ such that $\dim_{\mathbb{K}} E = n$ and $\dim_{\mathbb{K}} F = m$. Then, for any basis $\mathbf{B}_E$ of E and any basis $\mathbf{B}_F$ of F, the matrix associated to the zero transformation $0_{\mathscr{L}(E,F)}$ defined in Example 5.1 is

$$M(0_{\mathscr{L}(E,F)}) = \mathbf{0}_{m\times n},$$

where $\mathbf{0}_{m\times n}$ is the zero matrix introduced in Definition 1.1.9. ■

Now, a natural question is: *how to define matrices associated to the sum and composition of two linear transformations?* To answer this question, we may easily show some properties of these matrices:

Theorem 6.1.1
Assume that E and F are as above and let G be another vector space defined over the same field $\mathbb{K}$ with $\dim_{\mathbb{K}} G = r$. Let $\mathbf{B}_G = \{v_1, v_2, \ldots, v_r\}$ be a basis of G. Let f and g be elements of $\mathscr{L}(E,F)$, h be an element of $\mathscr{L}(F,G)$, and λ be an element of $\mathbb{K}$. Then, it holds that

1. $M(f+g) = M(f) + M(g)$.
2. $M(\lambda f) = \lambda M(f)$.
3. $M(h\circ f) = M(h)M(f)$.
4. *If $E = F$ and f is bijective, then $M(f)$ is invertible and $M(f^{-1}) = (M(f))^{-1}$.*

Proof
1. As above, assume that

$$f(u_j) = a_{1j}w_1 + a_{2j}w_2 + \cdots + a_{mj}w_m, \quad 1 \le j \le n$$

and

$$g(u_j) = d_{1j}w_1 + d_{2j}w_2 + \cdots + d_{mj}w_m, \quad 1 \le j \le n,$$

where a_{ij} and d_{ij}, $1 \le i \le m$, $1 \le j \le n$ are elements of $\mathbb{K}$. Then, we have

$$(f+g)(u_j) = f(u_j) + g(u_j) = (a_{ij} + d_{1j})w_1 + (a_{2j} + d_{2j})w_2 + \cdots + (a_{mj} + d_{mj})w_m.$$

Consequently,

$$\begin{aligned} M(f+g) &= \begin{bmatrix} a_{11}+d_{11} & a_{12}+d_{12} & a_{13}+d_{13} & \ldots & a_{1n}+d_{1n} \\ a_{21}+d_{21} & a_{22}+d_{22} & a_{23}+d_{23} & \ldots & a_{2n}+d_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1}+d_{m1} & a_{m2}+d_{m2} & a_{m3}+d_{m3} & \ldots & a_{mn}+d_{mn} \end{bmatrix} \\ &= M(f) + M(g). \end{aligned}$$

2. In the same way, we can show that $M(\lambda f) = \lambda M(f)$.
3. Assume that

$$h(w_i) = b_{1i}v_1 + b_{2i}v_2 + \cdots + b_{ri}v_r, \quad 1 \leq i \leq m.$$

We have $(h \circ f)(u_j) \in G$ for any $1 \leq j \leq n$. Hence,

$$(h \circ f) = h(f(u_j)) = c_{1j}v_1 + c_{2j}v_2 + \cdots + c_{rj}v_r, \quad 1 \leq j \leq n, \tag{6.1}$$

where c_{kj}, $1 \leq j \leq n$, $1 \leq k \leq r$ are elements of $\mathbb{K}$.

On the other hand, we have

$$\begin{aligned} h(f(u_j)) &= h(a_{1j}w_1 + a_{2j}w_2 + \cdots + a_{mj}w_m) \\ &= a_{1j}h(w_1) + a_{2j}h(w_2) + \cdots + a_{mj}h(w_m) \\ &= \sum_{i=1}^{m} a_{ij}h(w_i) \\ &= \sum_{i=1}^{m} a_{ij}(b_{1i}v_1 + b_{2i}v_2 + \cdots + b_{ri}v_r) \\ &= \left(\sum_{i=1}^{m} b_{1i}a_{ij}\right)v_1 + \left(\sum_{i=1}^{m} b_{2i}a_{ij}\right)v_2 + \cdots + \left(\sum_{i=1}^{m} b_{ri}a_{ij}\right)v_r. \end{aligned} \tag{6.2}$$

Comparing (6.1) and (6.2) and using Theorem 4.4.3, we find that

$$c_{kj} = \sum_{i=1}^{m} b_{ki}a_{ij}, \quad 1 \leq k \leq r,$$

which are exactly the entries of the matrix product $M(h)M(f)$ as introduced in Definition 1.1.11 (with some changes in the indices).

4. If f is an automorphism, then

$$f \circ f^{-1} = f^{-1} \circ f = \mathrm{Id}_{\mathscr{L}(E)},$$

whence, by using (3),

$$M(f \circ f^{-1}) = M(f)M(f^{-1}) = M(f^{-1})M(f) = I.$$

The uniqueness of the inverse (Theorem 1.2.3) shows that $M(f^{-1}) = (M(f))^{-1}$. □

Example 6.4

Consider the linear transformation f defined in Example 6.1 and the linear transformation $h : \mathbb{R}^3 \to \mathbb{R}^2$ defined by

$$h(w, y, z) = (w + y - z, y + z).$$

Find the matrix associated to $h \circ f$ relative to the standard basis of $\mathbb{R}^2$. ■

Solution

We need first to find the matrix $M(h)$. We have

$$h(1,0,0) = (1,0) = 1(1,0) + 0(0,1), \quad h(0,1,0) = (1,1) = 1(1,0) + 1(0,1)$$

and

$$h(0,0,1) = (-1,1) = -1(1,0) + 1(0,1),$$

so

$$M(h) = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix}.$$

Now, we need to find the matrix $M(h \circ f)$. We have

$$\begin{aligned} h \circ f : \mathbb{R}^2 &\to \mathbb{R}^2 \\ (u, v) &\mapsto (2u, u + 3v). \end{aligned}$$

Thus, we have

$$(h \circ f)(1,0) = (2,1) = 2(1,0) + 1(0,1) \quad \text{and} \quad (h \circ f)(0,1) = (0,3) = 0(1,0) + 3(0,1).$$

Consequently,

$$M(h \circ f) = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix}.$$

On the other hand, we have

$$M(h)M(f) = \begin{bmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 2 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 1 & 3 \end{bmatrix} = M(h \circ f).$$

◄

Remark 6.1.2 (Linear Transformation Associated to a Matrix) We have seen above how to find the matrix associated to a linear transformation. Conversely, assume that we have a matrix associated to a linear transformation f defined from E to F, where E and F are as above. Then, we can easily see that

$$f(u_j) = M(f)u_j.$$

Example 6.5

Assume, for example, that we know the matrix $M(f)$ defined in Example 6.1. Then

$$f(1,0) = M(f)\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}1\\1\\0\end{bmatrix}.$$

and

$$f(0,1) = M(f)\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}-1\\2\\1\end{bmatrix}.$$

Now, to find f, let (u, v) be an element of $\mathbb{R}^2$. Then, we have

$$(u, v) = u(1,0) + v(0,1).$$

Hence, by the linearity of f,

$$f(u, v) = uf(1,0) + vf(0,1) = u(1,1,0) + v(-1,2,1) = (u - v, u + 2v, v),$$

which is precisely the linear transformation given in Example 6.1. ■

So, we have seen above, that for any linear transformation f in $\mathscr{L}(E, F)$ we associate a unique matrix $M(f)$ in $\mathscr{M}_{m\times n}(\mathbb{K})$ and for any matrix in $\mathscr{M}_{m\times n}(\mathbb{K})$, we have a unique linear transformation associated to it. So, clearly, the transformation T defined as follows:

$$\begin{aligned} T : \mathscr{L}(E, F) &\to \mathscr{M}_{m\times n}(\mathbb{K}) \\ f &\mapsto T(f) = M(f) \end{aligned}$$

is a bijection from $\mathscr{L}(E, F)$ to $\mathscr{M}_{m\times n}(\mathbb{K})$. In addition, as we have seen in Theorem 6.1.1, this transformation is linear. Thus, it defines an isomorphism between $\mathscr{L}(E, F)$ and $\mathscr{M}_{m\times n}(\mathbb{K})$. Thus, we summarize these in the following theorem.

Theorem 6.1.3 ($\mathscr{L}(E,F)$ **and** $\mathscr{M}_{m\times n}$ **are Isomorphic)**
Let E and F be two finite-dimensional vector spaces over the same field $\mathbb{K}$, and denote $\dim_{\mathbb{K}} E = n$ and $\dim_{\mathbb{K}} F = m$. Then the spaces $\mathscr{L}(E,F)$ and $\mathscr{M}_{m\times n}(\mathbb{K})$ are isomorphic.

Remark 6.1.4 Since $\dim_{\mathbb{K}} \mathscr{M}_{m\times n}(\mathbb{K}) = m \times n$, Theorem 6.1.3 shows that,

$$\dim_{\mathbb{K}} \mathscr{L}(E,F) = m \times n = \dim_{\mathbb{K}} E \times \dim_{\mathbb{K}} F.$$

6.2 Change of Basis and Similarity

We have seen that a finite-dimensional vector space may has more than one basis. So, one question is: *what happens to the matrix associated to a linear transformation if we change bases?* Also, we have seen in ▶ Chap. 1 that it is much easier to deal with diagonal matrices, especially when dealing with differential equations or powers of a matrix or eigenvalues, as we will see in the coming chapters, since diagonal matrices enjoy nice properties. So, suppose that we have a linear operator f in $\mathscr{L}(E)$. *Can we choose two bases in E in which $M(f)$ is a diagonal matrix?* Thus, it is so important to investigate the effect of a change of bases on the matrix $M(f)$.

Let E be a vector space over a field $\mathbb{K}$. Denote $\dim_{\mathbb{K}} E = n$ and let

$$\mathbf{B}_1 = \{u_1, u_2, \ldots, u_n\} \qquad \text{and} \qquad \mathbf{B}_2 = \{v_1, v_2, \ldots, v_n\}$$

be two bases of E. We define the linear transformation

$$\begin{aligned} p : E &\to E \\ u_j &\mapsto p(u_j) = v_j, \quad 1 \le j \le n. \end{aligned}$$

As before, since $v_j \in E$, then it can be uniquely written as a linear combination of the elements of $\mathbf{B}_1$:

$$v_j = p(u_j) = \alpha_{1j}u_1 + \alpha_{2j}u_2 + \cdots + \alpha_{nj}u_n, \quad 1 \le j \le n,$$

where α_{ij}, $1 \le i,j \le n$, are elements of $\mathbb{K}$. Let $\Lambda = \Lambda(p, \mathbf{B}_1, \mathbf{B}_2)$ be the matrix associate to p. Then, we have

$$\Lambda = \begin{bmatrix} \alpha_{11} & \alpha_{12} & \ldots & \alpha_{1n} \\ \alpha_{21} & \alpha_{22} & \ldots & \alpha_{2n} \\ \vdots & \vdots & \vdots & \vdots \\ \alpha_{n1} & \alpha_{n2} & \ldots & \alpha_{nn} \end{bmatrix}.$$

It is quite obvious that p is an automorphism, so according to Theorem 6.1.1, Λ^{-1} exists and satisfies

$$\Lambda^{-1} = \Lambda(p, \mathbf{B}_2, \mathbf{B}_1) = \Lambda(p^{-1}, \mathbf{B}_1, \mathbf{B}_2,). \tag{6.3}$$

Thus, we have the following definition.

Definition 6.2.1 (Transition Matrix)

The matrix $\Lambda(p, \mathbf{B}_1, \mathbf{B}_2)$ defined above is called the *transition matrix* from the basis $\mathbf{B}_1$ to the basis $\mathbf{B}_2$.

Let us denote by $[v_j]_{\mathbf{B}_k}$ the components of v_j in the basis $\mathbf{B}_k$, $k = 1, 2$. As we have seen above, these components can be found using the formula

$$[v_j]_{\mathbf{B}_1} = \Lambda[v_j]_{\mathbf{B}_2}.$$

Similarly, if $[u_j]_{\mathbf{B}_k}$ are the components of u_j with respect to the basis $\mathbf{B}_k$, $k = 1, 2$, then

$$[u_j]_{\mathbf{B}_2} = \Lambda^{-1}[u_j]_{\mathbf{B}_1}.$$

Example 6.6

Consider the vector space $\mathbb{R}^3$ and the two bases

$$\mathbf{B}_1 = \{u_1, u_2, u_3\}, \quad u_1 = (1, 1, 0),\ u_2 = (-1, 1, 0),\ u_3 = (0, 2, 1).$$

and

$$\mathbf{B}_2 = \{v_1, v_2, v_3\}, \quad v_1 = (1, 2, 0),\ v_2 = (1, 1, 2),\ v_3 = (-1, 0, 3).$$

Find the transition matrix Λ from $\mathbf{B}_1$ to $\mathbf{B}_2$. ■

Solution

First, one can easily check that $\mathbf{B}_1$ and $\mathbf{B}_2$ are bases of $\mathbb{R}^3$. Now, we need to find the components of v_j, $j = 1, 2, 3$, in the basis $\mathbf{B}_1$. We easily see that

$$\begin{aligned} v_1 &= \frac{3}{2}u_1 + \frac{1}{2}u_2 + 0u_3, \\ v_2 &= -u_1 - 2u_2 + 2u_3, \\ v_3 &= -\frac{7}{2}u_1 - \frac{5}{2}u_2 + 3u_3. \end{aligned}$$

Thus, we obtain

$$\Lambda = \begin{bmatrix} 3/2 & -1 & -7/2 \\ 1/2 & -2 & -5/2 \\ 0 & 2 & 3 \end{bmatrix}.$$

Now, it is clear that

$$[v_1]_{\mathbf{B}_1} = \begin{bmatrix} 3/2 \\ 1/2 \\ 0 \end{bmatrix} = \Lambda[v_1]_{\mathbf{B}_2}, \quad [v_2]_{\mathbf{B}_1} = \Lambda[v_2]_{\mathbf{B}_2}, \quad [v_2]_{\mathbf{B}_1} = \Lambda[v_3]_{\mathbf{B}_2}.$$

Since

$$[v_1]_{\mathbf{B}_2} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad [v_2]_{\mathbf{B}_2} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \quad [v_3]_{\mathbf{B}_2} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix},$$

we have

$$\Lambda^{-1} = \begin{bmatrix} 2/7 & 8/7 & 9/7 \\ 3/7 & -9/7 & -4/7 \\ -2/7 & 6/7 & 5/7 \end{bmatrix}.$$

Now, the columns of Λ^{-1} are the components of u_j, $j = 1, 2, 3$, in the basis $\mathbf{B}_2$. ◀

Example 6.7

Consider $\mathscr{P}_2$ to be the set of all polynomials with real coefficients and of degree less than or equal to 2. We have seen in Example 4.19, that

$$\mathbf{B}_1 = \{1, x, x^2\}$$

is a basis of $\mathscr{P}_2$. Consider the set

$$\mathbf{B}_2 = \{1, 1 + x, 1 + x + x^2\}.$$

1. Show that $\mathbf{B}_2$ is a basis of $\mathscr{P}_2$.
2. Find the transition matrix Λ from $\mathbf{B}_1$ to $\mathbf{B}_2$.
3. Find the components of $q(x) = 3 + 2x + 4x^2$ relative to the basis $\mathbf{B}_2$.

■

Solution

1. By Theorem 4.6.5, it is enough to show that $\mathbf{B}_2$ is a linearly independent set. So, let α, β and γ be real numbers satisfying, for all $x \in \mathbb{R}$,

$$\alpha + \beta(1 + x) + \gamma(1 + x + x^2) = 0.$$

Taking some particular values of x, such as $(x = 0, x = -1, x = 1)$, we get the system of equations

$$\begin{cases} \alpha + \beta + \gamma = 0, \\ \alpha + \gamma = 0, \\ \alpha + 2\beta + 3\gamma = 0. \end{cases}$$

It is clear that this system has the unique solution $\alpha = \beta = \gamma = 0$. Thus, $\mathbf{B}_2$ is a linearly independent set and hence a basis of $\mathscr{P}_2$.

2. To find the transition matrix Λ from $\mathbf{B}_1$ to $\mathbf{B}_2$, we need to find the components of the elements of $\mathbf{B}_2$ with respect to $\mathbf{B}_1$. We have

$$1 = 1 + 0x + 0x^2,$$
$$1 + x = 1 + x + 0x^2,$$
$$1 + x + x^2 = 1 + 1x + 1x^2.$$

Thus,

$$\Lambda = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

We can also easily see that the transition matrix from $\mathbf{B}_2$ to $\mathbf{B}_1$ is

$$\Lambda^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

3. We can write the components of $q(x)$ with respect to $\mathbf{B}_1$ as

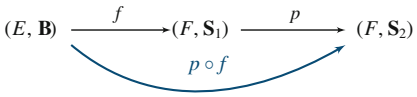
$$[q(x)]_{\mathbf{B}_1} = \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix}.$$

Then, the components of $q(x)$ with respect to $\mathbf{B}_2$ are

$$[q(x)]_{\mathbf{B}_2} = \Lambda^{-1}[q(x)]_{\mathbf{B}_1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 2 \\ 4 \end{bmatrix} = \begin{bmatrix} 1 \\ -2 \\ 4 \end{bmatrix}.$$

◀

Now, let E and F be two finite-dimensional vector spaces over the same field $\mathbb{K}$. Denote $\dim_{\mathbb{K}} E = n$ and $\dim_{\mathbb{K}} F = m$. Let $\mathbf{B}$ be a basis of E and let $\mathbf{S}_1$ and $\mathbf{S}_2$ be two bases of F. Let f be a linear transformation from E to F and let $M(f, \mathbf{B}, \mathbf{S}_1)$ be the corresponding matrix of f with respect to the bases $\mathbf{B}$ and $\mathbf{S}_1$. Then, the question is: *how to find* $M(f, \mathbf{B}, \mathbf{S}_2)$*?*



Let p be the linear operator that transform the elements of $\mathbf{S}_1$ into the element of $\mathbf{S}_2$ as shown above and $\Lambda(p, \mathbf{S}_1, \mathbf{S}_2)$ be its corresponding matrix. Then, Theorem 6.1.1, yields

$$M(f, \mathbf{B}, \mathbf{S}_2) = M(p \circ f, \mathbf{B}, \mathbf{S}_2) = \Lambda(p, \mathbf{S}_1, \mathbf{S}_2) M(f, \mathbf{B}, \mathbf{S}_1). \tag{6.4}$$

Example 6.8

Consider the linear transformation f defined as

$$f : \mathbb{R}^2 \to (\mathbb{R}^3, \mathbf{B}_1)$$
$$(x, y) \mapsto (x - y, x + y, -y),$$

where in $\mathbb{R}^2$ we consider the standard basis. Let p be the linear operator defined from $\mathbb{R}^3$ to $\mathbb{R}^3$ as

$$p(u_j) = v_j, \qquad j = 1, 2, 3,$$

where u_j, v_j and $\mathbf{B}_1$ are given in Example 6.6. Find $M(p \circ f)$, the matrix associated to $p \circ f$. ■

Solution

First, we need to find $M(f)$. So, we need to find the components of $f(1, 0)$ and $f(0, 1)$ with respect to the basis $\mathbf{B}_1$. We have

$$f(1, 0) = (1, 1, 0) = u_1 + 0u_2 + 0u_3$$

and

$$f(0, 1) = (-1, 1, -1) = u_1 + 2u_2 - u_3.$$

Thus,

$$M(f) = \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 0 & -1 \end{bmatrix}.$$

Using (6.4), we have

$$\begin{aligned} M(p \circ f) &= \Lambda M(f) \\ &= \begin{bmatrix} 3/2 & -1 & -7/2 \\ 1/2 & -2 & -5/2 \\ 0 & 2 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 2 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 3/2 & 3 \\ 1/2 & -1 \\ 0 & 1 \end{bmatrix}. \end{aligned}$$

◀

Now, let E and F be two finite-dimensional vector spaces over the same field $\mathbb{K}$, with $\dim_{\mathbb{K}} E = n$ and $\dim_{\mathbb{K}} F = m$. Let $\mathbf{B}_1$ and $\mathbf{B}_2$ be two bases of E and $\mathbf{S}_1$ and $\mathbf{S}_2$ be two bases of F. Let f be a linear transformation from E into F and $M(f, \mathbf{B}_1, \mathbf{S}_1)$ be the corresponding matrix to f with respect to the bases $\mathbf{B}_1$ and $\mathbf{S}_1$. We want now to find $M(f, \mathbf{B}_2, \mathbf{S}_2)$, the corresponding matrix of f with respect to $\mathbf{B}_2$ and $\mathbf{S}_2$. So, let $\Lambda(p_1, \mathbf{B}_1, \mathbf{B}_2)$ be the transition matrix from $\mathbf{B}_1$ to $\mathbf{B}_2$ and $\Lambda(p_2, \mathbf{S}_1, \mathbf{S}_2)$ be the transition matrix from $\mathbf{S}_1$ to $\mathbf{S}_2$, as shown in the following diagram:

$$(E, \mathbf{B}_2) \xrightarrow{p_1^{-1}} (E, \mathbf{B}_1) \xrightarrow{f} (F, \mathbf{S}_1) \xrightarrow{p_2} (F, \mathbf{S}_2), \qquad p_2 \circ f \circ p_1^{-1}$$

Then, it is clear that

$$\begin{aligned} M(f, \mathbf{B}_2, \mathbf{S}_2) &= M(p_2 \circ f \circ p_1^{-1}, \mathbf{B}_2, \mathbf{S}_2) \\ &= \Lambda(p_2, \mathbf{S}_1, \mathbf{S}_2)) \cdot M(f, \mathbf{B}_1, \mathbf{S}_1) \cdot \Lambda(p_1^{-1}, \mathbf{B}_1, \mathbf{B}_2), \\ &= \Lambda(p_2, \mathbf{S}_1, \mathbf{S}_2)) \cdot M(f, \mathbf{B}_1, \mathbf{S}_1) \cdot \Lambda^{-1}(p_1, \mathbf{B}_1, \mathbf{B}_2), \end{aligned} \tag{6.5}$$

where we have used (6.3). The above formula can be rewritten as

$$M(f, \mathbf{B}_1, \mathbf{S}_1) = \Lambda^{-1}(p_2, \mathbf{S}_1, \mathbf{S}_2)) \cdot M(f, \mathbf{B}_2, \mathbf{S}_2) \cdot \Lambda(p_1, \mathbf{B}_1, \mathbf{B}_2).$$

Two matrices satisfying (6.5) are called *equivalent* and we may give a general definition of two equivalent matrices as follows.

Definition 6.2.2 (Equivalent Matrices)

Let A and B be two matrices in $\mathscr{M}_{m\times n}(\mathbb{K})$. We say that A and B are *equivalent* if there exist two invertible matrices R in $\mathscr{M}_m(\mathbb{K})$ and S in $\mathscr{M}_n(\mathbb{K})$ such that

$$B = RAS.$$

Remark 6.2.1 It is clear that the relation A *is equivalent to* B is an *equivalence relation* on the set of matrices $\mathscr{M}_{m\times n}(\mathbb{K})$, since:

- A is equivalent to A *(reflexivity)*.
- If A is equivalent to B, then B is equivalent to A *(symmetry)*.
- If A is equivalent to B and B is equivalent to C, then C is equivalent to A *(transitivity)*.

One of the important cases in (6.5) is when $E = F$, $\mathbf{S}_1 = \mathbf{B}_1$ and $\mathbf{S}_2 = \mathbf{B}_2$; then f is an endomorphism, $p_1 = p_2$, and we have

$$M(f, \mathbf{B}_1, \mathbf{B}_1) = \Lambda^{-1}(p_1, \mathbf{B}_1, \mathbf{B}_2)) \cdot M(f, \mathbf{B}_2, \mathbf{B}_2) \cdot \Lambda(p_1, \mathbf{B}_1, \mathbf{B}_2).$$

In this case, we say that the matrix $M(f, \mathbf{B}_1, \mathbf{B}_1)$ is *similar* to the matrix $M(f, \mathbf{B}_2, \mathbf{B}_2)$ and we can give a more general definition as follows:

Definition 6.2.3 (Similarity)

Let A and B be two square matrices in $\mathscr{M}_n(\mathbb{K})$. Then, A and B are called *similar* or *conjugate* if there exists an invertible matrix P in $\mathscr{M}_n(\mathbb{K})$ such that

$$B = P^{-1}AP. \tag{6.6}$$

It is clear that two similar matrices are equivalent. As we have seen above, similar matrices represent the same endomorphism with respect to two different bases. The matrix P is also called a change of bases matrix.

As indicated in the beginning of ▶ Sect. 6.2, one of the main goals of the change of bases is to transform matrices to diagonal form. Formula (6.6) is a very important tool to obtain diagonal matrices from some square matrices called *diagonalizable matrices*; the process is known as *diagonalization* of matrices. We will see in the coming chapters that in this case A and B have something in common. For instance, similar matrices share the same eigenvalues. We will come back to this in details in the next chapter, but just to clarify the ideas, we give the following example.

Example 6.9

Consider in $\mathscr{M}_3(\mathbb{R})$ the matrix

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & 9 \end{bmatrix}.$$

Take two bases in $\mathbb{R}^3$:

$$\mathbf{B}_1=\{e_1,e_2,e_3\}, \quad \text{and} \quad \mathbf{B}_2=\{u_1,u_2,u_3\}, \quad u_1=(0,-2,1),\ u_2=(1,0,0),\ u_3=(0,1,2).$$

Find the matrix $B = P^{-1}AP$, where P is the transition matrix from $\mathbf{B}_1$ to $\mathbf{B}_2$. ■

Solution

It is clear that since $\mathbf{B}_1$ is the standard basis of $\mathbb{R}^3$,

$$P = [u_1, u_2, u_3] = \begin{bmatrix} 0 & 1 & 0 \\ -2 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix}.$$

Now, we may easily check, by using the methods in ▶ Chap. 1 (for instance), that

$$P^{-1} = \begin{bmatrix} 0 & -2/5 & 1/5 \\ 1 & -2/5 & -4/5 \\ 0 & 1/5 & 2/5 \end{bmatrix}.$$

Thus, we have by a simple computation

$$B = P^{-1}AP = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 11 \end{bmatrix}.$$

It is clear that B is a diagonal matrix. This method of obtaining B from the matrix A is a very important topic in linear algebra and linear programming. ◀

6.3 Rank of Matrices

We have defined in ▶ Sect. 5.2.1 the rank of a linear transformation to be the dimension of its image. Also, we have seen in ▶ Chap. 6 that any linear transformation between two finite-dimensional vector spaces can be represented by a matrix. So, now, in order to define the rank of a matrix, we have to define first a subspace corresponding to the image of a linear transformation. This subspace is known as the *column space* of the matrix.

So, let E and F be two finite-dimensional vector spaces over the same field $\mathbb{K}$ and let f be a linear transformation from E into F, as in Definition 6.1.1. Then, once bases are chosen in E and F, one can associate a unique matrix to this linear transformation. Conversely, we have also seen in Remark 6.1.2 that we can always associate a unique linear transformation to a given matrix. So, let $M(f)$ be the matrix given in Definition 6.1.1. We define the set

$$\mathscr{R}(M(f)) = \{M(f)u, \text{ with } u \in E\} = \text{Im}(f).$$

As we have seen in Theorem 5.2.8, this set is a subspace of F. Thus, the rank of $M(f)$ is the rank of f and it is the dimension of $\mathscr{R}(M(f))$, and we have the following definition.

Definition 6.3.1 (The Rank of a Matrix)

Let E and F be two vector spaces over the same field $\mathbb{K}$, with $\dim_{\mathbb{K}} E = n$ and $\dim_{\mathbb{K}} F = m$. The *rank* of A is defined to be the dimension of the column space $\mathscr{R}(A)$ defined as

$$\mathscr{R}(A) = \{Au, \text{ with } u \in E\}.$$

and we write

$$\text{rank}(A) = \dim_{\mathbb{K}} \mathscr{R}(A).$$

Remark 6.3.1 From above and since $\mathscr{R}(A)$ is a subspace of F, Theorem 4.6.7 shows that $\text{rank}(A) \leq \dim_{\mathbb{K}} F = m$.

In addition, recall from Example 4.12 that the null space $\mathscr{N}(A)$ of the matrix A is defined as

$$\mathscr{N}(A) = \{u \in E, \text{ such that } Au = 0_F\} = \text{Ker}(f).$$

Thus, using Theorem 5.2.10, we deduce that

$$n = \dim_{\mathbb{K}} E = \text{rank}(A) + \text{null}(A).$$

This shows that $\text{rank}(A) \leq n$. Consequently,

$$\text{rank}(A) \leq \min(n, m). \tag{6.7}$$

If $\text{rank}(A) = \min(n, m)$, then we say that A is a matrix of *full rank*.

One of the main goals in ▶ Chaps. 1 and 2 was to determine whether a matrix is invertible or not. One of the requirements for A to be invertible was $\det(A) \neq 0$, (see Theorem 2.4.8). Here we can easily deduce an equivalent invertibility criterion as given in the following theorem.

Theorem 6.3.2 (Rank Criterion for Invertibility)

Let A be a square matrix in $\mathscr{M}_n(\mathbb{K})$. Then, A is invertible if and only $\text{rank}(A) = n$.

Proof
The assertion is obvious since in this case, the linear transformation associated to A is surjective and therefore according to Theorem 5.3.3 it is an isomorphism, and thus invertible. □

6.3.1 Some Properties of the Rank of a Matrix

Next, we discuss some properties of the rank of a matrix. To clarify the ideas, we assume that $E = \mathbb{K}^n$, $F = \mathbb{K}^m$ and, for simplicity, take $\mathbb{K} = \mathbb{R}$, but the method works for any field $\mathbb{K}$. In $\mathbb{R}^n$ and $\mathbb{R}^m$ we use the standard bases, that is

$$\mathbf{B}_{\mathbb{R}^n} = \{e_1, e_2, \ldots, e_n\} \qquad \text{and} \qquad \mathbf{B}_{\mathbb{R}^m} = \{e_1, e_2, \ldots, e_m\}.$$

Let A be a matrix in $\mathscr{M}_{m\times n}(\mathbb{K})$ and f be the linear transformation associated to it. It is clear that the components of $f(e_j)$, $1 \leq j \leq n$, with respect to the basis $\mathbf{B}_{\mathbb{R}^m}$ form the jth column of A. Moreover, the rank of f is the number of the linearly independent columns of A and we have the following definition.

Definition 6.3.2
Let A be a matrix in $\mathscr{M}_{m\times n}(\mathbb{K})$. Then, the rank of A is equal to the largest number of columns of A which are linear independent.

So, now the question is: *how to find the linearly independent columns of a matrix?*

If $n = m$, then we have seen that a square matrix A in $\mathscr{M}_n(\mathbb{K})$ is invertible if and only if one of the following conditions is satisfied:

1. $\det(A) \neq 0$,
2. $\operatorname{rank}(A) = n$.

So, necessarily these two conditions should be equivalent and thus, we have the following theorem.

Theorem 6.3.3
Let A be a square matrix in $\mathscr{M}_n(\mathbb{K})$. Then,

$$\operatorname{rank}(A) = n$$

if and only if $\det(A) \neq 0$.

Proof
We have seen in Exercise 4.5 that the columns of A are linearly independent if and only if $\det(A) \neq 0$. This means that $\operatorname{rank}(A) = n$ if and only if $\det(A) \neq 0$. □

Theorem 6.3.3 shows that in the case where $\det(A) \neq 0$, the problem of finding the rank of A is completely solved: it is equal to the size of the square matrix A.

Combining Theorem 6.3.3 and Theorem 2.4.8, we easily deduce another invertibility criterion.

Theorem 6.3.4

Let A be a square matrix in $\mathscr{M}_n(\mathbb{K})$. Then, A is invertible if and only if

$$\operatorname{rank}(A) = n.$$

We have seen in Theorem 2.3.1 that $\det(A) = \det(A^T)$; in particular, $\det(A) \neq 0$ if and only if $\det(A^T) \neq 0$. In this case, and according to Theorem 6.3.3,

$$\operatorname{rank}(A) = \operatorname{rank}(A^T). \tag{6.8}$$

Since the columns of A^T are the rows of A, then we can say that the rank of A is also the maximal number of its linearly independent rows. In fact (6.8) is true for any matrix in $\mathscr{M}_{m\times n}(\mathbb{K})$ and the fact that the rank of a matrix A is equal to the rank of its transpose A^T is one of the important theorems in linear algebra. The space $\mathscr{R}(A^T)$ is also called the *row space* of A. We have the following assertion.

Theorem 6.3.5

Let A be a matrix in $\mathscr{M}_{m\times n}(\mathbb{K})$. Then we have

$$\operatorname{rank}(A) = \operatorname{rank}(A^T).$$

Proof

Actually, several proofs of the above theorem are available. Here we present a simple proof from [16], assuming that $\mathbb{K} = \mathbb{R}$ (the same proof also works for $\mathbb{K} = \mathbb{C}$). First, it is clear that if Y is a vector in $\mathbb{R}^n$, then

$$Y^T Y = \|Y\|^2 = 0, \tag{6.9}$$

if and only if $Y = 0_{\mathbb{R}^n}$, (Theorem 3.3.1). Hence, using (6.9), we deduce, by taking $Y = AX$, that

$$AX = 0_{\mathbb{R}^m}$$

if and only if $A^T AX = 0_{\mathbb{R}^n}$ for any vector X in $\mathbb{R}^n$. Using this last property, we see that the vectors $AX_1, AX_2, \ldots, AX_k$ are linearly independent if and only if the vectors $A^T AX_1, A^T AX_2, \ldots, A^T AX_k$ are linearly independent, where $X_1, X_2, \ldots, X_k$ are vectors in $\mathbb{R}^n$.

Consequently, we deduce that

$$\operatorname{rank}(A) = \operatorname{rank}(A^T A). \tag{6.10}$$

Now, we have

$$\operatorname{rank}(A) = \operatorname{rank}(A^T A) = \dim_{\mathbb{R}}\{A^T(AX) : X \in \mathbb{R}^n\} \leq \dim_{\mathbb{R}}\{A^T Y : Y \in \mathbb{R}^n\} = \operatorname{rank}(A^T).$$

On the other hand, since $(A^T)^T = A$, we have

$$\operatorname{rank}(A^T) \leq \operatorname{rank}((A^T)^T) = \operatorname{rank}(A).$$

This finishes the proof of Theorem 6.3.5. □

It is natural to expect that equivalent matrices share some algebraic properties. *So, do equivalent matrices have the same rank?* In fact this turns out to be true.

Theorem 6.3.6 (Equivalent Matrices and Rank)
Let A and B be two equivalent matrices. Then

$$\operatorname{rank}(A) = \operatorname{rank}(B).$$

Proof
As we have seen in (6.5), if two matrices are equivalent, then they represent the same linear transformation, with respect to different bases and so they have the same rank. □

This theorem has the following immediate corollary.

Corollary 6.3.7 (Similarity and Rank) *Let A and B be two similar matrices in $\mathscr{M}_n(\mathbb{K})$. Then*

$$\operatorname{rank}(B) = \operatorname{rank}(A).$$

Remark 6.3.8 The opposite of Corollary 6.3.7 is not true: The fact that two matrices have the same rank does not imply that they are similar. For example, the matrices

$$A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

have the same rank equal to 2, but they are not similar. Indeed, we can easily see that $A^2 \neq 0_{\mathscr{M}_4(\mathbb{R})}$, whereas $B^2 = 0_{\mathscr{M}_4(\mathbb{R})}$. So, if there exists an invertible matrix P such that $B = P^{-1}AP$, then

$$A^2 = PBP^{-1}PBP = PB^2P^{-1} = 0_{\mathscr{M}_4(\mathbb{R})},$$

which is a contradiction. Thus, A and B are not similar.

6.4 Methods for Finding the Rank of a Matrix

Definition 6.3.1 does not necessarily make the computation of the rank of a matrix easy because finding the dimension of the space spanned by the column vectors of a matrix could be difficult. However, there are several methods that can be applied to calculate the rank of a matrix. Here we introduced two of them, that we believe are the most effective and easy.

6.4.1 The Method of Elementary Row and Column Operations

Here we introduce a method for finding the rank of a matrix A in $\mathscr{M}_{m\times n}(\mathbb{K})$. Given A, we can change bases in $\mathbb{K}^n$ in such a way that there exists $1 \leq r \leq \min(m, n)$ such that the matrix A can be written (in the new basis) as

$$A = \begin{bmatrix} I_r & \mathbf{0}_{r\times(n-r)} \\ \mathbf{0}_{(m-r)\times r} & \mathbf{0}_{(m-r)\times(n-r)} \end{bmatrix}, \tag{6.11}$$

where $0_{p\times q}$ is the $p \times q$ matrix with all its entries equal to zero and I_r is the $r \times r$ identity matrix.

We have seen in ▶ Sect. 6.2 that the change of basis is equivalent to the multiplication of the original matrix by an invertible matrix. Since the form (6.11) can be obtained by a series of changes of bases, the procedure is equivalent to the multiplication of the original matrix by a series of invertible matrices. That is, for any matrix A in $\mathscr{M}_{m\times n}(\mathbb{K})$, we can show that there exists a finite sequence of invertible matrices $E_1, E_2, \ldots, E_{k+s}$ such that the matrix

$$E_kE_{k-1}\cdots E_1AE_{k+1}E_{k+2}\cdots E_{k+s} \tag{6.12}$$

has the form (6.11). It is clear that the matrices on the left-hand side of (6.12) belong to $\mathscr{M}_m(\mathbb{K})$ and the matrices in the right-hand side belong to $\mathscr{M}_n(\mathbb{K})$. Of course according to Theorem 6.3.6, this operation does not change the rank of the matrix, since all the matrices that we obtain by means of such a multiplication will be equivalent.

Now, once the form (6.11) is obtained, then it is clear that $\operatorname{rank}(A) = r$, and we can also show that each matrix of rank r is equivalent to the matrix in (6.11).

So, the question now is: *how to choose the matrices in* (6.12) *so as to reach the form* (6.11)*?*

To clarify this, consider the example

$$A = \begin{bmatrix} 1 & 3 & -1 \\ 0 & 1 & 7 \end{bmatrix}.$$

Multiplying A from the left by the matrix

$$E_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

we get

$$A_1 = E_1 A = \begin{bmatrix} 1 & 4 & 6 \\ 0 & 1 & 7 \end{bmatrix}.$$

It is clear that E_1 is invertible. Also, multiplication by E_1 from the left is clearly equivalent to the row operation $r_1 + r_2$, where r_1 is the first row in the matrix A and r_2 is the second row. Now, replacing the first row r_1 in A_1 by $r_1 - 4r_2$, we get

$$A_2 = \begin{bmatrix} 1 & 0 & -22 \\ 0 & 1 & 7 \end{bmatrix}.$$

This is equivalent to multiply A_1 by the matrix

$$E_2 = \begin{bmatrix} 1 & -4 \\ 0 & 1 \end{bmatrix}.$$

Thus, we have $A_2 = E_2 A_1$. Also, it is clear that E_2 is invertible.

Next, multiplying A_2 from the right by the matrix

$$E_3 = \begin{bmatrix} 1 & 0 & 22 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

we get

$$A_3 = A_2 E_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 7 \end{bmatrix}.$$

This last operation is equivalent to replace c_3 by $22c_1 + c_3$, where c_1, c_2, and c_3 are, respectively, the first, the second, and the third columns of A_2. Finally, replacing c_3 in

A_3 by $c_3 - 7c_2$, we get

$$A_4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}. \tag{6.13}$$

This last operation is also equivalent to multiply A_3 from the right by the matrix

$$E_4 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -7 \\ 0 & 0 & 1 \end{bmatrix}.$$

Summarizing all the above operations, we obtain

$$A_4 = E_2E_1AE_3E_4.$$

It is clear that all the above matrices E_i, $1 \leq i \leq 4$ are invertible. Now, if we put $R = E_2E_1$ and $S = E_3E_4$, then we obtain

$$A_4 = RAS.$$

This means that the matrices A_4 and A are equivalent, so they have the same rank. It is clear that A_4 is in the form (6.11), with $r = 2$. Consequently,

$$\text{rank}(A) = \text{rank}(A_4) = 2.$$

What we have done above is a series of changes of bases in the vector spaces $\mathbb{R}^2$ and $\mathbb{R}^3$ to find the appropriate bases in these spaces in which A has the final form (6.13). Also, as we have seen above, to find these appropriate bases, we need to perform some row and column operations on the matrix. Basically, these row and column operations are:

- Multiply a row (or a column) through by a nonzero constant.
- Interchange two rows (or two columns).
- Add a constant times one row to another row (or a constant times one column to another column).

These operations are called elementary row (respectively, column) operations.

Example 6.10

Find the rank of the matrix

$$A = \begin{bmatrix} 0 & 1 & 2 & -1 \\ -2 & 0 & 0 & 6 \\ 4 & -2 & -4 & -10 \end{bmatrix}.$$

■

Solution

Denote the columns of A and the matrices obtained by means of elementary operations on A by $c_j,\ 1 \le j \le 4$, and their rows by $r_i,\ 1 \le i \le 3$. First, keeping in mind (6.7), we deduce that $\operatorname{rank}(A) \le 3$. Now, our goal is to perform some row or column elementary operations so as to obtain a matrix of the form (6.11). Interchanging r_1 and r_2, we get

$$A_1 = \begin{bmatrix} -2 & 0 & 0 & 6 \\ 0 & 1 & 2 & -1 \\ 4 & -2 & -4 & -10 \end{bmatrix},$$

and replacing r_3 by $r_3 + 2r_2$ in A_1, we get

$$A_2 = \begin{bmatrix} -2 & 0 & 0 & 6 \\ 0 & 1 & 2 & -1 \\ 0 & -2 & -4 & 2 \end{bmatrix}.$$

Now, replacing r_3 by $r_3 + 2r_2$ in A_2, we obtain

$$A_3 = \begin{bmatrix} -2 & 0 & 0 & 6 \\ 0 & 1 & 2 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Next, replacing c_3 by $c_3 - 2c_2$ in A_3 we obtain

$$A_4 = \begin{bmatrix} -2 & 0 & 0 & 6 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

and replacing c_4 by $c_4 + 3c_1$ in A_4, we get

$$A_5 = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Now, replacing c_4 by $c_4 + c_2$ in A_5, we get

$$A_6 = \begin{bmatrix} -2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

and finally, replacing here c_1 by $-\frac{1}{2}c_1$, we get

$$A_7 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Thus, we have brought A to the form (6.11) with $r = 2$. Consequently,

$$\text{rank}(A) = \text{rank}(A_7) = 2.$$

◀

Remark 6.4.1 As we have seen in Example 6.10 it can be a long process to reach the form (6.11), especially if the size of the matrix is large. But if on the way we found the rank of the matrix, then we can stop even before finding the final form (6.11), since our main goal is to find the rank of the matrix, not writing it as in (6.11). For instance, in Example 6.10, we can easily determine the rank from A_2 since in A_2, r_2 and r_3 are linearly dependent and r_1 and r_2 and linearly independent. This gives $\text{rank}(A_2) = 2 = \text{rank}(A)$. So, in the process of finding the form (6.11), by applying elementary row (or column) operations, it is very helpful to check, after each step, for linearly independent columns and linearly independent rows, since this together with the above theorems may help us finding the rank of a matrix quickly without even reaching the final form (6.11). We illustrate this in the following example.

Example 6.11

Find the rank of the matrix

$$A = \begin{bmatrix} 2 & -3 & -4 \\ 3 & 1 & 5 \\ -1 & 0 & -1 \\ 0 & 2 & 4 \end{bmatrix}.$$

■

Solution

First, using (6.7), we deduce that $\text{rank}(A) \leq 3$. Our goal now is to perform some elementary row and column operations so as to obtain a matrix in the form (6.11). We interchange the second row r_2 and the fourth row r_4, and obtain

$$A_1 = \begin{bmatrix} 2 & -3 & -4 \\ 0 & 2 & 4 \\ -1 & 0 & -1 \\ 3 & 1 & 5 \end{bmatrix}.$$

Now, replacing the third column c_3 in A_1 by $c_1 - c_3$, we get

$$A_2 = \begin{bmatrix} 2 & -3 & 6 \\ 0 & 2 & -4 \\ -1 & 0 & 0 \\ 3 & 1 & -2 \end{bmatrix}.$$

Interchanging here r_1 and r_3, we get

$$A_3 = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 2 & -4 \\ 2 & -3 & 6 \\ 3 & 1 & -2 \end{bmatrix}.$$

We see immediately that $c_3 = -2c_2$. Thus, we deduce that $\text{rank}(A_3) \leq 2$. On the other hand, we see that c_1 and c_2 are linearly independent. Then $\text{rank}(A_3) \geq 2$ (since the rank is the maximal number of linearly independent columns). Thus, we deduce that $\text{rank}(A) = 2$. ◀

Example 6.12
Find the rank of the matrix

$$A = \begin{bmatrix} 2 & 4 & 1 \\ -1 & 2 & 0 \\ 0 & 5 & -3 \end{bmatrix}.$$

■

Solution
Since A is a square matrix, we may first calculate the determinant of A and get

$$\det(A) = -29.$$

Since $\det(A) \neq 0$, Theorem 6.3.3 implies that $\text{rank}(A) = 3$. ◀

6.4.2 The Method of Minors for Finding the Rank of a Matrix

In Example 6.12, we have seen that when $\det(A) \neq 0$, the problem of finding the rank of A is easily solved. This approach is more convenient than the elementary row and column operations in Example 6.10. But up until now, it seems to work only for invertible square matrices. So, it is natural to look for a possible extension of this approach to matrices that are not necessarily square or invertible. In fact, it turns out that this method can be applied to an arbitrary matrix and we have the following theorem.

Theorem 6.4.2 (Minors and Rank)
Let A be a matrix in $\mathscr{M}_{m\times n}(\mathbb{K})$. Then, the rank of A is equal to $r, 1 \leq r \leq \min(m, n)$, if and only if there exists a nonzero minor[a] of order r which is the largest nonzero minor.

[a] See Definition 2.12 for the definition of the minors of a matrix.

Proof

Let A be the matrix written in the standard bases of $\mathbb{K}^n$ and $\mathbb{K}^m$ as

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \dots & a_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{bmatrix}.$$

Assume first that $\operatorname{rank}(A) = r$. Then, there exist r linearly independent column vectors $v_1, v_2, \dots, v_r$ (without loss of generality, we assume that these vectors are the first r columns of A). Let $\mathbf{B}_1 = \{e_1, e_2, \dots, e_m\}$ be the standard basis of $\mathbb{K}^m$.

First, if $r = m \leq n$, then $\mathbf{B}_2 = \{v_1, v_2, \dots, v_m\}$ constitutes a basis of $\mathbb{K}^m$ and we have

$$v_j = \sum_{i=1}^{m} a_{ij} e_i, \qquad 1 \leq j \leq m;$$

hence, the transition matrix from the basis $\mathbf{B}_1$ to the basis $\mathbf{B}_2$ is

$$\Lambda = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1m} \\ a_{21} & a_{22} & \dots & a_{2m} \\ \vdots & \vdots & \vdots & \vdots \\ a_{m1} & a_{n2} & \dots & a_{mm} \end{bmatrix}.$$

It is clear that Λ is invertible, so $\det(\Lambda) \neq 0$. Also, $\det(\Lambda)$ is a minor of A since the matrix Λ can be obtained from A by removing the last $n - m$ columns.

Second, if $r < m$, since the elements of $\mathbf{B}_2$ are linearly independent, Theorem 4.6.8 shows that there exist $v_{r+1}, \dots, v_m$ such that (we are allowed to choose $v_j = e_j$, $r + 1 \leq j \leq m$)

$$\mathbf{S} = \{v_1, v_2, \dots, v_r, e_{r+1}, \dots, e_m\}$$

is a basis of $\mathbb{K}^m$. In this case, the transition matrix from $\mathbf{B}$ to $\mathbf{S}$ takes the form

$$\Lambda = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1r} & 0 \dots 0 \\ a_{21} & a_{22} & \dots & a_{2r} & 0 \dots 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots \vdots \vdots \\ a_{r1} & a_{n2} & \dots & a_{rr} & 0 \dots 0 \\ a_{(r+1)1} & a_{(r+1)2} & \dots & a_{(r+1)r} & 1 \dots 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \ddots \vdots \\ a_{m1} & a_{m2} & \dots & a_{mr} & 0 \dots 1 \end{bmatrix}.$$

Since $\mathbf{S}$ is a basis of $\mathbb{K}^m$, Λ is invertible and $\det(\Lambda) \neq 0$. This determinant can be computed by using the last columns ($m - r$ times) as

$$\det(\Lambda) = \det \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1r} \\ a_{21} & a_{22} & \dots & a_{2r} \\ \vdots & \vdots & \vdots & \vdots \\ a_{r1} & a_{n2} & \dots & a_{rr} \end{bmatrix}.$$

Therefore, this minor is not zero.

Conversely, assume that there exists a nonzero minor of order r of the matrix A and that the minor corresponding to the rows $i_1, i_2, \dots, i_r$ is nonzero, that is the determinant of the matrix

$$B = \begin{bmatrix} a_{i_1 1} & a_{i_1 2} & \dots & a_{i_1 r} \\ a_{i_2 1} & a_{i_2 2} & \dots & a_{i_2 r} \\ \vdots & \vdots & \vdots & \vdots \\ a_{i_r 1} & a_{i_r 2} & \dots & a_{i_r r} \end{bmatrix}$$

is not zero ($\det(B) \neq 0$). Now, to show that $v_1, v_2, \dots, v_r$ are linearly independent, we take $\lambda_1, \lambda_2, \dots, \lambda_r$ in $\mathbb{K}$ such that

$$\lambda_1 v_1 + \lambda_2 v_2 + \cdots + \lambda_r v_r = 0_{\mathbb{K}^m}, \tag{6.14}$$

and we need to show that $\lambda_1 = \lambda_2 = \cdots = \lambda_r = 0_{\mathbb{K}}$.

Now, as before, each v_j, $1 \leq j \leq r$ can be written as a linear combination of the elements of the basis $\mathbf{B}_1 = \{e_1, e_2, \dots, e_m\}$ as

$$v_j = \sum_{i=1}^{m} a_{ij} e_i, \qquad 1 \leq j \leq r. \tag{6.15}$$

Taking into account (6.15), then (6.14) can be expressed as a linear system of the form

$$\begin{cases} a_{11}\lambda_1 + a_{12}\lambda_2 + \cdots + a_{1r}\lambda_r = 0, \\ a_{21}\lambda_1 + a_{22}\lambda_2 + \cdots + a_{2r}\lambda_r = 0, \\ \vdots \\ a_{m1}\lambda_1 + a_{m2}\lambda_2 + \cdots + a_{mr}\lambda_r = 0. \end{cases}$$

This leads, by retaining the rows $i_1, i_2, \dots, i_r$,

$$\begin{cases} a_{i_1 1}\lambda_1 + a_{i_1 2}\lambda_2 + \cdots + a_{i_1 r}\lambda_r = 0, \\ a_{i_2 1}\lambda_1 + a_{i_2 2}\lambda_2 + \cdots + a_{i_2 r}\lambda_r = 0, \\ \vdots \\ a_{i_r 1}\lambda_1 + a_{i_r 2}\lambda_2 + \cdots + a_{i_r r}\lambda_r = 0. \end{cases}$$

This last system has a unique solution $\lambda_1 = \lambda_2 = \cdots = \lambda_r = 0_{\mathbb{K}}$, since the matrix B is invertible. This shows that $v_1, v_2, \ldots, v_r$ are linearly independent, and thus $\text{rank}(A) = r$. This completes the proof of Theorem 6.4.2. □

Example 6.13

Use Theorem 6.4.2 to find the rank of the matrix

$$A = \begin{bmatrix} 2 & -3 & -4 \\ 3 & 1 & 5 \\ -1 & 0 & -1 \\ 0 & 2 & 4 \end{bmatrix}.$$

■

Solution

First, according to (6.7), it is clear that $\text{rank}(A) \leq 3$. We may easily check that the determinants of all the 3×3 submatrices are zero. On the other hand, we have

$$\det \begin{bmatrix} 3 & 1 \\ -1 & 0 \end{bmatrix} = 1 \neq 0.$$

Thus, $\text{rank}(A) = 2$. ◄

Example 6.14

Consider the matrix

$$A = \begin{bmatrix} a & 2 & -1 & b \\ 3 & 0 & 1 & -4 \\ 5 & 4 & -1 & 2 \end{bmatrix},$$

where a and b are real numbers.

1. Show that for any real numbers a and b, we have $\text{rank}(A) \geq 2$.
2. Find the values a and b for which $\text{rank}(A) = 2$.

■

Solution

1. Using (6.7), we deduce that $\text{rank}(A) \leq 3$. To show that $\text{rank}(A) \geq 2$ and according to Theorem 6.4.2, it suffices to find a nonzero minor of order 2. Indeed, we have

$$\det \begin{bmatrix} 3 & 0 \\ 5 & 4 \end{bmatrix} = 12 \neq 0.$$

Then, $\text{rank}(A) \geq 2$ for all values a and b.

2. To find the values a and b for which $\text{rank}(A) = 2$ we need to look for all values a and b that make the first row a linear combination of the second and the third rows. So, let λ and μ be two real numbers such that

$$(a, 2, -1, b) = \lambda(3, 0, 1, -4) + \mu(5, 4, -1, 2).$$

This leads to the system of equations

$$\begin{cases} 3\lambda + 5\mu = a, \\ 4\mu = 2, \\ \lambda - \mu = -1, \\ -4\lambda + 2\mu = b, \end{cases}$$

which gives $\lambda = -1/2, \mu = 1/2, a = 1$ and $b = 3$. So, if $a = 1$ and $b = 3$, then $\text{rank}(A) = 2$, otherwise, the three rows will be linearly independent and $\text{rank}(A) = 3$. ◀

6.5 Exercises

Exercise 6.1
Consider in $\mathscr{M}_3(\mathbb{R})$ the matrix

$$A = \begin{bmatrix} a & 0 & b \\ b & a & 0 \\ 0 & b & a \end{bmatrix}.$$

Find the rank of A according to the values of a and b. ■

Solution
First, if $a = b = 0$, then $\text{rank}(A) = 0$. Also, by (6.7), $\text{rank}(A) \leq 3$. On the other hand, we have

$$\det(A) = a^3 + b^3.$$

Hence, if $a \neq -b$, then $\det(A) \neq 0$, and Theorem 6.3.3 shows that $\text{rank}(A) = 3$.

Now, if $a = -b \neq 0$, then A becomes

$$A = \begin{bmatrix} a & 0 & -a \\ -a & a & 0 \\ 0 & -a & a \end{bmatrix}.$$

Adding the first column to the last column, we get

$$A_1 = \begin{bmatrix} a & 0 & 0 \\ -a & a & -a \\ 0 & -a & a \end{bmatrix}.$$

Clearly the second and third columns are linearly dependent. Hence, $\text{rank}(A_1) \leq 2$. On the other hand, since $a \neq 0$, then the first and second columns are linearly independent. Thus, $\text{rank}(A_1) \geq 2$. Consequently, if $a = -b \neq 0$, then $\text{rank}(A) = \text{rank}(A_1) = 2$. ◀

6

Exercise 6.2 (A Property of a Matrix of Rank 1)

1. Let A and B be two matrices in $\mathscr{M}_{m\times n}(\mathbb{K})$. Show that if $\text{rank}(B) = 1$, then

$$|\,\text{rank}(A + B) - \text{rank}(A)| \leq 1. \tag{6.16}$$

2. Let A be a square matrix in $\mathscr{M}_n(\mathbb{K})$ and B be a matrix in $\mathscr{M}_{n\times m}(\mathbb{K})$. Show that if A is invertible, then

$$\text{rank}(AB) = \text{rank}(B).$$

■

Solution

1. Since any matrix can be represented by a linear transformation and its rank is the rank of this transformation, then all the properties for rank and nullity of linear transformations obtained in ▶ Chap. 5 remain true for matrices. So, it is clear from Exercise 5.7 that

$$\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B).$$

This means, since $\text{rank}(B) = 1$, that

$$\text{rank}(A + B) - \text{rank}(A) \leq 1. \tag{6.17}$$

On the other hand, Exercise 5.7 also shows that

$$\text{rank}(A) - \text{rank}(B) \leq \text{rank}(A + B).$$

Again, since $\text{rank}(B) = 1$, it follows that

$$\text{rank}(A + B) - \text{rank}(A) \geq -1. \tag{6.18}$$

Combining (6.17) and (6.18), we obtain inequality (6.16).

2. It is clear that (5.21) can be written (in matrix form) as

$$\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B)). \tag{6.19}$$

Since A is invertible, we have $B = A^{-1}AB$, so applying (6.19), we get

$$\operatorname{rank}(B) = \operatorname{rank}(A^{-1}AB) \leq \operatorname{rank}(AB).$$

Consequently, we deduce that

$$\operatorname{rank}(AB) \leq \operatorname{rank}(B) \leq \operatorname{rank}(AB),$$

which finally yields $\operatorname{rank}(AB) = \operatorname{rank}(B)$. ◀

Exercise 6.3 (Frobenius Inequality and Sylvester Law of Nullity)
Let A be a matrix in $\mathscr{M}_{m\times r}(\mathbb{K})$, B be a matrix in $\mathscr{M}_{r\times p}(\mathbb{K})$, and C be a matrix in $\mathscr{M}_{p\times n}(\mathbb{K})$.

1. Show that

$$\operatorname{rank}(AB)+\operatorname{rank}(BC) \leq \operatorname{rank}(B)+\operatorname{rank}(ABC) \qquad \text{(Frobenius' inequality).} \tag{6.20}$$

2. Deduce that for any two matrices A in $\mathscr{M}_{m\times r}(\mathbb{K})$ and K in $\mathscr{M}_{r\times n}(\mathbb{K})$

$$\operatorname{rank}(AK) \geq \operatorname{rank}(A) + \operatorname{rank}(K) - r. \tag{6.21}$$

3. Prove that if A and B are square matrices in $\mathscr{M}_n(\mathbb{K})$, then

$$\operatorname{null}(AB) \leq \operatorname{null}(A) + \operatorname{null}(B) \qquad \text{(Sylvester's law of nullity).} \tag{6.22}$$

■

Solution

1. Inequality (6.20) is a direct consequence of (5.27).

2. Applying the Frobenius inequality for $p = r$, $C = K$ and $B = I_r$, we obtain

$$\operatorname{rank}(A) + \operatorname{rank}(K) \leq \operatorname{rank}(B) + \operatorname{rank}(AK).$$

Since $\operatorname{rank}(B) = r$, (6.21) holds.

3. If A is square matrix of order n, then Theorem 5.2.10 implies that

$$\operatorname{null}(A) = n - \operatorname{rank}(A).$$

Thus, using (6.21) for $K = B$ and $r = n$, we get

$$n - \operatorname{rank}(AB) \leq n - \operatorname{rank}(A) - \operatorname{rank}(B) + n.$$

This leads to (6.22), by applying once again Theorem 5.2.10. ◀

Exercise 6.4 (Rank of Idempotent Matrices)
Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$ such that $\operatorname{rank}(A) = r$.
1. Prove that there exist matrices B in $\mathscr{M}_{n\times r}(\mathbb{K})$ and C in $\mathscr{M}_{r\times n}(\mathbb{K})$ such that

$$A = BC \qquad \text{and} \qquad \operatorname{rank}(B) = \operatorname{rank}(C) = r.$$

The matrix A in $\mathscr{M}_n(\mathbb{K})$ is called idempotent[a] (see Exercise 1.5) if $A^2 = A$.
2. Show that if A is an idempotent matrix, then

$$\operatorname{rank}(A) = \operatorname{tr}(A).^{b}$$

3. Deduce that if A is an idempotent matrix in $\mathscr{M}_n(\mathbb{K})$ with $\operatorname{rank}(A) = n$, then $A = I_n$.
4. Show that if A is idempotent with $\operatorname{rank}(A) = r$, then

$$\operatorname{rank}(I_n - A) = n - r.$$

5. Let A and B be two idempotent matrices in $\mathscr{M}_n(\mathbb{K})$. Show that if $AB = BA = \mathbf{0}$, then $A + B$ is idempotent and

$$\operatorname{rank}(A + B) = \operatorname{rank}(A) + \operatorname{rank}(B).$$

6. Find the rank of the matrix

$$A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}.$$

■

[a] Idempotent matrices are the matrices associated to projections.
[b] In fact if $A^2 = kA$, then we have $\operatorname{tr}(A) = k \cdot \operatorname{rank}(A)$.

Solution
1. Since $\operatorname{rank}(A) = r$, then there exists r linearly independent column vectors $B_1, B_2, \ldots, B_r$ of A. Then, these vectors form a basis of the column space $\mathscr{R}(A)$. We introduce the matrix B whose columns are the vectors $B_1, B_2, \ldots, B_r$ as

$$B = [B_1, B_2, \ldots, B_r].$$

It is clear that B is in $\mathscr{M}_{n\times r}(\mathbb{K})$. Since $B_1, B_2, \ldots, B_r$ form a basis, they are linearly independent. Hence, $\operatorname{rank}(B) = r$.

Now, it is clear that any column of A, say the ith column A_i, may be expressed as

$$A_i = BC_i$$

where C_i is the vector of the coefficients of the linear combination of $B_1, B_2, \ldots, B_r$ that gives A_i. Denoting

$$C = [C_1, C_2, \ldots, C_n]$$

and

$$A = [A_1, A_2, \ldots, A_n],$$

then we have

$$A = BC.$$

Finally, (6.19) shows that

$$r = \operatorname{rank}(A) = \operatorname{rank}(BC) \leq \operatorname{rank}(C) \leq r.$$

Hence, $\operatorname{rank}(C) = r$.

2. Now, since A is idempotent, we have

$$A^2 = BCBC = A = BC. \tag{6.23}$$

Using (6.19), we deduce that

$$\operatorname{rank}(CB) \geq \operatorname{rank}(A) = r.$$

Since CB is a square matrix in $\mathscr{M}_r(\mathbb{K})$, $\operatorname{rank}(CB) \leq r$. This shows that

$$\operatorname{rank}(CB) = r,$$

so CB is invertible (Theorem 6.3.2). Hence, multiplying (6.23) from the left by $(CB)^{-1}C$ and from the right by $B(CB)^{-1}$, we obtain

$$CB = I_r.$$

Now using Theorem 1.2.14, we have

$$\operatorname{tr}(A) = \operatorname{tr}(BC) = \operatorname{tr}(CB) = \operatorname{tr}(I_r) = r = \operatorname{rank}(A).$$

3. It is clear that, according to Theorem 6.3.4, A is invertible and we have

$$A = I_n A = A^{-1}AA = A^{-1}A^2 = A^{-1}A = I_n.$$

4. We have seen in Exercise 1.5 that if A is idempotent, then $I_n - A$ is also idempotent. Thus, using assertion (1), we get

$$\operatorname{rank}(I_n - A) = \operatorname{tr}(I_n - A) = \operatorname{tr}(I_n) - \operatorname{tr}(A) = n - r.$$

5. We have

$$(A + B)^2 = A^2 + B^2 + AB + BA = A^2 + B^2 = A + B,$$

since $AB = BA = \mathbf{0}$ and A and B are idempotent. This shows that $A + B$ is idempotent.

Now, using assertion (2) together with Theorem 1.2.14, we have

$$\operatorname{rank}(A + B) = \operatorname{tr}(A + B) = \operatorname{tr}(A) + \operatorname{tr}(B) = \operatorname{rank}(A) + \operatorname{rank}(B).$$

6. We may easily check that $A^2 = A$ and thus A is idempotent. Then applying (1), we get

$$\operatorname{rank}(A) = \operatorname{tr}(A) = 2 + 3 - 3 = 2.$$

◀

Exercise 6.5

Let f be an endomorphism of $\mathbb{R}^3$ whose matrix with respect to the standard basis $\mathbf{B} = \{e_1, e_2, e_3\}$ is

$$M(f) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & -3 & 3 \end{bmatrix}.$$

1. Show that f is an automorphism and find its inverse f^{-1}.
2. Find a basis $\mathbf{S} = \{s_1, s_2, s_3\}$ in $\mathbb{R}^3$ such that

$$f(s_1) = s_1, \quad f(s_2) = s_1 + s_2, \quad f(s_3) = s_2 + s_3.$$

3. Find the transition matrix Λ from the basis $\mathbf{B}$ to the basis $\mathbf{S}$ and find Λ^{-1}.

■

Solution

First, we have $\operatorname{rank}(f) = \dim_{\mathbb{R}} \operatorname{Im}(f) = \operatorname{rank}(M(f))$. Now, we have by a simple computation

$$\det(M(f)) = 1 \neq 0.$$

This immediately shows, by using Theorem 6.3.3, that $\operatorname{rank}(M(f)) = 3$. Consequently,

$$\dim_{\mathbb{R}} \operatorname{Im}(f) = \dim_{\mathbb{R}} \mathbb{R}^3.$$

Hence, since $\mathrm{Im}(f)$ is a subspace of $\mathbb{R}^3$, Theorem 4.6.7 implies that

$$\mathrm{Im}(f) = \mathbb{R}^3.$$

Now Theorem 5.2.9 shows that f is surjective and hence, applying Theorem 5.3.3, we deduce that f is bijective and therefore an automorphism.

Applying Theorem 6.1.1, we have

$$M(f^{-1}) = (M(f))^{-1}.$$

So, we can easily, by using the methods in ▶ Chap. 1 or ▶ Chap. 2, find that

$$(M(f))^{-1} = \begin{bmatrix} 3 & -3 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Consequently, we have (see Remark 6.1.2),

$$f^{-1}(u,v,w) = (M(f))^{-1}\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 3 & -3 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} 3u - 3v + w \\ u \\ v \end{bmatrix},$$

Thus, we deduce that

$$f^{-1}(u,v,w) = (3u - 3v + w, u, v).$$

2. It is clear that

$$f(u,v,w) = M(f)\begin{bmatrix} u \\ v \\ w \end{bmatrix} = \begin{bmatrix} v \\ w \\ u - 3v + 3w \end{bmatrix},$$

so

$$f(u,v,w) = (v, w, u - 3v + 3w).$$

Let $s_1 = (u_1, v_1, w_1)$ then $f(s_1) = s_1$ implies that

$$\begin{cases} v_1 = u_1, \\ w_1 = v_1, \\ u_1 - 3v_1 + 3w_1 = w_1. \end{cases}$$

This means that $u_1 = v_1 = w_1$. Consequently, $s_1 = (u_1, u_1, u_1) = u_1(1, 1, 1)$. So, we can choose $s_1 = (1, 1, 1)$. Similarly, we have $f(s_2) = f(u_2, v_2, w_2) = s_1 + s_2$, whence

$$\begin{cases} v_2 = 1 + u_2, \\ w_2 = 1 + v_2, \\ u_2 - 3v_2 + 3w_2 = 1 + w_2. \end{cases}$$

So, $s_2 = (u_2, u_2 + 1, 2 + u_2)$, and we can choose $s_2 = (0, 1, 2)$. By the same method and since $f(s_3) = f(u_3, v_3, w_3) = s_2 + s_3$, we obtain the system of equations

$$\begin{cases} v_3 = u_3, \\ w_2 = 1 + v_3, \\ u_3 - 3v_3 + 3w_3 = 2 + w_3. \end{cases}$$

Then, we get $s_3 = (u_3, u_3, 1 + u_3)$. As before, we can choose $s_3 = (0, 0, 1)$.

3. It is clear that the entries of the column $j, j = 1, 2, 3$ of the transition matrix Λ are the components of s_j with respect to the basis $\mathbf{B}$. That is,

$$\Lambda = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 2 & 1 \end{bmatrix}.$$

It is clear that this matrix is invertible. To find Λ^{-1}, we simply need to find the components of $e_j, j = 1, 2, 3$ with respect to the basis $\mathbf{S}$. We have

$$\begin{cases} s_1 = e_1 + e_2 + e_3, \\ s_2 = e_2 + 2e_3, \\ s_3 = e_3. \end{cases}$$

Then, this leads to

$$\begin{cases} e_1 = s_1 - s_2 + s_3, \\ e_2 = s_2 - 2s_3, \\ e_3 = s_3. \end{cases}$$

Consequently, we obtain

$$\Lambda^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 1 & -2 & 1 \end{bmatrix}.$$

◄

Exercise 6.6

Let A be a matrix in $\mathscr{M}_{m\times n}(\mathbb{K})$ and B be a matrix in $\mathscr{M}_{n\times m}(\mathbb{K})$. Show that if $m > n$, then

$$\det(AB) = 0.$$

■

Solution

First, using (5.21) together with Theorem 6.1.1, we deduce that

$$\operatorname{rank}(AB) \leq \min(\operatorname{rank}(A), \operatorname{rank}(B)). \tag{6.24}$$

Also, keeping in mind (6.7), we have $\operatorname{rank}(A) \leq \min(n, m)$. Since $m > n$, we deduce that $\operatorname{rank}(A) \leq n$. Consequently, applying (6.24), we obtain

$$\operatorname{rank}(AB) \leq \operatorname{rank}(A) \leq n < m.$$

Since AB is a square matrix in $\mathscr{M}_m(\mathbb{K})$, Theorem 6.3.3 and the fact that $\operatorname{rank}(AB) \neq m$ imply that $\det(AB) = 0$. ◄

Exercise 6.7 (Consistency and Rank)

Consider the system of m linear equations in n unknowns of the form

$$AX = b, \tag{6.25}$$

where A is a matrix in $\mathscr{M}_{m\times n}(\mathbb{K})$, X is a vector in $\mathscr{M}_{n\times 1}(\mathbb{K})$, and b is a vector in $\mathscr{M}_{m\times 1}(\mathbb{K})$. The system (6.25) is said to be consistent (or solvable) if it possesses at least one solution.

1. Show that system (6.25) is consistent if and only if

$$\operatorname{rank}\left[\begin{array}{c|c} A & b \end{array}\right] = \operatorname{rank}(A).$$

2. Show that the system

$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & -3 & 7 \\ 1 & 4 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 11 \\ -2 \\ -7 \end{bmatrix} \tag{6.26}$$

is inconsistent.

■

Solution

1. First, if the system is consistent, then it has at least one solution. Then, in this case b should be a linear combination of the columns of A, that is

$$b = x_1A_1 + x_2A_2 + \cdots + x_nA_n,$$

where $A_1, A_2, \ldots, A_m$ are the columns of A. Therefore, the rank of $\left[\begin{array}{c|c} A & b \end{array}\right]$ is the number of linearly independent elements from the set $\{A_1, A_2, \ldots, A_n\}$, which is exactly (by definition) the rank of A.

Conversely, assume that $\text{rank}\left[\begin{array}{c|c} A & b \end{array}\right] = \text{rank}(A)$, then b is a linear combination of the columns of A, and the coefficients of this combination provide a solution to (6.25). This means that (6.25) is consistent.

2. We have

$$\det\begin{bmatrix} 1 & 2 & 0 \\ 2 & -3 & 7 \\ 1 & 4 & -2 \end{bmatrix} = 0,$$

and

$$\det\begin{bmatrix} 1 & 2 \\ 2 & -3 \end{bmatrix} = -7 \neq 0.$$

Thus, $\text{rank}(A) = 2$. On the other hand, for the matrix

$$\begin{bmatrix} 1 & 2 & 0 & 11 \\ 2 & -3 & 7 & -2 \\ 1 & 4 & -2 & -7 \end{bmatrix}$$

we have

$$\det\begin{bmatrix} 1 & 2 & 11 \\ 2 & -3 & -2 \\ 1 & 4 & -7 \end{bmatrix} = 174 \neq 0.$$

Consequently, $\text{rank}\left[\begin{array}{c|c} A & b \end{array}\right] = 3$. Hence, the system (6.26) is inconsistent. ◄

Exercise 6.8 (Diagonally Dominant Matrix)

Let $A = (a_{ij})$, $1 \leq i, j \leq n$, be a matrix in $\mathscr{M}_n(\mathbb{C})$. Show that if for all $i = 1, \ldots, n$,

$$|a_{ii}| > \sum_{i \neq j} |a_{ij}|, \tag{6.27}$$

then A is invertible. Property (6.27) means that the magnitude of the diagonal entry in a row is larger than or equal to the sum of all the magnitudes of all the other, non-diagonal entries in that row. A matrix satisfying this property is called *diagonally dominant of its rows*.

As an application to the above result, show that the matrix

$$A = \begin{bmatrix} 3 & -2 & 0 \\ 1 & -3 & 1 \\ -1 & 2 & 4 \end{bmatrix}$$

is invertible. ■

Solution

To show that A is invertible it is equivalent to prove that the null space

$$\mathcal{N}(A) = \{X \in \mathbb{C}^n, \text{ such that } AX = 0_{\mathbb{C}^n}\}$$

is $\{0_{\mathbb{C}^n}\}$, since in this case $\text{rank}(A) = n$ and Theorem 6.3.2 shows that A is invertible. Assume that A is not invertible. Then, there exists a vector X in $\mathcal{N}(A)$ with

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \neq 0_{\mathbb{C}^n}.$$

Let x_{i_0} be the component in X satisfying

$$|x_{i_0}| = \max |x_i|, \quad i = 1, \ldots, n.$$

It is clear that since $X \neq 0_{\mathbb{C}^n}$, then $|x_{i_0}| > 0$. On the other hand, since X is in $\mathcal{N}(A)$, we have $AX = 0_{\mathbb{C}^n}$. This implies that for all $i = 1, \ldots, n$, we have

$$\sum_{j=1}^{n} a_{ij} x_j = 0.$$

Now, in particular, for $i = i_0$, we have

$$\sum_{j=1}^{n} a_{i_0 j} x_j = 0,$$

which yields

$$|a_{i_0 i_0} x_{i_0}| = \Big| - \sum_{j \neq i_0} a_{i_0 j} x_j \Big| \leq \sum_{j \neq i_0} |a_{i_0 j}| \cdot |x_j| \leq |x_{i_0}| \sum_{j \neq i_0} |a_{i_0 j}|.$$

Since $|x_{i_0}| \neq 0$, we have

$$|a_{i_0 i_0}| \leq \sum_{j \neq i_0} |a_{i_0 j}|.$$

Consequently, (6.27) is not satisfied for $i = i_0$. Thus, we deduce that if (6.27) holds, then A has to be invertible.

For the application, we have for the matrix A,

$$|3| > |1| + |0|, \quad |-3| > |1| + |1|, \quad |4| > |-1| + |2|.$$

Consequently, A is invertible. The converse of the above result is not true. For instance, the matrix

$$\begin{bmatrix} 3 & -2 & 0 \\ 1 & -4 & 4 \\ -1 & 2 & 4 \end{bmatrix}$$

is invertible, although (6.27) is not satisfied, since $|-4| < |1| + |4|$. ◀

Exercise 6.9 (Idempotent Matrix)

Let P be a matrix in $\mathscr{M}_n(\mathbb{K})$. We say that P is idempotent if $P^2 = P$.

Now, let A be a matrix in $\mathscr{M}_n(\mathbb{R})$ satisfying for α and β in $\mathbb{R}$ with $\alpha \neq \beta$,

$$(A - \alpha I_n)(A - \beta I_n) = \mathbf{0}. \tag{6.28}$$

1. Show that there exists two elements λ and μ in $\mathbb{K}$ such that the two matrices P and Q defined as:

   $$P = \lambda(A - \alpha I_n) \quad \text{and} \quad Q = \mu(A - \beta I_n)$$

   are idempotent. Show that $P + Q = I_n$.
2. Write the matrix A in terms of P and Q and deduce A^k for any positive integer k.
3. Show that if $\alpha\beta \neq 0$, then A is invertible and find A^{-1}.
4. Consider for $m \neq 0$ the matrix

   $$A = \begin{bmatrix} 0 & m & m^2 \\ 1/m & 0 & m \\ 1/m^2 & 1/m & 0 \end{bmatrix}. \tag{6.29}$$

   Find α and β such that A satisfies (6.28) and deduce A^k for any integer k.

■

Solution

1. We need to find λ and μ such that $P^2 = P$ and $Q^2 = Q$. Indeed, we have

$$\begin{aligned} P^2 &= \lambda(A - \alpha I_n)(\lambda(A - \alpha I_n)) \\ &= \lambda^2(A^2 - 2\alpha A + \alpha^2 I_n) \\ &= \lambda^2(\alpha - \beta)(-A + \alpha I_n), \end{aligned}$$

where we have used (6.28). Thus, $P^2 = P$ if and only if $\lambda = -\dfrac{1}{\alpha - \beta}$. By the same argument, we can show that $Q^2 = Q$ if and only if $\mu = \dfrac{1}{\alpha - \beta}$. Now, it is clear that

$$P + Q = -\frac{1}{\alpha - \beta}(A - \alpha I_n) + \frac{1}{\alpha - \beta}(A - \beta I_n) = I_n.$$

2. We can easily deduce from the definition of P that

$$\begin{aligned} A &= -(\alpha - \beta)P + \alpha I_n \\ &= \beta P + \alpha Q. \end{aligned}$$

Now, since $P^2 = P$, $Q^2 = Q$, and

$$PQ = \lambda\mu(A - \alpha I_n)(A - \beta I_n) = \mathbf{0},$$

by (6.28), then, it follows that for any positive integer k, we have

$$A^k = \beta^k P + \alpha^k Q.$$

3. Now, if $\alpha\beta \neq 0$, then, we may rewrite (6.28) as

$$A\left(\frac{\alpha + \beta}{\alpha\beta} I_n - \frac{1}{\alpha\beta} A\right) = \left(\frac{\alpha + \beta}{\alpha\beta} I_n - \frac{1}{\alpha\beta} A\right) A = I_n.$$

Hence A is invertible and the uniqueness of the inverse shows that

$$A^{-1} = \frac{\alpha + \beta}{\alpha\beta} I_n - \frac{1}{\alpha\beta} A.$$

In terms of P and Q, the above formula can be rewritten as

$$A^{-1} = \frac{1}{\beta} P + \frac{1}{\alpha} Q.$$

4. For A defined as in (6.29), we have

$$(A - \alpha I_n)(A - \beta I_n) = \mathbf{0},$$

whence

$$\begin{bmatrix} \alpha & m & m^2 \\ 1/m & \alpha & m \\ 1/m^2 & 1/m & \alpha \end{bmatrix} \begin{bmatrix} \beta & m & m^2 \\ 1/m & \beta & m \\ 1/m^2 & 1/m & \beta \end{bmatrix} = \begin{bmatrix} 0\ 0\ 0 \\ 0\ 0\ 0 \\ 0\ 0\ 0 \end{bmatrix}.$$

That is,

$$\begin{bmatrix} \alpha\beta + 2 & m(\alpha + \beta + 1) & m^2(\alpha + \beta + 1) \\ \dfrac{\alpha + \beta + 1}{m} & \alpha\beta + 2 & m(\alpha + \beta + 1) \\ \dfrac{\alpha + \beta + 1}{m^2} & \dfrac{\alpha + \beta + 1}{m} & \alpha\beta + 2 \end{bmatrix} = \begin{bmatrix} 0\ 0\ 0 \\ 0\ 0\ 0 \\ 0\ 0\ 0 \end{bmatrix}.$$

Since $m \neq 0$, this last equation leads to the two equations

$$\alpha\beta + 2 = 0 \qquad \text{and} \qquad \alpha + \beta + 1.$$

That is, $\alpha = 2$ and $\beta = -1$. Hence, the matrices P and Q now becomes

$$P = -\frac{1}{3}(A - 2I_3) = \begin{bmatrix} 2/3 & -m/3 & -m^2/3 \\ -1/(3m) & 2/3 & -m/3 \\ -1/(3m^2) & -1/(3m) & 2/3 \end{bmatrix}$$

and

$$Q = \frac{1}{3}(A + I_3) = \begin{bmatrix} 1/3 & m/3 & m^2/3 \\ 1/(3m) & 1/3 & m/3 \\ 1/(3m^2) & 1/(3m) & 1/3 \end{bmatrix}.$$

Thus $A = -P + 2Q$ and it is invertible, and therefore for any integer k, we have

$$A^k = (-1)^k P + 2^k Q = \begin{bmatrix} \dfrac{2(-1)^k + 2^k}{3} & m\dfrac{2^k - (-1)^k}{3} & m^2\dfrac{2^k - (-1)^k}{3} \\ \dfrac{2^k - (-1)^k}{3m} & \dfrac{2(-1)^k + 2^k}{3} & m\dfrac{2^k - (-1)^k}{3} \\ \dfrac{2^k - (-1)^k}{3m^2} & \dfrac{2^k - (-1)^k}{3m} & \dfrac{2(-1)^k + 2^k}{3} \end{bmatrix}.$$

◄

Eigenvalues and Eigenvectors

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7.1 Definitions

In the previous chapters, we have defined some numbers associated to a matrix, such as the determinant, trace, and rank. In this chapter, we focus on scalars and vectors known as *eigenvalues* and *eigenvectors*. The eigenvalues and eigenvectors have many important applications, in particular, in the study of differential equations.

Let E be a vector space over a field $\mathbb{K}$ with $\dim_{\mathbb{K}} E = n$, and f be an endomorphism in $\mathscr{L}(E)$. We have seen in Example 6.9 that it might be possible to find a basis of E in which the matrix $M(f)$ associated to f is diagonal. So, the question is: *Does such basis always exist, and if so, how can we find it? In addition, is the matrix $M(f)$ always diagonal in this basis?* One of the main goals here is to answer these questions.

Definition 7.1.1 (Eigenvector and Eigenvalue of an Endomorphism)
Let E be a vector space over a field $\mathbb{K}$ and f be an endomorphism in $\mathscr{L}(E)$. Let u be an element in E such that $u \neq 0_E$. Then we say that u is an *eigenvector* of f if there exists λ in $\mathbb{K}$ such that

$$f(u) = \lambda u. \tag{7.1}$$

In this case, we say that λ is an *eigenvalue* of f, and u is an associated eigenvector.

It is clear that if $u = 0_E$, then (7.1) is satisfied for any λ in $\mathbb{K}$.

Theorem 7.1.1 (Uniqueness of the Eigenvalue)
Let f be an endomorphism in $\mathscr{L}(E)$ and $u \neq 0_E$ be an eigenvector of f. Then, the eigenvalue λ of f associated to the eigenvector u is unique.[a]

[a] We can deduce here that two different eigenvalues of f cannot have the same associated eigenvector u.

Proof
Assume that there exists another eigenvalue μ associated to u and satisfying (7.1). Then

$$\lambda u = \mu u.$$

This gives, $(\lambda - \mu)u = 0_E$. This yields, by using Theorem 4.2.1, $\lambda - \mu = 0_{\mathbb{K}}$, which shows that $\lambda = \mu$. This gives the uniqueness of the eigenvalue. □

We have seen above that the eigenvalue λ associated to the eigenvector u is unique. On the other hand, the eigenvector u is not unique. In fact, if u is an eigenvector, then for any $\alpha \neq 0_{\mathbb{K}}$, αu is also an eigenvector, since

$$f(\alpha u) = \alpha f(u) = \alpha(\lambda u) = \lambda(\alpha u).$$

Moreover, if u and v are two eigenvectors associated to the same eigenvalue λ, then $u + v$ is also an eigenvector associated to λ, since

$$f(u + v) = f(u) + f(v) = \lambda u + \lambda v = \lambda(u + v).$$

Thus, if λ is an eigenvalue of f, then we may construct a whole subspace $V(\lambda) \neq \{0_E\}$ of E associated to λ and we have the following definition.

Definition 7.1.2
Let λ be an eigenvalue of f as in Definition 7.1.1. Then the *eigenspace* $V(\lambda)$ is defined to be the set of all u in E such that $f(u) = \lambda u$.

Now, we introduce an important property of the eigenspace.

Theorem 7.1.2
Let f be an endomorphism as in Definition 7.1.1. Let λ_1 and λ_2 be two eigenvalues of f such that $\lambda_1 \neq \lambda_2$. Then we have

$$V(\lambda_1) \cap V(\lambda_2) = \{0_E\}. \tag{7.2}$$

Proof
Let u be an element of $V(\lambda_1) \cap V(\lambda_2)$. Then we have

$$f(u) = \lambda_1 u = \lambda_2 u.$$

That is $(\lambda_1 - \lambda_2)u = 0_E$. Since $\lambda_1 \neq \lambda_2$, we deduce that $u = 0_E$, which completes the proof. □

Notation When there is no confusion, we write vectors in $\mathbb{K}^n$ as

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \quad \text{or simply as} \quad X = (x_1, x_2, \ldots, x_n),$$

where $x_1, x_2, \ldots, x_n$ are elements of $\mathbb{K}$.

Example 7.1
Consider the endomorphism f defined as follows:

$$\begin{aligned} f : \mathbb{R}^3 &\to \mathbb{R}^3, \\ (x, y, z) &\mapsto (2x + z, 4y + 5z, -z). \end{aligned}$$

We have

$$f(2, 0, 0) = (4, 0, 0) = 2(2, 0, 0).$$

Consequently, $\lambda = 2$ is an eigenvalue of f and the vector $(2, 0, 0)$ is an associated eigenvector. ■

Example 7.2
Consider the endomorphism f defined as follows:

$$\begin{aligned} f : \mathbb{R}^3 &\to \mathbb{R}^3, \\ (x, y, z) &\mapsto (x - y, -x + y + z, 3z). \end{aligned}$$

Find its eigenvalues and the corresponding eigenvectors and eigenspaces. ■

Solution
Let λ be an eigenvalue of f, i.e.,

$$f(x, y, z) = \lambda(x, y, z) = (\lambda x, \lambda y, \lambda z).$$

This yields the system of equations

$$\begin{cases} x - y = \lambda x, \\ -x + y + z = \lambda y, \\ 3z = \lambda z. \end{cases}$$

Solving this system gives

$$\begin{aligned} &\lambda = 0, && \text{and} && x = y,\ z = 0 \\ &\lambda = 2, && \text{and} && x = -y,\ z = 0, \\ &\lambda = 3, && \text{and} && y = -2x,\ z = -3x. \end{aligned}$$

Thus, the eigenvalues are $\lambda_1 = 0, \lambda_2 = 2$ and $\lambda_3 = 3$ and the eigenspaces are

$$V(\lambda_1) = \{(x, x, 0)\}, \quad V(\lambda_2) = \{(x, -x, 0)\}, \quad V(\lambda_3) = \{(x, -2x, -3x)\},$$

with $x \in \mathbb{R}$. Clearly, $V(\lambda_1)$ is spanned by the vector $X_1 = (1, 1, 0)$, $V(\lambda_2)$ by the vector $X_2 = (1, -1, 0)$, and $V(\lambda_3)$ by the vector $X_3 = (1, -2, -3)$. Thus, X_1, X_2 and X_3 are the eigenvectors associated to the eigenvalues λ_1, λ_2, and λ_3, respectively. ◀

7

Example 7.3 (Eigenvalues of a Projection)
Let E be a finite-dimensional vector space over a field $\mathbb{K}$. Show that if f is a projection in $\mathscr{L}(E)$, then its eigenvalues are $0_{\mathbb{K}}$ or 1. ■

Solution
We have defined in Exercise 5.1 a projection as an element f in $\mathscr{L}(E)$ that satisfies $f \circ f = f$. So, let λ be an eigenvalue of f, i.e.,

$$f(u) = \lambda u,$$

for some u in E with $u \neq 0_E$. Then

$$(f \circ f)(u) = f(f(u)) = f(\lambda u) = \lambda f(u) = \lambda^2 u.$$

On the other hand, since f is a projection, we have

$$f(f(u)) = f(u) = \lambda u.$$

Combining the two identities above, we obtain $\lambda u = \lambda^2 u$. Since $u \neq 0_E$, it follows that $\lambda = 1$ or $\lambda = 0_{\mathbb{K}}$. ◀

7.2 Properties of Eigenvalues and Eigenvectors

By (7.1) λ being an eigenvalue of the endomorphism f means that

$$f(u) = \lambda u, \quad \text{with} \quad u \neq 0_E.$$

This is equivalent to

$$(f - \lambda \operatorname{Id}_E)(u) = 0_E, \quad \text{with} \quad u \neq 0_E.$$

This is also equivalent to say that $\operatorname{Ker}(f - \lambda \operatorname{Id}_E)$ contains u and since $u \neq 0_E$, then we deduce that λ is an eigenvalue of f if and only if $\operatorname{Ker}(f - \lambda \operatorname{Id}_E) \neq \{0_E\}$. Hence, Theorem 5.2.4 shows that $f - \lambda \operatorname{Id}_E$ is not injective, and therefore is not bijective. Hence, $f - \lambda \operatorname{Id}_E$ is not invertible.

So, we have already proved the following statement.

Theorem 7.2.1 (Characterization of Eigenvalues of an Endomorphism)
Let E be a finite dimensional vector space over a field $\mathbb{K}$ and f be an endomorphism in $\mathscr{L}(E)$. Then the following statements are equivalent:

1. *λ is an eigenvalue of f.*
2. *The endomorphism $f - \lambda \operatorname{Id}_E$ is not invertible.*

We go on to investigate some properties of the eigenvalues and eigenvectors. First, looking at Example 7.2 one more time, we see that the eigenvectors X_1, X_2, and X_3 are linearly independent and thus according to Theorem 4.6.5, they form a basis of $\mathbb{R}^3$. In fact, this turns out to be always true if the eigenvalues are distinct:

Theorem 7.2.2 (Eigenvectors Associated to Distinct Eigenvalues Form a Basis)
Let E be a vector space over a field $\mathbb{K}$ such that $\dim_{\mathbb{K}} E = n$ and f be an endomorphism in $\mathscr{L}(E)$. Suppose that f has n distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ and let $u_1, u_2, \ldots, u_n$ be the associated eigenvectors, respectively. Then the set

$$\mathbf{B} = \{u_1, u_2, \ldots, u_n\}$$

is a basis of E.

Proof
Let $\alpha_1, \alpha_2, \ldots, \alpha_n$ be n elements of $\mathbb{K}$ satisfying

$$\alpha_1 u_1 + \alpha_2 u_2 + \cdots + \alpha_n u_n = 0_E. \tag{7.3}$$

Now, let

$$g_i = f - \lambda_i \operatorname{Id}_E, \qquad i = 1, 2, \ldots, n.$$

Applying g_1 to (7.3) and using the fact that $g_1(u_1) = 0_E$ (since λ_1 is an eigenvalue of f) and

$$g_1(u_j) = (\lambda_j - \lambda_1)u_j, \qquad j = 2, \dots n,$$

we obtain

$$\sum_{j=2}^{n} (\lambda_j - \lambda_1)\alpha_j u_j = 0_E. \tag{7.4}$$

Now, applying g_2 to (7.4), we obtain, as above,

$$\sum_{j=3}^{n} (\lambda_j - \lambda_1)(\lambda_j - \lambda_2)\alpha_j u_j = 0_E. \tag{7.5}$$

After $n-1$ such operations, by applying each time g_i, $i = 2, \dots, n$, to

$$\sum_{j=i}^{n} \left(\prod_{k=1}^{i-1} (\lambda_j - \lambda_k) \right) \alpha_j u_j = 0_E, \tag{7.6}$$

we obtain

$$(\lambda_n - \lambda_1)(\lambda_n - \lambda_2) \cdots (\lambda_n - \lambda_{n-1})\alpha_n u_n = 0_E.$$

This gives $\alpha_n = 0_{\mathbb{K}}$, since all the eigenvalues are distinct. Since the ordering of the eigenvalues and eigenvectors is arbitrary, we can easily verify that $\alpha_1 = \alpha_2 = \cdots = \alpha_{n-1} = 0_{\mathbb{K}}$. This shows that the set $\mathbf{B} = \{u_1, u_2, \dots, u_n\}$ is linearly independent, and since $\dim_{\mathbb{K}} E = n$, Theorem 4.6.5 implies that $\mathbf{B}$ is a basis of E. □

Corollary 7.2.3 *Let E be a vector space over a field $\mathbb{K}$ such that $\dim_{\mathbb{K}} E = n$ and f be an endomorphism in $\mathscr{L}(E)$. Then f has at most n distinct eigenvalues.*

Proof

Suppose that $\lambda_1, \lambda_2, \dots, \lambda_m$ are distinct eigenvalues of f. Let $u_1, u_2, \dots, u_m$ be their corresponding eigenvectors. Thus, Theorem 7.2.2 shows that $u_1, u_2, \dots, u_m$ are linearly independent. Hence, Lemma 4.6.4 shows that $m \leq n$. □

Now, consider the endomorphism f defined as

$$\begin{aligned} f : \mathbb{R}^3 &\to \mathbb{R}^3, \\ (x, y, z) &\mapsto (2x + 4y + 3z, -4x - 6y - 3z, 3x + 3y + z). \end{aligned} \tag{7.7}$$

We can easily check that f has three eigenvalues

$$\lambda_1 = 1, \qquad \lambda_2 = \lambda_3 = -2.$$

One of the eigenvalues has a multiplicity 2. We may also show that the eigenspace $V(\lambda_1)$ is spanned by the vector $X_1 = (1, -1, 1)$ and $V(\lambda_2) = V(\lambda_3)$ is spanned by the vector $X_2 = (-1, 1, 0)$. By Theorem 4.6.3, the set $\{X_1, X_2\}$ is not a basis of $\mathbb{R}^3$.

So, we have seen in Theorem 7.2.2, that if E is a vector space over a field $\mathbb{K}$ with $\dim_{\mathbb{K}} E = n$ and f is an endomorphism of E which has n distinct eigenvalues (all with multiplicity one), then the associated eigenvectors form a basis of E. On the other hand, we have shown that for the endomorphism defined in (7.7), the eigenvectors associated to the eigenvalues of f does not form a basis of $\mathbb{R}^3$, since not all the eigenvalues are of multiplicity one. Thus, the question now is: *when do the eigenvectors associated to eigenvalues with multiplicities not necessary equal to one form a basis of E?* To answer this question, we define what we call *algebraic multiplicity* and *geometric multiplicity* of an eigenvalue.

Definition 7.2.1 (Algebraic and Geometric Multiplicities)

Let E be a finite-dimensional vector space over a field $\mathbb{K}$ and f be an endomorphism of E. Let λ be an eigenvalue of f.

- The *algebraic multiplicity* of the eigenvalue λ is the number of times λ appears as an eigenvalue of f.
- The *geometric multiplicity* of the eigenvalue λ is the number of linearly independent eigenvectors associated to λ. Or the dimension of the eigenspace associated to λ.

For example for the endomorphism defined in (7.7), the eigenvalue $\lambda_2 = -2$ has algebraic multiplicity 2 and geometric multiplicity 1.

Definition 7.2.2 (Complete and Defective Eigenvalues)

Let E be a finite-dimensional vector space over a field $\mathbb{K}$ and f be an endomorphism of E. Let λ be a repeated eigenvalue of f with algebraic multiplicity ℓ.

- The eigenvalue λ is called *complete* if there are ℓ corresponding linearly independent associated eigenvectors. That is, if the geometric multiplicity of λ is equal to its algebraic multiplicity.
- The eigenvalue λ is *defective* if the geometric multiplicity is strictly less than the algebraic multiplicity.

Example 7.4

Show that all the eigenvalues of the endomorphism

$$\begin{aligned} f : \mathbb{R}^3 &\to \mathbb{R}^3, \\ (x, y, z) &\mapsto (-2x + y + z, x - 2y + z, x + y - 2z), \end{aligned} \tag{7.8}$$

are complete. ■

Solution

First, we can easily show that f has two eigenvalues,

$$\lambda_1 = 0, \qquad \lambda_2 = \lambda_3 = -3.$$

It is clear that λ_2 has algebraic multiplicity 2 (that is $\ell = 2$). Now, in order for λ_2 to be complete, we need to find two independent eigenvectors associated to $\lambda_2 = -3$. That is, we need to show that the geometric multiplicity is also equal to 2. So, let $X = (x, y, z)$ be an eigenvector associated to $\lambda_2 = -3$, i.e.,

$$f(X) = -3X.$$

Equivalently,

$$\begin{cases} x + y + z = 0, \\ x + y + z = 0, \\ x + y + z = 0. \end{cases}$$

This means that X is an eigenvector associated to $\lambda = -3$ if and only if its components satisfies

$$x + y + z = 0,$$

or

$$z = -x - y.$$

Therefore, we can write X as

$$X = (x, y, z) = x(1, 0, -1) + y(0, 1, -1).$$

Consequently,

$$X_1 = (1, 0, -1) \qquad \text{and} \qquad X_2 = (0, 1, -1)$$

are two linearly independent eigenvectors associated to $\lambda_2 = -3$. Therefore, the geometric multiplicity is equal to 2, so indeed $\lambda_2 = -3$ is a complete eigenvalue. ◀

Example 7.5

Show that the endomorphism

$$\begin{aligned} f : \mathbb{R}^2 &\to \mathbb{R}^2, \\ (x, y) &\mapsto (y, 0), \end{aligned} \tag{7.9}$$

has a defective eigenvalue. ■

Solution

It is easy to check that $\lambda = 0$ is an eigenvalue of f with algebraic multiplicity 2. If $X = (x, y)$ is an eigenvector associated to λ, then $y = 0$. Thus all the eigenvectors of f are multiple of the vector $X_1 = (1, 0)$. Therefore, the geometric multiplicity is 1, and so the eigenvalue $\lambda = 0$ is defective. ◀

Theorem 7.2.4

Let E be a vector space over a field $\mathbb{K}$ such that $\dim_{\mathbb{K}} E = n$ and f be an endomorphism in $\mathscr{L}(E)$ such that all its eigenvalues are complete. Then, the set of eigenvectors associated to the complete eigenvalues form a basis of E.

Proof

Let $\lambda_1, \lambda_2, \ldots, \lambda_\ell$ be the set of complete eigenvalues of f and let k_i, $i = 1, 2, \ldots, \ell$, be the algebraic multiplicities of the λ_i. Then,

$$k_1 + k_2 + \cdots + k_\ell = n.$$

Since the eigenvalues are complete,

$$\dim_{\mathbb{K}} V(\lambda_i) = k_i, \qquad i = 1, 2, \ldots, \ell,$$

where $V(\lambda_i)$ is the eigenspace associated to λ_i, $i = 1, 2, \ldots, \ell$. Using (4.16) together with (7.2), we deduce that

$$\dim_{\mathbb{K}}[V(\lambda_1) \oplus V(\lambda_2) \oplus \cdots \oplus V(\lambda_\ell)] = n.$$

Therefore, applying Theorem 4.6.7, we obtain

$$E = V(\lambda_1) \oplus V(\lambda_2) \oplus \cdots \oplus V(\lambda_\ell).$$

Hence, the union of the bases of $V(\lambda_i)$, $i = 1, 2, \ldots, \ell$ which consists of all the eigenvectors of f forms a basis of E. This completes the proof of Theorem 7.2.4. □

7.3 Eigenvalues and Eigenvectors of a Matrix

In this section, we can restate for matrices all the results concerning the eigenvalues of an endomorphism. As we saw in ▶ Chap. 6, to each linear transformation f, one can associate a unique matrix $A = M(f)$ and vice versa. The eigenvalues, eigenvectors, and eigenspaces of $A = M(f)$ are then, by definition, the eigenvalues, eigenvectors, and eigenspaces of f.

Definition 7.3.1 (Eigenvalues of a Matrix and Spectrum)

Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$. Let X be a vector in $\mathbb{K}^n$ such that $X \neq 0_{\mathbb{K}^n}$. Then we say that X is an *eigenvector* of A if there exists λ in $\mathbb{K}$ such that

$$AX = \lambda X. \tag{7.10}$$

In this case, we say that λ is an *eigenvalue* of A and X is called an associated eigenvector.
The set of all distinct eigenvalues of A is called the *spectrum* of A and it is denoted by $\sigma(A)$.

7

Example 7.6

Consider the matrix

$$A = \begin{bmatrix} 1 & b_1 & c_1 \\ 2 & b_2 & c_2 \\ 3 & b_3 & c_3 \end{bmatrix}.$$

Find the entries b_i, c_i, $i = 1, 2, 3$ such that A will have the following eigenvectors:

$$X_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad X_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad X_3 = \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}.$$

What are the eigenvalues of A? ■

Solution

By Definition 7.3.1, X_1 is an eigenvector of A if

$$AX_1 = \lambda_1 X_1,$$

for some λ_1 in $\mathbb{R}$. This gives the system

$$\begin{cases} 1 + c_1 - \lambda_1 = 0, \\ c_2 + 2 = 0, \\ c_3 - \lambda_1 + 3 = 0, \end{cases}$$

the solution of which is

$$c_2 = -2, \qquad \text{and} \qquad c_1 - c_3 = 2. \tag{7.11}$$

Similarly, X_2 and X_3 are eigenvectors of A, if

$$AX_2 = \lambda_2 X_2, \qquad \text{and} \qquad AX_3 = \lambda_3 X_3,$$

for some λ_2 and λ_3 in $\mathbb{R}$. Thus, we obtain the systems

$$\begin{cases} b_1 + \lambda_2 - 1 = 0, \\ b_2 - \lambda_2 - 2 = 0, \\ b_3 - 3 = 0. \end{cases} \quad \text{and} \quad \begin{cases} b_1 - c_1 = 0, \\ b_2 - c_2 - \lambda_3 = 0, \\ b_3 - c_3 + \lambda_3 = 0. \end{cases}$$

Consequently, we obtain

$$b_3 = 3, \quad b_1 = c_1, \quad b_1 + b_2 = 3, \quad b_2 + b_3 = c_2 + c_3. \tag{7.12}$$

Hence, (7.11) and (7.12) yield

$$b_1 = 5, \quad b_2 = -2, \quad b_3 = 3, \quad c_1 = 5, \quad c_2 = -2, \quad c_3 = 3.$$

Thus, the matrix A is

$$A = \begin{bmatrix} 1 & 5 & 5 \\ 2 & -2 & -2 \\ 3 & 3 & 3 \end{bmatrix}.$$

Now, from the above systems, we can easily see that $\lambda_1 = 6, \lambda_2 = -4$ and $\lambda_3 = 0$, which are the eigenvalues of A. ◀

We have seen in Theorem 7.2.1 that λ is an eigenvalue of an endomorphism f if and only if $f - \lambda \operatorname{Id}_E$ is not invertible. Now, similarly, λ is an eigenvalue of A if and only if $A - \lambda I_n$ is not invertible, and this according to Theorem 2.4.8 is equivalent to say that $\det(A - \lambda I_n) = 0$. Thus, we summarize these in the following theorem.

Theorem 7.3.1 (Characterization of Eigenvalues of a Matrix)
Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$ and λ be an element of $\mathbb{K}$. Then the following statements are equivalent:

1. *λ is an eigenvalue of A.*
2. *The matrix $A - \lambda I_n$ is not invertible i.e., it is singular.*
3. $\det(A - \lambda I_n) = 0$.

In Examples 7.1 and 7.6, we have found the eigenvalues and eigenvectors simultaneously. The third statement in Theorem 7.3.1 separates completely the problem of finding the eigenvalues of a matrix from that of finding the associated eigenvectors. So, we can easily find an eigenvalue of a matrix, without needing to know a corresponding eigenvector.

Example 7.7

Find the eigenvalues and the corresponding eigenvectors for the matrices:

$$A = \begin{bmatrix} -1 & 6 \\ 0 & 5 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 5 & 0 & 1 \\ 1 & 1 & 0 \\ -7 & 1 & 0 \end{bmatrix}.$$

■

Solution

1. To find the eigenvalues of A, we use Theorem 7.3.1 and we compute $\det(A - \lambda I_2)$. We get

$$\det(A - \lambda I_2) = \det \begin{bmatrix} -1-\lambda & 6 \\ 0 & 5-\lambda \end{bmatrix} = (-1-\lambda)(5-\lambda).$$

Therefore, λ is an eigenvalue of A is and only if $(-1-\lambda)(5-\lambda) = 0$. This gives

$$\lambda_1 = -1, \qquad \lambda_2 = 5.$$

Now, if

$$X_1 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

is an eigenvector associated to λ_1, then we have

$$(A - \lambda_1 I) \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

This gives $6x_2 = 0$, so $x_2 = 0$, and then

$$X_1 = \begin{bmatrix} x_1 \\ 0 \end{bmatrix} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

So, we can choose

$$X_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Similarly, if

$$X_2 = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

is an eigenvector associated to $\lambda_2 = 5$, then we have

$$(A - \lambda_2 I_2)\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

This leads to $-6x_1 + 6x_2 = 0$, that is, $x_1 = x_2$ and then

$$X_2 = x_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Thus,

$$X_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

2. For the matrix B, we have

$$\det(B - \lambda I_3) = \det \begin{bmatrix} 5-\lambda & 0 & 1 \\ 1 & 1-\lambda & 0 \\ -7 & 1 & -\lambda \end{bmatrix} = -\lambda^3 + 6\lambda^2 - 12\lambda + 8 = -(\lambda - 2)^3.$$

Thus, $\lambda = 2$ is an eigenvalue of algebraic multiplicity 3. By the same method as above, we can find one eigenvector associated to $\lambda = 2$:

$$Y_1 = \begin{bmatrix} 1 \\ 1 \\ -3 \end{bmatrix}.$$

◀

Example 7.8

Find the eigenvalues and the corresponding eigenvectors of the matrix

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & 9 \end{bmatrix}.$$

■

Solution

As above, we compute

$$\begin{aligned} \det(A - \lambda I) &= -\lambda^3 + 14\lambda^2 - 35\lambda + 22 \\ &= -(\lambda - 11)(\lambda - 2)(\lambda - 1). \end{aligned}$$

Thus, the eigenvalues of A are the zeros of this polynomial, namely

$$\lambda_1 = 1, \qquad \lambda_2 = 2, \qquad \lambda_3 = 11.$$

Now, if

$$X_1 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

is an eigenvector associated to $\lambda_1 = 1$, that is,

$$(A - \lambda_1 I) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

then $x_1 = 0$ and $x_2 = -2x_3$. Hence,

$$X_1 = \begin{bmatrix} 0 \\ -2x_3 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}.$$

So, we can take

$$X_1 = \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}.$$

Similarly, if

$$X_2 = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

is an eigenvector associated to $\lambda_2 = 2$, i.e.,

$$(A - \lambda_2 I) \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix},$$

then we get $x_2 = x_3 = 0$. Hence,

$$X_2 = x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

Thus,

$$X_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

By the same method, we find that

$$X_3 = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

is an eigenvector associated to $\lambda_3 = 11$. ◄

Example 7.9
Consider the matrix A in $\mathscr{M}_3(\mathbb{C})$ given by

$$A = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad 0 < \theta < 2\pi.$$

Find the eigenvalues of A and the corresponding eigenvectors. ■

Solution
We have

$$\det(A - \lambda I) = -(\lambda - 1)\left(-2\lambda\cos\theta + \lambda^2 + 1\right).$$

Thus, $\det(A - \lambda I) = 0$ implies that

$$\lambda_1 = 1, \quad \lambda_2 = \cos\theta + i\sin\theta, \quad \text{and} \quad \lambda_3 = \cos\theta - i\sin\theta.$$

Alternatively, using the Euler formula

$$e^{i\theta} = \cos\theta + i\sin\theta,$$

we have

$$\lambda_1 = 1, \quad \lambda_2 = e^{i\theta}, \quad \text{and} \quad \lambda_3 = e^{-i\theta}.$$

By the same method as before, we easily find eigenvectors X_1, X_2 and X_3 associated to λ_1, λ_2, and λ_3, respectively, as

$$X_1 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \qquad X_2 = \begin{bmatrix} 1 \\ -i \\ 0 \end{bmatrix}, \qquad \text{and} \qquad X_3 = \begin{bmatrix} 1 \\ i \\ 0 \end{bmatrix}.$$

◀

7

Definition 7.3.2 (Characteristic Polynomial)

Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$. Then, for any λ in $\mathbb{K}$,

$$p(\lambda) = \det(A - \lambda I_n), \tag{7.13}$$

is a polynomial of degree n in λ called *the characteristic polynomial* of A.

So, it is clear from Theorem 7.3.1 that λ is an eigenvalue of A if and only if $p(\lambda) = 0$. That is, if and only if λ is a zero of $p(\lambda)$.

Example 7.10

The characteristic polynomial associated to the matrix A in Example 7.8 is

$$p(\lambda) = -\lambda^3 + 14\lambda^2 - 35\lambda + 22.$$

■

We proved in Exercise 1.4 that if A is a matrix in $\mathscr{M}_2(\mathbb{K})$, then

$$p(\lambda) = \lambda^2 - \operatorname{tr}(A) + \det(A).$$

Or equivalently,

$$p(\lambda) = (-1)^n\lambda^2 + (-1)^{n-1}\operatorname{tr}(A) + \det(A), \qquad n = 2.$$

In fact this formula can be generalized to any matrix A in $\mathscr{M}_n(\mathbb{K})$ and we have the following theorem.

Theorem 7.3.2
Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$. Then, its characteristic polynomial has the form

$$p(\lambda) = (-1)^n\lambda^n + (-1)^{n-1}\operatorname{tr}(A)\lambda^{n-1} + a_{n-2}\lambda^{n-2} + a_{n-3}\lambda^{n-3} + \cdots + a_1\lambda + \det(A), \tag{7.14}$$

where $a_1, \ldots, a_{n-2}$ are elements of $\mathbb{K}$.

Proof
Let $A = (a_{ij})$, $1 \le i, j \le n$. Then, we have

$$p(\lambda) = \det(A - \lambda I_n) = \det \begin{bmatrix} a_{11} - \lambda & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} - \lambda & \ldots & a_{2n} \\ \vdots & & \ddots & \vdots \\ a_{n1} & a_{n2} & \ldots & a_{nn} - \lambda \end{bmatrix}.$$

Computing this determinant using the cofactor expansion along the first row, we get

$$\det(A - \lambda I_n) = (a_{11} - \lambda) \det \begin{bmatrix} a_{22} - \lambda & a_{23} & \ldots & a_{2n} \\ a_{32} & a_{33} - \lambda & \ldots & a_{3n} \\ \vdots & & \ddots & \vdots \\ a_{n2} & a_{n3} & \ldots & a_{nn} - \lambda \end{bmatrix} + Q_{n-2}(\lambda),$$

where Q_{n-2} is a polynomial of degree $n-2$ with coefficients in $\mathbb{K}$. The determinant appearing here is the characteristic polynomial of an $(n-1) \times (n-1)$ matrix. So, by induction, we find

$$p(\lambda) = \det(A - \lambda I_n) = (a_{11} - \lambda)(a_{22} - \lambda) \cdots (a_{nn} - \lambda) + Q_{n-2}(\lambda), \tag{7.15}$$

We compute

$$(a_{11} - \lambda)(a_{22} - \lambda) \cdots (a_{nn} - \lambda) = (-\lambda)^n + (-\lambda)^{n-1} \underbrace{(a_{11} + a_{22} + \cdots + a_{nn})}_{:= \operatorname{tr}(A)} + \widetilde{Q}_{n-2}(\lambda), \tag{7.16}$$

where $\widetilde{Q}_{n-2}$ is a polynomial of degree at most $n-2$. Thus, combining (7.15) and (7.16), we get

$$p(\lambda) = (-1)^n\lambda^n + (-1)^{n-1}\operatorname{tr}(A)\lambda^{n-1} + P_{n-2},$$

where P_{n-2} is a polynomial of degree $n-2$ and has the form

$$P_{n-2}(\lambda) = a_{n-2}\lambda^{n-2} + a_{n-3}\lambda^{n-3} + \cdots + a_1\lambda + a_0. \tag{7.17}$$

The last coefficient in (7.17) can be obtained easily. Indeed, from (7.14), we have

$$p(0) = \det(A).$$

On the other hand, using (7.13), we have

$$p(0) = a_0.$$

This gives $a_0 = \det(A)$ and concludes the proof of Theorem 7.3.2. □

Corollary 7.3.3 *Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$ and let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be the eigenvalues of A, not necessary distinct. Then,*

$$\operatorname{tr}(A) = \sum_{i=1}^{n} \lambda_i \qquad \textit{and} \qquad \det(A) = \prod_{i=1}^{n} \lambda_i.$$

Proof

The eigenvalues of A are the roots of its characteristic polynomial $p(\lambda)$. Hence, from elementary algebra, the polynomial $p(\lambda)$ can be factored as

$$p(\lambda) = (-1)^n \prod_{i=1}^{n} (\lambda - \lambda_i).$$

Expanding this last formula, we find that the coefficient of λ^{n-1} is $(-1)^{n-1} \sum_{i=1}^{n} \lambda_i$ and the constant term is $\prod_{i=1}^{n} \lambda_i$. Comparing this with (7.14), then the result follows. □

Theorem 7.3.4 (Spectrum of the Transpose)
Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$. Then, we have

$$\sigma(A^T) = \sigma(A).$$

Proof

This is clear, since

$$\det(A^T - \lambda I_n) = \det(A - \lambda I_n)^T = \det(A - \lambda I_n),$$

where we have used Theorem 2.3.1. □

We have seen in Theorems 2.3.2 and 2.3.4 that if A is a diagonal or triangular matrix, then it is easy to find its determinant, because it is the product of the entries of the main diagonal. As we show next, it is also not difficult to find the eigenvalues and eigenvectors of a diagonal or triangular matrix and we have the following theorem.

Theorem 7.3.5 (Eigenvalues and Eigenvectors of Diagonal and Triangular Matrices)
Let A be a diagonal or triangular matrix in $\mathscr{M}_n(\mathbb{K})$. Then, the eigenvalues of A are the entries of the main diagonal.

Proof
We prove the statement for diagonal matrices; the same argument applies to triangular matrices. So, let

$$A = D = \begin{bmatrix} d_1 & 0 & 0 & \cdots & 0 \\ 0 & d_2 & 0 & \cdots & 0 \\ 0 & 0 & d_3 & \cdots & 0 \\ \vdots & & & \ddots & \\ 0 & 0 & 0 & \cdots & d_n \end{bmatrix},$$

Then,

$$D - \lambda I_n = \begin{bmatrix} d_1 - \lambda & 0 & 0 & \cdots & 0 \\ 0 & d_2 - \lambda & 0 & \cdots & 0 \\ 0 & 0 & d_3 - \lambda & \cdots & 0 \\ \vdots & & & \ddots & \\ 0 & 0 & 0 & \cdots & d_n - \lambda \end{bmatrix}.$$

Consequently, the characteristic polynomial

$$p(\lambda) = \det(D - \lambda I_n) = (d_1 - \lambda)(d_2 - \lambda) \cdots (d_n - \lambda),$$

has the roots

$$\lambda_i = d_i, \qquad i = 1, \ldots, n.$$

This yields the desired result. □

Example 7.11
Consider the two matrices

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} -1 & 2 & -5 \\ 0 & 0 & 9 \\ 0 & 0 & 5 \end{bmatrix}.$$

Then, since A is diagonal, the eigenvalues of A are $\lambda_1 = 1$, $\lambda_2 = -3$ and $\lambda_3 = 4$. Also, since B is triangular, the eigenvalues of B are $\lambda_1 = -1$, $\lambda_2 = 0$ and $\lambda_3 = 5$ ■

We have seen in Theorem 6.3.6 that similar matrices have the same rank. So, one can ask: *do similar matrices share the same spectrum?* The answer turns out to be affirmative:

Theorem 7.3.6 (Spectrum of Similar Matrices)
Let A and B be two similar matrices in $\mathscr{M}_n(\mathbb{K})$. Then we have

$$\sigma(A) = \sigma(B). \tag{7.18}$$

In addition, A and B have the same characteristic polynomial.

Proof
To show (7.18), we need to prove that $\sigma(A) \subset \sigma(B)$ and $\sigma(B) \subset \sigma(A)$. Since A and B are similar matrices, (see Definition 6.2.3), there exists an invertible matrix P such that

$$B = P^{-1}AP \quad \text{and} \quad A = PBP^{-1}.$$

Let λ be an eigenvalue of A, i.e., there exists X in $\mathbb{K}^n$ with $X \neq 0_{\mathbb{K}^n}$ such that $AX = \lambda X$. This implies that $PBP^{-1}X = \lambda X$. This gives

$$B(P^{-1}X) = \lambda(P^{-1}X).$$

Now, since $X \neq 0_{\mathbb{K}^n}$ and P^{-1} is invertible, then $Y = P^{-1}X \neq 0_{\mathbb{K}^n}$ and we have $BY = \lambda Y$. Hence λ is an eigenvalue of B and Y is its associated eigenvector. This shows that $\sigma(A) \subset \sigma(B)$. Conversely, let λ be an eigenvalue of B, then there exists Y in $\mathbb{K}^n$ such that $Y \neq 0_{\mathbb{K}^n}$ and $BY = \lambda Y$. This gives $P^{-1}APY = \lambda Y$. This yields

$$A(PY) = \lambda(PY).$$

Hence, λ is an eigenvalue of A and $X = PY$ is its corresponding eigenvector. Thus, $\sigma(B) \subset \sigma(A)$.

It remains to show that A and B have the same characteristic polynomial. We have

$$p_A(\lambda) = \det(A - \lambda I_n) = \det(P(B - \lambda I_n)P^{-1}) = \det(B - \lambda I_n) = p_B(\lambda),$$

where we have used the fact that $I_n = PI_nP^{-1}$, (2.26), and (2.30). □

Remark 7.3.7 The converse of Theorem 7.3.6 is not true. For example, the two matrices

$$A = I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

have the same eigenvalues, but they are not similar. Indeed, assuming that there exists an invertible matrix P such that

$$B = P^{-1}AP,$$

we would have

$$B = P^{-1}AP = P^{-1}I_2P = I_2$$

which a contradiction since $B \neq I_2$.

Remark 7.3.8 If λ is an eigenvalue of A, i.e., there exists an X in $\mathbb{K}^n$, $X \neq 0_{\mathbb{K}^n}$, with

$$AX = \lambda X,$$

this gives

$$A^2X = A(AX) = \lambda(AX) = \lambda^2X,$$

i.e., λ^2 is an eigenvalue of A^2. It easily follows, by induction, that if λ is an eigenvalue of A, then λ^n is an eigenvalue of A^n for any positive integer n. In addition, if A is invertible, then the statement is also true for negative integers n.

Example 7.12
The matrix

$$A = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix}$$

has the eigenvalues $\lambda_1 = -3$ and $\lambda_2 = -1$.

Then

$$A^2 = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}$$

which has the eigenvalues $\lambda_1 = 9 = (-3)^2$ and $\lambda_2 = 1 = (-1)^2$. ■

7.4 Diagonalization

We have seen before the importance of diagonal matrices. In this section, we look for necessary conditions for a square matrix A to be diagonalizable (similar to a diagonal matrix). We start with the following definition.

Definition 7.4.1 (Diagonalizable Matrix)

Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$. We say that A is *diagonalizable* if it is similar to a diagonal matrix. That is, if there exists an invertible matrix P in $\mathscr{M}_n(\mathbb{K})$ such that the matrix

$$B = P^{-1}AP$$

is diagonal.

Example 7.13

The matrix A in Example 7.8 is diagonalizable, since the matrix

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 11 \end{bmatrix} = \begin{bmatrix} 0 & -2/5 & 1/5 \\ 1 & 0 & 0 \\ 0 & 1/5 & 2/5 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 4 \\ 0 & 4 & 9 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ -2 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix} = P^{-1}AP$$

with

$$P^{-1} = \begin{bmatrix} 0 & -2/5 & 1/5 \\ 1 & 0 & 0 \\ 0 & 1/5 & 2/5 \end{bmatrix}, \qquad P = \begin{bmatrix} 0 & 1 & 0 \\ -2 & 0 & 1 \\ 1 & 0 & 2 \end{bmatrix},$$

is a diagonal matrix. ■

The diagonalization of a matrix has many important applications. We give here three of them.

Powers of Diagonalizable Matrices

Suppose we have a square matrix A in $\mathscr{M}_n(\mathbb{K})$ and we want to compute A^k for any $k \geq 0$. If A is diagonalizable, i.e., there exist a diagonal matrix B and an invertible matrix P such that

$$B = P^{-1}AP, \qquad \text{or} \qquad A = PBP^{-1},$$

then, we have

$$\begin{aligned} A^k &= \underbrace{AA\cdots A}_{k \text{ times}} \\ &= (PBP^{-1})(PBP^{-1})\cdots(PBP^{-1}) \\ &= PB(P^{-1}P)\cdots(P^{-1}P)BP^{-1} \\ &= PB^kP^{-1}. \end{aligned}$$

Since B is a diagonal matrix, computing B^k is trivial, see Theorem 1.2.12

Decoupling of a System of Differential Equations

Another very important application of diagonalization is the decoupling technique for systems of differential equations. So, consider the system of differential equations

$$\frac{dX(t)}{dt} = AX(t), \tag{7.19}$$

where $A = (a_{ij})$, $1 \leq i,j \leq n$, with a_{ij} in $\mathbb{R}$ and $X(t)$ a vector in $\mathbb{R}^n$,

$$X(t) = \begin{bmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{bmatrix}.$$

That is, (7.19) has the form

$$\begin{cases} \dfrac{dx_1(t)}{dt} = a_{11}x_1(t) + \cdots + a_{1n}x_n(t), \\ \vdots \qquad\qquad \cdots\cdots\cdots\cdots\cdots \\ \dfrac{dx_n(t)}{dt} = a_{n1}x_1(t) + \cdots + a_{nn}x_n(t). \end{cases}$$

Suppose A is a diagonalizable matrix, then there exist a diagonal matrix B and an invertible matrix P such that $B = P^{-1}AP$. Performing the change of variables

$$Y(t) = P^{-1}X(t), \qquad X(t) = PY(t),$$

we obtain

$$\begin{aligned}\frac{dY(t)}{dt} &= P^{-1}\frac{dX(t)}{dt}\\ &= P^{-1}AX(t)\\ &= P^{-1}APY(t)\\ &= BY(t),\end{aligned}$$

where $B = (\lambda_i), 1 \leq i \leq n$, is the diagonal matrix which has on its main diagonal the eigenvalues of A. The last system can be rewritten as

$$\begin{cases}\dfrac{dy_1(t)}{dt} = \lambda_1 y_1(t)\\ \vdots \qquad\qquad \cdots\cdots\\ \dfrac{dy_n(t)}{dt} = \lambda_n y_n(t).\end{cases}$$

This system is decoupled and each of its equations can be solved separately.

The Solution of a System of Recurrence Sequences

In numerical analysis, we sometimes end up with a system of recurrence sequences, and if the matrix representing this system is diagonalizable, then it is easy to write each term of the resulting sequences as a function of n, as we show in the following example.

Example 7.14

Consider the two sequences (u_n) and (v_n) defined for all n in $\mathbb{N}$ by the relations

$$\begin{cases}u_{n+1} = 2u_n + v_n,\\ v_{n+1} = u_n + 2v_n\end{cases} \tag{7.20}$$

with $u_0 = 1$ and $v_0 = 0$. Find u_n and v_n as functions of n for all n in $\mathbb{N}$. ■

Solution

We can write system (7.20) as

$$X_{n+1} = AX_n, \quad \text{with} \quad A = \begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}, \quad X_n = \begin{bmatrix}u_n\\ v_n\end{bmatrix}, \quad X_{n+1} = \begin{bmatrix}u_{n+1}\\ v_{n+1}\end{bmatrix}.$$

The eigenvalues of A are $\lambda_1 = 1$ and $\lambda_2 = 3$. Thus A is diagonalizable and we have

$$A = PBP^{-1}, \quad \text{with} \quad B = \begin{bmatrix}1 & 0\\ 0 & 3\end{bmatrix}, \qquad P = \begin{bmatrix}-1 & 1\\ 1 & 1\end{bmatrix}, \quad P^{-1} = \begin{bmatrix}-1/2 & 1/2\\ 1/2 & 1/2\end{bmatrix}.$$

Consequently, for any n in $\mathbb{N}$, we have

$$A^n = PB^nP^{-1} = \begin{bmatrix} -1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 3^n \end{bmatrix} \begin{bmatrix} -1/2 & 1/2 \\ 1/2 & 1/2 \end{bmatrix}$$
$$= \begin{bmatrix} \frac{1}{2} + \frac{3^n}{2} & -\frac{1}{2} + \frac{3^n}{2} \\ -\frac{1}{2} + \frac{3^n}{2} & \frac{1}{2} + \frac{3^n}{2} \end{bmatrix}.$$

Consequently,

$$X_n = A^nX_0 = \begin{bmatrix} \frac{1}{2} + \frac{3^n}{2} & -\frac{1}{2} + \frac{3^n}{2} \\ -\frac{1}{2} + \frac{3^n}{2} & \frac{1}{2} + \frac{3^n}{2} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} + \frac{3^n}{2} \\ -\frac{1}{2} + \frac{3^n}{2} \end{bmatrix}.$$

Finally, for all n in $\mathbb{N}$, we have

$$u_n = \frac{1}{2} + \frac{3^n}{2}, \qquad v_n = -\frac{1}{2} + \frac{3^n}{2}.$$

◀

Now we state the following result on diagonalizability.

Theorem 7.4.1 (Necessary and Sufficient Conditions for Diagonalization)
Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$. Then, A is a diagonalizable if and only if the matrix A has n linearly independent eigenvectors.

Proof

First, assume that A has n linearly independent eigenvectors $u_1, u_2, \ldots, u_n$. Then the matrix P whose columns are these eigenvectors, that is

$$P = [u_1, u_2, \ldots, u_n] \tag{7.21}$$

is invertible, (see Exercise 4.5). We have

$$P^{-1}AP = P^{-1}[Au_1, Au_2, \ldots, Au_n]$$
$$= P^{-1}[\lambda_1 u_1, \lambda_2 u_2, \ldots, \lambda_n u_n]$$
$$= P^{-1}[u_1, u_2, \ldots, u_n]B$$
$$= P^{-1}PB = B,$$

where

$$B = \begin{bmatrix} \lambda_1 & 0 & 0 & \cdots & 0 \\ 0 & \lambda_2 & 0 & \cdots & 0 \\ 0 & 0 & \lambda_3 & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_n \end{bmatrix}.$$

Consequently, A is similar to a diagonal matrix B i.e., A is diagonalizable.

Conversely, assume that A is diagonalizable, then there exists an invertible matrix P such that the matrix $B = P^{-1}AP$ is diagonal. This yields

$$AP = PB. \tag{7.22}$$

Let $v_1, v_2, \ldots, v_n$ be the column vectors of P. Thus,

$$P = [v_1, v_2, \ldots, v_n].$$

Formula (7.22) shows that

$$Av_i = b_i v_i, \qquad i = 1, 2, \ldots, n \tag{7.23}$$

where b_i is the ith diagonal entry of B. Since P is invertible, $v_i \neq 0_{\mathbb{K}^n}$, $i = 1, 2, \ldots, n$. Consequently, (7.23) shows that b_i is an eigenvalue of A and v_i is an associated eigenvector. □

Combining Theorems 7.2.4 and 7.4.1, we deduce the following corollary.

Corollary 7.4.2 *Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$, all of whose eigenvalues are complete. Then A is diagonalizable.*

Example 7.15

Show that the matrix

$$A = \begin{bmatrix} 1 & 2 & 0 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

is defective. ■

Solution

The matrix A has one eigenvalue $\lambda = 1$ with algebraic multiplicity equal to 4: this is easily seen because A is a triangular matrix. By Definition 7.2.2 the geometric multiplicity of $\lambda = 1$

is not equal to the algebraic multiplicity. Or, more importantly, we can use Corollary 7.4.2 to show that A is not diagonalizable, and thus is defective. Indeed, since $\lambda = 1$ is the only eigenvalue of A, then assuming that A is diagonalizable, there exists and invertible matrix P such that

$$A = PI_4P^{-1} = I_4.$$

This is a contradiction, since $A \neq I_4$, and so A is not diagonalizable and therefore is defective. ◀

Remark 7.4.3 The transition matrix P defined in (7.21) is also called the *eigenvector matrix* and it is not unique, since the eigenvectors are not unique. For instance, in Example 7.13, if we multiply the first column by 2, then we get

$$Q = \begin{bmatrix} 0 & 1 & 0 \\ -4 & 0 & 1 \\ 2 & 0 & 2 \end{bmatrix},$$

and Q also satisfies

$$B = Q^{-1}AQ.$$

The following result is important in applications.

Theorem 7.4.4 (Diagonalization and Commuting Matrices)
Let A and B be two diagonalizable matrices in $\mathscr{M}_n(\mathbb{K})$. Then, A and B share the same eigenvector matrix P if and only if they commute, that is $AB = BA$.

Proof

First, assume that A and B are diagonalizable, with the same eigenvector matrix P. Then there exist two diagonal matrices, D_1 and D_2, such that

$$A = PD_1P^{-1} \qquad \text{and} \qquad B = PD_2P^{-1}.$$

Hence,

$$AB = (PD_1P^{-1})(PD_2P^{-1}) = PD_1D_2P^{-1}.$$

On the other hand, we have

$$BA = (PD_2P^{-1})(PD_1P^{-1}) = PD_2D_1P^{-1}.$$

Since D_1 and D_2 are diagonal matrices, $D_1D_2 = D_2D_1$ (diagonal matrices always commute), and therefore $AB = BA$.

Conversely, assume that A and B are diagonalizable and $AB = BA$. Let X be an eigenvector of A, that is

$$AX = \lambda X \tag{7.24}$$

for some λ in $\mathbb{K}$ (we assume that $\lambda \neq 0_K$, since for $\lambda = 0_K$ the result is trivial). We need to show that X is also an eigenvector of B. Indeed, we have

$$ABX = BAX = B\lambda X = \lambda BX. \tag{7.25}$$

From (7.24) and (7.25) we deduce that both X and BX are eigenvectors of A sharing the same eigenvalue λ (unless $BX = 0_{\mathbb{K}^n}$, which is not the case since $\lambda \neq 0_K$ and $X \neq 0_{\mathbb{K}^n}$). If λ is a simple eigenvalue, then the eigenspace $V(\lambda)$ is a one-dimensional vector space, so necessarily $BX = \mu X$, for some μ in $\mathbb{K}$ with $\mu \neq 0_{\mathbb{K}}$ (since X and BX must be linearly dependent). Therefore, X is an eigenvector of B with corresponding eigenvalue μ. We leave the case where the eigenvalues are not simple to the reader, since it requires some additional work.

□

7.4.1 Spectrum of Symmetric Matrices

In this section, we study some important properties of symmetric matrices. First, we show that for real symmetric matrices all the eigenvalues are real. In this case, it is easy to determine the signs of the eigenvalues (which is very important, for instance, in the stability theory of differential equations). We also prove that symmetric matrices are diagonalizable. Therefore, all the properties of diagonalizable matrices can be carried over to symmetric matrices.

Theorem 7.4.5 (Eigenvalues of Real Symmetric Matrices)
Let A be a symmetric matrix in $\mathscr{M}_n(\mathbb{C})$, with all its entries real numbers. Then all the eigenvalues of A are real or, equivalently, its characteristic polynomial splits over $\mathbb{R}$.

Proof

By Definition 1.4.2, the matrix A is symmetric if and only if $A = A^T$. Now, let λ be an eigenvalue of A. To show that λ is real, we need just to prove that $\bar{\lambda} = \lambda$, where $\bar{\lambda}$ is the complex conjugate of λ. Since λ is an eigenvalue of A, there exists u in $\mathbb{C}^n$ with $u \neq 0_{\mathbb{C}^n}$ such that $Au = \lambda u$. Then, we can take the conjugate to get $\bar{A}\bar{u} = \bar{\lambda}\bar{u}$. Since the entries of A are

real, we have $\bar{A} = A$. Hence,

$$A\bar{u} = \bar{\lambda}\bar{u}.$$

Now, by taking the transpose of the this equation and using Theorem 1.4.1, we get

$$\bar{u}^T A^T = \bar{u}^T A = \bar{\lambda}\bar{u}^T.$$

Next, we multiply both sides of the above equation from the right by u, we get

$$\bar{u}^T Au = \bar{\lambda}\bar{u}^T u.$$

This yields

$$\bar{u}^T(\lambda u) = \lambda \bar{u}^T u = \bar{\lambda}\bar{u}^T u.$$

Since $u \neq 0_{\mathbb{C}^n}$, then $\bar{u}^T u \neq 0$. This gives $\bar{\lambda} = \lambda$. □

The following theorem is extremely important since it can be used in many applications.

Theorem 7.4.6 (Eigenvalues of a Symmetric Matrix)
If A is a symmetric matrix in $\mathscr{M}_n(\mathbb{K})$, then all its eigenvalues are complete. Therefore, it is diagonalizable.

The proof of Theorem 7.4.6 is beyond the scope of this book and can be found in advanced linear algebra textbooks.

Example 7.16
Consider the matrix

$$\begin{bmatrix} 13 & -4 & 2 \\ -4 & 13 & -2 \\ 2 & -2 & 10 \end{bmatrix}.$$

Show that A is diagonalizable. ■

Solution
It is not hard to check that A has the eigenvalues $\lambda_1 = 9, \lambda_2 = 9$, and $\lambda_3 = 18$. Consequently, λ_1 has algebraic multiplicity 2. Now, since A is symmetric, then according to Theorem 7.4.6, the geometric multiplicity of λ_1 equal to 2. This can be easily seen by computing the

eigenvectors of A associated to $\lambda_1 = 9$, which yields

$$u_1 = \begin{bmatrix} -1 \\ 0 \\ 2 \end{bmatrix}, \quad u_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}.$$

The two vectors u_1 and u_2 are linearly independent, showing that the geometric multiplicity of λ_1 is equal to 2. Hence, A is diagonalizable. ◀

7.5 Triangularization and the Jordan Canonical Form

In this section, we discuss the case of non diagonalizable matrices (endomorphisms). So, we try to find simple forms for those matrices (endomorphisms) that can be easily handled.

7.5.1 Triangularization of an Endomorphism

We have seen above that if a matrix A is complete (that is, all its eigenvalues are complete), then it is similar to a diagonal matrix (diagonalizable), with the eigenvector matrix providing the transition matrix. In this case, the matrix satisfies some nice properties that are inherited from the properties of diagonal matrices. Now, if A is not diagonalizable (as in the defective case), *is it possible for A to be similar to a triangular matrix, for instance?* That is to say, is there an invertible matrix M such that the matrix $B = M^{-1}AM$ is triangular? If such a matrix M exists, then we say that the matrix A is *triangularizable*. Of course in this case we do not expect to have all the nice properties of diagonalizable matrices, but at least we can keep some of them. So, in this section, we will be looking for the necessary assumptions that a defective matrix A should satisfy in order to be triangularizable. We start with the following definition.

Definition 7.5.1 (Triangulability)

1. Let E be a finite-dimensional vector space over a field $\mathbb{K}$. Let f be an endomorphism of E. Then, f is said to be *triangularizable* if there exists a basis $\mathbf{B}$ in E such that the matrix $M(f)$ associated to f in the basis $\mathbf{B}$ is triangular.[a]
2. A matrix A in $\mathscr{M}_n(\mathbb{K})$ is said to be *triangularizable* if it is similar to a triangular matrix. That is, if there exists an invertible matrix M such that the matrix $B = M^{-1}AM$ is triangular.

[a] See Definition 1.2.5.

Now, we want to find a traingularizability criterion for matrices. We state the theorem for matrices, but the same thing can be said for endomorphisms.

Theorem 7.5.1 (Necessary and Sufficient Traingularizability Conditions)
Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$. Then, A is triangularizable if and only if its characteristic polynomial has n zeros (counted with their algebraic multiplicity) in $\mathbb{K}$.

Proof

First assume that the characteristic polynomial $p_A(\lambda)$ of A has n zeros in $\mathbb{K}$. We prove by induction that A is similar to a triangular matrix.

For $n = 1$, it is clear that a matrix of order 1 is triangular, and thus triangularizable.

Now, assume that all matrices of order n whose characteristic polynomials have n zeros in $\mathbb{K}$ are triangularizable. Let A be a matrix of order $n+1$ for which p_A has $n+1$ zeros in $\mathbb{K}$. Then there exist at least an eigenvalue λ in $\mathbb{K}$ and an eigenvector Y in $\mathbb{K}^{n+1}$ associated to it, i.e., $Y \neq 0_{\mathbb{K}^{n+1}}$ and $AY = \lambda Y$. Now, according to Theorem 4.6.8, there exist vectors $X_1, X_2, \ldots, X_n$ in $\mathbb{K}^{n+1}$ such that the set $\{Y, X_1, \ldots, X_n\}$ forms a basis in $\mathbb{K}^{n+1}$. Consider the matrix P_1 with $Y, X_1, \ldots, X_n$ as its column vectors. That is,

$$P_1 = [Y, X_1, \ldots, X_n].$$

This matrix P_1 is the transition matrix from the standard basis of $\mathbb{K}^{n+1}$ to the basis $\{Y, X_1, \ldots, X_n\}$. It is clear that P_1 is invertible and we have

$$A_1 = P_1^{-1} A P_1 = \left[\begin{array}{c|c} \lambda & * \ldots * \\ \hline \begin{matrix} 0 \\ \vdots \\ 0 \end{matrix} & B \end{array}\right] = \begin{bmatrix} \lambda & V^T \\ \mathbf{0}_{n\times 1} & B \end{bmatrix},$$

where B is a matrix in $\mathscr{M}_n(\mathbb{K})$ and V is a vector in $\mathbb{K}^n$. It is clear from the hypothesis that p_B has n zeros in $\mathbb{K}$. Thus, applying the induction hypothesis, we deduce that B is triangularizable. Thus, there exists an invertible matrix Q in $\mathscr{M}_n(\mathbb{K})$ such that the matrix $A_2 = Q^{-1}BQ$ is triangular. We consider the matrix P_2 defined as

$$P_2 = \left[\begin{array}{c|c} 1 & 0 \ldots 0 \\ \hline \begin{matrix} 0 \\ \vdots \\ 0 \end{matrix} & Q \end{array}\right] = \begin{bmatrix} 1 & \mathbf{0}_{n\times 1}^T \\ \mathbf{0}_{n\times 1} & Q \end{bmatrix}.$$

It is clear that P_2 is invertible and we have

$$P_2^{-1} = \left[\begin{array}{c|ccc} 1 & 0 & \dots & 0 \\ \hline 0 & & & \\ \vdots & & Q^{-1} & \\ 0 & & & \end{array}\right] = \begin{bmatrix} 1 & \mathbf{0}_{n\times 1}^T \\ \mathbf{0}_{n\times 1} & Q^{-1} \end{bmatrix}.$$

We have

$$T = P_2^{-1}A_1P_2 = \begin{bmatrix} 1 & \mathbf{0}_{n\times 1}^T \\ \mathbf{0}_{n\times 1} & Q^{-1} \end{bmatrix}\begin{bmatrix} \lambda & V^T \\ \mathbf{0}_{n\times 1} & B \end{bmatrix}\begin{bmatrix} 1 & \mathbf{0}_{n\times 1}^T \\ \mathbf{0}_{n\times 1} & Q \end{bmatrix} = \begin{bmatrix} \lambda & V^TQ \\ \mathbf{0}_{n\times 1} & A_2 \end{bmatrix}.$$

This last matrix is triangular and we have

$$T = M^{-1}AM, \quad \text{with} \quad M = P_1P_2.$$

Conversely, suppose A is triangularizable, i.e., there exist a triangular matrix K and an invertible matrix Λ such that

$$K = \Lambda^{-1}A\Lambda = \begin{bmatrix} \lambda_1 & * & * & \cdots & * \\ 0 & \lambda_2 & * & \cdots & * \\ 0 & 0 & \lambda_3 & \cdots & * \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & \lambda_n \end{bmatrix},$$

where $\lambda_1, \lambda_2, \dots, \lambda_n$ are the eigenvalues of A. Then

$$p_K(\lambda) = (\lambda - \lambda_1)(\lambda - \lambda_2)\cdots(\lambda - \lambda_n).$$

Hence, since $p_K(\lambda) = p_A(\lambda)$ (Theorem 7.3.6), we deduce that $p_A(\lambda)$ has n zeros in $\mathbb{K}$. This finishes the proof of Theorem 7.5.1. □

Example 7.17
Consider the matrix

$$A = \begin{bmatrix} -2 & -1 & 2 \\ -15 & -6 & 11 \\ -14 & -6 & 11 \end{bmatrix}.$$

Show that A is not diagonalizable, but it is triangularizable. ■

Solution

We compute the eigenvalues of A. We have

$$\det(A - \lambda I_3) = p_A(\lambda) = -(\lambda - 1)^3.$$

Thus, A has one eigenvalue $\lambda = 1$ with algebraic multiplicity 3. We have only one eigenvector (up to a constant multiple) associated to λ, namely

$$X = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}.$$

Thus, the geometric multiplicity of A is equal to 1. Hence, A is defective and therefore, not diagonalizable. But since p_A has three zeros (one zero with algebraic multiplicity equal to 3) in $\mathbb{R}$, then A is triangularizable. One can also verify that

$$A = M^{-1}BM, \quad \text{with } M = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 3 & 2 \\ 2 & 2 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}, \quad M^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -3 & -1 & 2 \\ 4 & 2 & -3 \end{bmatrix}.$$

There are several methods for finding the matrix M. We will not discuss those methods now. ◀

According to the theorem of d'Alembert, that says that each polynomial of order n with coefficients in $\mathbb{C}$ has n zeros in $\mathbb{C}$, we deduce from Theorem 7.5.1, the following corollary.

Corollary 7.5.2 *Let A be a matrix in $\mathscr{M}_n(\mathbb{C})$. Then A is triangularizable.*

7.5.2 The Jordan Canonical Form

We have defined in (1.45) the exponential of a matrix A in $\mathscr{M}_n(\mathbb{K})$, as

$$e^A = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \cdots + \frac{A^k}{k!} + \ldots \tag{7.26}$$

One of the challenging problems for instance, in the study of differential equations is to compute (7.26). For a diagonal matrix $D = \operatorname{diag}(d_1, d_2, \ldots, d_n)$, this is simple and we obtain

$$e^D = \begin{bmatrix} e^{d_1} & 0 & 0 & \cdots & 0 \\ 0 & e^{d_2} & 0 & \cdots & 0 \\ 0 & 0 & e^{d_3} & \cdots & 0 \\ \vdots & & & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & e^{d_n} \end{bmatrix} = \operatorname{diag}(e^{d_1}, e^{d_2}, \ldots, e^{d_n}).$$

If A is not a diagonal matrix, but is diagonalizable, that is

$$A = PDP^{-1},$$

then we can also compute e^A as

$$e^A = Pe^D P^{-1}.$$

Another case is when we can write the matrix A as the sum of two matrices,

$$A = D + N,$$

where D is a diagonal matrix and N (called a *nilpotent* matrix, see Exercise 1.2) satisfies

$$N^{k_0} = \mathbf{0},$$

(here zero is the $n \times n$ matrix with all its entries equal to zero) for some positive integer $k_0 > 0$. The above decomposition is also known as the *Dunford decomposition.*

Here the simplicity of the method depends on the smallness of k_0. In this case, D commutes with any other square matrix of the same order. In particular, $DN = ND$. Thus, applying the formula $e^{D+N} = e^D e^N = e^N e^D$,[1] we obtain

$$\begin{aligned} e^A &= e^N e^D \\ &= \left(I + N + \frac{N^2}{2!} + \cdots + \frac{A^{k_0-1}}{(k_0-1)!} \right) \operatorname{diag}(e^{d_1}, e^{d_2}, \cdots, e^{d_n}). \end{aligned} \tag{7.27}$$

Example 7.18
Find e^A for

$$A = \begin{bmatrix} a & b \\ 0 & a \end{bmatrix},$$

where a and b are scalars. ■

Solution
The matrix A can be written as $A = D + N$ with

$$D = \begin{bmatrix} a & 0 \\ 0 & a \end{bmatrix} \quad \text{and} \quad N = \begin{bmatrix} 0 & b \\ 0 & 0 \end{bmatrix}.$$

[1] This formula holds only for commuting matrices.

It is clear that $N^2 = \mathbf{0}$. Thus, applying formula (7.27) we get ◄

$$\begin{aligned} e^A &= e^a (I + N) \\ &= e^a \begin{bmatrix} 1 & b \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} e^a & be^a \\ 0 & e^a \end{bmatrix}. \end{aligned}$$

Example 7.19
Find the Dunford decomposition of the matrix

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

■

Solution
We can simply write A as

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} = I_3 + N.$$

Clearly,

$$N^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

and $N^3 = \mathbf{0}$. Thus, N is nilpotent and $NI_3 = I_3N$. Hence, the above decomposition is the Dunford decomposition of the matrix A. ◄

The problem now is to find e^A when A is not diagonalizable. It turns out that there are several methods of how to do this (at least 19 of them, see [19]). One of these methods is to reduce the matrix to the so-called *Jordan canonical form.*

We can easily see that a matrix of the form

$$J(\lambda) = \begin{bmatrix} \lambda & 1 & 0 & \cdots & 0 \\ 0 & \lambda & 1 & \ddots & 0 \\ 0 & 0 & \lambda & \ddots & 0 \\ \vdots & & & \ddots & 1 \\ 0 & 0 & 0 & \cdots & \lambda \end{bmatrix} \tag{7.28}$$

has the Dunford decomposition

$$J(\lambda) = \lambda I_n + N, \tag{7.29}$$

where I_n is the identity matrix and N is the matrix

$$N(\lambda) = \begin{bmatrix} 0\ 1\ 0 & \cdots & 0 \\ 0\ 0\ 1 & \ddots & 0 \\ 0\ 0\ 0 & \ddots & 0 \\ \vdots & \ddots & 1 \\ 0\ 0\ 0 & \cdots & 0 \end{bmatrix}.$$

Clearly, the matrix N is nilpotent and thus we can compute the exponential of the matrix J as we have shown above. The matrix $J(\lambda)$ is called a *Jordan block*.

Definition 7.5.2 (Jordan Block)

A *Jordan block* with value λ is a square, upper triangular matrix whose entries are all equal to λ on the diagonal, all equal to 1 immediately above the diagonal, and equal to 0 elsewhere as in (7.28).

Thus, we know how to compute the exponential of a Jordan block using the Dunford decomposition. Thus if we can show that there exists an invertible matrix P such that

$$A = PJP^{-1}, \tag{7.30}$$

where J is the block diagonal matrix

$$J = \begin{bmatrix} J(\lambda_1) & 0 & \cdots & 0 \\ 0 & J(\lambda_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & J(\lambda_\ell) \end{bmatrix} \tag{7.31}$$

with $J_i = J(\lambda_i)$, $1 \leq i \leq \ell \leq n$, are Jordan blocks, then we can compute e^A. For example, every diagonal matrix is a matrix in the Jordan canonical form with each Jordan block a 1×1 block.

Writing A in the form (7.30) leads eventually to the computation of e^A by simply computing e^J as

$$e^J = \begin{bmatrix} e^{J(\lambda_1)} & 0 & \cdots & 0 \\ 0 & e^{J(\lambda_2)} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & e^{J(\lambda_\ell)} \end{bmatrix}$$

and hence it remains to compute the exponential of each Jordan block. However, the method of obtaining the matrix J requires detailed knowledge of the *generalized* eigenvectors for each eigenvalue of A, which are vectors X satisfying

$$(A - \lambda I)^k X = 0_{\mathbb{K}^n}$$

for some positive integer k, and also requires some knowledge of the *generalized* eigenspace of λ,

$$V(\lambda) = \left\{ V \middle| \, (A - \lambda I)^k X = 0_{\mathbb{K}^n}, \quad \text{for some } k \right\}.$$

So now, the question is *when does the matrix J exists?* Here is a theorem that answers this question.

Theorem 7.5.3 (Existence of the Jordan Form Matrix)
Let A be a matrix in $\mathscr{M}_n(\mathbb{C})$ with ℓ distinct eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_\ell$. For each λ_i, $1 \leq i \leq \ell$, denote its algebraic multiplicity by m_i and its geometric multiplicity by μ_i. Then A is similar to a Jordan form matrix of the form (7.31), where

- $\ell = \mu_1 + \mu_2 + \cdots + \mu_\ell$.
- *For each λ_i, the number of Jordan blocks in J with value λ_i is equal to μ_i.*
- *λ_i appears on the diagonal of J exactly m_i times.*

Furthermore, the matrix J is unique, up to re-ordering the Jordan blocks on the diagonal.

The proof of Theorem 7.5.3 is very technical, and we omit it here. In fact several proofs are available; for a discussion of these proofs we refer the reader to the paper [31] and references therein.

Next, we formulate one of the important theorems in linear algebra.

Theorem 7.5.4 (Cayley–Hamilton Theorem)
Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$ and let

$$p_A(\lambda) = a_0 + a_1\lambda + \cdots + a_n\lambda^n$$

be the characteristic polynomial of A. Then, we have

$$p_A(A) = a_0 I_n + a_1 A + \cdots + a_n A^n = \mathbf{0}.$$

Proof

First, if A is a diagonal matrix, that is $A = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$, then

$$p_A(\lambda) = \det(A - \lambda I_n) = (-1)^n(\lambda - \lambda_1)(\lambda - \lambda_2)\cdots(\lambda - \lambda_n).$$

Hence,

$$\begin{aligned} p_A(A) &= (-1)^n(A - \lambda_1 I_n)(A - \lambda_2 I_n)\cdots(A - \lambda_n I_n) \\ &= (-1)^n \operatorname{diag}(0, \lambda_2 - \lambda_1, \ldots, \lambda_n - \lambda_1)\cdots \operatorname{diag}(\lambda_1 - \lambda_n, \lambda_2 - \lambda_n, \ldots, \lambda_{n-1} - \lambda_n, 0) \\ &= \mathbf{0}. \end{aligned}$$

Second, if A is diagonalizable, i.e., there exists a diagonal matrix B such that $A = PBP^{-1}$, then

$$p_A(A) = P p_B(B) P^{-1} = \mathbf{0},$$

since $p_B(B) = \mathbf{0}$.

Third, in general, if A is not diagonalizable, then, according to Theorem 7.5.3, A is similar to a matrix J of Jordan form. In this case, we have $J = PAP^{-1}$ and the characteristic polynomial of J can be written as

$$p_J(\lambda) = (-1)^n(\lambda - \lambda_1)^{m_1}(\lambda - \lambda_2)^{m_2}\cdots(\lambda - \lambda_\ell)^{m_\ell}.$$

Since, J is a block diagonal matrix, it suffices to show that $p_J(J_i) = \mathbf{0}$. Indeed, we have

$$p_J(J_i) = (-1)^n(J_i - \lambda_1 I)^{m_1}(J_i - \lambda_2 I)^{m_2}\cdots(J_i - \lambda_\ell I)^{m_\ell}.$$

We can easily see that the matrix $J_i - \lambda_i I$ has the form

$$J_i - \lambda_i I = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \ddots & 0 \\ 0 & 0 & 0 & \ddots & 0 \\ \vdots & & & \ddots & 1 \\ 0 & 0 & 0 & \cdots & 0 \end{bmatrix}.$$

This is an $m_i \times m_i$ nilpotent matrix. Thus, $(J_i - \lambda_i I)^{m_i} = \mathbf{0}$. Consequently, we deduce that

$$p_J(J_i) = \mathbf{0}.$$

Hence, we obtain as before $p_A(A) = \mathbf{0}$. This completes the proof of Theorem 7.5.4. □

Example 7.20
Consider the matrix

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}.$$

Then, we have

$$p_A(\lambda) = \lambda^2 - 5\lambda - 2.$$

Moreover,

$$A^2 = \begin{bmatrix} 7 & 10 \\ 15 & 22 \end{bmatrix}.$$

Now, we can easily check that

$$A^2 - 5A - 2I_2 = \mathbf{0}.$$

■

7.6 Exercises

Exercise 7.1 (Spectrum of Commuting Matrices)
Let A and B be matrices in $\mathscr{M}_n(\mathbb{K})$. We say that A and B commute if $AB = BA$. Show that if A and B commute, then

$$\sigma(AB) - \{0_{\mathbb{K}}\} = \sigma(BA) - \{0_{\mathbb{K}}\},$$

meaning that AB and BA have the same nonzero eigenvalues. ■

Solution
We need to show that

$$\sigma(AB) - \{0_{\mathbb{K}}\} \subset \sigma(BA) - \{0_{\mathbb{K}}\} \quad \text{and} \quad \sigma(BA) - \{0_{\mathbb{K}}\} \subset (AB) - \{0_{\mathbb{K}}\}.$$

So, let λ be an eigenvalue of AB with $\lambda \neq 0_{\mathbb{K}}$. Then there exists an eigenvector X in $\mathbb{K}^n$, $X \neq 0_{\mathbb{K}^n}$, such that

$$(AB)X = \lambda X.$$

We have

$$BA(BX) = B(AB)X = B(\lambda X) = \lambda BX.$$

Thus, BX is an eigenvector of BA associated to λ, with $BX \neq 0_{\mathbb{K}^n}$ (since $\lambda \neq 0_{\mathbb{K}}$ and $X \neq 0_{\mathbb{K}^n}$). This yields

$$\sigma(AB) - \{0_{\mathbb{K}}\} \subset \sigma(BA) - \{0_{\mathbb{K}}\}.$$

Since A and B play a symmetric role, we also have

$$\sigma(BA) - \{0_{\mathbb{K}}\} \subset (AB) - \{0_{\mathbb{K}}\}.$$

7

◀

Exercise 7.2 (Matrix of Rank 1)
Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$.

1. Show that if $\text{rank}(A) = k$, $k \leq n$, then A has the eigenvalue $\lambda = 0$ at least of multiplicity $n - k$.
2. Deduce that if $\text{rank}(A) = 1$ and $\text{tr}(A) \neq 0$, then $\lambda = \text{tr}(A)$ is an eigenvalue of multiplicity 1 and if $\text{tr}(A) = 0$, then $\lambda = 0$ is the only eigenvalue of A of multiplicity n.

■

Solution
1. Since $\text{rank}(A) = k$, then it is clear from Theorem 6.4.2, that all minors of A of order strictly bigger than k are zero. Hence, in this case, the characteristic polynomial of A reads as

$$p(\lambda) = (-1)^n\lambda^n + (-1)^{n-1}\,\text{tr}(A)\lambda^{n-1} + a_{n-2}\lambda^{n-2} + \cdots + a_{n-k}\lambda^{n-k},$$

which has $\lambda = 0$ as a root of order at least $n - k$.

2. Applying (1) for $k = 1$, we deduce that the characteristic polynomial is

$$p(\lambda) = (-1)^n\lambda^n + (-1)^{n-1}\,\text{tr}(A)\lambda^{n-1}.$$

Hence, if $\text{tr}(A) \neq 0$, then $\lambda = \text{tr}(A)$ is a root of $p(\lambda)$, otherwise $\lambda = 0$ is a root of multiplicity n. ◀

Exercise 7.3 (Properties of Nilpotent Matrix)
Let N be a square matrix in $\mathscr{M}_n(\mathbb{K})$.

1. Show that N is nilpotent if and only if all its eigenvalues are equal to 0.
2. Show that if N is a nilpotent matrix, then $\text{tr}(N) = 0$.

3. Show that if N is a nilpotent matrix, then N is similar to a strictly upper or strictly lower triangular matrix (a triangular matrix with the entries of the main diagonal are all zero).
4. Show that a nilpotent matrix remains nilpotent with respect to any basis.

■

Solution

1. First, if all the eigenvalues of N are equal to 0, then its characteristic polynomial is

$$p_N(\lambda) = \det(N - \lambda I_n) = (-1)^n \lambda^n.$$

Then using the Cayley–Hamilton theorem (Theorem 7.5.4), we have

$$p_N(N) = (-1)^n N^n = \mathbf{0}.$$

Hence, $N^n = \mathbf{0}$, i.e., N is nilpotent.

Conversely, assume that N is nilpotent, then, we have $N^k = \mathbf{0}$ and $N^{k-1} \neq \mathbf{0}$, for some positive integer $k \geq 1$. Let λ be an eigenvalue of N, i.e., there exists X in $\mathbb{K}^n$, $X \neq 0_{\mathbb{K}^n}$ such that

$$NX = \lambda X.$$

Hence,

$$N^k X = \lambda^k X.$$

This means that λ^k is an eigenvalue of N^k and since N^k is the zero matrix, $\lambda^k = 0_{\mathbb{K}}$. This yields $\lambda = 0_{\mathbb{K}}$.

2. Since N is nilpotent, all its eigenvalues are equal to zero. Then, using (7.3.3), we have

$$\mathrm{tr}(N) = \sum_{i=1}^{n} \lambda_i = 0.$$

3. Since N is nilpotent, its characteristic polynomial has one root equal to 0, with algebraic multiplicity n. Then according to Theorem 7.5.1, N is triangularizable. That is, there exist an invertible matrix P and a triangular matrix T such that $N = P^{-1}TP$. Since N and T are similar matrices, they share the same eigenvalues, (Theorem 7.3.6). Since, the eigenvalues of a triangular matrix are on its main diagonal, the entries of the main diagonal of T are all equal to zero (because the eigenvalues of N are all equal to zero). Thus, the matrix T must be strictly upper or lower triangular.

4. Let $\mathbf{B}_1$ be a basis of $\mathbb{K}^n$ and N be a nilpotent matrix with respect to the basis $\mathbf{B}_1$. Let $\mathbf{B}_2$ be another basis of $\mathbb{K}^n$ and P be the transition matrix from $\mathbf{B}_1$ to $\mathbf{B}_2$. Then, there exists a matrix B such that

$$B = P^{-1}NP.$$

Since N is nilpotent, there exists a positive integer k such that $N^k = \mathbf{0}$. Then

$$B^k = P^{-1}N^kP = \mathbf{0},$$

so B is nilpotent. ◀

Exercise 7.4 (Nilpotent Matrix and Determinant)
Let A and N be two matrices in $\mathscr{M}_n(\mathbb{K})$. Assume that N is nilpotent.

1. Show that if A is invertible, then $A + N$ is invertible.
2. Assume that $AN = NA$. Show that

$$\det(A + N) = \det(A).$$

Study first the case where A is invertible.

■

Solution

1. Assume that A is invertible. Then, $A^{-1}N$ is nilpotent, since if for some k_0 we have $N^{k_0} = \mathbf{0}$, then (since A^{-1} also commute with N)

$$(A^{-1}N)^{k_0} = A^{-k_0}N^{k_0} = \mathbf{0}.$$

Consequently (see Exercise 1.2), $I_n + A^{-1}N$ is invertible and

$$A + N = A(I_n + A^{-1}N)$$

is invertible, since it is the product of two invertible matrices.

2. If A is invertible, then from above, we have that

$$\det(A + N) = \det(A)\det(I_n + A^{-1}N).$$

We claim that

$$\det(I_n + A^{-1}N) = 1.$$

Indeed, we proved in Exercise 7.3 that a nilpotent matrix is similar to a triangular matrix in which all the entries of the main diagonal are equal to zero. We denote this triangular matrix by T. Thus, there exists an invertible matrix P such that $A^{-1}N = PTP^{-1}$. Then, we have

$$I_n + A^{-1}N = PI_nP^{-1} + PTP^{-1} = P(I_n + T)P^{-1}.$$

Consequently, the matrix $I_n + A^{-1}N$ is similar to the matrix $I_n + T$, and thus $\det(A^{-1}N) = \det(I_n + T)$ (similar matrices have the same determinant). Since $I_n + T$ is a triangular matrix

with all the entries on the main diagonal are equal to 1, then, $\det(I_n + T) = 1$. This proves the claim.

If A is not invertible, then $\det(A) = 0$ (see Theorem 2.4.8). If $\det(A + N) \neq 0$, then $A + N$ is invertible. In addition, $(A + N)$ commute with $-N$, which is a nilpotent matrix. So, applying what we have proved above, we find that

$$\det(A) = \det(A + N - N) = \det(A + N) \neq 0.$$

This is a contradiction. Thus, one necessarily has $\det(A + N) = \det(A) = 0$. ◄

Exercise 7.5

Let $E = \mathscr{P}_n$ be the vector space of all polynomials with real coefficients and of degree less or equals to n. Consider the endomorphism f in $\mathscr{L}(E)$ defined for any p in E as

$$f(p) = (x + 1)(x - 3)p' - xp.$$

Find the eigenvalues and eigenvectors of f. ■

Solution

Let p be an element of E such that $p \neq 0_E$. Then, for λ in $\mathbb{R}$,

$$f(p) = \lambda p,$$

implies that

$$(x + 1)(x - 3)p' - (x + \lambda)p = 0. \tag{7.32}$$

By looking at the higher-order terms in this relation, we deduce that $n = 1$. Therefore, $p(x) = ax + b$. Plugging this into (7.32), we obtain

$$\begin{cases} 2a + \lambda a + b = 0, \\ 3a + \lambda b = 0, \end{cases}$$

or equivalently,

$$\begin{cases} b = -(2 + \lambda)a, \\ (\lambda^2 + 2\lambda - 3)a = 0. \end{cases}$$

Since we assumed that $p \neq 0_E$, we deduce that $\lambda_1 = 1$ and $\lambda_2 = -3$. Now, for $\lambda_1 = 1$, we get $b = -3a$ and the eigenvector associated to λ_1 is $p_1(x) = x - 3$. For $\lambda_2 = -3$, we have $b = a$ and the eigenvector associated to λ_2 is $p_2(x) = x + 1$. ◄

Exercise 7.6

Let A and B be two matrices in $\mathscr{M}_n(\mathbb{R})$. Assume that $AB - BA = A$.

1. Show that for any k in $\mathbb{N}$, we have

$$A^k B - BA^k = kA^k.$$

2. Deduce that A is nilpotent.

■

Solution

1. It is clear that the above identity is true of $k = 0$. Now, for any k in $\mathbb{N} - \{0\}$, we have

$$\begin{aligned} A^k B - BA^k &= A^k B - A^{k-1}BA + A^{k-1}BA - A^{k-2}BA^2 + A^{k-2}BA^2 \\ &\quad - \cdots - ABA^{k-1} + ABA^{k-1} - AB^k \\ &= \sum_{i=0}^{k-1} (A^{k-i}BA^i - A^{k-i-1}BA^{i+1}) \\ &= \sum_{i=0}^{k-1} A^{k-i-1}(AB - BA)A^i \\ &= \sum_{i=0}^{k-1} A^{k-i-1}AA^i \\ &= kA^k. \end{aligned}$$

2. Define the endomorphism f as follows:

$$\begin{aligned} f : \mathscr{M}_n(\mathbb{R}) &\to \mathscr{M}_n(\mathbb{R}) \\ K &\mapsto KB - BK. \end{aligned}$$

We have, for any k in $\mathbb{N} - \{0\}$,

$$f(A^k) = kA^k.$$

Now, if $A^k \neq \mathbf{0}$, then A^k is an eigenvector of f associated to the eigenvalue $\lambda = k$. Consequently, if $A^k \neq \mathbf{0}$ for any k, then f has an infinite number of distinct eigenvectors. This is impossible, since $\dim_{\mathbb{R}} \mathscr{M}_n(\mathbb{R}) = n^2 < \infty$. ◄

Exercise 7.7 (Minimal Polynomial)

Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$.

1. Show that there exists a polynomial p such that $p(A) = \mathbf{0}$. In this case we say that the polynomial p *annihilates* the matrix A.

2. The *minimal polynomial* of A is the smallest monic (the term of highest degree equals 1) polynomial m_A that satisfies $m_A(A) = \mathbf{0}$. Show that the minimal polynomial of A is unique.
3. Show that the degree of m_A is less than or equal to n.
4. Show that if p is a polynomial such that $p(A) = \mathbf{0}$, then m_A divides p.
5. Prove that λ is an eigenvalue of A if and only if λ is a root of m_A.
6. Show that if the characteristic polynomial p_A of A has the form

$$p_A(\lambda) = \prod_{i=1}^{\ell} (\lambda - \lambda_i)^{a_i},$$

with $a_1 + a_2 + \cdots + a_\ell = n$, then the minimal polynomial m_A of A has the form

$$m_A(\lambda) = \prod_{i=1}^{\ell} (\lambda - \lambda_i)^{m_i}, \tag{7.33}$$

where $1 \leq m_i \leq a_i$.
7. Use the result in (6) to find the minimal polynomial of the matrix

$$A = \begin{bmatrix} 3 & -1 & 1 \\ 1 & 0 & 1 \\ 1 & -1 & 2 \end{bmatrix}.$$

8. Show that if A and B are similar matrices in $\mathscr{M}_n(\mathbb{K})$, then they have the same characteristic polynomial.

■

Solution

1. We have seen that $\dim_{\mathbb{K}} \mathscr{M}_n(\mathbb{K}) = n^2$. Consequently, any set of $n^2 + 1$ elements in $\mathscr{M}_n(\mathbb{K})$ is a linearly dependent set (Lemma 4.6.4). So, consider the set $I, A, A^2, \ldots, A^{n^2}$. This set contains $n^2 + 1$ elements, so it is linearly dependent. Therefore, there exist $a_0, a_1, a_2, \ldots, a_{n^2+1}$ elements in $\mathbb{K}$, not all zero, such that

$$a_0 I + a_1 A + a_2 A^2 + \cdots + a_{n^2+1} A^{n^2+1} = \mathbf{0}. \tag{7.34}$$

So, if we consider the polynomial $p(x)$ defined by

$$p(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_{n^2+1} x^{n^2+1},$$

then, by (7.34), $p(A) = \mathbf{0}$.

2. If there are two minimal polynomials $m_1(x)$ and $m_2(x)$ of A, then they should have the same degree r and both have the coefficient of the leading term (the term of the highest

degree) equal to 1. Thus,

$$m_1(x) = a_0 + a_1x + \cdots + a_{r-1}x^{r-1} + x^r$$

and

$$m_2(x) = b_0 + b_1x + \cdots + b_{r-1}x^{r-1} + x^r.$$

Then $m(x) = m_1(x) - m_2(x)$ is a polynomial of degree $r-1$ and it can be written as

$$m(x) = c_0 + c_1x + \cdots + c_{r-1}x^{r-1} = 0_{\mathbb{K}}, \quad \text{with} \quad c_i = a_i - b_i, \quad i = 1, \ldots r-1.$$

Thus, we have

$$m(A) = c_0I + c_1A + \cdots + c_{r-1}A^{r-1} = \mathbf{0}.$$

Hence, $c_i = 0_{\mathbb{K}}$ for all $i = 1, \ldots, r-1$, otherwise $m(x)$ would be a minimal polynomial of A, which is a contradiction.

3. We have seen that the characteristic polynomial p_A of A is of degree n. Also, the Cayley–Hamilton theorem (Theorem 7.5.4) gives $p_A(A) = \mathbf{0}$. This gives us the desired result, by the definition of m_A.

4. Let p be a polynomial such that $p(A) = \mathbf{0}$. Then the degree of m is less than or equal to the degree of p. In this case we can write p as $p = qm_A + r$, where q and r are two polynomials with the degree of r strictly less than the degree of m_A. Hence,

$$r(A) = p(A) - q(A)m_A(A) = \mathbf{0}.$$

This contradicts the minimality of m_A unless r is the zero polynomial. Thus, m_A divides p.

5. First, assume that λ is an eigenvalue of A, then we have $AX = \lambda X$, for some $X \neq 0_{\mathbb{K}^n}$. Thus, as we know, that $A^kX = \lambda^kX$, for any integer $k \geq 0$. Hence, for any polynomial p, we have $p(A) = p(\lambda)X$. In particular, $m_A(A) = m_A(\lambda)X$. But since $m_A(A) = \mathbf{0}$ and $X \neq 0_{\mathbb{K}^n}$, we deduce that $m_A(\lambda) = 0_{\mathbb{K}}$.

Conversely, if λ is a root of m_A, then according to (4), λ is also a root of p_A, since m_A divides p_A. Thus, λ is an eigenvalue of A.

6. Since $p_A(A) = \mathbf{0}$, from (4), we deduce that m_A divides p_A. Therefore, the roots of p_A should be the roots of m_A with different multiplicity. Indeed, since $p_A = qm_A$, if λ is a root of m_A, then it is clear that λ is also a root of p_A. Conversely, if λ is a root of p_A, then λ is an eigenvalue of A, hence, according to (5), λ is a root of m_A. Thus, the only possibility for m_A is to have the form (7.33).

7. The characteristic polynomial of A is

$$p_A(\lambda) = -(\lambda - 2)^2(\lambda - 1).$$

Thus, we have two possibilities for the minimal polynomial $m_A(\lambda)$:

$$m_A(\lambda) = (\lambda - 2)(\lambda - 1) \qquad \text{or} \qquad m_A(\lambda) = (\lambda - 2)^2(\lambda - 1).$$

First, we compute $(A - 2I)(A - I)$. If this matrix is the zero matrix, then the minimal polynomial is $m_A(\lambda) = (\lambda - 2)(\lambda - 1)$. Otherwise, $m_A(\lambda) = (\lambda - 2)^2(\lambda - 1)$. We may easily check that $(A - 2I)(A - I) \neq \mathbf{0}$, so indeed $m_A(\lambda) = (\lambda - 2)^2(\lambda - 1)$.

8. Since A and B are similar matrices, there exists an invertible matrix S such that $B = S^{-1}AS$. Then

$$\begin{aligned} m_A(B) &= m_A(S^{-1}AS) \\ &= S^{-1}m_A(A)S \\ &= \mathbf{0}. \end{aligned}$$

So, if there is a minimal polynomial of B of a smaller degree, say m_B, then we have, by the same argument, $m_B(A) = \mathbf{0}$ which contradicts the minimality of m_A. Thus, we conclude that m_A is a minimal polynomial for B, and since the minimal polynomial is unique, we deduce that $m_A = m_B$. ◀

Exercise 7.8 (Minimal Polynomial and Jordan Canonical Form)
Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$. Let λ_0 be an eigenvalue of A. We define the Jordan block of order m_0 associated to λ_0 as

$$J(\lambda_0) = \begin{bmatrix} \lambda_0 & 1 & 0 & \cdots & 0 \\ 0 & \lambda_0 & 1 & \ddots & 0 \\ 0 & 0 & \lambda_0 & \ddots & 0 \\ \vdots & & & \ddots & 1 \\ 0 & 0 & 0 & \cdots & \lambda_0 \end{bmatrix}. \tag{7.35}$$

1. Show that the minimal polynomial of $J(\lambda_0)$ is:

$$m_{J(\lambda_0)}(\lambda) = (\lambda - \lambda_0)^{m_0}.$$

2. Show that the minimal polynomial of the matrix A is

$$m_A(\lambda) = \prod_{i=1}^{\ell} (\lambda - \lambda_i)^{m_i}, \tag{7.36}$$

where m_i, $1 \leq i \leq \ell$, is the order (the size) of the largest Jordan block $J(\lambda_i)$ in the Jordan canonical form of A.

■

Solution

1. It is clear from the Dunford decomposition (7.29), that $N = J(\lambda_0) - \lambda_0 I_{m_0}$ is a nilpotent matrix of order m_0. Thus, we have

$$N^{m_0} = (J(\lambda_0) - \lambda_0 I_{m_0})^{m_0} = \mathbf{0},$$

and $N^{m_0-1} \neq \mathbf{0}$. Hence, we have shown that $m_{J(\lambda_0)}$ is the polynomial of the smallest degree which satisfies $m_{J(\lambda_0)}(J(\lambda_0)) = \mathbf{0}$. Thus, $m_{J(\lambda_0)}$ is the minimal polynomial of $J(\lambda_0)$.

2. Let J be the block diagonal matrix

$$J = \begin{bmatrix} J(\lambda_1) & 0 & \cdots & 0 \\ 0 & J(\lambda_2) & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & J(\lambda_\ell) \end{bmatrix} \tag{7.37}$$

where $J(\lambda_i)$, $1 \leq i \leq \ell \leq n$, are the Jordan blocks. We have

$$A = PJP^{-1},$$

for some invertible matrix P. Now, it is clear that

$$m_A(\lambda) = (\lambda - \lambda_1)^{m_1} \cdots (\lambda - \lambda_\ell)^{m_\ell}$$

is the minimal polynomial of J and since A and J are similar matrices, they have the same minimal polynomial (Exercise 7.7). ◀

Exercise 7.9 (Minimal Polynomial and Diagonalization[a])

Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$ and let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be the eigenvalues of A.

1. Show that A is diagonalizable if and only if its minimal polynomial has the form

$$m_A(\lambda) = \prod_{i=1}^{n} (\lambda - \lambda_i).$$

2. Consider the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Is A diagonalizable?

■

[a] As this exercise shows, the minimal polynomial provides another criterion for diagonalizability.

Solution

1. In this case, according to Exercise 7.8, each Jordan block is of order 1. Hence, the matrix J defined in (7.37) is diagonal, and it is similar to A. Hence, A is diagonalizable.

2. We compute the characteristic polynomial of A and find that

$$p_A(\lambda) = -(\lambda - 1)^3.$$

So, according to Exercise 7.7, its minimal polynomial is

$$m_A(\lambda) = (\lambda - 1)^{m_0}, \qquad m_0 \leq 3.$$

We may easily check that $A - I \neq \mathbf{0}$, and $(A - I)^2 \neq \mathbf{0}$. Thus $m_0 = 3$. Consequently, the matrix A is not diagonalizable. ◀

Exercise 7.10 (Nonderogatory Matrix)

A matrix A in $\mathscr{M}_n(\mathbb{K})$ is said to be *nonderogatory* if its characteristic polynomial and minimal polynomial coincide (up to a multiplicative constant). Otherwise it is called *derogatory*.

1. Show that if A is nonderogatory, then every eigenvalue of A has geometric multiplicity 1; equivalently, A has only one Jordan block for each eigenvalue.
2. Show that the companion matrix C_p, defined in Exercise 2.8 as

$$C_p = \begin{bmatrix} 0 & 1 & 0 & \dots & 0 \\ 0 & 0 & 1 & \vdots & 0 \\ \vdots & \vdots & & \ddots & \vdots \\ 0 & 0 & 0 & \dots & 1 \\ -a_0 & -a_1 & -a_2 & \dots & -a_{n-1} \end{bmatrix},$$

where $a_0, a_1, \dots, a_{n-1}$ are in $\mathbb{K}$, is a nonderogatory matrix.

■

Solution

1. Suppose that there exists one eigenvalue λ_k of A such that to λ_k there correspond two Jordan blocks. Let the order of the first block be n_1 and that of the second block be n_2, with $n_1 \geq n_2$. Then the characteristic polynomial of A is

$$p_A(A) = (\lambda - \lambda_1)^{m_1}(\lambda - \lambda_2)^{m_2} \cdots (\lambda - \lambda_k)^{n_1}(\lambda - \lambda_k)^{n_2} \cdots (\lambda - \lambda_\ell)^{m_\ell}.$$

Hence, the polynomial

$$(\lambda - \lambda_1)^{m_1}(\lambda - \lambda_2)^{m_2} \cdots (\lambda - \lambda_k)^{n_1} \cdots (\lambda - \lambda_\ell)^{m_\ell},$$

where we removed $(\lambda - \lambda_k)^{n_2}$, annihilates A. This contradicts the minimality of m_A, since, we assumed that $m_A = p_A$.

2. We need to show that the minimal polynomial of C_p is exactly its characteristic polynomial

$$p(\lambda) = a_0 + a_1\lambda + \cdots + a_{n-1}\lambda^{n-1} + \lambda^n.$$

Let m_{C_p} be the minimal polynomial of C_p. By the preceding results, m_{C_p} divides p. If m_{C_p} is a polynomial of degree $r < n$, then

$$m_{C_p}(\lambda) = \alpha_0 + \alpha_1\lambda + \cdots + \alpha_{r-1}\lambda^{r-1} + \lambda^r, \quad r < n.$$

7

We have

$$m_{C_p}(C_p) = \mathbf{0}$$

Moreover,

$$C_p^T e_i = e_{i+1}, \qquad 1 \leq i \leq n-1,$$

where e_i is the vector with all components equal to $0_{\mathbb{K}}$, except for the ith component, which is $1_{\mathbb{K}}$. Since C_p and C_p^T have the same characteristic and minimal polynomials, we have for $e_1 = (1_{\mathbb{K}}, 0_{\mathbb{K}}, \ldots, 0_{\mathbb{K}})^T$ that

$$\begin{aligned}\mathbf{0} &= m_{C_p^T}(C_p^T)e_1 = \alpha_0 e_1 + \alpha_1 e_2 + \cdots + \alpha_{r-1}e_r + e_{r+1} \\ &= (\alpha_0, \alpha_1, \ldots, \alpha_{r-1}, 1, 0, \ldots, 0)^T \\ &\neq \mathbf{0}.\end{aligned}$$

This is a contradiction. Hence, $r = n$ and therefore the matrix C_p is nonderogatory.

◀

Exercise 7.11

Consider the matrix

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ a_1 & 1 & 0 & 0 \\ a_2 & b_1 & 2 & 0 \\ a_3 & b_2 & c_1 & 2 \end{bmatrix},$$

where a_1, a_2, a_3, b_1, b_2, and c_1 are real numbers. Study the diagonalizability of the matrix A. ■

Solution

Actually, there are several methods that we can use to study the diagonalizability of A. Here we use the one based on the minimal polynomial. Since the matrix A is triangular, the eigenvalues are the elements of the main diagonal. Thus, A has two eigenvalues $\lambda_1 = 1$ and $\lambda_2 = 2$, each of algebraic multiplicity 2. Thus, the characteristic polynomial of A is

$$p_A(\lambda) = (\lambda - 1)^2(\lambda - 2)^2.$$

According to the first question in Exercise 7.9, A is diagonalizable if and only if its minimal polynomial has the form

$$m_A(\lambda) = (\lambda - 1)(\lambda - 2).$$

That is, if and only if

$$(A - I_4)(A - 2I_4) = \mathbf{0},$$

or equivalently,

$$\begin{bmatrix} 0 & 0 & 0 & 0 \\ -a_1 & 0 & 0 & 0 \\ a_1 b_1 & 0 & 0 & 0 \\ a_1 b_2 + a_2 c_1 & b_1 c_1 & c_1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

This gives $a_1 = c_1 = 0$. Consequently, A is diagonalizable if and only if $a_1 = c_1 = 0$. ◀

Exercise 7.12 (Circulant Matrices[a])

An $n \times n$ circulant matrix is a matrix formed from any vector in $\mathbb{C}^n$ by cyclically permuting the entries. For example, if $v = (a, b, c)$ is a vector in $\mathbb{R}^3$, then the associated 3×3 circulant matrix is

$$C = \begin{bmatrix} a & b & c \\ c & a & b \\ b & c & a \end{bmatrix}.$$

We see that circulant matrices have constant values on the main diagonal.

1. Find the eigenvalues and the corresponding eigenvectors of the matrix C.
2. Show that if $\omega_j, j = 1, 2, 3$ are the cubic roots of the unity, then the eigenvalues of C are

$$\lambda_j = q(\omega_j), \qquad j = 1, 2, 3,$$

where

$$q(t) = a + bt + ct^2,$$

with the coefficients being the entries of the first row of C.

3. Consider the polynomial

$$p(t) = t^2 + \alpha t + \beta.$$

Show that there exists a 2×2 circulant matrix such that $p(t)$ is its characteristic polynomial and find a polynomial $q(t)$ such that the eigenvalues λ_1 and λ_2 are

$$\lambda_1 = q(1) \quad \text{and} \quad \lambda_2 = q(-1)$$

(1 and -1 are the square roots of the unity 1).

■

[a]The goal of this exercise is to exhibit the beautiful unity of the solutions of the quadratic and cubic equations, in a form that is easy to remember, which is based on the circulant matrices. This exercise is based on a result in [12].

Solution

1. By a simple computation, the eigenvalues of C are

$$\lambda_1 = a + b + c, \quad \lambda_2 = a + b\omega + c\omega^2, \quad \lambda_3 = a + b\bar{\omega} + c\bar{\omega}^2,$$

where $\omega = -\frac{1}{2} + \frac{\sqrt{3}}{2}i$ with $\omega^2 = \bar{\omega}$, $\omega^3 = 1$. The corresponding eigenvectors are

$$V_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad V_2 = \begin{bmatrix} 1 \\ \omega \\ \omega^2 \end{bmatrix}, \quad V_3 = \begin{bmatrix} 1 \\ \bar{\omega} \\ \bar{\omega}^2 \end{bmatrix}.$$

We see that ω is the cubic root of the unity and satisfies $\omega = e^{2\pi i/3}$.

2. The cubic roots of the unity are $1, \omega$, and $\bar{\omega}$. Hence, we have

$$q(1) = a+b+c = \lambda_1, \quad q(\omega) = a+b\omega+c\omega^2\lambda_2 \quad \text{and} \quad q(\bar{\omega}) = a+b\bar{\omega}+c\bar{\omega}^2 = \lambda_3.$$

3. Consider the circulant matrix

$$C = \begin{bmatrix} a & b \\ b & a \end{bmatrix}.$$

The characteristic polynomial of C is

$$\det \begin{bmatrix} a-\lambda & b \\ b & a-\lambda \end{bmatrix} = \lambda^2 - 2a\lambda + a^2 - b^2.$$

This characteristic polynomial equals p if and only if

$$a = -\frac{\alpha}{2} \quad \text{and} \quad b = \pm\sqrt{\frac{\alpha^2}{4} - \beta}.$$

Hence, we can take for C the matrix

$$C = \begin{bmatrix} -\alpha/2 & \sqrt{\alpha^2/4 - \beta} \\ \sqrt{\alpha^2/4 - \beta} & -\alpha/2 \end{bmatrix}.$$

Now, we can construct the polynomial $q(t)$ whose coefficients are the entries of the first row of C, as:

$$q(t) = -\frac{\alpha}{2} + t\sqrt{\frac{\alpha^2}{4} - \beta}.$$

Now, we have

$$q(1) = -\frac{\alpha}{2} + \sqrt{\frac{\alpha^2}{4} - \beta} \qquad \text{and} \qquad q(-1) = -\frac{\alpha}{2} - \sqrt{\frac{\alpha^2}{4} - \beta},$$

so $q(1)$ and $q(-1)$ are the roots of the polynomial $p(t)$.

As we have seen here, to find the roots of a polynomial p, we first need to find a circulant matrix C having $(-1)^n p$ as its characteristic polynomial. The first row of C then defines a different polynomial q and the roots of p are the eigenvalues of C and are obtained by applying q to the nth roots of the unity. The same ideas can be applied for cubic and quartic polynomials.

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Orthogonal Matrices and Quadratic Forms

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8.1 Orthogonal Matrices

In Definition 1.4.2 we introduced the symmetric matrices as those square matrices that are invariant under the transpose operation. That is, those matrices A in $\mathscr{M}_n(\mathbb{K})$ satisfying $A = A^T$. This class of matrices has very important properties: for instance, they are diagonalizable (Theorem 7.4.6) and if the entries of a symmetric matrix are real, then it has only real eigenvalues (Theorem 7.4.5). Here we will study another class of matrices, whose inverses coincide with their transpose. These matrices are called the *orthogonal matrices*. In this section we restrict ourselves to the case of matrices with real entries. But all the results can be easily extended to the matrices with complex entries.

Definition 8.1.1 (Orthogonal Matrix)
Let A be a matrix in $\mathscr{M}_n(\mathbb{R})$. Then, A is said to be *orthogonal* if its columns and rows are orthogonal unit vectors (i.e., orthonormal vectors). See Definition 3.4.1 for orthogonal vectors in $\mathbb{R}^n$.

Remark 8.1.1 We have a similar situation if $\mathbb{K} = \mathbb{C}$, the matrix will be then called a *unitary* matrix and it enjoys properties similar to those of an orthogonal matrix, but with respect to the inner product in $\mathbb{C}^n$ defined by $(u_1, u_2, \ldots, u_n) \cdot (v_1, v_2, \ldots, v_n) = u_1\bar{v}_1 + \cdots + u_n\bar{v}_n$. We will not discuss this here, since all the results on orthogonal matrices can be easily adapted to the case of unitary matrices.

Example 8.1
The matrices

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}, \quad \begin{bmatrix} 3/7 & 2/7 & 6/7 \\ -6/7 & 3/7 & 2/7 \\ 2/7 & 6/7 & -3/7 \end{bmatrix}$$

are orthogonal. ■

Example 8.2 (Permutation Matrix)
A permutation matrix is a square matrix obtained from the same size identity matrix by a permutation of rows. There are $n!$ permutation matrices of size n. For example, if $n = 2$, the permutation matrices are

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

It is clear that a permutation matrix is orthogonal since all its row vectors and column vectors are orthonormal vectors. ■

One of the important properties of orthogonal matrices is given next.

Theorem 8.1.2 (Characterization of Orthogonal Matrices)
Let A be a matrix in $\mathscr{M}_n(\mathbb{R})$. Then, A is orthogonal if it is invertible and its inverse is the same as its transpose. That is, if

$$A^{-1} = A^T, \tag{8.1}$$

or equivalently, if

$$AA^T = A^TA = I_n. \tag{8.2}$$

Proof
To prove the theorem it is enough to show that (8.2) holds. Then, the uniqueness of the inverse (Theorem 1.2.3) gives (8.1). Let $v_1, v_2, \ldots, v_n$ be the row vectors of the matrix A. Thus

$$A = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix};$$

hence, the columns of A^T are $v_1^T, v_1^T, \ldots, v_n^T$. Then, we have

$$AA^T = \begin{bmatrix} v_1 \\ v_2 \\ \cdots \\ v_n \end{bmatrix} [v_1^T, v_2^T, \ldots, v_n^T] = [e_1, e_2, \ldots, e_n] = I_n,$$

where $e_i, 1 \leq i \leq n$, are the standard unit vectors in $\mathbb{R}^n$. Here we used the fact that the row vectors of A are orthonormal, that is $v_i v_j^T = 0$ for $i \neq j$ and $v_i v_i^T = 1$. By the same argument, we may show that $A^T A = I_n$. Thus, (8.2) holds and the proof of Theorem 8.1.2 is complete. □

Theorem 8.1.3 (Determinant of an Orthogonal Matrix)
Let A be an orthogonal matrix in $\mathscr{M}_n(\mathbb{R})$. Then

$$\det(A) = \pm 1.$$

Proof
Since A is orthogonal, then according to Theorem 8.1.2,

$$AA^T = A^T A = I_n.$$

Thus, using Theorems 2.3.1 and 2.4.7, we have

$$\det(AA^T) = \det(A)\det(A^T) = (\det(A))^2 = \det(I_n) = 1,$$

as claimed. □

In the following theorem we show and important property of the orthogonal matrices in $\mathscr{M}_2(\mathbb{R})$.

Theorem 8.1.4 (Characterization of Orthogonal Matrices in $\mathscr{M}_2(\mathbb{R})$)
Let A be an orthogonal matrix in $\mathscr{M}_2(\mathbb{R})$. Then, A has one of the following two forms:

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \quad or \quad A = \begin{bmatrix} \cos\theta & \sin\theta \\ \sin\theta & -\cos\theta \end{bmatrix}, \tag{8.3}$$

for some angle θ.

Proof

The first matrix in (8.3) is called the counterclockwise *rotation* matrix and if $\theta = \pi/2$, the second one is called the *reflection* matrix. First, it is clear that the matrices defined in (8.3) are orthogonal. Now, let

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

be a matrix in $\mathscr{M}_2(\mathbb{R})$. By definition, A is orthogonal if and only if its columns and row vectors of A are orthonormal vectors, i.e., if and only if the following holds:

$$\begin{cases} a^2 + b^2 = 1, \\ a^2 + c^2 = 1, \\ c^2 + d^2 = 1, \\ b^2 + d^2 = 1 \end{cases}$$

and

$$\begin{cases} ac + bd = 0, \\ ab + cd = 0. \end{cases}$$

From the first system, we deduce that there exists and angle θ such that

$$a = d = |\cos\theta|, \quad b = c = |\sin\theta|.$$

The second system gives

$$a = \cos\theta, \quad b = -\sin\theta, \quad c = \sin\theta, \quad d = \cos\theta$$

or

$$a = \cos\theta, \quad b = -\sin\theta, \quad c = \sin\theta, \quad d = \cos\theta.$$

Hence, the matrix A has one of the forms in (8.3). □

Now, we can easily show that the inverse of any orthogonal matrix is orthogonal. Indeed, if A is orthogonal, then

$$AA^T = A^TA = I_n,$$

so taking here the inverse, we get

$$(A^{-1})^TA^{-1} = A^{-1}(A^{-1})^T = I_n.$$

Also, the product of two orthogonal matrices is also orthogonal: if A and B are orthogonal, then

$$AA^T = A^TA = I_n \quad \text{and} \quad BB^T = B^TB = I_n.$$

Hence, by Theorem 1.4.1.

$$(AB)(AB)^T = A(BB^T)A = I_n \quad \text{and} \quad (AB)^T(AB) = B^T(A^TA)B = I_n.$$

Note that the set of orthogonal matrices is not empty, since it contains at least the matrix I_n. Consequently, we have here the algebraic structure of a subgroup (see Exercise 1.11) and thus we have already proved the following theorem.

Theorem 8.1.5 (The Group $O(n)$)
The orthogonal matrices form a subgroup of the group of matrices $(\mathscr{M}_n(\mathbb{R}), \cdot)$. This subgroup is called the orthogonal subgroup (or group) and it is denoted by $O(n)$.

We have proved in Theorem 7.4.5, that the eigenvalues of a real symmetric matrix are all real. Now, we introduce a very important property of the eigenvectors of a symmetric matrix.

Theorem 8.1.6 (Eigenvectors of a Real Symmetric Matrix)
Let A be a symmetric matrix in $\mathscr{M}_n(\mathbb{R})$. Then the eigenvectors of A associated to distinct eigenvalues are orthogonal.

Proof
Let λ_1 and λ_2 be two eigenvalues of A with $\lambda_1 \neq \lambda_2$ and let X_1 and X_2 be corresponding associated eigenvectors, i.e.,

$$AX_1 = \lambda_1 X_1 \qquad \text{and} \qquad AX_2 = \lambda_2 X_2. \tag{8.4}$$

Multiplying the first equation in (8.4) by X_2^T, we obtain

$$\begin{aligned} \lambda_1 X_2^T X_1 &= X_2^T(AX_1) \\ &= (X_2^TA)X_1 \\ &= (A^TX_2)^TX_1 \\ &= (AX_2)^TX_1 \\ &= \lambda_2 X_2^TX_1, \end{aligned}$$

where we have used the fact that $A^T = A$. It follows that

$$(\lambda_1 - \lambda_2)X_2^T X_1 = 0.$$

Since $\lambda_1 \neq \lambda_2$, we deduce that $X_2^T X_1 = 0$, i.e., X_1 and X_2 are orthogonal. □

Since the eigenvectors associated to distinct eigenvalues of a real symmetric matrix are orthogonal, then in order to build an orthogonal matrix of these eigenvectors, it remains to normalize them. This process of normalizing orthogonal vectors is known as the *the Gram–Schmidt process.*

8.1.1 The Gram–Schmidt Process

More generally, the Gram–Schmidt process is a method of orthonormalizing a set $\mathbf{S}_0 = \{v_1, v_2, \ldots, v_k\}$, $k \leq n$ of linearly independent vectors in $\mathbb{R}^n$. Basically, it takes the set $\mathbf{S}_0$ and generates a new set $\mathbf{S}_1 = \{u_1, u_2, \ldots, u_k\}$ of orthonormal vectors that spans the same k-dimensional subspace of $\mathbb{R}^n$ as $\mathbf{S}_0$. To make the set $\mathbf{S}_0$ orthonormal, there are two main steps. First, one has to make $\mathbf{S}_0$ orthogonal, and second, once it is orthogonal, one has to make it orthonormal. The main idea of this method is based on the projection formula (3.33). To explain the process, let us, for example, take three linearly independent vectors v_1, v_2, and v_3 of $\mathbb{R}^n$. We want to find three linearly independent and orthonormal vectors u_1, u_2, and u_3. So, the first vector u_1 can go with the direction of v_1, it just has to be normalized. So, we put

$$u_1 = \frac{v_1}{\|v_1\|}.$$

Now, the task is to choose u_2 such that it is orthogonal to u_1 and has norm equal to 1. We proceed exactly as we did in Theorem 3.4.1. If we choose

$$w_2 = v_2 - \frac{v_2 \cdot u_1}{\|u_1\|^2} u_1 = v_2 - \text{proj}_{u_1} v_2,$$

then since w_2 is orthogonal to v_1, we need to take

$$u_2 = \frac{w_2}{\|w_2\|},$$

since u_2 is required to be a unit vector. We see that u_1 has the same direction as v_1, but for u_2 we subtracted from v_2 the component in the direction of u_1 (which is the direction of v_1). Now, at this point, the vectors u_1 and u_2 are set. Now, we need to choose u_3 such that it will not lie in the plane of u_1 and u_2, which is exactly the plane of v_1 and v_2. So, we simply need to subtract from v_3 any component of u_3 in the plane of u_1 and u_2. Thus, we take w_3 to be the vector

$$w_3 = v_3 - \text{proj}_{u_1} v_3 - \text{proj}_{u_2} v_3.$$

It is clear that w_3 is orthogonal to u_1 and u_2. Since u_3 is required to be a unit vector, we choose

$$u_3 = \frac{w_3}{\|w_3\|}.$$

This process of choosing u_1, u_2, and u_3 is called the *the Gram–Schmidt process* and we may apply the same ideas to any finite number of vectors. So, suppose now that we have the set $\mathbf{S}_0 = \{v_1, v_2, \ldots, v_k\}$, $k \leq n$, of linearly independent vectors. Hence, to construct the set $\mathbf{S}_1 = \{u_1, u_2, \ldots, u_k\}$ described above, we use the following algorithm:

$$\begin{aligned}
w_1 &= v_1, & u_1 &= \frac{w_1}{\|w_1\|},\\
w_2 &= v_2 - \text{proj}_{u_1} v_2, & u_2 &= \frac{w_2}{\|w_2\|},\\
w_3 &= v_3 - \text{proj}_{u_1} v_3 - \text{proj}_{u_2} v_3, & u_3 &= \frac{w_3}{\|w_3\|},\\
&\vdots & &\vdots\\
w_k &= v_k - \sum_{j=1}^{k-1} \text{proj}_{u_j} v_k, & u_k &= \frac{w_k}{\|w_k\|}.
\end{aligned} \tag{8.5}$$

We may easily show that the vectors $u_1, u_2, \ldots, u_k$ are orthonormal. We see that at the step k, we subtracted from v_k its components in the directions that are already settled.

Now, if $\mathbf{S}_0$ contains n vectors, then according to Theorem 4.6.2, it forms a basis of $\mathbb{R}^n$. Therefore, the set $\mathbf{S}_1$ is also a basis of $\mathbb{R}^n$, but it is already an orthonormal basis. Thus, we have already proved the following theorem.

Theorem 8.1.7 (Orthonormal Basis)
In any finite dimensional Euclidean vector space there exists an orthonormal basis.

Example 8.3
Apply the Gram–Schmidt process to the following vectors in $\mathbb{R}^4$:

$$v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix} \quad \text{and} \quad v_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix}.$$

■

Solution

We follow the Gram–Schmidt process described in (8.5) and define

$$u_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{14}} v_1 = \begin{bmatrix} \frac{1}{\sqrt{14}} \\ \sqrt{\frac{2}{7}} \\ \frac{3}{\sqrt{14}} \\ 0 \end{bmatrix}.$$

Now, to find u_2, we need first to find w_2, so, compute

$$\text{proj}_{u_1} v_2 = \frac{v_2 \cdot u_1}{\|u_1\|^2} u_1 = (v_2 \cdot u_1) u_1.$$

We have $v_2 \cdot u_1 = 5/\sqrt{14}$. Thus,

$$\text{proj}_{u_1} v_2 = \frac{5}{14} u_1 = \begin{bmatrix} \frac{5}{14} \\ \frac{5}{7} \\ \frac{15}{14} \\ 0 \end{bmatrix},$$

and then

$$w_2 = v_2 - \text{proj}_{u_1} v_2 = \begin{bmatrix} \frac{9}{14} \\ \frac{9}{7} \\ -\frac{15}{14} \\ 0 \end{bmatrix}.$$

Finally,

$$u_2 = \frac{w_2}{\|w_2\|} = \begin{bmatrix} \frac{27\sqrt{\frac{5}{14}}}{14} \\ \frac{27\sqrt{\frac{5}{14}}}{7} \\ -\frac{45}{14}\sqrt{\frac{5}{14}} \\ 0 \end{bmatrix}.$$

To be convinced, one can easily verify that u_1 and u_2 are orthonormal vectors. ◀

8.1.2 The *QR* Factorization

One of the important ideas in linear algebra is to write a real square matrix as the product of two matrices, one orthogonal and the other one upper triangular. This process is called

the *QR factorization* or *QR decomposition*. So, if A is a square matrix in $\mathscr{M}_n(\mathbb{R})$, we will show that one can write

$$A = QR,$$

where Q is an orthogonal matrix and R is an upper triangular matrix. We will also show that if A is invertible, then the above decomposition is unique. Actually, several methods of finding the QR decomposition are available; here we discuss the method based on the Gram–Schmidt process. We state the following result.

Theorem 8.1.8 (The QR Decomposition)
Let A be a matrix in $\mathscr{M}_n(\mathbb{R})$. Then, A can be factorized as

$$A = QR, \tag{8.6}$$

where Q is an orthogonal matrix and R is an upper triangular matrix. In addition, if A is invertible and if the diagonal entries of R are positive, then the decomposition (8.6) is unique.

Proof
First, let us prove the uniqueness. So, assume that A is invertible and assume that there exists two orthogonal matrices Q_1 and Q_2 and two upper triangular matrices R_1 and R_2, such that

$$A = Q_1R_1 = Q_2R_2.$$

Then multiplying from the left by Q_2^{-1} and from the right by R_1^{-1}, we obtain

$$Q = Q_2^{-1}Q_1 = R_2R_1^{-1} = R.$$

Now, since Q is orthogonal, we have

$$Q^T = Q^{-1} = R^{-1}.$$

Since the inverse of a triangular matrix is triangular of the same type, we deduce that Q and Q^T are both upper triangular matrices. Hence, Q is an upper as well as a lower triangular matrix, and so it is a diagonal matrix. In addition its diagonal entries are strictly positive, then we have

$$Q^2 = QQ^T = I_n.$$

This means that $Q = I_n$ and therefore the uniqueness of the inverse gives $Q_1 = Q_2$ and $R_1 = R_2$.

The existence of the decomposition (8.6) follows from the Gram–Schmidt process. Indeed, let $v_1, v_2, \ldots, v_n$ be the column vectors of the matrix A. Then, according to the Gram–Schmidt process, we can form a set of orthogonal vectors $u_1, u_2, \ldots, u_n$ as described in (8.5). Thus, the matrix Q defined as

$$Q = [u_1, u_2, \ldots, u_n]$$

is an orthogonal matrix. Thus, we need to find the matrix $R = [r_1, r_2, \ldots, r_n]$ with the column vectors $r_1, r_2, \ldots, r_n$ such that

$$A = [v_1, v_2, \ldots, v_n] = QR = [u_1, u_2, \ldots, u_n][r_1, r_2, \ldots, r_n].$$

To find R, we simply need to write (8.5) in matrix form, which yields

$$R = \begin{bmatrix} v_1 \cdot u_1 & v_2 \cdot u_1 & \cdots & v_n \cdot u_1 \\ 0 & v_2 \cdot u_2 & \cdots & v_n \cdot u_2 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & v_n \cdot u_n \end{bmatrix}.$$

It is clear that R is an upper triangular matrix. This completes the proof of Theorem 8.1.8. □

Example 8.4

Find the QR decomposition of the matrix

$$A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}.$$

■

Solution

Let v_1, v_2, and v_3 be the column vectors of A. Now, we need to find u_1, u_2, and u_3 using the Gram–Schmidt process. Indeed, we have

$$u_1 = \frac{v_1}{\|v_1\|} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{bmatrix}.$$

Now, we need to find w_2, thus, we first take

$$\text{proj}_{u_1} v_2 = \frac{v_2 \cdot u_1}{\|u_1\|^2} u_1 = (v_2 \cdot u_1) u_1 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 0 \end{bmatrix}.$$

Thus,

$$w_2 = v_2 - \text{proj}_{u_1} v_2 = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{bmatrix}.$$

Hence,

$$u_2 = \frac{w_2}{\|w_2\|} = \begin{bmatrix} \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{bmatrix}.$$

Now, to find w_3, we need first to compute

$$\text{proj}_{u_1} v_3 \qquad \text{and} \qquad \text{proj}_{u_2} v_3.$$

We have

$$\text{proj}_{u_1} v_3 = (v_3 \cdot u_1)u_1 = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ 0 \end{bmatrix},$$

and similarly,

$$\text{proj}_{u_2} v_3 = (v_3 \cdot u_2)u_2 = \begin{bmatrix} \frac{1}{6} \\ -\frac{1}{6} \\ \frac{2}{6} \end{bmatrix}.$$

Hence,

$$w_3 = v_3 - \text{proj}_{u_1} v_3 - \text{proj}_{u_2} v_3 = \begin{bmatrix} -\frac{2}{3} \\ \frac{2}{3} \\ \frac{2}{3} \end{bmatrix}.$$

Then,

$$u_3 = \frac{w_3}{\|w_3\|} = \begin{bmatrix} -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}.$$

Consequently, the matrix Q has the form

$$Q = [u_1, u_2, u_3] = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}.$$

Also, a simple computation yields

$$R = \begin{bmatrix} v_1 \cdot u_1 & v_1 \cdot u_2 & v_1 \cdot u_3 \\ 0 & v_2 \cdot u_2 & v_2 \cdot u_3 \\ 0 & 0 & v_3 \cdot u_3 \end{bmatrix} = \begin{bmatrix} \frac{2}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \frac{3}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{2}{\sqrt{3}} \end{bmatrix}.$$

◀

8

Now, we state another important theorems of linear algebra. We have seen in Theorem 7.4.6 that every symmetric matrix is diagonalizable. Thus, if A is a symmetric matrix, then there exists an invertible matrix S such that the matrix

$$D = S^{-1}AS$$

is diagonal. Moreover, the matrix S is the eigenvector matrix (the matrix whose column vectors are the eigenvectors of A). So, in the light of Theorem 8.1.6, we may ask *whether one can choose these eigenvectors to be orthonormal?* In other words, *is there an orthogonal matrix S such that the identity*

$$S^{-1}AS = S^TAS$$

holds true? The answer to this question is affirmative and in this case the matrix A is called *orthogonally diagonalizable* .

Theorem 8.1.9 (Spectral Theorem)
Let A be a matrix in $\mathscr{M}_n(\mathbb{R})$. *Then, A is* orthogonally diagonalizable *(that is there exits an orthogonal matrix S such that the matrix*

$$D = S^{-1}AS = S^TAS \tag{8.7}$$

is diagonal), if and only if A is symmetric.

Proof
First, if (8.7) is satisfied, then, A can be written as

$$A = SDS^{-1} = SDS^T.$$

Hence,

$$A^T = (SDS^T)^T = SD^T S^T = SDS^T = A,$$

since D is a diagonal matrix, and therefore symmetric. Consequently, A is symmetric.

Now, assume that A is symmetric. To show that A is orthogonally diagonalizable, we first show that there exists an orthogonal matrix S such that the matrix

$$T = S^T AS \tag{8.8}$$

is upper triangular. (This result is known as Schur's lemma).

Since A is a symmetric matrix, T is also symmetric matrix; indeed

$$(S^T AS)^T = S^T A^T S = S^T AS = T^T.$$

Since T is symmetric and triangular, it is automatically diagonal. Thus, $T = D$. It remains to prove (8.8) which will be done in the next lemma. □

Lemma 8.1.10 (Schur's Lemma) *Let A be a matrix in $\mathscr{M}_n(\mathbb{R})$ with real eigenvalues.*[a] *Then, there exists an orthogonal matrix S such that the matrix*

$$T = S^T AS \tag{8.9}$$

is upper triangular.

[a] If A is a matrix in $\mathscr{M}_n(\mathbb{C})$, then the assumption of real eigenvalues is not needed and we need to use a unitary matrix instead of the orthogonal one.

Proof

We proceed by induction on the size of the matrix n. For $n = 1$, $A = [a]$, therefore, we can take $S = [1]$. Now suppose that the lemma holds true for any $(n-1) \times (n-1)$ (with $n \geq 2$) matrix. Let A be an $n \times n$ matrix. Let λ_1 be an eigenvalue of A and v_1 be an eigenvector associated to λ_1. By dividing v_1 by $\|v_1\|$, one can assume that v_1 is a unit vector. Thus, according to Theorem 4.6.8, we can extend the set $\{v_1\}$ to a basis $\{v_1, u_2, \ldots, u_n\}$ of $\mathbb{R}^n$. Using the Gram–Schmidt process, we transform this basis into an orthonormal basis $\mathbf{B} = \{v_1, v_2, \ldots, v_n\}$ of $\mathbb{R}^n$. Consider the matrix Q whose columns are $v_1, v_2, \ldots, v_n$,

$$Q = [v_1, v_2, \ldots, v_n].$$

It is obvious that Q is an orthogonal matrix. Now, set $A_1 = Q^T AQ$. Then for

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

we have

$$QA_1e_1 = AQe_1 = Av_1 = \lambda_1 v_1 = \lambda_1 Qe_1.$$

Hence, we obtain $A_1e_1 = \lambda_1 e_1$. In other words, A_1 has the block-triangular form

$$A_1 = \left[\begin{array}{c|c} \lambda_1 & * \ \ldots \ * \\ \hline 0 & \\ \vdots & B \\ 0 & \end{array}\right],$$

where B is an $(n-1) \times (n-1)$ matrix. Applying the induction hypothesis, there exists an $(n-1) \times (n-1)$ orthogonal matrix W such that the matrix W^TBW is upper triangular. Now we consider the matrix (block-diagonal) $W_1 = \text{diag}(1, W)$. Then the matrix $T = W_1^T A_1 W_1$ is upper triangular. Set

$$S = QW_1.$$

Then it is clear that S is orthogonal (since the product of two orthogonal matrices is orthogonal) and we have

$$S^TAS = W_1^T Q^T AQW_1 = W_1^T A_1 W_1 = T.$$

This finishes the proof of Lemma 8.1.10. □

8.1.3 The *LU* Factorization

In many applications where linear systems arise, one needs to solve the equation $AX = b$, where A is an invertible matrix in $\mathscr{M}_n(\mathbb{K})$[1] and X and b are vectors in $\mathbb{K}^n$. The best way to solve this equation (system) is to replace the coefficient matrix A (through some row operations) with another matrix that is triangular. This procedure is known as the Gauss elimination method and is basically equivalent of writing the matrix A in the form

$$A = LU,$$

where L is a lower triangular matrix and U is an upper triangular matrix. The question now is the following: *does such decomposition always exist, and if so, is it unique?* To provide an answer, we start with the following definitions.

[1] $\mathbb{K}$ here is not necessary $\mathbb{R}$ or $\mathbb{C}$.

Definition 8.1.2 (Principal Minor)

Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$. The *principal submatrix* of order k $(1 \leq k \leq n)$ is the submatrix formed by deleting from A $n-k$ rows and the $n-k$ columns with the same indices (for example, delete rows $1, 2$ and 5 and columns $1, 2$ and 5). The determinant of this submatrix is called the *principal minor* of order k of A.

Definition 8.1.3 (Leading Principal Minor)

Let A be a matrix in $\mathscr{M}_n(\mathbb{K})$. A minor of A of order k, $(1 \leq k \leq n)$ is called a *leading principal minor* of order k if it is the determinant of the submatrix obtained by deleting from A the last $n-k$ rows and columns. This submatrix is called the *leading principal submatrix*.

Example 8.5

Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 3 & 4 \\ 2 & 2 & 5 \end{bmatrix}.$$

Then, the leading principal minors of A are

$$A^{(1)} = \det[1] = 1, \quad A^{(2)} = \det \begin{bmatrix} 1 & 2 \\ -1 & 3 \end{bmatrix} = 5, \quad A^{(3)} = \det(A) = 33.$$

■

We will see in ▶ Sect. 8.2 that the leading principal minors can be used as a test for the definiteness of symmetric matrices.

Theorem 8.1.11 (LU Decomposition)

Let A be an invertible matrix in $\mathscr{M}_n(\mathbb{K})$. Then A admits a unique decomposition of the form

$$A = LU, \tag{8.10}$$

where L is a lower triangular matrix with 1's on the diagonal and U is an upper triangular matrix, if and only if all its leading principal minors are nonzero.

Proof

We first prove the uniqueness. Assume that there exist L_1, L_2, U_1, and U_2 satisfying the assumptions in Theorem 8.1.11, such that

$$A = L_1U_1 = L_2U_2.$$

This gives, by multiplying from the left by L_2^{-1} and from the right by U_2^{-1} (recall that all the four matrices above are invertible, since A is invertible)

$$L_2^{-1}L_1U_1U_2^{-1} = I_n.$$

Now, we put $L = L_2^{-1}L_1$ and $U = U_1U_2^{-1}$. It is clear that L is a lower triangular matrix and U is an upper triangular matrix. Thus, since $L = U$, they necessarily are diagonal matrices, with the diagonal entries of L being 1's. Then, we have $L = U = I_n$. This shows that $L_1 = L_2$ and $U_1 = U_2$, so, the decomposition (8.10) is unique.

The proof of the existence of the factorization (8.10), is done by induction on n (the size of the matrix A). Trivially, if $n = 1$, then $A = [a] = [1][a]$, so (8.10) is satisfied for $n = 1$. Now, assume that (8.10) holds for square matrices of size $(n-1) \times (n-1)$ and partition A as

$$A = \begin{bmatrix} A_{n-1} & a_1 \\ a_2^T & a_{nn} \end{bmatrix}$$

where A_{n-1} is an $(n-1) \times (n-1)$ matrix, a_1 and a_2 are vectors of $\mathbb{K}^n$, and a_{nn} is in $\mathbb{K}$. Now, it is clear from the hypothesis of the theorem that all the minors of A_{n-1} are not zero and so A_{n-1} is invertible. Hence, we may apply the induction assumption to A_{n-1} and write it as

$$A_{n-1} = L_{n-1}U_{n-1},$$

where L_{n-1} and U_{n-1} are $(n-1) \times (n-1)$ matrices with L_{n-1} a lower triangular matrix whose diagonal elements are equal to 1 and U_{n-1} an upper triangular matrix.

Now, consider the two $n \times n$ matrices

$$L = \begin{bmatrix} L_{n-1} & \mathbf{0}_{(n-1)\times 1} \\ d^T & 1 \end{bmatrix} \quad \text{and} \quad U = \begin{bmatrix} U_{n-1} & c \\ \mathbf{0}_{1\times(n-1)} & a_{nn} - d^Tc \end{bmatrix} \tag{8.11}$$

where c and d are vectors in $\mathbb{K}^n$ to be determined. Now, the formula $A = LU$, gives

$$LU = \begin{bmatrix} L_{n-1}U_{n-1} & L_{n-1}c \\ d^TU_{n-1} & a_{nn} \end{bmatrix} = \begin{bmatrix} A_{n-1} & a_1 \\ a_2^T & a_{nn} \end{bmatrix}.$$

This yields

$$L_{n-1}c = a_1, \quad \text{and} \quad d^TU_{n-1} = a_2^T.$$

Since L_{n-1} and U_{n-1} are invertible matrices, then c and d are uniquely determined as

$$c = L_{n-1}^{-1} a_1, \qquad \text{and} \qquad d^T = a_2^T U_{n-1}^{-T}.$$

Thus, we have shown that $A = LU$, with L and U unique and given as in (8.11).

The converse is clear, since if A has the factorization (8.11), then, we may easily show that if $A^{(k)}$ is the leading principal minor of A of order k, then $A^{(k)} = L^{(k)}U^{(k)}$, where $L^{(k)}$ and $U^{(k)}$ are the principal leading minors of order k of L and U, respectively. Consequently,

$$A^{(k)} = \prod_{1 \le j \le p} u_{jj},$$

which is nonzero, since U is an invertible matrix. This finishes the proof of Theorem 8.1.11. □

Example 8.6

Find the LU decomposition of the matrix

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{bmatrix}.$$

■

Solution

First, we compute the leading principal minors of A, obtaining

$$A^{(1)} = \det[1] = 1, \quad A^{(2)} = \det \begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix} = 2, \quad A^{(3)} = \det(A) = 6.$$

Thus, according to Theorem 8.1.11, the LU decomposition exists and we have

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

$$= \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ l_{21}u_{11} & l_{21}u_{12} + u_{22} & l_{21}u_{13} + u_{23} \\ l_{31}u_{11} & l_{31}u_{12} + l_{32}u_{22} & l_{31}u_{13} + l_{32}u_{23} + u_{33} \end{bmatrix}$$

Solving the above system, we obtain

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 1 & 1 \end{bmatrix} \qquad \text{and} \qquad U = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}. \tag{8.12}$$

◄

The LU Algorithm

Computers usually solve square linear systems using the LU decomposition, since it is simpler and less costly (see Remark 8.2.9 below). When such a decomposition is available, then solving the system

$$AX = b \tag{8.13}$$

is relatively fast and simple. We first write $A = LU$ and solve

$$LY = b \tag{8.14}$$

for Y. Of course here we get a unique solution Y, since L is invertible. Also, since L is a lower triangular matrix, then system (8.14) should be solved in the "forward" direction. That is, if $y_1, y_2, \ldots, y_n$ are the components of the vector Y, then these components are found successively in the same order. Once Y is obtained, one solves for X the system

$$UX = Y. \tag{8.15}$$

Once again the triangular form of the matrix U makes the computations of X from (8.15) easy. In this case, we solve the system (8.15) in the "backward" direction. That is, the components of X are found in the order $x_n, x_{n-1}, \ldots, x_2, x_1$. Consequently, (8.14) and (8.15) yield

$$AX = L(UX) = LY = b.$$

Hence, the solution X obtained in (8.15) is the solution of (8.13).

Example 8.7

Solve the system of equations

$$\begin{cases} x_1 + 2x_2 + 4x_3 = 1, \\ 3x_1 + 8x_2 + 14x_3 = 0, \\ 2x_1 + 6x_2 + 13x_3 = 1. \end{cases} \tag{8.16}$$

■

Solution

The system (8.16) can be written as

$$AX = b$$

with

$$A = \begin{bmatrix} 1 & 2 & 4 \\ 3 & 8 & 14 \\ 2 & 6 & 13 \end{bmatrix}, \qquad X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \qquad \text{and} \qquad b = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}.$$

We have seen in Example 8.6 that the matrix A can be written as $A = LU$, with L and U as in (8.12). Next, we solve the system $LY = b$, where Y is the vector in $\mathbb{R}^3$ with the components y_1, y_2, and y_3. This gives

$$\begin{cases} y_1 = 1, \\ 3y_1 + y_2 = 0, \\ 2y_1 + y_2 + y_3 = 1. \end{cases}$$

Therefore, $y_1 = 1, y_2 = -3$ and $y_3 = 2$. Finally, we solve the system $UX = Y$, that is

$$\begin{cases} x_1 + 2x_2 + 4x_3 = 1, \\ 2x_2 + 2x_3 = -3, \\ 2x_3 = 2. \end{cases}$$

This yields $x_3 = 2/3, x_2 = -13/6$ and $x_1 = 8/3$. Hence, the solution of (8.16) is

$$X = \begin{bmatrix} \frac{8}{3} \\ -\frac{13}{6} \\ \frac{2}{3} \end{bmatrix}.$$

◀

8.2 Positive Definite Matrices

We have seen in Theorem 7.4.5, that real symmetric matrices have only real eigenvalues; these, of course can be positive or negative. Thus, the question now is *what happen if all the eigenvalues of a symmetric matrix are positive? Does the matrix enjoy some particular properties?* The signs of the eigenvalues of a matrix are important in applications, for instance in the stability theory of differential equations. So, it is quite important to determine which matrices have positive eigenvalues. These matrices are called *positive definite* matrices. Symmetric positive definite matrices have rather nice properties; for example, as we will see later on, every positive definite matrix is invertible. Also, by studying these matrices we will bring together many things that we have learned about determinants, eigenvalues,... We restrict our discussion here to the case $\mathbb{K} = \mathbb{R}$, but all the properties can be easily extended to the case $\mathbb{K} = \mathbb{C}$.

Now, we start with the definition.

Definition 8.2.1 (Positive Definite Matrix)

Let A be a symmetric matrix in $\mathscr{M}_n(\mathbb{R})$. Then A is said to be *positive definite* if all its eigenvalues are positive.

Example 8.8
The matrix

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 5 \end{bmatrix}$$

is positive definite, since it is symmetric and its eigenvalues λ_1 and λ_2 satisfy (see Corollary 7.3.3)

$$\lambda_1 + \lambda_2 = \text{tr}(A) = 6, \qquad \text{and} \qquad \det A = \lambda_1 \lambda_2 = 1.$$

Thus, λ_1 and λ_2 have to be positive. ■

Notation Here we introduce a new notation which is very useful in this chapter. If X and Y are two vectors in $\mathbb{R}^n$, we denote

$$\langle X, Y \rangle = X^T Y.$$

Then for any matrix A in $\mathscr{M}_n(\mathbb{R})$ it holds that

$$\langle AX, Y \rangle = \langle X, A^T Y \rangle.$$

As we have seen in the previous chapters, computing the eigenvalues of a matrix can be a challenging problem, especially for matrices of high order. So, it seems difficult to use Definition 8.2.1 in general. Hence, our goal is to find a test that can be applied directly to a symmetric matrix A, without going through the computation of its eigenvalues, which will guarantee that the matrix is positive definite. This test is given in the following theorem.

Theorem 8.2.1 (Characterization of Positive Definite Matrices)
Let A be a symmetric matrix in $\mathscr{M}_n(\mathbb{R})$. Then A is positive definite if and only if

$$\langle X, AX \rangle = X^T AX > 0,^a \tag{8.17}$$

for all nonzero vectors X in $\mathbb{R}^n$.

[a] Since A is symmetric, we have $\langle AX, Y \rangle = \langle X, A^T Y \rangle = \langle X, AY \rangle$.

Proof
First assume that A is positive definite. Then, by Theorem 8.1.9, the eigenvectors $v_1, v_2, \ldots, v_n$ of A form an orthonormal basis of $\mathbb{R}^n$. Thus, if λ_i is an eigenvalue of A and v_i is an associated eigenvector, then we have

$$\langle v_i, Av_i \rangle = \langle v_i, \lambda_i v_i \rangle = \lambda_i \langle v_i, v_i \rangle = \lambda_i \|v_i\|^2 = \lambda_i > 0, \quad i = 1, 2, \ldots, n.$$

Since (8.17) is satisfied for the elements of a basis of $\mathbb{R}^n$, it is satisfied for any nonzero vector X of $\mathbb{R}^n$.

Second, assume that (8.17) is satisfied and let λ be an eigenvalue of A, i.e., $AX = \lambda X$, for some eigenvector X associated to λ. Then

$$\langle X, AX\rangle = \langle X, \lambda X\rangle = \lambda \|X\|^2 > 0.$$

This shows that $\lambda > 0$. Thus, A is positive definite. □

Example 8.9

Show that the matrix

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

is positive definite. ■

Solution

To do so, we apply Theorem 8.2.1. So, let

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \tag{8.18}$$

be a nonzero vector in $\mathbb{R}^3$. Then,

$$\langle X, AX\rangle = 2\left(x_1^2 - x_2x_1 + x_2^2 + x_3^2 - x_2x_3\right).$$

Since

$$-x_2x_1 \geq -\frac{1}{2}(x_1^2 + x_2^2) \quad \text{and} \quad -x_2x_3 \geq -\frac{1}{2}(x_2^2 + x_3^2),$$

it is clear that

$$\langle X, AX\rangle \geq x_1^2 + x_3^2 > 0,$$

if x_1 and x_3 are not both zero. Otherwise, if $x_1 = x_3 = 0$, then $x_2 \neq 0$ and hence we have

$$\langle X, AX\rangle = 2x_2^2 > 0.$$

Consequently, the matrix A is positive definite. ◄

Now, we can show an important property of positive definite matrices.

Theorem 8.2.2 (Invertibility and Positive Definite Matrices)
Let A be a positive definite matrix in $\mathscr{M}_n(\mathbb{R})$. Then A is invertible and A^{-1} is also positive definite.

Proof
If A is positive definite, then all its eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$ are necessarily positive. Then, by Corollary 7.3.3, we have

$$\det(A) = \prod_{i=1}^{n} \lambda_i \neq 0,$$

and Theorem 2.4.8 shows that A is invertible.

Now, A^{-1} is symmetric since $(A^{-1})^T = (A^T)^{-1} = A^{-1}$. Also, if λ is a positive eigenvalue of A, then $\frac{1}{\lambda}$ is a positive eigenvalue of A^{-1}. Thus, A^{-1} is positive definite. □

Remark 8.2.3 It is clear that if A is positive definite, then $\det(A) > 0$. However, if $\det(A) > 0$, then A is not necessarily positive definite. For, example, the determinant of the matrix

$$A = \begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$$

is positive, but A is not positive definite.

Next, we introduce another important property of positive definite matrices.

Theorem 8.2.4 (Leading Principal Minors and Positive Definite Matrices)
Let A be a symmetric matrix in $\mathscr{M}_n(\mathbb{R})$. Then, A is positive definite if an only if all its leading principal minors are positive.

To prove Theorem 8.2.4, we need the following lemma, known as the Rayleigh–Ritz theorem, which gives the relation between the eigenvalues of the matrix A and those of its principal submatrices. We do not prove this lemma; we need it in order to show Theorem 8.2.4. The reader is refereed to [13, Theorem 8.5.1].

Lemma 8.2.5 (Rayleigh–Ritz) *Let A be a symmetric matrix in $\mathscr{M}_n(\mathbb{R})$ and let A_k be its principal submatrix of order k. Let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be the eigenvalues of A and let $\mu_1, \mu_2, \ldots, \mu_k$ be the eigenvalues of A_k. Then, we have*

$$\lambda_i \leq \mu_i \leq \lambda_{i+n-k}, \quad i = 1, \ldots, k.$$

Proof of Theorem 8.2.4

First assume that A is positive definite. Then we can prove that all its principal minors are positive. Let A_k be the principal submatrix of order k, and let $\mu_1, \mu_2, \ldots, \mu_k$ be the eigenvalues of A_k. Since A_k is symmetric, its eigenvalues are real (Theorem 7.4.5). Hence, $\mu_1, \mu_2, \ldots, \mu_k$ can be ordered as $\mu_1 \leq \mu_2 \leq \cdots \leq \mu_k$, for instance. Applying Lemma 8.2.5, we deduce that

$$0 < \lambda_1 \leq \mu_1 \leq \mu_2 \leq \cdots \leq \mu_k.$$

This shows that

$$\det(A_k) = \prod_{1 \leq i \leq k} \mu_i > 0.$$

(Recall that $\lambda_1 > 0$, since A is positive definite). Hence, all the principal minors of A are positive. Consequently, in particular its leading principal minors are positive.

Conversely, assume that all the leading principal minors of A are positive. We denote by A_k the principal leading submatrix of order k. Then, we prove by induction on $k (k = 1, 2, \ldots, n)$ that the matrix $A = A_n$ is positive definite. For $k = 1$, $A_1 = [a]$ is positive definite, since in this case $a = \det(A) > 0$ (by assumption) and at the same time a is the eigenvalue of A_1. Now, for $k \geq 2$ we assume that A_{k-1} is positive definite and show that A_k is positive definite. Let

$$0 < \mu_1 \leq \mu_2 \leq \cdots \leq \mu_{k-1} \tag{8.19}$$

be the eigenvalues of A_{k-1} and $\alpha_1 \leq \alpha_2 \leq \cdots \leq \alpha_k$ be the eigenvalues of A_k. Since A_{k-1} is a principal submatrix of A_k, then applying Lemma 8.2.5 we get

$$\alpha_1 \leq \mu_1 \leq \alpha_2 \leq \cdots \leq \alpha_{k-1} \leq \mu_k \leq \alpha_k.$$

The above formula together with (8.19) show the positivity of $\alpha_2, \ldots, \alpha_k$. Since all the leading principal minors of A are positive, $\det(A_k)$ is positive and we have

$$\det(A_k) = \alpha_1 \left(\prod_{2 \leq i \leq k} \alpha_i \right) > 0.$$

Thus, $\alpha_1 > 0$. Hence, all the eigenvalues of A_k are positive, therefore A_k is positive definite. We conclude that A is positive definite. □

Example 8.10

Use Theorem 8.2.4 to show that the matrix

$$A = \begin{bmatrix} 3 & 0 & 3 \\ 0 & 1 & -2 \\ 3 & -2 & 8 \end{bmatrix}$$

is positive definite. ■

Solution

We just need to verify that the principal minors of A are positive. Indeed, we have

$$A^{(1)} = 3, \quad A^{(2)} = \det \begin{bmatrix} 3 & 0 \\ 0 & 1 \end{bmatrix} = 3, \quad A^{(3)} = \det A = 3.$$

Thus, A is positive definite. ◄

Next, we investigate the relationship between the eigenvalues of a symmetric matrix and its rank.

Theorem 8.2.6 (Characterization of the Rank of a Symmetric Matrix)
Let A be a symmetric matrix in $\mathscr{M}_n(\mathbb{R})$. Then, the rank of A is the total number of nonzero eigenvalues of A. In particular, A has full rank if and only if A is positive definite.

To prove the above result, we need to show the following lemma.

Lemma 8.2.7 *Let A and C be two invertible matrices. Then for any matrix B, we have*

$$\operatorname{rank}(ABC) = \operatorname{rank}(B),$$

provided that the size of the above matrices are chosen such that the product ABC makes sense.

Proof

We define the matrix $K = ABC$. Now, our goal is to show that $\operatorname{rank}(K) = \operatorname{rank}(B)$. Applying (6.24), we deduce that

$$\operatorname{rank}(K) \leq \operatorname{rank}(AB) \leq \operatorname{rank}(B).$$

On the other hand, $B = A^{-1}KC^{-1}$. Hence, applying (6.24) again, we get

$$\operatorname{rank}(B) \leq \operatorname{rank}(A^{-1}K) \leq \operatorname{rank}(K).$$

The two inequalities above show that $\operatorname{rank}(K) = \operatorname{rank}(B)$. □

Proof of Theorem 8.2.6

Let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be the eigenvalues of A. Using the spectral theorem (Theorem 8.1.9), then we can write A as $A = SDS^T$, where $D = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$ and S is an invertible matrix. Hence, applying Lemma 8.2.7, we deduce that $\operatorname{rank}(A) = \operatorname{rank}(D)$, which is clearly the total number of nonzero eigenvalues of A. If A is positive definite, then all its eigenvalues are positive, and therefore A has full rank. □

Now, if we relax the condition (8.17) a little bit and allow the inequality $\langle X, AX\rangle \geq 0$, then we obtain a new class of symmetric matrices, called *positive semi-definite* matrices.

Definition 8.2.2 (Semi-Definite Matrix)

Let A be a symmetric matrix in $\mathscr{M}_n(\mathbb{R})$. Then A is called *positive semidefinite* if all its eigenvalues are nonnegative, or equivalently, if

$$\langle X, AX\rangle = X^T AX \geq 0, \tag{8.20}$$

for all nonzero vectors X in $\mathbb{R}^n$.

Example 8.11

Show that the matrices

$$A = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

are positive semi-definite. ■

Solution

First, the matrix A is positive semi-definite since its eigenvalues are $\lambda_1 = 0$ and $\lambda_2 = 1$.

Second, for the matrix B, we have

$$\langle BX, X\rangle = \langle X, BX\rangle = (x_1 + x_2 + x_3)^2 \geq 0$$

for any nonzero vector X given as in (8.18). Thus, B is positive semi-definite. ◄

Example 8.12

Show that for any rectangular matrix A in $\mathscr{M}_{m\times n}(\mathbb{R})$, the matrices $A^T A$ and AA^T are positive semi-definite. ■

Solution

First, it is clear that $A^T A$ is a square symmetric matrix in $\mathscr{M}_n(\mathbb{R})$. For any nonzero vector X in $\mathbb{R}^n$,

$$\langle X, A^T AX\rangle = X^T A^T AX = (AX)^T (AX) = \langle AX, AX\rangle = \|AX\|^2 \geq 0.$$

Thus, the matrix $A^T A$ is positive semi-definite.

Second, it is also clear that AA^T is a symmetric matrix in $\mathscr{M}_m(\mathbb{R})$. Thus, for any nonzero vector Y in $\mathbb{R}^m$, we have

$$\langle AA^T Y, Y\rangle = \langle A^T Y, A^T Y\rangle = \|A^T Y\|^2 \geq 0.$$

Consequently, the matrix AA^T is positive semi-definite. ◄

8.2.1 The Cholesky Decomposition

We have seen in Theorem 8.1.8 that if A is a square matrix in $\mathscr{M}_n(\mathbb{R})$, then we can write A as

$$A = QR,$$

where Q is an orthogonal matrix and R is an upper triangular matrix.

Also, we have seen in Theorem 8.1.11 that if A is an invertible matrix and if all its leading principal minors are nonzero, then we can write A as

$$A = LU,$$

where L and U are as before. Now, if we assume that the matrix A is positive definite, then U has to be equal to L^T and we write A as the product of a lower triangular matrix L and its transpose L^T. This product is known as the Cholesky decomposition (or factorization) and it is very useful in numerical analysis. See Remark 8.2.9 below.

Theorem 8.2.8 (Cholesky's Decomposition)
Let A be a positive definite matrix in $\mathscr{M}_n(\mathbb{R})$. Then there exists a unique[a] lower triangular matrix L in $\mathscr{M}_n(\mathbb{R})$, with strictly positive diagonal entries, such that

$$A = LL^T. \tag{8.21}$$

[a] We obtain uniqueness only if we assume that the diagonal entries of L are positive.

Proof
Let us first prove the uniqueness. Assume that there exist two lower triangular matrices L_1 and L_2 satisfying (8.21). Then,

$$L_1 L_1^T = L_2 L_2^T.$$

This gives, upon multiplying from the left by L_2^{-1} and from the right by $(L_2^{-1})^T = (L_2^T)^{-1}$,

$$I_n = LL^T, \quad \text{with} \quad L = L_2^{-1} L_1.$$

This means that

$$L^{-1} = L^T. \tag{8.22}$$

Since L is a lower triangular matrix (as the product of two lower triangular matrices), then L^{-1} is a lower triangular matrix and L^T is an upper triangular matrix. Then (8.22) shows that L is a diagonal matrix and satisfies $L^2 = I_n$. Since its diagonal entries are positive (keep in mind that the diagonal entries of L_1 and L_2 are positive). Then we obtain $L = I_n$, and thus $L_1 = L_2$. This shows the uniqueness of the decomposition (8.21).

To establish the existence of the decomposition (8.21), we proceed by induction on the size of the matrix A. The statement is trivial for $n = 1$, since if $A = [a]$, with $a > 0$, we can take $L = L^T = [\sqrt{a}]$.

Now, assume that the decomposition (8.21) exists for any $(n-1) \times (n-1)$ matrix. The matrix A can be written as

$$A = \begin{bmatrix} A_{n-1} & b \\ b^T & a_{nn} \end{bmatrix},$$

where A_{n-1} is a leading principal submatrix of A which is positive definite (Theorem 8.2.4), b is a vector in $\mathbb{R}^{n-1}$, and a_{nn} is a real positive number. By the induction hypothesis, A_{n-1} satisfies (8.21). Thus, there exists a unique lower triangular matrix L_{n-1} with strictly positive diagonal entries, such that

$$A_{n-1} = L_{n-1} L_{n-1}^T.$$

Next, we look for the desired matrix L in the form

$$L = \begin{bmatrix} L_{n-1} & \mathbf{0}_{(n-1)\times 1} \\ c^T & \alpha \end{bmatrix},$$

where c a vector in $\mathbb{R}^{n-1}$ and $\alpha > 0$ are to be determined. Now, the desired identity

$$A_{n-1} = LL^T$$

leads to

$$\begin{bmatrix} A_{n-1} & b \\ b^T & a_{nn} \end{bmatrix} = \begin{bmatrix} L_{n-1} & \mathbf{0}_{(n-1)\times 1} \\ c^T & \alpha \end{bmatrix} \begin{bmatrix} L_{n-1}^T & c \\ \mathbf{0}_{1\times(n-1)} & \alpha \end{bmatrix}.$$

This equation gives

$$L_{n-1} c = b, \quad \text{and} \quad \alpha^2 + \|c\|^2 = a_{nn}.$$

Since L_{n-1} is invertible, it follows that $c = L_{n-1}^{-1} b$. Also, $\alpha = \sqrt{a_{nn} - \|c\|^2}$. It is clear that $a_{nn} - \|c\|^2 > 0$, since $0 < \det(A) = \alpha^2 (\det(L_{n-1}))^2$. This shows that (8.21) holds for the matrix A. Thus the proof of Theorem 8.2.8 is complete. □

Example 8.13
Find the Cholesky decomposition of the matrix

$$A = \begin{bmatrix} 25 & 15 & -5 \\ 15 & 18 & 0 \\ -5 & 0 & 11 \end{bmatrix}.$$

■

Solution
First, it is clear that A is symmetric. To show that A is positive definite, we need to compute $\langle X, AX\rangle$ for any nonzero vector

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

of $\mathbb{R}^3$. We have

$$\langle X, AX\rangle = 25x_1^2 + 30x_2x_1 - 10x_3x_1 + 18x_2^2 + 11x_3^2.$$

Next, using the inequalities

$$x_2x_1 \geq -\frac{1}{2}(x_1^2 + x_2^2) \quad \text{and} \quad -x_1x_3 \geq -\frac{1}{2}(x_1^2 + x_3^2),$$

we obtain form above that

$$\langle X, AX\rangle \geq 5x_1^2 + 3x_2^2 + 6x_3^2 > 0.$$

Now, according to Theorem 8.2.8, there exists a unique lower triangular matrix

$$L = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix}$$

with $l_{11}, l_{22}, l_{33} > 0$, such that $A = LL^T$. That is

$$\begin{bmatrix} 25 & 15 & -5 \\ 15 & 18 & 0 \\ -5 & 0 & 11 \end{bmatrix} = \begin{bmatrix} l_{11} & 0 & 0 \\ l_{21} & l_{22} & 0 \\ l_{31} & l_{32} & l_{33} \end{bmatrix} \begin{bmatrix} l_{11} & l_{21} & l_{31} \\ 0 & l_{22} & l_{32} \\ 0 & 0 & l_{33} \end{bmatrix}$$

$$= \begin{bmatrix} l_{11}^2 & l_{11}l_{21} & l_{11}l_{31} \\ l_{11}l_{21} & l_{21}^2 + l_{22}^2 & l_{21}l_{31} + l_{22}l_{32} \\ l_{11}l_{31} & l_{21}l_{31} + l_{22}l_{32} & l_{31}^2 + l_{32}^2 + l_{33}^2 \end{bmatrix}.$$

Hence, we obtain the following system of equations

$$\begin{cases} l_{11}^2 = 25 \\ l_{11}l_{21} = 15, \\ l_{11}l_{31} = -5, \\ l_{21}^2 + l_{22}^2 = 18, \\ l_{21}l_{31} + l_{22}l_{32} = 0, \\ l_{31}^2 + l_{32}^2 + l_{33}^2 = 11. \end{cases}$$

Solving this system, we obtain

$$l_{11} = 5, \quad l_{21} = 3, \quad l_{22} = 3, \quad l_{31} = -1, \quad l_{32} = 1, \quad l_{33} = 3.$$

Consequently,

$$L = \begin{bmatrix} 5 & 0 & 0 \\ 3 & 3 & 0 \\ -1 & 1 & 3 \end{bmatrix}.$$

◀

Remark 8.2.9 The Cholesky decomposition can be used to solve linear systems of the form

$$AX = b,$$

where A is a matrix in $\mathscr{M}_n(\mathbb{R})$ and X and b are vectors in $\mathbb{R}^n$. If A is invertible, then to solve the above system by computing A^{-1} is costly. First, we compute A^{-1}, which costs $2n^3$ flops, and then compute $X = A^{-1}b$, which costs $2n^2$ flops.

However, if we use the LU decomposition in Theorem 8.1.11, then the cost is $(2/3)n^3$ flops, which is three times less than the cost in the first method.

Now, if A is positive definite, then we use the Cholesky decomposition and write $A = LL^T$ which costs $(1/3)n^3$ flops, and then we solve $LY = b$ for Y by forward substitution at the cost of n^2 flops. Then, we solve $L^TX = Y$ for X by back substitution and as above the cost for this is n^2. We see that the first method costs six times as much as the Cholesky decomposition, and the Cholesky decomposition costs two times less than the LU method.

As we see, there are large differences in the computation cost between the first method, the LU method, and the Cholesky decomposition method, especially if n is large.

8.3 Quadratic Forms

Quadratic forms arise in various areas of application; for example, in mechanics a quadratic form can describe the angular momentum of a solid body rotating about an axis. Quadratic forms are also used in optimization problems. Now, we introduce the definition of a quadratic form. Recall that a symmetric matrix is positive definite if for any nonzero vector X in $\mathbb{R}^n$, the quantity X^TAX is a positive real number. Define the function f as

$$\begin{aligned} f : \mathbb{R}^n &\to \mathbb{R} \\ X &\mapsto X^TAX = \langle X, AX \rangle. \end{aligned}$$

This function is called the *quadratic form* associated to the symmetric matrix A. In Example 8.9, we have

$$f(X) = f(x_1, x_2, x_3) = 2\left(x_1^2 - x_2x_1 + x_2^2 + x_3^2 - x_2x_3\right), \tag{8.23}$$

which is the quadratic form associated to the symmetric matrix A considered in Example 8.9. We see in (8.23) that each term has degree two. So, there are no linear or constant terms.

Definition 8.3.1 (Quadratic Form)

Let A be a real symmetric matrix. Then the *quadratic form* associated to A is the function f that maps each vector X in $\mathbb{R}^n$ to the real number

$$f(X) = X^TAX = \langle X, AX \rangle. \tag{8.24}$$

It is clear that the function in (8.24) is a polynomial of degree two and it can be written as

$$f(X) = a_{11}x_1^2 + a_{22}x_2^2 + \cdots + a_{nn}x_n^2 + 2a_{12}x_1x_2 + 2a_{13}x_1x_3 + \cdots + 2a_{n-1,n}x_{n-1}x_n, \tag{8.25}$$

where $x_1, x_2, \ldots, x_n$, are the components of the vector X and $a_{ij}, 1 \leq i, j \leq n$ are the entries of the matrix A. The terms involving the products x_ix_j are called the *mixed products*, and the matrix A is called the *coefficient matrix* of the quadratic form $f(X)$.

We have seen above that the symmetric matrix A is positive definite if the function $f(X)$ defined in (8.25) is positive for each nonzero vector X in $\mathbb{R}^n$. One way to see this is to write the quadratic form (8.25) as the sum of squares. This can be accomplished by using the spectral theorem (Theorem 8.1.9), as shown in the following theorem.

Theorem 8.3.1 (The Principal Axes Theorem)
Let A be a symmetric matrix in $\mathscr{M}_n(\mathbb{R})$ and $\lambda_1, \lambda_2, \ldots, \lambda_n$ be its eigenvalues. Let S be the orthogonal matrix given in Theorem 8.1.9. Then, the change of variable $X = SY$ transforms the quadratic form (8.25) into the standard form

$$f(Y) = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2,$$

where $y_1, y_2, \ldots, y_n$ are the components of the vector Y in $\mathbb{R}^n$.

Proof
Putting $X = SY$, we have

$$\begin{aligned} f(X) = X^T AX &= (SY)^T A(SY) \\ &= Y^T (S^T AS) Y \\ &= Y^T DY \\ &= \lambda_1 y_1^2 + \lambda_2 y_2^2 + \cdots + \lambda_n y_n^2, \end{aligned}$$

where D is the diagonal matrix defined in (8.7). This process is also called the diagonalization of the quadratic form $f(X)$. □

Example 8.14
Consider the quadratic form

$$f(x_1, x_2, x_3) = 4x_1^2 + 4x_2^2 + x_3^2 - 2x_1 x_2. \tag{8.26}$$

Find its standard form. ■

Solution
The quadratic form can be written as $f(X) = X^T AX$, with

$$X = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \text{and} \quad A = \begin{bmatrix} 4 & -1 & 0 \\ -1 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

We see that the diagonal entries of A are the coefficients of the squared terms in (8.26) and the off-diagonal entries are half the coefficient of the mixed product.

The eigenvalues of A are $\lambda_1 = 1, \lambda_2 = 5$, and $\lambda_3 = 3$. Thus, f can be written in the standard form as

$$f(y_1, y_2, y_3) = y_1^2 + 5y_2^2 + 3y_3^2.$$

◀

Definition 8.3.2 (Positive Definite Quadratic Form)

Let A be a symmetric matrix in $\mathscr{M}_n(\mathbb{R})$ and let $f(X)$ be the quadratic form associated to it. Then f is said to be positive definite (respectively, semi-definite) if A is positive definite (respectively, semi-definite).

8.3.1 Congruence and Sylvester's Law of Inertia

We have seen in the principal axes theorem (Theorem 8.3.1) that if $f(X)$ is the quadratic form associated to a symmetric matrix A, then we can write $f(X)$ as

$$\begin{aligned} f(X) = f(SY) &= Y^T \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n) Y \\ &= (S^T X)^T \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)(S^T X) \end{aligned}$$

where $\lambda_1, \lambda_2, \ldots, \lambda_n$ are the eigenvalues of A and S is the matrix of eigenvectors of A. Now, we have the so-called *Sylvester's law of inertia,* which says that if there exists another matrix R such that

$$f(X) = (R^T X)^T \operatorname{diag}(\mu_1, \mu_2, \ldots, \mu_n)(R^T X)$$

then if we consider the two sets

$$\Lambda = \{\lambda_1, \lambda_2, \ldots, \lambda_n\} \quad \text{and} \quad M = \{\mu_1, \mu_2, \ldots, \mu_n\},$$

the number of positive elements in Λ is equal to the number of positive elements in M, the number of negative elements in Λ is equal to the number of negative elements in M and the number of zeros in Λ is equal to the number of zeros in M. This means that regardless of how we diagonalize a quadratic form, it will have the same number of positive coefficients, negative coefficients, and zero coefficients.

Now, in order to prove the above result, we start with this some definitions.

Definition 8.3.3 (Inertia of a Symmetric Matrix)

The *inertia* of a real symmetric matrix A is defined to be the triple (p, ν, q), where p, ν, and q are, respectively, the numbers of positive, negative, and zero eigenvalues of A, counted with the algebraic multiplicity. We denote it by $\operatorname{In}(A)$ and write

$$\operatorname{In}(A) = (p, \nu, q).$$

Example 8.15
Find the inertia of the matrix

$$A = \begin{bmatrix} 20 & 6 & 8 \\ 6 & 3 & 0 \\ 8 & 0 & 8 \end{bmatrix}.$$

■

Solution
We compute the eigenvalues of A, finding

$$\lambda_1 = \frac{1}{2}\left(31 + \sqrt{385}\right), \quad \lambda_2 = \frac{1}{2}\left(31 - \sqrt{385}\right), \quad \lambda_3 = 0.$$

Hence, we have $p = 2, v = 0$, and $q = 1$: $\text{In}(A) = (2, 0, 1)$. ◄

Definition 8.3.4 (Congruent Matrices)
Let A and B be two symmetric matrices in $\mathscr{M}_n(\mathbb{R})$. Then, A and B are called *congruent* if there exists an invertible matrix S such that

$$A = S^T B S. \tag{8.27}$$

Now, we establish the following important result.

Theorem 8.3.2
Let A be a symmetric matrix in $\mathscr{M}_n(\mathbb{R})$. Then, A is congruent to the matrix

$$D_0 = \begin{bmatrix} I_p & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -I_v & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0}_{q\times q} \end{bmatrix} \tag{8.28}$$

with

$$p + v + q = n, \quad v = r - p, \quad \text{and} \quad q = n - r,$$

where $r = \text{rank}(A)$. The matrix D_0 is the canonical form of A with respect to congruence.

In terms of the quadratic form, this means that if there exists an orthonormal basis $\{e_1, e_2, \ldots, e_n\}$ of $\mathbb{R}^n$ such that the quadratic form associated to the symmetric matrix A has the form

$$f(X) = \sum_{i=1}^{n} a_i x_i^2,$$

then there exists an orthogonal basis $\{v_1, v_2, \ldots, v_n\}$ with $v_i = \sqrt{|a_i|}e_i$, $i = 1, 2, \ldots, n$, in which the quadratic form can be written as

$$f(X) = \sum_{i=1}^{p} x_i^2 - \sum_{j=1}^{v} x_j^2.$$

8

Proof of Theorem 8.3.2
Since A is symmetric, then using the spectral theorem (Theorem 8.1.9), we write A as

$$A = SDS^T, \tag{8.29}$$

where $D = \text{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$ is the diagonal matrix of the eigenvalues of A and S is an orthogonal matrix. Without loss of generality we may assume that the first p eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_p$ of A are positive and the next $r - p$ eigenvalues $\lambda_{p+1}, \lambda_{p+1}, \ldots, \lambda_r$ of A are negative (if this is not the case, we may simply order these eigenvalues by using a permutation matrix, which is an orthogonal matrix as shown in Example 8.2). Now, let D_1 be the diagonal matrix

$$D_1 = \text{diag}\left(\sqrt{\lambda_1}, \sqrt{\lambda_2}, \ldots, \sqrt{\lambda_p}, \sqrt{|\lambda_{p+1}|}, \sqrt{|\lambda_{p+2}|}, \sqrt{|\lambda_r|}, 0, \ldots, 0\right).$$

Thus, the matrix D can be written as

$$D = D_1 D_0 D_1,$$

where D_0 is the matrix defined in (8.28). Now, by substituting into (8.29), we obtain

$$A = SD_1 D_0 D_1 S^T = QD_0 Q^T,$$

with $Q = SD_1$, which is orthogonal. This completes the proof of Theorem 8.3.2. □

Theorem 8.3.2 exhibits an important connection between congruent matrices and positive definite matrices, as shown in the next corollary.

Corollary 8.3.3 *Let A be a symmetric matrix in $\mathscr{M}_n(\mathbb{R})$. Then A is positive definite if and only if it is congruent to the identity matrix.*

Proof

First, assume that A is positive definite. Then, all its eigenvalues are positive. Hence, $p = n$ in (8.28) and therefore D_0 becomes the identity matrix. Conversely, if A is congruent to the identity matrix, then

$$A = SI_nS^T.$$

Let X be a nonzero vector in $\mathbb{R}^n$. The quadratic form associated to A is

$$\begin{aligned} f(X) &= X^TAX \\ &= X^TSI_nS^TX \\ &= Y^TI_nY \\ &= \sum_{i=1}^{n} y_i^2 > 0, \end{aligned}$$

for any nonzero vector $Y = S^TX$ in $\mathbb{R}^n$. Hence, A is positive definite. □

Some properties are preserved under the congruence transformation, among them, the rank.

Theorem 8.3.4 (Rank of Congruent Matrices)
Let A and B be two congruent matrices in $\mathscr{M}_n(\mathbb{R})$. Then

$$\operatorname{rank}(A) = \operatorname{rank}(B).$$

Proof

Since A and B are congruent, there exists an invertible matrix S such that $A = S^TBS$. Applying the result of Exercise 5.3 for two matrices C and D, we have

$$\operatorname{Ker}(D) \subset \operatorname{Ker}(CD), \tag{8.30}$$

with equality if C is invertible (which is very easy to check), and

$$\operatorname{Im}(CD) \subset \operatorname{Im}(C), \tag{8.31}$$

with equality if C is invertible.

Now, since S is invertible, S^T is also invertible and applying (8.31), we deduce that

$$\operatorname{Im}(B) = \operatorname{Im}(BS).$$

Hence,

$$\text{rank}(B) = \text{rank}(BS).$$

Likewise, since S^T is invertible, then applying (8.30), we deduce that

$$\text{Ker}(S^T(BS)) = \text{Ker}(BS).$$

Now, applying Theorem 5.2.10, we have

$$\begin{aligned} \text{rank}(A) &= \text{rank}(S^T BS) \\ &= n - \text{null}(S^T BS) \\ &= n - \text{null}(BS) \\ &= n - \text{null}(B) \\ &= \text{rank}(B). \end{aligned}$$

This gives the desired result. □

Finding a matrix S satisfying (8.27) can be quite difficult in practice. So, we need a test that allows us to quickly determine whether two matrices are congruent or not without computing the matrix S. This test is given by Sylvester's law of inertia. This law asserts that two congruent matrices have the same number of positive eigenvalues, the same number of negative eigenvalues, and the same number of zero eigenvalues. Moreover, this law is very useful in the study of the stability of solutions of differential equations, which usually requires the knowledge of the signs of the eigenvalues of some symmetric matrices.

First, we start with the following definition.

Definition 8.3.5 (Index and Signature of a Diagonal Matrix)

Let D be a diagonal matrix. The *index* of D is the number of positive entries in D and the *signature* of D is the number of positive entries of D minus the number of negative entries in D. We denote the index of D by p and the number of negative entries by ν.

Since D is a diagonal matrix, the rank r of D is the number of nonzero entries in D. Thus,

$$r = p + \nu.$$

If s is the signature of D, then we have

$$s = p - \nu.$$

Hence,

$$s = 2p - r.$$

Theorem 8.3.5 (Sylvester's Law of Inertia)
Let A and B be two symmetric matrices in $\mathscr{M}_n(\mathbb{R})$. Then, A and B are congruent if and only if the diagonal representations for A and B have the same rank, index, and signature. That is, if and only if $\text{In}(A) = \text{In}(B)$.

Proof
First, assume that A and B are congruent. Then Theorem 8.3.4 implies that $\text{rank}(A) = \text{rank}(B)$. Now, let p_1 be the number of positive eigenvalues of A and p_2 be the number of positive eigenvalues of B. Then to show that $\text{In}(A) = \text{In}(B)$ it is enough to prove that $p_1 = p_2$, since the two matrices have the same rank. Now, since A and B are congruent, there exists an invertible matrix S such that

$$A = SBS^T. \tag{8.32}$$

Now, using Theorem 8.3.2, we deduce that there exist two matrices $D_0^{(1)}$ and $D_0^{(2)}$ such that

$$D_0^{(1)} = \begin{bmatrix} I_{p_1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -I_{r-p_1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0}_{(n-r)\times(n-r)} \end{bmatrix} = PAP^T$$

and

$$D_0^{(2)} = \begin{bmatrix} I_{p_2} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -I_{r-p_2} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0}_{(n-r)\times(n-r)} \end{bmatrix}$$

with $B = QD_0^{(2)}Q^T$, where P and Q are invertible matrices. Plugging these into (8.32), we get

$$D_0^{(1)} = PAP^T = PSQD_0^{(2)}Q^TS^TP^T = RD_0^{(2)}R^T, \tag{8.33}$$

with $R = PSQ$. Now, assume that $p_2 \neq p_1$, for instance $p_2 < p_1$. Then, we have to reach a contradiction. Let X be the nonzero vector in $\mathbb{R}^n$, with its first p_1 components are not all zero, but with its last $n - p_1$ components all equal to zero. That is

$$X = \begin{bmatrix} X_1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad \text{with} \quad X_1 = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_{p_1} \end{bmatrix}$$

and $X_1 \neq 0_{\mathbb{R}^{p_1}}$. Hence, the quadratic form associated to $D_0^{(1)}$ reads

$$X^T D_0^{(1)} X = \sum_{i=1}^{p_1} x_i^2 > 0. \tag{8.34}$$

Partition R^T in the form

$$R^T = \begin{bmatrix} R_1 & R_2 \\ R_3 & R_4 \end{bmatrix},$$

where R_1, R_2, R_3 and R_4 are sub-matrices with R_1 a $p_2 \times p_1$ matrix, R_2 an $p_2 \times (n-p_1)$ matrix, R_3 an $n-p_2 \times p_1$ matrix and R_4 an $(n-p_2) \times (n-p_1)$ matrix. Since $p_2 < p_1$, we can choose X_1 such that $X_1 \neq 0_{\mathbb{R}^{p_1}}$ and $R_1 X_1 = 0_{\mathbb{R}^{p_2}}$. Now, we define the vector $Y = R_3 X_1$ in $\mathbb{R}^{n-p_2}$ and we have

$$R^T X = \begin{bmatrix} R_1 & R_2 \\ R_3 & R_4 \end{bmatrix} \begin{bmatrix} X_1 \\ \mathbf{0}_{(n-p_1)\times 1} \end{bmatrix} = \begin{bmatrix} R_1 X_1 \\ R_3 X_1 \end{bmatrix} = \begin{bmatrix} \mathbf{0}_{p_2 \times 1} \\ Y \end{bmatrix}.$$

Consequently, using (8.33), we get

$$X^T D_0^{(1)} X = (R^T X)^T D_0^{(2)} (R^T X) = -\sum_{j=1}^{r-p_2} y_i^2 \leq 0.$$

This contradicts (8.34). Similarly, interchanging the roles of $D_0^{(1)}$ and $D_0^{(2)}$, we can prove that it is impossible to have $p_1 < p_2$. Consequently, $p_1 = p_2$.

Conversely, if A and B have the same inertia, $\text{In}(A) = \text{In}(B) = (p, \nu, q)$, then both matrices are congruent to the matrix

$$D_0 = \begin{bmatrix} I_p & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & -I_\nu & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0}_{q\times q} \end{bmatrix},$$

and then they are congruent to each other (since congruence is an equivalence relation). □

Theorem 8.3.5, is very interesting since it tells us that we can determine whether two symmetric matrices are congruent by just computing their eigenvalues as shown in the following example.

Example 8.16

Show that the matrices

$$A = \begin{bmatrix} 2 & 0 & 2 \\ 0 & 6 & 2 \\ 2 & 2 & 4 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & -1 & -1 \\ -1 & 5 & 1 \\ -1 & 1 & 5 \end{bmatrix}$$

are congruent. ■

Solution

The eigenvalues of A are

$$\lambda_1 = 2(2 + \sqrt{3}), \quad \lambda_2 = 4, \quad \lambda_3 = 2(2 - \sqrt{3}).$$

Hence, $\text{In}(A) = (3, 0, 0)$.

Now, the eigenvalues of B are

$$\mu_1 = \frac{1}{2}(7 + \sqrt{33}), \quad \mu_2 = 4, \quad \mu_3 = \frac{1}{2}(7 - \sqrt{33}).$$

Thus, $\text{In}(B) = (3, 0, 0)$. Since $\text{In}(A) = \text{In}(B)$, A and B are congruent. ◀

8.4 Exercises

Exercise 8.1 (Square Root of a Definite Matrix)

Let A be a positive semi-definite matrix. A matrix A_0 such that $A_0^2 = A$ is called the *square root* of A.

1. Show that a matrix A in $\mathscr{M}_n(\mathbb{R})$ is positive definite (respectively, semi-definite) if and only if it has a positive definite (respectively, semi-definite) square root A_0.
2. Show that if A_0 is the square root of A, then $\text{rank}(A) = \text{rank}(A_0)$.
3. Prove that if A is positive semi-definite, then the positive semi-definite square root A_0 of A is unique.
4. Use the above results to find the square root of the matrix

$$A = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix}.$$

■

Solution

1. We give the proof for the case of a positive definite matrix (the positive semi-definite case can be proved by the same method). So, assume that A is a positive definite matrix. Since A is symmetric, then according to the spectral theorem (Theorem 8.1.9), there exists an orthogonal matrix S such that $A = SDS^T$, where D is the diagonal matrix $D = \text{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$. Now, since A is positive definite, all its eigenvalues are positive. Introduce the matrix

$$D_0 = \text{diag}(\sqrt{\lambda_1}, \sqrt{\lambda_2}, \ldots, \sqrt{\lambda_n}).$$

Hence, $D_0^2 = D$ and for

$$A_0 = SD_0S^T, \tag{8.35}$$

we have

$$A_0^2 = A_0 A_0 = (SD_0S^T)(SD_0S^T) = SD_0S^T = A.$$

Consequently, A_0 is the square root of A. In addition, it is clear from above that the eigenvalues of A_0 are $\sqrt{\lambda_1}, \sqrt{\lambda_2}, \ldots, \sqrt{\lambda_n}$, which are positive. Hence, A_0 is positive definite.

Conversely, if A_0 is positive definite, then its eigenvalues are positive. Since, the eigenvalues of A are the squares of those of A_0, they are also positive, hence, A is positive definite.

2. If A is positive definite, then according to Theorem 8.2.2, A is invertible. Also, according to (1) A_0 is positive definite, hence A_0 is also invertible. Consequently, Theorem 6.3.2 yields

$$\text{rank}(A) = \text{rank}(A_0) = n.$$

Now, if A is positive semi-definite and $\text{rank}(A) = r$, then according to Theorem 8.2.6, A has exactly r positive eigenvalues. Hence the matrix A_0 introduced above also has r positive eigenvalues and therefore its rank is also equal to r.

3. Assume that there exists a positive semi-definite matrix A_1 such that $A_1^2 = A$. Hence, the eigenvalues of A_1 are the square root of those of A, and since A_1 is positive semi-definite, its eigenvalues must be nonnegative. Consequently, the eigenvalues of A_1 and of the matrix A_0 given in (8.35) coincide. Now, since A_1 is symmetric, the spectral theorem (Theorem 8.1.9) implies that

$$A_1 = QD_0Q^T,$$

for some orthogonal matrix Q. Since $A_0^2 = A_1^2 = A$, we have

$$SD_0^2S^T = QD_0^2Q^T,$$

whence

$$(S^TQ)D_0^2 = D_0^2(S^TQ).$$

This implies, since D_0 is positive semi-definite,

$$(S^TQ)D_0 = D_0(S^TQ).$$

This yields, upon multiplying from the left by S and from the right by Q^T, $A_1 = A_0$.

4. The eigenvalues of A are $\lambda_1 = 9$ and $\lambda_2 = 1$. Hence, A is positive definite, and A has a positive definite square root matrix A_0 satisfying $A_0^2 = A$. To find A_0, we see first that A can be written as

$$A = SDS^T,$$

where D is the diagonal matrix

$$D = \begin{bmatrix} 9 & 0 \\ 0 & 1 \end{bmatrix}$$

and S is the orthogonal matrix

$$S = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}.$$

Hence, we have from above that

$$A_0 = SD_0S^T = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

It is clear that A_0 is also positive definite, since its eigenvalues are $\mu_1 = 3$ and $\mu_2 = 1$. ◀

Exercise 8.2 (The Gram Matrix)
Let $u_1, u_2, \ldots, u_\ell$ be vectors in $\mathbb{R}^n$. The Gram matrix (or Gramian matrix) is the matrix defined as

$$G = (g_{ij}), \quad \text{with} \quad g_{ij} = u_i \cdot u_j, \quad 1 \le i, j \le \ell.$$

That is,

$$G = \begin{bmatrix} u_1 \cdot u_1 & u_1 \cdot u_2 & \cdots & u_1 \cdot u_\ell \\ u_2 \cdot u_1 & u_2 \cdot u_2 & \cdots & u_2 \cdot u_\ell \\ \vdots & \vdots & \ddots & \vdots \\ u_\ell \cdot u_1 & u_\ell \cdot u_2 & \cdots & u_\ell \cdot u_\ell \end{bmatrix}.$$

1. Prove that there exists a matrix A such that $G = A^TA$.
2. Show that G is symmetric and positive semi-definite.
3. Show that G is positive definite if and only if the vectors $u_1, u_2, \ldots, u_\ell$ are linearly independent.
4. Prove that

$$\operatorname{rank}(G) = \dim_{\mathbb{R}} \operatorname{span}\{u_1, u_2, \ldots, u_\ell\}.$$

■

Solution
1. Let A be the $n \times \ell$ matrix whose columns are the vectors $u_1, u_2, \ldots, u_\ell$. Then we can easily check that $G = A^TA$.

2. It is clear that G is symmetric, since $u_i \cdot u_j = u_j \cdot u_i$ for all $1 \leq i, j \leq \ell$. Now, to show that G is positive semi-definite, we can use two approaches. First, using (1) and Example 8.12, we deduce immediately that G is positive semi-definite.

Second, we can use a direct approach. Indeed, let X be a nonzero vector in $\mathbb{R}^\ell$. Then, by using the properties of the dot product in $\mathbb{R}^\ell$ we have

$$\begin{aligned} X^T G X &= \sum_{i,j=1}^{\ell} (u_i \cdot u_j) x_i x_j \\ &= \sum_{i,j=1}^{\ell} ((x_i u_i) \cdot (x_j u_j)) \\ &= \left(\sum_{i=1}^{\ell} x_i u_i \right) \cdot \left(\sum_{j=1}^{\ell} x_j u_j \right) \\ &= \left\| \sum_{i=1}^{\ell} x_i u_i \right\|^2 \geq 0, \end{aligned} \tag{8.36}$$

where x_i, $i = 1, 2, \ldots \ell$ are the components of the vector X. Hence G is positive semi-definite.

3. We see that the inequality (8.36) is an equality if and only if

$$\sum_{i=1}^{\ell} x_i u_i = 0_{\mathbb{R}^n}.$$

This is not the case if X is a nonzero vector and the vectors $\{u_1, u_2, \ldots, u_\ell\}$ are linearly independent. Hence, in this case (8.36) is a strict inequality and therefore G is positive definite.

Conversely, if G is positive definite, then $X^T G X > 0$ whenever $X \neq 0_{\mathbb{R}^\ell}$. Hence,

$$\sum_{i=1}^{\ell} x_i u_i \neq 0_{\mathbb{R}^\ell}.$$

Thus, the vectors $\{u_1, u_2, \ldots, u_\ell\}$ are linearly independent.

4. This can be seen from the rank identity

$$\operatorname{rank}(G) = \operatorname{rank}(A^T A) = \operatorname{rank}(A A^T) = \operatorname{rank}(A) = \operatorname{rank}(A^T) \tag{8.37}$$

Since $\{u_1, u_2, \ldots, u_\ell\}$ are the column vectors of A, we have (see Definition 6.3.1)

$$\operatorname{rank}(G) = \operatorname{rank}(A) = \dim_{\mathbb{R}} \operatorname{span}\{u_1, u_2, \ldots, u_\ell\}.$$

We leave it to the reader to check (8.37). See (6.10) for instance. ◄

Exercise 8.3 (Minimization of Quadratic Functions[a])
Let A be a symmetric matrix in $\mathscr{M}_n(\mathbb{R})$. Let X and b be two vectors in $\mathbb{R}^n$ and c be a real number. Consider the function

$$F(X) = X^T AX - 2X^T b + c. \tag{8.38}$$

We say that f has a *global minimum point* X_0 if $f(X_0) \leq f(X)$ for all X in $\mathbb{R}^n$.

1. Show that if A is positive definite, then the quadratic function $F(X)$ has a unique global minimum X_0, which is the solution of the linear system

$$AX = b, \qquad \text{namely} \qquad X_0 = A^{-1}b.$$

2. Prove that the minimum value of F is

$$F(X_0) = c - X_0^T AX_0.$$

3. Show that if A is positive definite, then the function F is strictly convex, that is

$$F(\lambda X_1 + (1-\lambda)X_2) < \lambda F(X_1) + (1-\lambda)F(X_2), \tag{8.39}$$

for all X_1 and X_2 in $\mathbb{R}^n$ with $X_1 \neq X_2$ and for any λ in $[0, 1]$. If we have "$\leq$" instead of "$<$" in (8.39), then we say that F is convex.

4. We define the function F by

$$F : \mathbb{R}^2 \to \mathbb{R},$$
$$(x_1, x_2) \mapsto F(x_1, x_2) = 4x_1^2 - 2x_1x_2 + 3x_2^2 + 3x_1 - 2x_2 + 1.$$

Find the unique minimum of F.

■

[a] The quadratic optimization problems appear frequently in applications. For instance, many problems in physics and engineering can be stated as the minimization of some energy functions.

Solution
1. First, since A is positive definite, Theorem 8.2.2 implies that A is invertible. Hence the linear system $AX = b$ has a unique solution $X_0 = A^{-1}b$. Now, by replacing b by AX_0, we write the function $F(X)$ as

$$\begin{aligned} F(X) &= X^T AX - 2X^T b + c \\ &= X^T AX - 2X^T AX_0 + c \\ &= (X - X_0)^T A(X - X_0) + (c - X_0^T AX_0). \end{aligned} \tag{8.40}$$

It is clear that if $X \neq X_0$, then the first term on the right-hand side of (8.40) is strictly positive, since the matrix A is positive definite. Also, this term is zero for $X = X_0$. Since, the second term in the right-hand side of (8.40) does not depend on X, the minimum of $F(X)$ occurs at $X = X_0 = A^{-1}b$.

2. We have from the above that

$$\begin{aligned} F(X_0) &= F(A^{-1}b) \\ &= (A^{-1}b)^T A(A^{-1}b) - 2(A^{-1}b)^T b + c \\ &= c - b^T A^{-1} b \\ &= c - b^T X_0 \\ &= c - X_0^T A X_0, \end{aligned}$$

since $b^T = X_0^T A^T = X_0^T A$.

3. According to Exercise 8.5, it is enough to show that the Hessian matrix of f,

$$H(X) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(X) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(X) & \dots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(X) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(X) & \frac{\partial^2 f}{\partial x_2^2}(X) & \dots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(X) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(X) & \frac{\partial^2 f}{\partial x_n \partial x_2}(X) & \dots & \frac{\partial^2 f}{\partial x_n^2}(X) \end{bmatrix},$$

is positive definite. Hence,

$$H(X) = 2A.$$

Since A is positive definite, $H(X)$ is positive definite and consequently, the function f is strictly convex.

4. The function F can we written in the form (8.38), with

$$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad A = \begin{bmatrix} 4 & -1 \\ -1 & 3 \end{bmatrix}, \quad b = \begin{bmatrix} -\frac{3}{2} \\ 1 \end{bmatrix}, \quad c = 1.$$

It is clear that A is positive definite, since its eigenvalues are

$$\lambda_1 = \frac{1}{2}\left(7 + \sqrt{5}\right), \quad \lambda_2 = \frac{1}{2}\left(7 - \sqrt{5}\right).$$

Hence, applying the above results, we deduce that F has a global minimum, which is

$$X_0 = A^{-1}b = \begin{bmatrix} -\frac{7}{22} \\ \frac{5}{22} \end{bmatrix}.$$

Now, the value of F at X_0 is

$$F(X_0) = c - X_0^T A X_0 = \frac{13}{44}.$$

Hence, we deduce that for any X in $\mathbb{R}^2$ we have $F(X) > \frac{13}{44}$. ◄

Exercise 8.4

Let A and B be two matrices in $\mathscr{M}_n(\mathbb{R})$.

1. Show that the following two statements are equivalent:

   (i) for all X and Y in $\mathbb{R}^n$, we have

   $$X^T A Y = X^T B Y. \tag{8.41}$$

   (ii) $A = B$.

2. Prove that if A and B are symmetric, then the following statements are equivalent:

   (i) for all X in $\mathbb{R}^n$, we have

   $$X^T A X = X^T B X. \tag{8.42}$$

   (ii) $A = B$.

■

Solution

1. First, it is clear that (ii) implies (i). Now, let us assume that (i) holds and show that $A = B$. So, (8.41) implies that for any vector X in $\mathbb{R}^n$,

$$X^T(AY - BY) = 0.$$

In particular, this equality also holds for the vector $AY - BY$, which is also a vector in $\mathbb{R}^n$. Hence, we have

$$(AY - BY)^T(AY - BY) = \|(AY - BY)\|^2 = 0.$$

This yields (see Theorem 3.3.1)

$$AY - BY = 0_{\mathbb{R}^n}.$$

Since the last equality is true for all vectors Y of $\mathbb{R}^n$, then necessarily $A - B = \mathbf{0}_{\mathscr{M}_n(\mathbb{R})}$.

2. As above, it is trivial to see that (ii) implies (i). Now, if A and B are symmetric and (8.42) holds, then, applying (8.42) for the vectors X, Y, and $X + Y$ in $\mathbb{R}^n$, we get

$$X^T A X = X^T B X, \quad Y^T A Y = Y^T B Y, \quad \text{and} \quad (X+Y)^T A (X+Y) = (X+Y)^T B (X+Y).$$

This implies

$$Y^TAX + X^TAY = Y^TBX + X^TBY.$$

Since A and B are symmetric, we have

$$Y^TAX = (X^TAY)^T = X^TAY \quad \text{and} \quad Y^TBX = X^TBY.$$

Hence, we obtain $X^TAY = X^TBY$, for any X and Y in $\mathbb{R}^n$. Thus, applying (1), we obtain $A = B$. ◀

Exercise 8.5 (The Hessian Matrix)

Let f be a function $f : \mathbb{R}^n \to \mathbb{R}$. If all the second derivatives of f exist and are continuous on the domain of f, the *Hessian* matrix of f at a point $X = (x_1, x_2, \ldots, x_n)$ of $\mathbb{R}^n$ is the $n \times n$ matrix

$$H(X) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(X) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(X) & \ldots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(X) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(X) & \frac{\partial^2 f}{\partial x_2^2}(X) & \ldots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(X) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(X) & \frac{\partial^2 f}{\partial x_n \partial x_2}(X) & \ldots & \frac{\partial^2 f}{\partial x_n^2}(X) \end{bmatrix}.$$

1. Show that the Hessian matrix is symmetric.
2. Show that if f is differentiable, then f is convex if and only if for any two vectors X and Y in $\mathbb{R}^n$,

$$f(Y) \geq f(X) + \nabla f(X)^T(Y - X), \tag{8.43}$$

where $\nabla f(X)^T$ is the vector

$$\nabla f(X)^T = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ldots, \frac{\partial f}{\partial x_n} \right).$$

3. Prove that if $H(X)$ is positive definite (respectively, semi-definite) for all X in $\mathbb{R}^n$, then f is strictly convex (respectively, convex) in $\mathbb{R}^n$.
4. Deduce that the function

$$f(x_1, x_2, x_3) = x_1^2 + 2x_2^2 + 3x_3^2 + 2x_1x_2 + 2x_1x_3 + 3$$

is strictly convex on $\mathbb{R}^3$.

■

Solution

1. Since $\frac{\partial^2 f}{\partial x_i \partial x_j} = \frac{\partial^2 f}{\partial x_j \partial x_i}$, we deduce that $H(X)$ is symmetric.

2. First, assume that (8.43) is satisfied and let X and Y be two vectors in $\mathbb{R}^n$. Let λ be in $[0, 1]$. We put $Z = \lambda Y + (1 - \lambda)X$. Then,

$$f(Y) \geq f(Z) + \nabla f(Z)^T (Y - Z), \tag{8.44}$$

and

$$f(X) \geq f(Z) + \nabla f(Z)^T (X - Z). \tag{8.45}$$

Hence, multiplying (8.44) by λ and (8.45) by $1 - \lambda$ and adding the results, we obtain

$$\begin{aligned} \lambda f(Y) + (1 - \lambda) f(X) &\geq f(Z) + \nabla f(Z)^T (\lambda Y + (1 - \lambda)X - Z) \\ &= f(\lambda Y + (1 - \lambda)X), \end{aligned}$$

since $Z = \lambda Y + (1 - \lambda)X$. Hence, f is convex.

Conversely, assume that f is convex. Then, for any X and Y in $\mathbb{R}^n$ and λ in $\mathbb{R}$, let $Z = \lambda Y + (1 - \lambda)X$. By the convexity of f,

$$f(Z) = f(\lambda Y + (1 - \lambda)X) \leq \lambda f(Y) + (1 - \lambda) f(X).$$

Hence, we get

$$\begin{aligned} f(Z) - f(X) &\leq \lambda f(Y) \leq \lambda f(Y) + (1 - \lambda) f(X) - f(X) \\ &= \lambda f(Y) - \lambda f(X). \end{aligned}$$

Recall that

$$\nabla f(X)^T d = \lim_{\lambda \to 0+} \frac{f(X + \lambda d) - f(X)}{\lambda},$$

where d is in $\mathbb{R}^n$. Therefore, we deduce (by taking $d = X - Y$) that,

$$\begin{aligned} \nabla f(X)^T (Y - X) &= \lim_{\lambda \to 0+} \frac{f(X + \lambda(Y - X)) - f(X)}{\lambda} \\ &\leq f(Y) - f(X). \end{aligned}$$

Hence, (8.43) holds.

It is also clear, by the same method, that the inequality in (8.43) is strict if and only if f is strictly convex.

3. Let X and Y be two vectors in $\mathbb{R}^n$ with $X \neq Y$. By using Taylor's theorem and the Lagrange form of the remainder, we have

$$F(Y) = F(X) + \nabla f(X)^T (Y - X) + (Y - X)^T H(X + \alpha(Y - X))(Y - X),$$

for some α in $[0, 1]$. Since H is positive definite, we have

$$(Y - X)^T H(X + \alpha(Y - X))(Y - X) > 0.$$

Consequently, we obtain from above

$$F(Y) > F(X) + \nabla f(X)^T (Y - X).$$

Therefore, from question (2), we deduce that f is strictly convex. By the same argument, if H is positive semi-definite, then f is convex.

4. To show that f is strictly convex, then it is enough to show that its Hessian matrix is positive definite. Indeed, we have

$$H(X) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(X) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(X) & \frac{\partial^2 f}{\partial x_1 \partial x_3}(X) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(X) & \frac{\partial^2 f}{\partial x_2^2}(X) & \frac{\partial^2 f}{\partial x_2 \partial x_3}(X) \\ \frac{\partial^2 f}{\partial x_3 \partial x_1}(X) & \frac{\partial^2 f}{\partial x_3 \partial x_2}(X) & \frac{\partial^2 f}{\partial x_3^2}(X) \end{bmatrix} = \begin{bmatrix} 2 & 2 & 2 \\ 2 & 4 & 0 \\ 2 & 0 & 6 \end{bmatrix}.$$

We see that the leading principal minors of $H(X)$ are

$$A^{(1)} = 2, \quad A^{(2)} = 4 \quad \text{and} \quad A^{(3)} = 8.$$

Hence, A is positive definite (see Theorem 8.2.4). Therefore, according to question (3), we deduce that f is strictly convex. ◄

Exercise 8.6

1. Let $v_1, v_2, \ldots, v_n$ be a basis of a vector space E over a field $\mathbb{K}$. Let W be a k-dimensional subspace of E. Show that if $m < k$, then there exists a nonzero vector in W which is a linear combination of the vectors $v_{m+1}, \ldots, v_n$.
2. Let A be a symmetric matrix in $\mathscr{M}_n(\mathbb{R})$. Show that if $Y^T AY > 0$, for all nonzero vectors Y in a k-dimensional subspace W of $\mathbb{R}^n$, then A has at least k positive eigenvalues (counting the multiplicity).[a]

■

[a] There are several proofs of this result, here we adapt the one in [7].

Solution

1. Consider the subspace F defined as

$$F = \text{span}\{v_{m+1}, \ldots, v_n\}.$$

Then, $\dim_{\mathbb{K}} F = n - m$. Hence, applying (4.16), we have

$$\dim_{\mathbb{K}} W + \dim_{\mathbb{K}} F = \dim_{\mathbb{K}}(F + W) - \dim(F \cap W).$$

That is, $k+n-m>n$, since $m<k$. Hence, we deduce that the subspace $F\cap W$ contains at least one nonzero vector v. Otherwise, we have $\dim_{\mathbb{K}}(F+W)>\dim_{\mathbb{K}}E$, which is impossible according to Theorem 4.6.7. Thus, $v\in F\cap W$. This means that v is a linear combination of $v_{m+1},\ldots,v_n$ since $v\in F$. ◀

2. Let $v_1,v_2,\ldots,v_n$ be an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of A. (Such a basis exists according to Theorem 8.1.7). Let $\lambda_1,\lambda_2,\ldots,\lambda_n$ be the corresponding eigenvalues. Without loss of generality, we may assume that the first m ($m\leq n$) eigenvalues of A are positive and the rest are not. If $m<k$, then from (1), we deduce that there exists a nonzero vector Y in W such that

$$Y=c_{m+1}v_{m+1}+\cdots+c_nv_n.$$

Hence, since $v_1,v_2,\ldots,v_n$ are orthonormal, we have

$$Y^TAY=c_{m+1}^2\lambda_{m+1}+\cdots+c_n^2\lambda_n\leq 0.$$

This is a contradiction, since we assumed that $Y^TAY>0$, for all nonzero vectors Y in W. Consequently, $m\geq k$.

Exercise 8.7 (The Submultiplicative Norm of a Matrix)
Let $A=(a_{ij})$, $1\leq i,j\leq n$, be a matrix in $\mathscr{M}_n(\mathbb{R})$.[a] We define the norm

$$\|A\|_F^2=\sum_{j=1}^{n}\sum_{i=1}^{n}|a_{ij}|^2=\operatorname{tr}(A^TA)\qquad\text{Frobenius' norm.}^{[b]}$$

1. Find $\|A\|_F$ for the matrix

$$A=\begin{bmatrix}1&2&-3\\0&5&-1\\6&-2&4\end{bmatrix}.$$

2. Show that

$$\|AB\|_F\leq\|A\|_F\|B\|_F\qquad\text{submultiplicativity property,}\tag{8.46}$$

and $\|I\|_F=\sqrt{n}$, where I is the identity matrix in $\mathscr{M}_n(\mathbb{R})$.
3. Show that the two norms

$$\|A\|_\infty=\max_{1\leq i\leq n}\sum_{j=1}^{n}|a_{ij}|\qquad\text{and}\qquad\|A\|_1=\max_{1\leq j\leq n}\sum_{i=1}^{n}|a_{ij}|\ \text{ satisfy (8.46).}$$

4. Show that if Q is an orthogonal matrix, then

$$\|AQ\|_F = \|QA\|_F = \|A\|_F.$$

That is, the Frobenius norm is invariant under orthogonal transformation.
5. We define the norm

$$\|A\|_2 = \sup_{X \neq 0_{\mathbb{R}^n}} \frac{\|AX\|}{\|X\|} = \max_{\|X\|=1} \|AX\|, \tag{8.47}$$

where $\|\cdot\|$ is the Euclidean norm in $\mathbb{R}^n$ introduced in Definition 3.3.1. Show that $\|\cdot\|_2$ is well defined and satisfies the property (8.46), and $\|I\|_2 = 1$.
6. Show that for any matrix A in $\mathscr{M}_n(\mathbb{R})$,

$$\|A\|_2 \leq \|A\|_F \leq \sqrt{n}\|A\|_2. \tag{8.48}$$

■

[a] All the results here remain true if we replace $\mathbb{R}$ by $\mathbb{C}$. See Remark 8.1.1.
[b] Sometimes referred to as the Hilbert–Schmidt norm and defined as the usual Euclidean norm of the matrix A when it is regarded as a vector in $\mathbb{R}^{n^2}$.

Solution
1. We have

$$A^TA = \begin{bmatrix} 1 & 0 & 6 \\ 2 & 5 & -2 \\ -3 & -1 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 & -3 \\ 0 & 5 & -1 \\ 6 & -2 & 4 \end{bmatrix} = \begin{bmatrix} 37 & -10 & 21 \\ -10 & 33 & -19 \\ 21 & -19 & 26 \end{bmatrix}.$$

Hence, $\operatorname{tr}(A^TA) = 96$, so $\|A\|_F = \sqrt{96}$.

2. We put $A = (a_{ij})$, $B = (b_{ij})$, and $C = AB = (c_{ij})$. Hence, we have (see Definition (1.1.11))

$$c_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj}.$$

Therefore, we obtain

$$|c_{ij}|^2 = \Big|\sum_{k=1}^{n} a_{ik}b_{kj}\Big|^2 \leq \Big(\sum_{k=1}^{n} |a_{ik}||b_{kj}|\Big)^2.$$

Now, applying the Cauchy–Schwarz inequality (see (3.16)), we obtain

$$\Big(\sum_{k=1}^{n} |a_{ik}|b_{kj}|\Big)^2 \leq \Big(\sum_{k=1}^{n} |a_{ik}|^2\Big)\Big(\sum_{k=1}^{n} |b_{kj}|^2\Big).$$

This yields

$$\|AB\|_F^2 = \sum_{i,j} |c_{ij}|^2 \le \Big(\sum_{i,k} |a_{ik}|^2 \Big) \Big(\sum_{l,j} |b_{lj}|^2 \Big) = \|A\|_F^2 \|B\|_F^2,$$

which gives the desired result. As a side note, property (8.46) can be seen as a generalization of the Cauchy–Schwarz inequality.

It is obvious that $\|I\|_F^2 = \mathrm{tr}(I^T I) = \mathrm{tr}(I) = n$. This yields $\|I\|_F = \sqrt{n}$.

3. First, we need to show that $\|A\|_\infty$ has the submultiplicativity property (8.46). Indeed, we have from above

$$|c_{ij}| \le \sum_{k=1}^{n} |a_{ik}||b_{kj}|.$$

Hence,

$$\sum_{j=1}^{n} |c_{ij}| \le \sum_{j,k} |a_{ik}||b_{kj}| = \sum_{k=1}^{n} \Big(|a_{ik}| \sum_{j=1}^{n} |b_{kj}| \Big).$$

Since

$$\sum_{j=1}^{n} |b_{kj}| \le \|B\|_\infty,$$

we obtain

$$\sum_{j=1}^{n} |c_{ij}| \le \Big(\sum_{k=1}^{n} |a_{ik}| \Big) \|B\|_\infty \le \|A\|_\infty \|B\|_\infty.$$

Thus,

$$\|C\|_\infty = \|AB\|_\infty \le \|A\|_\infty \|B\|_\infty.$$

By the same reasoning, we can show that the norm $\|\cdot\|_1$ has the submultiplicativity property.

4. Since Q is orthogonal, we have (see (8.2)) $Q^T Q = QQ^T = I$. Hence, we get

$$\|QA\|_F^2 = \mathrm{tr}((QA)^T (QA)) = \mathrm{tr}(A^T Q^T Q A) = \mathrm{tr}(A^T A) = \|A\|_F^2.$$

Similarly,

$$\|AQ\|_F^2 = \mathrm{tr}((AQ)^T (AQ)) = \mathrm{tr}(Q^T A^T A Q) = \mathrm{tr}(QQ^T A^T A) = \mathrm{tr}(A^T A) = \|A\|_F^2,$$

where we have used the cyclic property of the trace, that is, $\mathrm{tr}(ABC) = \mathrm{tr}(CAB)$. (The reader should be careful, $\mathrm{tr}(ABC) \ne \mathrm{tr}(ACB)$ in general).

5. We need to show that the supremum in (8.47) is well defined. Let

$$X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

be a vector in $\mathbb{R}^n$. Then, we have

$$\|AX\|^2 = \sum_{i=1}^{n} \Big| \sum_{j=1}^{n} a_{ij} x_j \Big|^2.$$

Applying the Cauchy–Schwarz inequality, we get

$$\|AX\|^2 \leq \sum_{i=1}^{n} \left(\sum_{j=1}^{n} |a_{ij}|^2 \right) \left(\sum_{j=1}^{n} |x_j|^2 \right),$$

whence

$$\|AX\|^2 \leq \|A\|_F^2 \|X\|^2$$

and therefore

$$\frac{\|AX\|}{\|X\|} \leq \|A\|_F, \qquad \text{for any } X \neq 0_{\mathbb{R}^n}.$$

This implies that the set of real numbers $\left\{ \frac{\|AX\|}{\|X\|},\ X \neq 0_{\mathbb{R}^n} \right\}$ is bounded and therefore is has a supremum and we have

$$\|A\|_2 \leq \|A\|_F.$$

Now, we need to show that $\|\cdot\|_2$ satisfies (8.46). Indeed, we have for any vector $X \neq 0_{\mathbb{R}^n}$.

$$\|(AB)X\| = \|A(BX)\| \leq \|A\|_2 \|BX\| \leq \|A\|_2 \|B\|_2 \|X\|.$$

Hence, we obtain,

$$\frac{\|(AB)X\|}{\|X\|} \leq \|A\|_2 \|B\|_2,$$

and so

$$\|AB\|_2 \leq \|A\|_2 \|B\|_2.$$

If $A = I$, then we have

$$\frac{\|AX\|}{\|X\|} = \frac{\|X\|}{\|X\|} = 1, \qquad X \neq 0_{\mathbb{R}^n}.$$

Hence, $\|I\|_2 = 1$.

6. We have already proved in (5) that $\|A\|_2 \leq \|A\|_F$. So, we just need to show that $\|A\|_F \leq \sqrt{n}\|A\|_2$. This inequality can be easily obtained due to the fact that $\|A\|_F = \operatorname{tr}(A^T A) \leq n\rho(A^T A) = \|A\|_2$ (see Exercise 8.8). ◀

Exercise 8.8 (Spectral Radius)

Let A be a matrix in $\mathscr{M}_n(\mathbb{C})$ and let $\lambda_1, \lambda_2, \ldots, \lambda_n$ be its eigenvalues. We define the spectral radius $\rho(A)$ of A as

$$\rho(A) = \max_{1 \leq i \leq n} |\lambda_i|.$$

1. Show that $\rho(A) \leq \|A\|_2$, where $\|\cdot\|_2$ is defined in (8.47) (with $\mathbb{C}$ instead of $\mathbb{R}$).[a]
2. Prove that

$$\lim_{m \to \infty} A^m = \mathbf{0} \iff \rho(A) < 1.$$

3. Show that

$$\rho(A) = \lim_{m \to \infty} \|A^m\|_2^{1/m} \quad \text{spectral radius formula.}^b \tag{8.49}$$

■

[a] In fact, this holds for any matrix norm.
[b] This formula yields a technique for estimating the top eigenvalue of A.

Solution

1. Let λ_i be an eigenvalue of A, i.e.,

$$AX_i = \lambda_i X_i, \qquad X_i \neq 0_{\mathbb{C}^n}.$$

On the other hand, we have, for any $X \neq 0_{\mathbb{C}^n}$,

$$\|AX\| \leq \|A\|_2 \|X\|.$$

In particular, for $X = X_i$, we have

$$\|AX_i\| = |\lambda_i| \|X_i\| \leq \|A\|_2 \|X_i\|.$$

This yields $|\lambda_i| \leq \|A\|_2$, which conclude the proof.

2. First, let λ be an eigenvalue of A. Assume that $\lim_{m\to\infty} A^m = \mathbf{0}$, and let X be an eigenvector corresponding to λ. Then

$$AX = \lambda X \qquad \text{and} \qquad A^m X = \lambda^m X.$$

Hence, since $\lim_{m\to\infty} A^m = \mathbf{0}$, then we deduce that $\lim_{m\to\infty} \lambda^m X = \mathbf{0}$. Since $X \neq 0_{\mathbb{C}^n}$, then we have $\lim_{m\to\infty} \lambda^m = 0$. This shows that $|\lambda| < 1$. This last inequality is satisfied for all the eigenvalues of A. Consequently, $\rho(A) < 1$. We leave it to the reader to show $\rho(A) < 1$ implies that $\lim_{m\to\infty} A^m = \mathbf{0}$.

3. To show the spectral radius formula, we have first from (1) that $\rho(A) \leq \|A\|_2$, and since $\rho(A) = (\rho(A^m))^{1/m}$, we deduce that

$$\rho(A) \leq \|A^m\|_2^{1/m}, \qquad \text{for all } m.$$

8

Now, to prove (8.49), we need to show that for any $\epsilon > 0$, there exists a positive integer $N = N(\epsilon)$ such that for any $m \geq N$,

$$\|A^m\|_2^{1/m} \leq \rho(A) + \epsilon.$$

Let $\epsilon > 0$ be given. Consider the matrix

$$A_\epsilon = \frac{1}{\rho(A) + \epsilon} A.$$

Thus, $\rho(A_\epsilon) < 1$. Therefore, we deduce from (2) that

$$\lim_{m\to\infty} A_\epsilon^m = \mathbf{0}.$$

Consequently, there exists a positive integer $l(\epsilon)$ such that for $m \geq l(\epsilon)$, we have

$$\|A_\epsilon^m\| = \frac{1}{(\rho(A) + \epsilon)^m} \|A^m\| < 1.$$

Now, it is enough to choose $N(\epsilon) = l(\epsilon)$. This finishes the proof of (8.49). ◄

Servicepart

References – 379

Index – 381



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References

1. H. Anton, C. Rorres, *Elementary Linear Algebra: with Supplemental Applications*, 11th edn. (Wiley, Hoboken, 2011)
2. M. Artin, *Algebra*, 2nd edn. (Pearson, Boston, 2011)
3. S. Axler, *Linear Algebra Done Right*. Undergraduate Texts in Mathematics, 2nd edn. (Springer, New York, 1997)
4. E.F. Beckenbach, R. Bellman, *Inequalities*, vol. 30 (Springer, New York, 1965)
5. F. Boschet, B. Calvo, A. Calvo, J. Doyen, *Exercices d'algèbre, 1er cycle scientifique, 1er année* (Librairie Armand Colin, Paris, 1971)
6. L. Brand, Eigenvalues of a matrix of rank *k*. Am. Math. Mon. **77**(1), 62 (1970)
7. G.T. Gilbert, Positive definite matrices and Sylvester's criterion. Am. Math. Mon. **98**(1), 44–46 (1991)
8. R. Godement, *Algebra* (Houghton Mifflin Co., Boston, MA, 1968)
9. J. Grifone, *Algèbre linéaire*, 4th edn. (Cépaduès–éditions, Toulouse, 2011)
10. G.N. Hile, Entire solutions of linear elliptic equations with Laplacian principal part. Pac. J. Math **62**, 127–140 (1976)
11. R.A. Horn, C.R. Johnson, *Matrix Analysis*, 2nd edn. (Cambridge University Press, Cambridge, 2013)
12. D. Kalman, J.E. White, Polynomial equations and circulant matrices. Am. Math. Mon. **108**(9), 821–840 (2001)
13. P. Lancaster, M. Tismenetsky, *The Theory of Matrices*, 2nd edn. (Academic Press, Orlando, FL, 1985)
14. S. Lang, *Linear Algebra*. Undergraduate Texts in Mathematics, 3rd edn. (Springer, New York, 1987)
15. L. Lesieur, R. Temam, J. Lefebvre, *Compléments d'algèbre linéaire* (Librairie Armand Colin, Paris, 1978)
16. H. Liebeck, A proof of the equality of column and row rank of a matrix. Am. Math. Mon. **73**(10), 1114 (1966)
17. C.D. Meyer, *Matrix Analysis and Applied Linear Algebra* (SIAM, Philadelphia, PA, 2000)
18. D.S. Mitrinović, J.E. Pečarić, A.M. Fink, *Classical and New Inequalities in Analysis*. Mathematics and Its Applications (East European Series), vol. 61 (Kluwer Academic, Dordrecht, 1993)
19. C. Moler, C. Van Loan, Nineteen dubious ways to compute the exponential of a matrix, twenty-five years later. SIAM Rev. **45**(1), 3–49 (2003)
20. J.M. Monier, *Algèbre et géométrie, PC-PST-PT*, 5th edn. (Dunod, Paris, 2007)
21. P.J. Olver, Lecture notes on numerical analysis, http://www.math.umn.edu/~olver/num.html. Accessed Sept 2016
22. F. Pécastaings, *Chemins vers l'algèbre, Tome 2* (Vuibert, Paris, 1986)
23. M. Queysanne, *Algebre*, 13th edn. (Librairie Armand Colin, Paris, 1964)
24. J. Rivaud, *Algèbre linéaire, Tome 1*, 2nd edn. (Vuibert, Paris, 1982)
25. S. Roman, *Advanced Linear Algebra*. Graduate Texts in Mathematics, vol. 135 (Springer, New York, 2008)
26. H. Roudier, *Algèbre linéaire: cours et exercices*, 3rd edn. (Vuibert, Paris, 2008)
27. B. Said-Houari, *Differential Equations: Methods and Applications*. Compact Textbook in Mathematics (Springer, Cham, 2015)

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B. Said-Houari, *Linear Algebra*, Compact Textbooks in Mathematics,
DOI 10.1007/978-3-319-63793-8

28. D. Serre, *Matrices. Theory and Applications*. Graduate Texts in Mathematics, vol. 216, 2nd edn. (Springer, New York, 2010)
29. G. Strang, *Linear Algebra and Its Applications*, 3rd edn. (Harcourt Brace Jovanovich, San Diego, 1988)
30. V. Sundarapandian, *Numerical Linear Algebra* (PHI Learning Pvt. Ltd., New Delhi, 2008)
31. H. Valiaho, An elementary approach to the Jordan form of a matrix. Am. Math. Mon. **93**(9), 711–714 (1986)

Index

Block matrix, 118
Cofactor
– matrix, 92
Fibonacci
– sequence, 116
Matrix
– skew–symmetric, 111

Abelian
– group, 11
Abelian group, 160
Addition
– of matrices, 8
Adjoint
– of a matrix, 92
Apollonius' identity, 149
Automorphism, 200
– of vector spaces, 211

Basis
– of a vector space, 176
bijective
– tarnsformation, 215
Binomial formula, 20

cardinality
– of a set, 178
Cauchy–Schwarz
– inequality, 136, 372
Cayley–Hamilton
– theorem, 305
Characteristic
– polynomial, 57
Characteristic polynomial, 284
Cholesky
– decomposition, 348
Cofactor
– expansion, 74
– of a matrix, 73
Column space, 241
Commutative
– group, 12
Complement
– of a subspace, 194
Components
– of a vector, 124
Congruent
– matrices, 355
Consistent
– system of linear equations, 263
Convex
– function, 365
Cramer's rule, 100
Cyclic property
– of the trace, 373

Determinant, 74
– of Vandermonde, 108
Determinant , 57
Diagonalizable
– matrix, 240, 290
Diagonally
– dominant matrix, 264
Dimension
– of a direct sum , 186
– of a subspace, 182
– of a vector space, 178
Direct sum
– of vector spaces, 170
Distance, 131
Dot product, 6, 132
Dunford
– decomposition, 302
Dunkl–Williams inequality, 151

Eigenspace, 270
Eigenvalue
– complete, 275
– defective, 275
– of a matrix, 278
– of a projection, 272
– of an endomorphism, 269
Eigenvector
– generalized, 305
– matrix, 295
– of a matrix, 278
– of an endomorphism, 269

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DOI 10.1007/978-3-319-63793-8

Elementary
– matrix, 82
Elementary row operation, 246
Endomorphism, 200
Equation
– linear, 4
Equivalent
– matrices, 240
Euclidean
– norm, 129
– vector space, 126
Euler
– formula, 283
Exponential
– of a matrix, 32

Factorization
– LU, 336
– QR, 330
Fibonacci
– matrix, 116
Frobenius inequality, 257
– for rank, 223

Gauss elimination, 336
Gauss–Jordan
– elimination method, 41
Global minimum
– of a function, 365
Gram's
– matrix, 363
Gram–Schmidt process, 328
Group, 11, 126
– $GL(n, \mathbb{K})$, 28
– Abelian, 126
– of invertible matrices, 65
– of matrices, 12
– orthogonal, 327

Hölder's
– inequality, 152
Hankel
– matrix, 110
Hessian
– matrix, 368
Homogeneous
– system, 5

Idempotent
– matrix, 258, 266
Identity
– operator, 200
Image
– of a linear transformation, 207
Inconsistent
– system of linear equations, 263
inconsistent, 264
Index
– of a matrix, 358
Inertia
– of a matrix, 354, 359
Injective
– linear transformation, 205
Inverse
– of an isomorphism, 212
Isomorphic
– vector spaces, 212
Isomorphism, 200
– of vector spaces, 211

Jordan
– block, 304
– canonical form, 303

Kernel
– of a linear transformation, 204

Lagrange form
– of the reminder, 369
Lagrange's identity, 157
Linear
– combination, 127, 165
– dependence, 172
– independence, 172
– operator, 200
– transformation, 200
Linear transformation
– associated to a matrix, 233

Matrix
– associated to a linear transformation, 227
– augmented, 40
– circulant, 319
– companion, 112, 317
– diagonal, 30, 166
– idempotent, 58, 60
– identity, 21
– inverse, 22
– involutory , 60
– nilpotent, 55, 302

– non derogatory, 317
– of full rank, 242
– of Vandermonde, 108, 110
– orthogonally diagonalizable, 334
– positive definite, 341
– positive semi-definite, 347
– skew-symmetric, 189
– square, 7, 19
– symmetric, 52, 189, 297
– transpose, 50
– triangular, 35, 166
– tridiagonal, 116
Matrix inverse
– determinant of, 90
Maximal
– linearly independent set, 179
Method
– of elimination, 2
– of substitution , 2
Minimal
– polynomial, 312
Minkowski
– inequality, 153
Minor
– of a matrix, 251
– leading principal , 337
– of a matrix, 72
– principal, 337
Multiplication
– of matrices, 13
Multiplicity
– algebraic, 275
– geometric, 275

Nilpotent
– matrix, 308
– linear transform, 222
– matrix, 310
Norm
– of a matrix, 371
– of a vector, 128
– of Frobenius, 371
– submultiplicative, 371
Null space, 167
Nullity
– of a linear transform, 209

Orthogonal
– complement, 196
– matrix, 145, 323
– projection, 142, 143
– subspace, 165
– vectors, 141
Orthonormal
– basis, 329
– vectors, 328

Parallelogram identity, 140
Polarization identity, 140
Positive definite
– quadratic form, 354
Principal axes
– theorem, 353
Product
– of two vector space, 192
Projection
– transformation, 215
Pythagoras theorem in $\mathbb{R}^n$, 144

Quadratic
– form, 352
– function, 365

Rank
– of a linear transformation, 209
– of a matrix, 242
– of a symmetric matrix, 346
Rank-nullity theorem, 209
Rayleigh–Ritz
– theorem, 344
Reflection matrix, 326
Ring, 20
– of matrices, 22
Rotation matrix, 59, 326
Row operation, 39
Row reduction method, 79
Row space
– of a matrix, 244

Schur's
– formula, 118
– lemma, 335
Semi-definite
– matrix, 347
– quadratic form, 354
Signature
– of a matrix, 358

Similar
– matrices, 240, 288
Singular
– matrix, 279
Spectral
– radius, 375
– theorem, 334
Spectrum, 278, 296
Square root
– of a matrix, 361
Submatrix
– leading principal, 337
– principal, 337
Subspace, 163
Surjective
– linear transformation, 208
Sylvester's law
– of nullity, 257
Sylvester's law of inertia, 359

Taylor's
– theorem, 369
Trace, 57
– of a matrix, 37
Transition matrix, 235, 252
Triangle inequality, 139
– for rank, 223
Triangularization
– of a matrix, 298
– of an endomorphism, 298

Unit
– vector, 130

Vector space, 127, 159

Young's inequality, 138, 153